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European business statistics compiler's manual for ICT usage and e-commerce in enterprises, 2024 survey

2025 edition



MANUALS AND GUIDELINES

European business statistics compiler's manual for ICT usage and e-commerce in enterprises, 2024 survey

2025 edition

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Abbreviations

ADSL	Asymmetric Digital Subscriber Line
AI	Artificial Intelligence
B2B	Business To Business
B2C	Business To Consumers
B2G	Business To Government
CDMA	Code Division Multiple Access
CRM	Customer Relationship Management
DDoS	Distributed Denial Of Service
DSL	Digital Subscriber Line
EBS	European Business Statistics
eDAMIS	electronic Dataflow Administration And Management Information System
EDI	Electronic Data Interchange
EEA	European Economic Area
EFTA	European Free Trade Association
ERP	Electronic Resource Planning
ESS	European Statistical System
ESS-MH	European Statistical System Metadata Handler
EU	European Union
FTE	Full-Time Equivalent
FTTP	Fiber optics technology
GIA	General Implementing Act
HDSL	High-Bit-Rate Digital Subscriber Line
ICT	Information And Communication Technology
IDS	Intrusion Detection System
IEEE	Institute Of Electrical And Electronics Engineers
INFOSOC	Information Society
ISO	International Organization for Standardization
MH	Metadata Handler
MS	Member States
NACE	Statistical Classification Of Economic Activities In The European Community (Nomenclature Generale Des Activites Economiques Dans Les Communautes Europeennes
NGFW	Next Generation Firewall

NGIPS	Next Generation Intrusion Prevention System
NSIs	National Statistical Institutes
NUTS	Nomenclature Of Territorial Units For Statistics
OECD	Organisation for Economic Co-Operation And Development
OTP	One-Time Password
PIN	Personal Identification Number
R&D	Research And Development
SBR	Statistical Business Register
SDMX	Statistical Data and Metadata Exchange
SDSL	Symmetric Digital Subscriber Line
SIEM	Security Information And Event Management
SIMS	Single Integrated Metadata Structure
SME	Small and Medium-Sized Enterprises
SSL	Secure Sockets Layer
UMTS	Universal Mobile Telephone System
UNECE	United Nations Economic Commission for Europe
VAT	Value-Added Tax
VPN	Virtual Private Network
VTL	Validation and Transformation Language
XML	Extensible Markup Language

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1 Introduction

1.1. What are statistics on ICT usage and e-commerce in enterprises?

Information and communications technologies (**ICT**) usage and e-commerce in enterprises statistics measure the uptake of technologies and the digitalisation of the European Union (**EU**) economy. These statistics are gathered through the annual 'ICT usage and e-commerce in enterprises' survey. The 2024 survey measured enterprises' access to and use of the internet and e-commerce (i.e. sale of goods and services online), ICT specialists and skills, ICT security, use of artificial intelligence.

This exercise has been carried out since 2002 when the European Commission (EC) established annual 'information society' surveys to benchmark ICT-driven development in enterprises and by individuals.

Eurostat is responsible for coordinating the 'ICT usage and e-commerce in enterprises' survey which is conducted at national level. Survey questions are developed every year, in close collaboration with Member States (MS) and the Organisation for Economic Cooperation and Development (OECD), in line with the changing needs of data users and policy makers. To facilitate the data collection a model questionnaire is provided, which is accompanied by methodological guidelines for its implementation.

1.2. Usage and importance of statistics on ICT usage and e-commerce in enterprises

ICT account for a significant part of EU productivity and growth and are transforming our societies and economies in a profound and unprecedented way. Official statistics are indispensable for an informed understanding of the implications of the transformations under way. Selecting appropriate indicators is a crucial step. Measuring the development of the information society with relevant statistics on business processes and the digital economy requires continuous revision and improvement.

The current compilers' manual refers to the model questionnaire used for collecting part of the

statistical data for monitoring progress towards the EU's digital targets for 2030, set by the Digital Decade Policy Programme. Four of the key performance indicators of this policy programme are based on the statistics on ICT usage and e-commerce in enterprises: artificial intelligence (AI), cloud computing, big data (data analytics) and the digital intensity index - a composite indicator reflecting the digital transformation of business.

1.3. What is the purpose of this compilers' manual

This compilers' manual (hereafter referred to as the 'Manual') is meant to serve as a practical reference document for all National Statistical Institutes (NSIs) involved in the compilation of data on the use of ICT and e-commerce in enterprises. As such, its main objectives are:

- to help NSIs translate Eurostat model questionnaires into national languages and to ensure that the same methodology is used by all countries when conducting national surveys;
- to set out the concepts, definitions and compilation methods, guiding the compilation of data;
- to explain the validation and quality rules, and metadata reporting;
- to explain the concepts and methods of data transmission to Eurostat.

In order to do so, each chapter of this manual describes a step of the production process of statistics on ICT usage and e-commerce in enterprises.

The second chapter is on data compilation. Data compilation is carried out by the NSIs on the basis of the model questionnaire. This chapter introduces the legislative background underpinning the process and explains how to interpret the model questionnaire to ensure the comparability of data between MS.

The third chapter is on data transmission. Once the data has been collected by the NSIs, they must transmit them to Eurostat. To that effect, this chapter sets out how to transfer the data to Eurostat (codification, transmission channel, deadlines, flags, confidentiality…).

The fourth chapter is on data quality. Once the data are transmitted, Eurostat applies validation rules to assess their quality. This chapter sets out the quality framework and the validation rules used by Eurostat.

Finally, the fifth chapter is on data dissemination. Once the whole data collection process is complete, Eurostat publishes the data. This chapter describes the type of data that is published and how to access them.

Note that this edition of the Manual serves as a reference for the compilation and transmission to Eurostat of data relating to 2024 as reference year.

To keep up with technological advances in digitalisation and changes in data requirements, this Manual will be updated every year.

This Manual focuses on issues relevant to ICT and e-commerce in enterprises. It does not provide an exhaustive list of all concepts and tools underpinning European statistics. For this information, please refer to the manuals, guidelines and other references listed in Section 1.4 below. Notably, the European Business Statistics Manual, 2021 edition (**EBS Manual**) covers the

concepts that are shared across the business statistics domains.

1.4. Where can I find further guidance?

Further guidance is available from the following sources:

- The Digital economy and society webpage provides a global and regularly updated overview of Digital economy and society statistics, including statistics on the use of ICT and e-commerce in enterprises.
- The European Business Statistics Manual provides an overview of business statistics while highlighting the features of the regulatory framework. The manual also describes the various statistical tools and activities supporting EBS production such as statistical units and profiling, classifications, data processing and statistical disclosure control.
- The European Statistical System handbook for quality and metadata reports sets out guidelines for the preparation of producer and user reports for the full range of statistical processes and their outputs within Member States, EFTA countries and Eurostat.
- The ESS Handbook Methodology for data validation 2.0 provides a generic reference framework and metrics for data validation.
- The Business Architecture for ESS Validation manual sets out a common understanding of how ESS validation should be conducted.
- The European Statistics Code of Practice sets out principles that aim to ensure that statistics produced within the European Statistical System are relevant, timely and accurate, and that they comply with the principles of professional independence, impartiality and objectivity.
- The Quality Assurance Framework of the European Statistical System provides a collection of methods, tools and good practices on how to implement the European Statistics Code of Practice.

2

Data compilation

Data compilation is done by the NSIs on the basis of the model questionnaire. This chapter explains the legislative background and the data requirements, and provides guidance on the interpretation of the model questionnaire.

2.1. Legislative background

Statistics on the usage of ICT and e-commerce in enterprises are based on EU legislation to ensure a harmonised approach for the production of statistics by all reporting countries. Three regulations currently form the legal basis of the survey on ICT usage and e-commerce in enterprises for the 2024 survey:

- Regulation (EU) 2019/2152 of the European Parliament and of the Council of 27 November 2019 on European business statistics, repealing 10 legal acts in the field of business statistics ('**EBS Regulation**');
- Commission Implementing Regulation (EU) 2020/1197 of 30 July 2020 laying down technical specifications and arrangements pursuant to Regulation (EU) 2019/2152 of the European Parliament and of the Council on European business statistics, repealing 10 legal acts in the field of business statistics ('**General Implementing Act**' or '**EBS GIA Regulation**'); and
- Commission Implementing Regulation (EU) 2023/1507 of 20 July 2023 laying down the technical specifications of data requirements for the topic ICT usage and e-commerce for the reference year 2024, pursuant to Regulation (EU) 2019/2152 of the European Parliament and of the Council ('**Implementing Act**').

Regulation (EU) 2019/2152 is a framework regulation. It therefore provides flexibility to adapt the survey on ICT usage and e-commerce in enterprises to the evolving needs of users and decision-makers. Annual implementing measures, such as Commission Implementing Regulation (EU) 2023/1507, are the basis for the Eurostat surveys. This ensures harmonised data for all EU-27 MS. The annual implementing Regulation is of relevance for the European Economic Area (EEA).

2.2. Data requirements

2.2.1. Introduction

The statistical product is the clear and precise definition of the statistical information to be produced. It is distinct from the production methodology. The production methodology is the way or method of doing, while the statistical product is its direct result. Different methodologies can produce the same statistical product, being only different ways of doing the same thing. This means that as long as it is guaranteed that two figures concern the same statistical product – for instance for two different countries – they are comparable. Distinguishing between the statistical product and the statistical methodology thus helps to focus on the aspects that are more important for ensuring comparability between national statistics and for producing new ones at EU level. NSIs may choose the most appropriate statistical methodology to be applied taking into account national particularities.

Elements that make up the statistical product, at input level, are the statistical unit, the target population and the observation variables. Elements at output level are the periodicity, variables, aggregates and tabulation. Covering all the elements of the statistical product, the statistical concepts and the nomenclatures are the additional elements ensuring that statistics are harmonised and comparable.

This chapter provides a detailed description of data requirements on ICT usage and e-commerce in enterprises. These are based on the EBS Regulation and the Implementing Act.

2.2.2. Statistical unit

The statistical unit is the base type of the elements of a group (also called population) that is intended for observation or analysis. The basic statistical operations of classification, aggregation and ordering are done on the statistical unit.

The choice of the statistical unit is a matter of both the data collection process (namely the operational restrictions associated with collecting data from each type of statistical unit) and the conceptual framework chosen to observe and analyse the phenomenon. The statistical unit is the bearer of statistical characteristics or attributes that are ultimately intended to be measured.

There are several types of statistical units, according to their usage. An observation unit represents an identifiable entity, about which data can be obtained. During the collection of data, this is the unit for which data are recorded. It should be noted that this may, or may not be, the same as the reporting unit. The reporting unit is the unit that reports to the survey authority. It reports information for the observation unit(s). In certain cases, it may be different from the observation unit. A reporting unit is a unit that supplies the data for a given survey instance.

The observation unit in the survey on ICT usage and e-commerce in enterprises is the enterprise, as defined in Council Regulation (EEC) No 696/93 of 15 March 1993 on the statistical units for the observation and analysis of the production system in the Community. The Regulation defines the enterprise as follows:

'The enterprise is the smallest combination of legal units that is an organisational unit producing goods or services, which benefits from a certain degree of autonomy in decision-making, especially for the allocation of its current resources. An enterprise carries out one or more activities at one or more locations. An enterprise may be a sole legal unit.'

The enterprise, therefore, is an economic entity that can, under certain circumstances, correspond to a grouping of several legal units. Some legal units, in fact, perform activities exclusively for other legal units and their existence can only be explained by administrative factors (e.g. tax reasons), without them being of any economic significance. A large proportion of legal units with no employees or self-employed persons also belong to this category. In many cases, the activities of these legal units should be seen as ancillary activities of the parent legal unit they serve, to which they belong and to which they must be attached to form an enterprise used for economic analysis.

However, the definition of the enterprise as the appropriate statistical unit poses some limitations. Some enterprises, especially larger ones, are composed of several local units (establishments). Because of this, a geographical breakdown of the results (although still possible using the location of the enterprise's main headquarters) is of limited use. Nevertheless, ICT usage is not easily attributable to an enterprise's various establishments, and for this reason the enterprise is the statistical unit adopted.

Implementation of the statistical unit 'enterprise' in business statistics requires specific considerations for the delineation of enterprises in business registers (profiling), the consolidation of the data on legal units, etc. Nevertheless, for surveys with qualitative variables specifically, the consolidation is more challenging.

Specific guidelines have been developed for this survey and are described in Section 2.2.3 below.

2.2.3. Statistical unit enterprise

2.2.3.1. The concept of enterprise in European business statistics

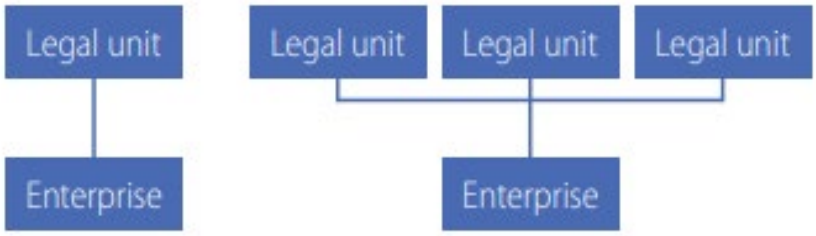
The enterprise is the most important statistical unit in EBS. The enterprise concept is applied in almost all domains of business statistics. It is also the core unit, given that the other statistical units, such as the kind-of-activity unit (KAU), the local kind-of-activity unit (LKAU)[1] and the local

[1] The kind-of-activity unit is a part of an enterprise. It groups together all the offices, production facilities etc. of an enterprise that contribute to the performance of a specific economic activity defined at class level (four digits) of the European classification of economic activities (NACE Rev. 2.) For example, a kind-of-activity unit might be the combination of all parts of a metal-producing enterprise that produce copper (class 24.44 in NACE Rev. 2); within the same enterprise, there might be another kind-of-activity unit consisting of those parts that produce aluminium (class 24.42 in NACE Rev. 2). The local part of kind of activity units is called local kind-of-activity unit; the term establishment is common as well, e.g. in SNA or ISIC (Glossary: Kind-of-activity unit and local-kind-of-activity unit - Statistics Explained (europa.eu)).

unit[2], are defined in relation to the enterprise. An enterprise may carry out one or more economic activities at one or more locations. The enterprise is thus not a homogeneous unit, neither with respect to its activity nor to its geographical location. However, in most enterprises, the principal activity accounts for quite a large share of the value added generated, and thus these enterprises come closer to the homogeneity of the economic activity. Also, most of the enterprises only have one location, which means that, for those enterprises, the regional attribution would be close to correct. Larger enterprises, however, perform various activities and have more than one location.

In business statistics, the term enterprise should always be used in the sense of a 'statistical enterprise'. This means, by definition, one aims to create a statistical unit that would allow to compile harmonised data, irrespective of the actual organisational and legal structures of the economic agents. The criteria that a statistical enterprise should satisfy are: operating the necessary factors of production; having an organisational and managerial structure; autonomy of decision-making; and producing and selling goods and services on the market (see European business statistics methodological manual for statistical business registers, 2021 edition).

The relationships between legal units and enterprises may be like in the two examples below.



An enterprise consisting of more than one legal unit under the same management can be referred to as a 'complex enterprise'.

2.2.3.2. Recommendation for the correct implementation of the statistical unit enterprise in the ICT domain

In general, the share of complex enterprises or enterprises consisting of more than one legal unit is low, at around 5% or even lower, depending on the country. However, complex enterprises are usually those with a high turnover or/and employment. So, an effort should be made to provide data for those enterprises in a harmonised way.

Consistency is especially encouraged with structural business statistics but also with other annual business statistics at national level.

[2] The local unit is an enterprise or part thereof (e.g. a workshop, factory, warehouse, office, mine or depot) situated in a geographically identified place. At or from this place, economic activity is carried out for which - save for certain exceptions - one or more persons work (even if only part-time) for one and the same enterprise (Glossary: Local unit - Statistics Explained (europa.eu)).

2.2.3.2.1. Target population

In the case of complex enterprises, the attributes for the complex enterprise as a whole should be relevant for including the enterprise and its underlying legal units in the sample, even if one or more legal units are out of scope.

Therefore, the target population should comprise enterprises, including complex ones, having the main attributes (e.g. given NACE[3] code, number of employees and self-employed persons) in the scope of the ICT survey.

2.2.3.2.2. Sampling

The sample should be drawn at the level of enterprise. The attributes of the enterprise should be used as they are registered in the statistical business registers (SBR).

Unit consistency with structural business statistics and preferably also other national annual business statistics at the national level should be ensured.

If the variables 'employees and self-employed persons' and 'turnover' from existing data sources prove to be outdated, it could be considered to collect those variables from the enterprises, pre-fill them for verification by the enterprise, or check against the latest information of the SBR when processing the data. It should be considered to ask for intra-enterprise flows on a voluntary basis.

2.2.3.2.3. Data collection

It is up to NSIs to collect the data either from legal units or from enterprises.

[3] NACE is the acronym for Nomenclature statistique des activités économiques dans la Communauté européenne. For more information on NACE Rev. 2.

Data collection		
At enterprise level	**Contact person**	If an enterprise consists of more than one legal unit, the NSI should make an agreement with the enterprise as to who should be the contact person. If that is not possible, the legal unit that is best suited to reply should be chosen. If it is not known which legal unit is best suited to reply, the group head or top management should be contacted. If they cannot reply, they should forward the questionnaire for the individual modules to the person who is best suited to reply.
At enterprise level	**Measures to ensure quality**	The contact person should be given a list of legal units for which they should respond. If it turns out that the information in the SBR as regards the legal units belonging to the enterprise is not correct, it should be reported back to colleagues in the responsible unit for profiling or the SBR.
From legal units	**Contact person**	Ideally, data should be collected from all legal units belonging to the enterprise in the sample. If data cannot be collected from all legal units, the legal units that report should report for all legal units belonging to the enterprise; they should be explicitly made aware for which legal units they should provide the data.
	Measures to ensure quality	Special care should be taken to include the legal units having e-commerce in order not to underestimate the e-commerce. These would include, for example, legal units that, in previous years, reported having e-commerce or websites with functionalities to order goods or services, or legal units known from other sources to have e-commerce. For non-additive variables, e.g. variables with intra-enterprise flows, it might be better to collect them from a contact person/legal unit nominated or agreed or legally best-suited to reply for the whole enterprise, or the group head or top management. For non-additive variables, legal units should be asked to deduct the intra-enterprise flows.

2.2.3.2.4. Consolidation of variables

Variable type	Consolidation
Qualitative dichotomous questions (yes/no questions)	As a general rule, if one answer is 'yes', the answer should be 'yes' for the whole enterprise.
Qualitative questions with different answer options	In the 2024 survey, this concerns the question on internet speed, where the fastest connection in the enterprise should be chosen, and the question on the definition of the most recent review of documents on measures, practices or procedures on ICT security, where it should be the most recent update in the enterprise.
Quantitative additive variables	**Additive variables** are those for which the simple sum of the amounts of the legal units yields the consolidated amount for the enterprise. This must be the case for all enterprises, regardless of the way in which the legal units are combined to form the enterprise. In principle, variables such as 'number of persons employed having access to the internet for business purposes' and 'number of persons employed using a portable device provided by the enterprise, that allows internet connection via mobile telephone networks', are additive variables. However, in line with the Summary note on additive and non-additive SBS variables, the number of persons employed are headcounts and not additive. For example, if one person (employee) is part-time employed by different legal units of the enterprise, that person should be counted once. If further information is not available, the variable should be treated as additive for practical reasons.
Quantitative non-additive variables	**Non-additive variables** are those that cannot be simply added up to calculate the total amount at enterprise level, if the enterprise consists of several legal units. However, a consolidation of the amounts of the variable in question (e.g. turnover) of the underlying legal units must be carried out, involving the elimination of values that are related to internal flows. Non-additive **Data collection from enterprises** If data are collected from enterprises, they should be asked to deduct the intra-enterprise flows.

Variable type	Consolidation	
Percentages	variables in the survey on ICT usage and e-commerce in enterprises are variables with internal flows, such as e-commerce or turnover. Non-additive variables should be collected from the contact person, the legal unit best suited to reply for the whole enterprise or the group head/top management, or if the consolidation is done in the national statistical institute, an attempt should be made to deduct the intra-enterprise flows in the case of web sales or EDI-type (electronic data interchange) sales.	**Data collection from legal units** If data are collected from legal units, information on the existence of intra-enterprise flows from the (statistical) business register should be used if available. Legal units could be asked to deduct the intra-enterprise flows.
	Data collection from enterprises	If data are collected from the enterprise, the contact person, the legal unit best suited to reply for the whole enterprise or the group head/ top management in the enterprise should provide the percentage or an estimate for the whole enterprise.
	Data collection from legal units	If data are collected from legal units, the NSI should consolidate by dividing the absolute values (e.g. persons employed having access to the internet for business purposes) from all legal units by the total value for the denominator (e.g. persons employed in all legal units). If data from any legal unit is not available, it should be imputed. Information from SBR can be used, if available or useful. Care should be taken if the percentages are for non-additive variables, where, for example, intra-enterprise flows should be excluded if possible.

Variable type	Consolidation
Mutually exclusive questions	**Mutually exclusive questions** are, for example, questions about using AI, where there is a different set of questions for those who are using AI and for those who don't (such as the reasons for not using AI). In that case, the answers for such a question (e.g. reasons for not using AI) should only be counted if no legal unit in the enterprise has used the relevant technology (AI technologies). Therefore, the replies should be counted for the enterprise as whole; a 'yes' answer in at least one legal unit should be counted as a 'yes' for the whole enterprise.

2.2.4. Statistical population

A population is a collection of objects of the same class. In statistical terms, this means a group of elements of the same statistical unit. There are two types of populations to be considered when producing statistics: the target population and the frame population.

The **target population** is the population of interest. It is identified by clearly delimiting the group of statistical elements about which information is desired. That delimitation is based on one or more attributes of the statistical unit. For example, for the enterprise, some commonly used attributes to delimit the target population are the size (e.g. number of employees and self-employed persons), the economic activity and its location. An example of a target population could be 'enterprises with 10 or more employees and self-employed persons, classified in Divisions 41–43 of NACE Rev. 2 (Construction), localised in the EU'.

The target population of the survey on ICT usage and e-commerce in enterprises is the group of enterprises delimited by the following attributes:

- **Economic activity**

NACE is the standard for the statistical classification of economic activities in the EU. The present NACE Rev. 2, which is the new revised version of the NACE Rev. 1 and of its minor update NACE Rev. 1.1, was implemented for the first time in the 2009 survey.

Enterprises are classified in the following categories of NACE Rev. 2:

NACE Rev. 2	Description
Section C	Manufacturing
Section D, E	Electricity, gas and steam and air-conditioning supply, water supply, sewerage, waste management and remediation activities
Section F	Construction
Section G	Wholesale and retail trade; repair of motor vehicles and motorcycles
Section H	Transportation and storage
Section I	Accommodation and food service activities
Section J	Information and communication
Section L	Real estate activities
Section M	Professional, scientific and technical activities
Section N	Administrative and support service activities
Group 95.1	Repair of computers and communication equipment

Enterprises are classified in one of these categories based on their principle economic activity. Since 2021, in accordance with the provisions of the EBS Regulation, Division 75 Veterinary activities is also included in the target population.

- **Enterprise size**

The target population consists of enterprises with 10 or more employees and self-employed persons.

Optionally, the target population can be extended to enterprises with numbers of employees and self-employed persons between 0 and 9.

With the adoption of the EBS Regulation, the variable 'persons employed' was replaced by the variable 'employees and self-employed persons'. This change of name of the variable does not imply any change in the scope. The two variables represent exactly the same concept. For the sake of user-friendliness, the term 'employees and self-employed persons' is only used in the introductory part of the model questionnaire and in Module X; in the rest of the questionnaire, the term 'persons employed' is still used.

The variable number of employees and self-employed persons is defined in Commission Implementing Regulation (EC) No 2020/1197 of 30 July 2020 (p. 92, Variable 120101: Number of employees and self-employed persons) and should not be confused with the number of

employees or with full-time equivalents (FTE)[4]. In what follows, from the statistical definition point of view, even though the word 'employees' is used, it always refers to 'employees and self-employed persons'.

- **Geographic scope**

Enterprises located in any part of the territory of the country.

The **frame population** is an operationalisation of the target population, consisting ideally of the complete list of the target population elements. A target population can be easily defined but, in practice, a list of all its elements is needed for its complete or partial observation (if a sample is used). This can be very difficult to obtain. The list should be complete and include each element of the target population only once. However, it will usually suffer from both under-coverage and over-coverage. Generally, files of statistical elements (registers) are maintained and updated, containing lists of statistical elements and also information on some attributes, usually used for delimiting target populations. Normally, frame populations are extracted from those registers.

2.2.5. Accuracy

The accuracy of statistical information refers to the closeness of estimates to the unknown true values. In practice, it is the degree to which the information correctly describes the phenomena it was designed to measure. The accuracy of statistical information is broken down into bias (systematic error) and variance (random error).

Sampling error is one of the quality indicators related to accuracy and, for most of the sampling surveys, is the most indicative quality information. The quality aspect is covered in Article 17 of the EBS Regulation, Articles 10 and 11 of EBS Regulation General Implementing Act, and Article 2 of the Implementing Act. The quantitative element for evaluating the data accuracy, established in agreement with MS, has been the standard error. The maximum standard error (commonly used for evaluating quality) has been set to 2 percentage points for the whole coverage and 5 percentage points for the breakdowns, for the proportions and ratios that are calculated on the basis of the aggregate data that MS transmit to Eurostat. It is an obligation of the NSIs, as producers of national official statistics, and of Eurostat, being the provider of official statistics on the EU, to estimate and present these statistical errors to the users.

2.2.6. Periodicity

The periodicity is annual. The data are collected and compiled once a year.

[4] A full-time equivalent (FTE) is a unit to measure employed persons or students in a way that makes them comparable although they may work or study a different number of hours per week. The unit is obtained by comparing an employee's or student's average number of hours worked to the average number of hours of a full-time worker or student. A full-time person is therefore counted as one FTE, while a part-time worker / student gets a score in proportion to the hours worked or studied. For example, a part-time worker employed for 20 hours a week where full-time work consists of 40 hours is counted as 0.5 FTE (Glossary: Full-time equivalent (FTE) - Statistics Explained (europa.eu)).

However, the content of the survey and questionnaire can be adapted to accommodate the evolving needs of data users. In practice, the content is changed every year. In addition, to minimise the burden on NSIs and respondents, some variables can be observed with a lower frequency, e.g. variables that tend to be stable over time.

The coverage of the survey is 2023-2024. Most variables of the 2024 survey are collected for the year 2024. However, the 2024 survey also collects certain variables for the year 2023. For example, questions of the 2024 survey relating to web sales of goods and services, ICT specialists and skills, ICT security incidents and background information refer to variables for the year 2023.

This periodicity is laid down in Article 6 and Annex II to the EBS Regulation.

2.2.7. Variables

Ultimately, the attributes of the statistical unit are intended to be observed; the observation variables hold the numerical measures of these attributes. Attributes and observation variables should not be confused. An attribute is some property of the statistical unit and each attribute may have one or more observation variables with qualitative or quantitative information.

For example, for the statistical unit ‘enterprise’, an attribute can be its size. The observation variable is then the number of employees and self-employed persons in its service.

There are many ways of classifying observation variables; the most relevant one for this survey is the distinction between qualitative and quantitative variables.

Qualitative variables concern non-numerical information. They serve merely as labels or names for identifying attributes of the statistical unit. An example is the ‘use (Yes/No) of any type of fixed line connection to the internet’ by the enterprise. Sometimes qualitative variables can be turned into numerical ones by coding the non-numeric values. Binary (or dichotomous) variables are an important type of qualitative variable. Binary variables assume only two different values, which are usually turned into numerical ones by attributing the values ‘0’ and ‘1’ in a meaningful way.

Quantitative variables contain information as to how much or how many. An example is the number of persons employed who have access to the internet.

In the survey on ICT usage and e-commerce in enterprises, the observation variables are mainly qualitative (binary variables).

The operational version of the observation variables are the questions in the 2024 model questionnaire that can be found in Annex 1 – Model questionnaire.

The definition of the observation variables or the model questions are discussed in more detail in Section 2.3 below.

2.3. Model questionnaire

2.3.1. Development of model questionnaires

The legal background presented in Section 2.1 provides the context for developing the model questionnaires. The legal basis is output oriented, i.e. the Implementing Act defines the variables and their characteristics but does not impose a specific method for data collection.

In order to reach a higher degree of harmonisation, Eurostat - together with a task force of experts in the survey on ICT usage and e-commerce in enterprises – prepares an annual model questionnaire that is recommended to the NSIs.

The EBS Regulation limits the response burden of the annual Implementing Act for the topic of ICT usage and e-commerce to 73 variables, including background variables. On the other hand, the need for measuring digitalisation is ever increasing. To collect relevant statistics measuring the significant technological changes of the ICT landscape, but at the same time not to increase the burden on respondents, topics and/or variables are replaced or updated to collect more relevant or more in-depth information every year.

In addition to the compulsory data collection, some variables are proposed to be collected by the NSIs on a voluntary basis.

2.3.2. Modules and questions

This chapter refers to the questions in the 2024 model questionnaire (**MQ**) (See Annex 1 – Model questionnaire). The structure of the chapter follows the model questionnaire, i.e. the explanatory notes are grouped by module (title) and question. It is recommended to have the model questionnaire at hand while reading this section.

2.3.2.1. General remark: ‘use’, ‘have’ or ‘have another enterprise use for you’

In many of the questions and the corresponding notes, reference is made to the use of networks, systems, software, etc. The term ‘use’ does not refer to the ownership of such goods and infrastructure. For example, AI technologies may belong to the enterprise, may be leased or may be shared with another organisation. In the case where AI technologies are used by the responding enterprise, but provided or maintained by another enterprise, they should be considered as **used** by the responding enterprise.

Additionally, enterprises often buy ICT services or services which relate to the use of ICT in the context of the survey (like accounting). In cases where ICT services or other services which include use of ICT are provided totally by another enterprise and the responding enterprise itself is not using ICT for that function, then it should not be counted as use of ICT of the responding enterprise.

In some cases there will be a mixture of ICT usage by the responding enterprise and the ICT

service provider (another enterprise). In these cases it should be made clear from the question what kind of activities will be counted as ICT usage of the responding enterprise.

Examples on the interpretation of special cases: In some questions the ICT involvement of the enterprise is not about ICT usage, but rather about whether the enterprise has or offers some digital solutions to its partners (customers or business partners). One example is the question about having a website (or websites). Here the question is about having, not using, and the subject is the solution that is offered to customers regardless of how the website(s) is technically maintained. The main issue about having a website is about control and responsibility for the contents in a separate area on the web. The enterprise's web pages can be maintained and designed by a service provider, but if the responding enterprise 'owns' the contents, it is considered to be its own website. However, not all presence in the web means that the respondent has a website. Presence in certain service catalogues, e.g. yellow pages, address lists etc., are not counted as the enterprise's website. E-commerce systems can also be provided and maintained by service providers. Similarly to the case of the website, if it is the responding enterprise's e-commerce facility, the responding enterprise has control over the content of the site and is the one who is selling. If it is the owner of the products sold, then it is the respondent's e-commerce. Internet marketplaces — where enterprises can sell their products — are a special case. Here the respondent does not own or control the site, but it does control the sale and own the products being sold and therefore it is their e-commerce (if e-commerce is a valid definition in the situation).

2.3.2.2. Module A: Access and use of the internet

Question A1: *How many persons employed have access to the internet for business purposes? (including fixed line, fixed wireless and mobile telephone network connection)*

If you can't provide this value,

Please indicate an estimate of the percentage of the total number of persons employed who have access to the internet for business purposes.

[Scope: all enterprises]

[Type: numerical, absolute or percentage values; filter question]

This variable refers to persons employed (employees and self-employed), including people working remotely, who have access to the World Wide Web (WWW) for business purposes from their workstation or from a computer to which they have free access. The following definitions applies: *Computers include personal computers, portable computers, tablets, and other portable devices such as smartphones*.

Remote workers are persons employed who work from a remote location outside of the traditional office environment. This means that they telework and carry out work remotely with the help of digital tools. In most cases it refers to working from home, but it is not necessary, and based on enterprise's remote work policy, which set of guidelines or an agreement between an employer and person employed.

The aim is to identify how many persons employed have access to the WWW for business

purposes. This also includes the usage of private internet connections where the enterprise may reimburse fully or part of the costs. The reason for applying this specific internet service is that, from the several services that can be run on the internet, the WWW (as well as email) is the most common. Accessing the internet only through the email service is not considered in this variable, because email is already very common in Europe and it doesn't effectively represent the potential in terms of access to information as much as the use of the WWW.

This variable can be collected in **absolute** or **percentage** values. Eurostat recommends using a combination of both, giving the respondent the opportunity to choose to answer in absolute or percentage values.

Independently of how this variable is collected, the background variable 'Average number of persons employed, during the previous year' (**X2**) is needed for grossing up the overall percentage of persons employed having access to the World Wide Web for business purposes.

It can be observed in smaller enterprises that – in some cases – the owners of the enterprises are not considering themselves as "persons employed". Make sure that those are included as well.

Question A2: *Does your enterprise use any type of fixed connection to the internet? (e.g. ADSL, SDSL, VDSL, fibre optics technology (FTTP), cable technology, fixed wireless) (add national examples)*

[Scope: enterprises with access to the internet, i.e. A1 > 0]

[Type: single answer (i.e. Tick only one); binary (Yes/No); filter question]

'Connection' means the type of the last mile connection of the enterprise (e.g. enterprise's computer-mediated network) to the network of the internet service provider (ISP). The **last mile** is the final segment between an ISP's infrastructure and a subscriber's location for delivering communications (connectivity) to the enterprise.

Note: From the 2020 model questionnaire onwards, the question does not explicitly mention 'broadband connection' as few countries mentioned the marginal use of 'narrowband' for the connection to the internet. Narrowband connections are not within the scope of this question (all examples mentioned refer to broadband connections). Enterprises that still use only narrowband connections to the internet should tick 'No' to this question.

Type of fixed internet connection:

- **DSL connection e.g. xDSL, ADSL, SDSL, VDSL**

xDSL digital subscriber line (DSL) is a family of technologies that provide digital data transmission over the wires of a local telephone network. DSL service is delivered simultaneously with a regular telephone service on the same telephone line as it uses a higher frequency band that is separated by filtering. A DSL line can carry both data and voice signals and the data part of the line is continuously connected.

The asymmetric digital subscriber line (ADSL), where more bandwidth is allocated to download than upload, and the high-bit-rate digital subscriber line (HDSL) are considered dominant DSL technologies. Symmetric digital subscriber line (SDSL) refers to either a DSL technology that offers symmetric bandwidth for upload and download or to a particular DSL variant where data

are only supported on a single line that does not support analogue calls. Very-high-bit-rate digital subscriber line (VDSL) is a DSL technology that offers faster data transmission. VDSL is capable of supporting high-definition television, telephone services (voice over IP (internet protocol)) and internet access over a single connection, among others.

- **Fibre optics technology (FTTP), cable technology, etc.**

FTTP refers to 'Fibre to the Premises', also called Fibre to the Home (FTTH). It may be more correct to use FTTP: 'fibre to the premises' referring to both 'home' and 'business'.

- **Other high capacity 'speed' fixed (wired or wireless) connection includes the following types of internet connections:**
  - Cable modem 'cable TV network connection';
  - High capacity leased lines 'frame relay, automatic teller machine (ATM), digital multiplex';
  - Ethernet LAN connection;
  - Optical fibre connection;
  - Fixed wireless access (FWA) connections, e.g. satellite connection, public Wi-Fi connection, WiMax.

Cable modem uses modems attached to cable television networks (cable TV lines) for permanent fixed access to the internet. The term cable internet (or simply cable) refers to the delivery of an internet service over this infrastructure. A cable modem is a device that enables to hook up a personal computer (PC) to a local cable TV line. It is considered as one of the high-capacity speed permanent fixed internet connections.

A **leased line** (dedicated line) is a telephone line that has been leased for private use. A leased line is usually contrasted with a switched line or dial-up line. Leased lines are usually available at speeds of 64k, 128k, 256k, 512k, 2 Mb (megabyte) and provided to the customer on X.21 presentation. Frame relay protocol and T-1 and T-3 (in Europe called E1 and E3) lines are used for internet connection via leased lines. Higher speeds are available on alternative interfaces.

High-capacity leased line is a permanent telephone connection between two points set up by a telecommunications common carrier. Typically, leased lines are used by businesses to connect geographically distant offices. Unlike normal dial-up connections, a leased line is always active. Because the connection doesn't carry anybody else's communications, the carrier can provide a given level of quality. For example, a T-1 channel is a type of leased line that provides a maximum transmission speed of 1 544 Mbit/s (megabits per second). The connection can be divided into different lines for data and voice communication or the channel can be used for one high-speed data circuit. Dividing the connection is called multiplexing. Increasingly, leased lines are being used by enterprises, and even individuals, for internet access because they support faster data transfer rates and are cost-effective if the internet connection is heavily used.

Fixed wireless internet connection (FWA) is a technology which uses radio frequency, infrared, microwave, or other types of electromagnetic or acoustic waves in place of wires, cables, or fibre optics to transmit signals or data (provide internet access) between stationary (fixed) points. It includes for example a satellite internet connection (long-range wireless transmission) or public Wi-Fi (medium-range wireless transmission).

Wi-Fi (or Wi-fi, WiFi, Wifi, wifi), short for 'Wireless Fidelity', is a set of Ethernet standards for wireless local area networks (WLAN) currently based on the IEEE[5] 802.11 specifications. New standards beyond the 802.11 specifications, such as 802.16 have been developed. They offer many improvements, from longer range to greater transfer speeds. Wi-Fi was intended to be used for wireless devices and local area networks (LAN), but is now often used for internet access (one of the main international standards for wireless broadband internet access and networking, with widespread use in business, homes and public spaces). It is based on radio signals with a frequency of 2.4 GHz and theoretically capable of speeds of over 54 Mbit/s (at close range, some versions of Wi-Fi, running on suitable hardware, can achieve speeds of over 1 Gbit/s). It enables a person with a wireless-enabled computer or personal digital assistant to connect to the internet when close to an access point called a hotspot. Internet connection via mobile telephone networks is not included under this category.

By public Wi-Fi, essentially refers not to the enterprise's Wi-Fi (that may be xDSL, cable or fibre optic) but to public Wi-Fi, hotspots, hotzones that have different names in different countries. For example, in Luxembourg it is called HotCity and the coverage is very wide. National examples (i.e. WiFi or WiMax with their national brand names) would help respondents.

Note: In case the enterprises are using only a mobile internet connection, e.g. 4G or 5G technology as the only connection in the office (instead of a fixed line connection), then they should answer NO in question A2, because it is mobile broadband technology even though it is used like fixed line connection to the internet only on one address. However, it was considered that a separate question on the use of mobile internet connections was not necessary, as the information can be derived from the information provided by enterprises in questions A1 and A2.

Question A3: *What is the maximum contracted download speed of the fastest fixed line internet connection of your enterprise?*

(additional categories at national level can be added, if needed)

[Scope: enterprises with fixed line connection to the internet, i.e. A2 = Yes]

[Type: single answer (i.e. Tick only one)]

Maximum contracted download speed means the maximum theoretical speed — according to the ISP's contractual obligations — at which data can be downloaded. The five options offered are measured in Mbit/s (Mb/s or Mbps) or Gbit/s (Gb/s or Gbits). These stand for megabits per second or gigabits per second and are a measure of bandwidth (the total information flow over a given time) on a telecommunications medium. Mbps is not to be confused with MBps (megabytes per second). Often the problem is that speed tests and ISP's use 'bits per second' while download agents/programs use 'bytes per second'. Note that 1 Byte = 8 bits.

The five options offered are: a) less than 30 Mbit/s; b) at least 30 but less than 100 Mbit/s; c) at least 100 but less than 500 Mbit/s; d) at least 500 but less than 1 Gbit/s; and e) at least 1 Gbit/s.

Additional categories can be added at national level, if needed.

Note that the actual bandwidth and download speeds experienced are dependent on a

[5] IEEE is the acronym for Institute of Electrical Electronics Engineers.

combination of factors including the ISP, the equipment and the software used, the internet traffic and the destination server. Hence, it may differ from the contracted download speed which is requested in this question. Respondents are not requested to run a speed test on their equipment in order to respond to the question.

Enterprises (respondents) can obtain the information about the maximum contracted download speed of the fastest fixed line internet connection from their monthly invoices for telecommunication services (internet) or in the contract with the telecommunication services (internet) providers.

Question A4: *Is the speed of your fixed line connection(s) to the internet usually sufficient for the actual needs of the enterprise?*

[Scope: enterprises with fixed broadband connection to the internet, i.e. A2 = Yes]

[Type: single answer (i.e. Tick only one); binary (Yes/No)]

The objective for enterprises to fully satisfy their needs in terms of adequate internet connection is a key focus of Commission policies. Investment is necessary for improving the infrastructure for high-speed broadband connections. The purpose of the question is to 'count' how many enterprises consider that their connection speed to the internet adequately serves their needs. Essentially question A4 aims at measuring the degree of demand for improved connectivity to the internet as well as the enterprise's overall appreciation of their current fixed internet connection. Responses to question A4 will be combined with respondents' choices in A3 on contracted download speed. Since some enterprises may have premises in various geographic locations, and in addition, the speed of the connection to the internet may fluctuate (e.g. depending on the day or the time of the day) it is proposed to use the formulation 'usually sufficient'. This would mean that the Yes/No possible responses would imply 'Yes, in most of the cases speed is sufficient' or 'No, in most of the cases speed is not sufficient'. Although the question is linked to Commission policies, the results cannot be isolated and used for taking any kind of decision on investment which might be needed for improving the infrastructure for providing high-speed broadband connections.

Use of a mobile connection to the internet for business purposes

(Scope: enterprises with access to the internet, i.e. if A1 >0)

The following question refers to the usage of portable devices connecting to the internet through mobile telephone networks for business purposes. This includes the use of devices where the enterprise provides the devices and pays for the subscription and the costs of use fully or at least up to a limit.

In principle, these are portable devices that are provided allowing mobile connection to the internet via mobile telephone networks. In the scope of the survey are enterprises that provide portable devices as part of the employee's equipment, provided that the conditions in the introduction of the module (payment of subscription and usage costs) are fulfilled and the portable devices allow mobile connection to the internet via mobile telephone networks. Additionally, 'business use' means that the purpose of the use is related to work.

Mobile and wireless connection to the internet: Question A5 clearly refers to 'mobile[6] connection to the internet' through mobile telephone networks (either within or outside the premises of the enterprise) which should not be confused with 'wireless connection to the internet' (e.g. Wi-Fi, Bluetooth, to a certain extent WiMAX). The exclusive use of wireless connection to the internet is excluded either within the premises of the enterprise or within the range of a hotspot or hot-zone outside the premises of the enterprise.

From a practical point of view, portable devices can switch between mobile (telephone networks) and wireless connections (Wi-Fi, Bluetooth, and WiMax). However, the intention is to measure enterprise's take up of 'mobile connection of the internet' when portable devices connect to the internet over mobile telephone networks, regardless of the fact that devices may switch/ connect to wireless networks and regardless of the technology used (4G or more advanced, but also less than 3G in the absence of adequate signal) for accessing the mobile telephone networks.

Portable devices: The portable devices that allow a mobile connection to the internet refer to laptops, tablet computers or smartphones.

'Smartphones' qualify as computers and were developed to provide functionalities beyond those of cellular phones (cell phones or mobile phones) which served only one purpose: voice communications and exchange of SMS/ MMS. Smartphones and similar devices are equipped with customised software, increased bandwidth efficiency, internet access, digital cameras, portable music players, global positioning system (GPS) functions and many more — business or entertainment — options despite their relatively small size.

However, growth in demand for portable devices requiring processors that are more powerful, abundant memory and larger screens has placed low-power portable devices at the centre of interest. Netbooks, notebooks, and tablet computers equipped with built-in card modem (SIM-card) or external dongles (USB modem), have considerably extended the family of 'mobile connectivity' devices to compact size 'portable computers'.

The use of handsets exclusively for voice/SMS/MMS communications (e.g. enterprises requiring that persons employed are available to take business calls at all times and anywhere) are not in the scope of questions A5. Additionally, M2M (machine-to-machine) communications should not be considered in the scope of the mobile use of the internet.

In certain cases, a portable computer (e.g. laptop or tablet computer) can be connected to a handset (e.g. via Bluetooth, cable) and the handset can be connected to the internet through a mobile connection (via the mobile telephone network). The portable computer is considered the device connected to the internet. In this case, the handset is the equivalent to a USB modem or a card modem. Moreover, the portable computer is the device used for accessing the business software application (or any application) via the internet and not the handset.

Question A5: *How many persons employed use a portable device provided by the enterprise,*

[6] The network type referring to the 'Mobile technology' is the wireless wide area network (WWAN). It provides high-speed internet access through portable devices and coverage of the national territory in each country. It is considered to provide full telephone network coverage. The exclusive use of the Wireless Local Area Networks is not in the scope of questions A5.

that allows connection to the internet via mobile telephone networks, for business purposes?

(e.g. portable computers or other portable devices such as smartphones)

If you can't provide this value,

please indicate an estimate of the percentage of the total number of persons employed who use a portable device provided by the enterprise, that allows connection to the internet via mobile telephone networks, for business purposes (e.g. portable computers or other portable devices like smartphones).

[Scope: enterprises with access to the internet, i.e. A1 > 0; optional question]

[Type: numerical, absolute or percentage values]

The question refers to enterprises providing portable devices that allow (through a contract/subscription) mobile connection to the internet. Portable computers using mobile telephone networks may refer to notebooks, netbooks, laptops, tablet computers, etc. while other portable devices may refer to smartphones using mobile telephone networks. Mobile connection to the internet refers to connection over mobile telephone networks.

Technically, a connection via high capacity 'speed' mobile telephone networks refers to accessing the internet using a long range wireless transmission of the 3rd / 4th or even 5th generation (3G, 4G, 5G) mobile network technologies based on the code division multiple access ('CDMA') as UMTS (universal mobile telephone system — Wideband W-CDMA); CDMA2000x; CDMA 2000 1xEV-DO; CDMA 2000 1xEV-DV).

- 3G (or 3-G) is short for third-generation mobile telephone technology. The services associated with 3G provide the ability to transfer both voice data (a telephone call) and non-voice data (such as downloading information, exchanging e-mail and instant messaging). It includes high-speed mobile networks (e.g. CDMA2000 1X, W-CDMA, CDMA2000 1xEV-DO).
- 4G is the fourth generation of cellular wireless standards (mobile telephone technology). It is a successor to the 3G and 2G families of standards. The International Telecommunication Union (ITU) Radiocommunication Sector specified the international mobile telecommunications advanced requirements for 4G standards, setting peak speed requirements for 4G service at 100 Mbit/s for high mobility communication (such as from trains and cars) and 1 Gbit/s for low mobility communication (such as pedestrians and stationary users).
- 5G is the fifth generation technology standard for broadband cellular networks, which cellular phone companies began deploying worldwide in 2019, and is the successor to the 4G. The main advantage of the new networks is that they will have greater bandwidth, giving higher download speeds, eventually up to 10 gigabits per second (Gbit/s).
- UMTS is one of the 3G mobile phone technologies. It uses W-CDMA as the underlying standard. It is standardised by the 3GPP, and represents the European answer to the ITU IMT-2000 requirements for 3G cellular radio systems. It presently delivers packet switched data transmission speeds up to 384 Kbit/s and up to 2 Mbit/s when fully implemented.
- CDMA2000 1x is an IMT-2000 3G mobile network technology, based on CDMA that delivers

packet switched data transmission speeds of up to 144 kbps. It is also referred to as 1XRTT.

- CDMA2000 1xEV-DO is an IMT-2000 3G mobile network technology, based on CDMA that delivers packet switched data transmission speeds of up to 2.4 Mbps.

The concept of 'persons employed' to be used in this variable is the one described in the explanatory note of variable X2 ('Average Number of number of employees and self-employed persons (Persons Employed)').

This variable can be collected in absolute or in percentage values. Eurostat recommends using a combination of both, giving the respondent the opportunity to choose either to answer in absolute values or percentage values. The absolute value might be more appropriate for smaller enterprises and the percentage for bigger ones. When the number of persons employed is small, it is easier for respondents to identify which ones use a portable device provided by the enterprise, that allows internet connection via mobile telephone networks, for business purposes, instead of having to calculate the percentages afterwards. Collection in percentage terms may introduce measurement errors, because of the rounding that is necessary to provide an answer in percentages made of integer numbers between 1 and 100. Collection in percentage terms may also introduce other measurement errors as it leads the respondent to make an educated guess of its value introducing the bias usually associated with this type of answer. On the other hand, for bigger enterprises, with high numbers of persons employed, the identification of every worker using a portable device provided by the enterprise, that allows internet connection via mobile telephone networks, for business purposes can be burdensome or not possible and result in non-response. In this case, the collection of this information in percentage terms may be preferable.

Independently of how this variable is collected, the background variable 'number of employees and self-employed' is needed for grossing up the overall percentage of persons employed who use portable devices provided by the enterprise, that allows internet connection via mobile telephone networks, for business purposes. In order to compute the overall percentage, both the total number of employees and self-employed persons and the total number of persons employed using a portable device provided by the enterprise that allows internet connection via mobile telephone networks, for business purposes need to be grossed up. If collected in percentage terms, the corresponding absolute value needs to be calculated afterwards multiplying the answer with the background variable to gross up the number of persons employed using a portable device provided by the enterprise that allows internet connection via mobile telephone networks, for business purposes.

This question refers to persons employed using portable devices (within or outside the premises) that allow internet access via mobile telephone networks for business purposes, provided by the enterprise permanently or temporary (e.g. when needed or upon request). The question refers to persons (not to devices), hence it is suggested to request an estimate of persons provided with devices. Portable devices provided when needed or upon request e.g. meetings away from office premises, business trips, participation at conferences, provision of maintenance and on-site support to customers, are in the scope of the question. Cases in which persons employed use portable devices without accessing the internet (but devices allow for it) are in scope too.

A mobile internet connection for the enterprise owner(s) should be included in A5.

Remote access and meetings via the internet

(Scope: enterprises with access to the internet, i.e. if A1 >0)

Remote access

This set of questions aims to measure enterprises' readiness, capacity or willingness to make it possible for their employees to work remotely by allowing them a remote access to enterprise's resources (e.g. remote e-mail access, remote access to documents and ICT systems of the enterprise).

The policy interest for indicators relating to the technological readiness of the enterprises to apply teleworking and/or use virtual meetings originates from the experience due to the Covid-19 pandemic. However, remote working and virtual meetings are not temporary phenomena. The recent experience has proven that the ever-increasing capabilities of ICT have made in many cases teleworking (full or partial) as efficient as working in the office even in businesses where this was not considered before. Therefore, the voices highlighting the economic (less costs for enterprises), environmental (less commuting) and social (work-life balance) benefits of teleworking have increased substantially. Therefore, there is a need to understand the remote working readiness of European enterprises (in terms of facilities provided to their employees), the percentage of employees that could potentially telework and the uptake of this new way of working in Europe.

Question A6: *Do any of the persons employed have remote access to the following? (via computers or portable devices such as smartphones)*

[Scope: enterprises with access to the internet, i.e. A1 > 0]

[Type: single answer per item (i.e. Tick only one); binary (Yes/No); multiple items may be expected]

Question A6 measures the technological readiness of the enterprises to apply teleworking by providing their employees with remote access to the enterprise's resources.

The question does not ask about the use, but about the access, so the possibility to remotely connect to enterprise's resources.

The access may be via any device, desktop computer or any portable device, including smartphones. There is no limitation in the scope concerning the ownership of the device, it covers access through a private device (owned by the employee) as well as a corporate device (provided by the enterprise).

a) E-mail system of the enterprise

Enterprises, which provide a remote access to enterprise's e-mail system for their employees, should answer 'Yes' to this answer option. If the official e-mail system of the enterprise is for instance Gmail (or other), which can be accessed anywhere, then the enterprise should tick 'Yes' to question A6a).

b) Documents of the enterprise (e.g. files, spreadsheets, presentations, charts, photos)

This answer option covers a remote access to enterprise's documents, including spreadsheets, presentations, or any other files. This item should also be ticked 'Yes' if only a limited number of documents can be accessed remotely. The item does not include in scope cases of accessing the documents (e.g. files, spreadsheets, presentations, charts, photos) via e-mail mailbox (saved in a sent or received email).

c) **Business applications or software of the enterprise (e.g. access to accounting, sales, orders, CRM)**

Please exclude applications used for internal communication, e.g. Skype, Teams, Viva Engage (old Yammer).

Answer option c) includes remote access to enterprise's business applications or software used by the enterprise, such as application or software related to accounting, sales or other business software. This item should also be ticked 'Yes' if only a limited number of business applications or software can be accessed remotely. The use of applications for internal communication, such as Skype, MS Teams, Viva Engage (old Yammer) are excluded from the scope of answer option c).

Meetings via the internet

The question aims to collect information about conducting meetings via the internet (remote meetings) by enterprises.

Telework and in particular virtual meetings are very relevant topics, also when looking at the Covid-19 pandemic which has forced many enterprises globally to turn to teleworking and/or virtual meetings, in order to maintain their businesses while respecting confinement and social distancing measures imposed by national governments.

This indicator is also relevant in the context of the use of digital technologies to build a smarter and more sustainable economy.

Question A7: *Does your enterprise conduct remote meetings (via e.g. Skype, Zoom, MS Teams, WebEx)?*

[Scope: enterprises with access to the internet, i.e. A1 > 0]

[Type: single answer (i.e. Tick only one); binary (Yes/No)]

Question A7 aims measuring whether enterprises conduct remote (online) meetings via internet using tools such as Zoom, Skype, MS Teams or others. The remote meeting can be internal (between persons in the enterprise, whether they are located in the office or outside) or external (with persons from outside the enterprise e.g. business partners). Remote meetings can be conducted either via video, via audio only or via both, video and audio. Both, remote meetings conducted via dedicated apps or via browsers are included. The video meetings may be conducted via any device, desktop or any portable device, including smartphones.

Other use of the internet

Question A8: *Does your enterprise pay to advertise on the internet? (e.g. adverts on search engines, on social media, on other websites or apps)*

[Scope: enterprises with access to the internet, i.e. A1 = Yes]

[Type: single answer per item (i.e. Tick only one); binary (Yes/No); filter question; optional question]

The question refers to enterprises using paid adverts (ads) on the internet to advertise their

goods or services. For example, enterprises may pay for their ads to appear at the top of the search results of various search engines, on social media banners, etc. when relevant search terms were entered.

Question A9: *Does your enterprise pay to advertise on the internet using any of the following targeted advertising methods?*

[Scope: enterprises that pay to advertise on the internet, i.e. A8 = Yes]

[Type: single answer per item (i.e. Tick only one); binary (Yes/No); optional question]

The following responses refer to targeted advertisement methods. In general, advertisement methods that are used over the internet offer enterprises more possibilities to reach their target audience. It is possible that enterprises reply to all four answer options with “no”, meaning that an enterprise pays for ads (A8=Yes) without using any of the proposed targeting methods.

a) Based on content or keywords searched by internet users

Response a) refers to the use of technologies that adapt adverts to the contents of the displayed pages (contextual advertising). Internet users are presented with adverts that automated systems have selected for them according to the content of the webpage or based on keywords that users have searched on the internet. For example, enterprises may pay for having their ads at the top of results for relevant search terms of various search engines, on social media banners, etc. Adverts based on the context of the webpage may be "push" type advertisements or "pull" advertisements. For the former, adverts are shown regardless of the need of the user while for the latter the user was actively seeking marketing content (e.g. through keywords) and thus the advert is shown.

b) Based on the tracking of internet users' past activities or profile

Response b) refers to the use of technologies for collecting information about the online (browsing) activity of internet users (behavioural advertising, web browsing habits). Tracking cookies (files stored on internet users’ computers when visiting a website) are used to help determine the users' interests based on the web pages that were visited, the content that users browsed and other actions that were taken online. The aim is to deliver adverts tailored to the user's interests and preferences.

c) Based on the geolocation of internet users

In the context of using the internet for ads, advertising based on the geolocation of internet users, refers to the use of technology (geo targeting) that regardless of the device (portable or not) and the connection to the internet (fixed incl. wireless or mobile) may be used in order to deliver advertisements based on the geographical location (e.g. hotels in the proximity). Determination of geolocation of the user is used by classical Google advertisement and Facebook advertisement. Geolocation is particularly used in cases of "push" advertisement via apps on mobile devices. However, it is reminded that the current scope is broader than mobile advertisement that may be implied by the use of apps on mobile devices according to the terminology that is used in the questionnaire. Technically, the important issue would be the proper identification of the internet protocol (IP) address of the connection to the internet through the closest tower antenna or router. The use of geo targeting technologies for advertising can be combined with the use of cookies (behavioural advertising, web browsing habits) in order to precise the need of a potential customer.

d) Any other method of targeted advertising on the internet not specified above

Other methods of internet advertising refer to the use of paid ads for delivering marketing messages to consumers that are not covered by the above mentioned dynamic methods (such as advertisements targeting specific websites that have different demographic characteristics – e.g. general population, younger or older persons, men or women or retargeting – targeting audiences who have visited website or social media profiles, paid advertising articles or videos, paid cooperation with You Tubers or well-known personalities (influencers)).

A combined use of the above mentioned technologies is possible.

2.3.2.3. Module B: e-Commerce sales

This module covers e-commerce sales (received orders) which are conducted via a website or apps or as EDI (electronic data interchange)-type sales. Since the 2011 survey, the measurement of e-commerce sales is done as a split between web sales and EDI-type sales which are separate submodules and comprise mandatory reporting variables.

One important difference in this module, compared with most other variables in the questionnaire, is that systematically the reference period is the previous calendar year, instead of the current situation. Flow economic variables like turnover and purchases – the main variables to measure in e-commerce – need to be measured for a longer period instead of at just one point in time. In order to keep comparability with the main business statistics, the calendar year is taken as reference period.

Definition of e-commerce

In order to ensure international comparability of the ICT usage in enterprises statistics, the OECD definition of e-commerce is used as a basis throughout this module but referring merely to sales:

The 2009 definition of e-commerce	
OECD definition of e-commerce	**Guidelines for the interpretation**
An **e-commerce transaction** is the sale or purchase of goods or services, conducted over computer networks by methods specifically designed for the purpose of receiving or placing of orders. The goods or services are ordered by those methods, but the payment and the ultimate delivery of the goods or services do not have to be conducted online. An e-commerce transaction can be between enterprises, households, individuals, governments, and other public or private organisations.	**Include**: orders made in web pages, extranet or EDI. The type is defined by the method of making the order. **Exclude**: orders made by telephone calls, facsimile, or manually typed e-mail.

The framework for measurement	
Type	Definition of the type of e-commerce
Web e-commerce	Orders made at an online store (web shop) or via web forms on the Internet or extranet regardless of how the web is accessed (computer, laptop, mobile phone, etc.)
EDI e-commerce	Orders initiated with EDI. EDI is an e-business tool for exchanging different kinds of business messages. EDI is here used as a generic term for sending or receiving business information in an agreed format which allows its automatic processing (e.g. EDIFACT, XML, etc.) and without the individual message being manually typed. “EDI e-commerce” is limited to EDI messages placing an order.

E-commerce sales may also include renewal orders (standing orders/recurrent payments/subscriptions) received in the reference period, which are associated with the initial order made via e-commerce channels before the reference year. It has been agreed that such standing orders are within the scope of the e-commerce module. However, experiences provided by statistical compilers show that measuring this would be challenging, because many enterprises cannot track the information needed to distinguish if such standing orders are linked to an initial order made via an e-commerce channel. Therefore, it has been decided not to add any explicit clarification on that matter in the questionnaire (neither model questionnaire, nor national questionnaires). If a respondent asks whether they should include such order/payments in the reference period, which are resulting from an initial order made before the reference period, they should be advised that standing orders/recurrent payments/subscriptions received in the reference period should be reported by them within the e-commerce module.

Defining e-commerce sales in the questionnaire

The e-commerce definition is implemented as follows:

In e-commerce sales of goods or services, the order is placed via web sites, apps or EDI-type messages by methods specifically designed for the purpose of receiving orders.

The payment may be done online or offline.

e-Commerce **does not include** orders written in e-mail.

Defining web sales in the questionnaire

Web sales cover the following:

- **Online store (web shop)** is the most obvious and clear example of web-based e-commerce. It is a separate site in the web or a separate part of a website where products are presented and typically ordered via shopping cart functionality.

- **Web forms** are simple forms integrated in the enterprises' website where goods and services can be ordered. In these forms, the order can be typed or clicked/selected and then sent it by clicking the "send" button on the web site.

- **Extranet** is a closed environment for agreed partners or customers where other information between those parties can be accessed or exchanged. If there are sales done in extranet, it is counted as web sales. In extranet, the actual shopping can be done either in a web shop or via web forms, as explained above.

- **Bookings and binding/fulfilled reservations** are considered similar to orders; for some economic sectors (e.g. NACE Rev. 2, sector 55 Accommodation), these terms are more commonly used to describe the order. The questions in this module are about sales (and generated turnover), therefore the scope is by default limited to the fulfilled sales, orders, and reservations. Reservations and orders that are not fulfilled are outside the scope of this module.

- Sales via **apps for mobile devices or computers** are also counted as web sales. App is a short name for web application. There are two types: the browser-based and the client-based web applications. The former are web applications accessible over a webpage (html and JavaScript) and run within the web browser. The latter are installed on a device (e.g. smartphone), run without going through a browser but using web protocols.

- Sales via external **e-commerce marketplace websites or apps** are also counted as enterprise web sales. E-Commerce marketplaces are external websites or apps used by several enterprises for trading (selling) their goods or services to customers.

Defining EDI-type sales in the questionnaire

In the beginning of the submodule on EDI-type sales, the concept of EDI-type sales is explained for the respondents.

EDI can be defined as the transfer of structured data, by agreed message standards, from one computer system to another without human intervention. EDI provides a technical basis for automated commercial 'conversations' between two entities, either internal or external. The term EDI encompasses the entire electronic data interchange process, including the transmission, message flow, document format, and software used to interpret the documents.

Demand-driven orders concern situations where, for example, a certain minimum level of inventory has been specified in the system of the buying enterprise; when the inventory drops below the specified minimum level, the system transmits an EDI message to the selling enterprise for more goods to be delivered.

This is an example of an advanced, but in certain activities common, way of business automation through the automated integration of systems from two trading partners.

Questions on web sales of goods or services

Web sales cover orders, bookings and reservations placed by your customers via

- your enterprise's **websites or apps**:
  - online store (web shop)
  - web forms
  - extranet (web shop or web forms)
  - booking/reservation applications for services
  - apps for mobile devices or computers
- **e-commerce marketplace websites or apps** (used by several enterprises for trading goods or services).

Orders written in e-mail are **not** counted as web sales.

Question B1: *During 2023, did your enterprise have web sales of goods or services via:*

[Scope: enterprises with access to the internet, i.e. A1 > 0]

[Type: single answer per item (i.e. Tick only one); binary (Yes/No); multiple items may be expected]

a) your enterprise's websites or apps? (including extranets)

b) e-commerce marketplace websites or apps used by several enterprises for trading goods or services? (e.g. e-Bookers, Booking, hotels.com, eBay, Amazon, Amazon Business, Alibaba, Rakuten, TimoCom)

[Please add national examples of e-commerce marketplaces including government marketplaces]

The question aims to quantify the use of electronic marketplaces by enterprises (intermediary e-commerce websites or apps) for receiving orders via various web platforms in the context of enterprise web sales. The question is relevant for developing policy initiatives addressing market competition issues, dominant positions of e-commerce players that are active in marketplaces, and (discriminatory) business models in various e-commerce marketplaces.

The term 'e-commerce marketplace' refers to websites or apps used by several enterprises for trading products (e.g. e-Bookers, Booking, hotels.com, eBay, Amazon, Amazon Business, Alibaba, Rakuten, TimoCom, Glovo, Wolt). *Amazon Business* is a marketplace on *Amazon*.com that is addressed to business-to-business (B2B) commercial transactions.

TimoCom runs the largest transport platform in Europe (it operates in 44 European countries). Services on this platform include: transport orders (to request orders digitally and manage transport orders); freight exchange (to avoid empty runs); vehicle offers; fixed contracts (transport tenders platform); and warehousing exchange.

Amadeus is not an example of an e-commerce marketplace. It is a platform that connects travel providers and travel sellers all over the globe, offering search, pricing, booking, ticketing, transaction and servicing capabilities. It mainly provides services designed for enterprises within the travel sector and it builds the technology that enables travel agents and websites to book airline, rail, cruise and ferry tickets, hotel rooms, car hire, tour packages and more. Travel agencies can build their services on top of Amadeus, e.g. their own platforms. Amadeus is not a marketplace, but rather an infrastructure.

Note that **e-commerce marketplaces** are different from **e-commerce platforms**, e.g. Shopify, WooCommerce, Magento, Bigcommerce, that provide scalable, self-made online solutions for

business that would like to set up their own e-commerce website.

The following should **not be considered as e-commerce marketplaces**:

a) a website or app of an enterprise selling the enterprise's own products;

b) a website or app of one seller acting as a distributor, selling other enterprises' specific products;

c) a website that provides e-commerce solutions for other enterprises to install for the enterprises' own e-commerce functionality; and

d) a website that focuses on non-trading activities like collaborative design.

The issue of translating accurately the term 'e-commerce marketplaces' in other languages is important. Other alternatives might work better (e.g. internet-based trade platforms) in other languages.

Question B2: *What was the value of your web sales? (please refer to the provided definition of web sales)*

[Scope: enterprises which had web sales of goods or services, i.e. B1a = Yes or B1b = Yes]

[Type: numerical, absolute or percentage values]

Please answer to a) OR b)

Please answer to a) OR b)	
a) What was the value of your web sales of goods or services, in 2023?	(National currency, excluding VAT) _____
OR	
b) What percentage of total turnover was generated by web sales of goods or services, in 2023?	
If you cannot provide the exact percentage an approximation will suffice.	␣ ␣ ␣ , ␣ %

For turnover, the structural business statistics (SBS) definition is to be used. See background variable X3 in this document.

Collecting electronic orders in percentage values has the advantage of allowing to obtain an estimate from respondents who don't have any records in the enterprise that can provide this value. Therefore, unless specific records of all these transactions are kept in a centralised form, it may be difficult for the enterprise to provide the value of orders received that were placed via a website or app. Another way of coping with this problem is to allow enterprises to estimate the web sales turnover in monetary terms by indicating that 'an approximation will suffice'.

For enterprises with a very small share of e-commerce in turnover, there is evidence that respondents round this significantly to around 1%. Therefore, when collected in this way as a

percentage figure, a relatively large proportion of the values collected are 1%. Another challenge of this method is that, for big enterprises, a value of less than 1% can still be a significant amount in value terms. When answers from big enterprises are involved, this can bring a significant instability to the results.

In addition, larger percentage shares are often also reported rounded to multiples of 10% or 5%.

For these reasons the answer should be provided in decimals.

Due to challenges in collecting data in percentage figures, the preferred way to collect value of sales via a website or app is in absolute values. Unfortunately, for many enterprises, this value is not available. For this reason, the recommended method is to ask the enterprise to answer in precise absolute values if it has the information. Or as an alternative, if such information is not available, an estimate could be asked, either in monetary terms or as a percentage of the total turnover.

Currently some countries ask for the absolute value while others ask for the percentage value. The current formulation aims to provide an alternative in order to avoid the bias that might exist from asking only one of the two ways, i.e. underestimation in the case of the absolute value and overestimation in the case of percentage values. **It may be preferable to give the option to enterprises for using the one for which they can provide the more accurate answers.**

Eurostat' preferred practice is to get the grossed-up value of the percentage, i.e. the percentage of turnover resulting from orders received that were placed via a website or apps.

In a situation where an enterprise (e.g. airline) gets commissions from sales (e.g. car rental) made from another enterprise (e.g. car rental enterprise) via its (air enterprise) website or app, the commissions should be counted as e-commerce.

Question B3*: What was the percentage breakdown of the value of web sales in 2023 for the following: (Please refer to value of web sales you reported in B2)*

If you cannot provide the exact percentages an approximation will suffice.

a) via your enterprise's websites or apps? (including extranets)

b) via e-commerce marketplace websites or apps used by several enterprises for trading goods or services? (*e.g. e-Bookers, Booking, hotels.com, eBay, Amazon, Amazon Business, Alibaba, Rakuten, TimoCom*)

[*Please add national examples of e-commerce marketplaces including government marketplaces*.]

[Scope: enterprises which had web sales of goods or services via own websites or apps and via e-commerce marketplaces, i.e. B1a = Yes and B1b = Yes]

[Type: numerical, percentage values that add up to 100%]

In the case of a web questionnaire, it is recommended that B1 serves as a 'smart filter'. **B3** should only be answered if both **B1 a) and B1 b)** have been answered with 'Yes'. If only one of the two has been answered with 'Yes', then the respective value for **B3** should be stored as **100 %** and the respondent should go to **B4**.

The sum of answer options a) and b) should always result in 100%.

Question B4: *What was the percentage breakdown of the value of web sales in 2023 by type of customer: (Please refer to value of web sales you reported in B2)*

If you cannot provide the exact percentages, an approximation will suffice.

a) Sales to private consumers (B2C)

b) Sales to other enterprises (B2B) and Sales to public sector (B2G)

Total: 100%

[Scope: enterprises which had web sales of goods or services, i.e. B1a = Yes or B1b = Yes]

[Type: numerical, percentage values that add up to 100%]

For respondents having received orders that were placed via a website or app, a percentage breakdown of the turnover by type of customer is requested. The two most important types of e-commerce occur in business-to-consumer (B2C) and business-to-business (B2B) markets.

a) Sales to private consumers (B2C)

The term B2C stands for business-to-consumer and refers to electronic commerce transactions between enterprises and the individuals as the end consumer. Business-to-consumer electronic commerce typically takes the form of websites or apps that offer the possibility to individuals to place orders for products.

b) Sales to other enterprises (B2B) and Sales to public sector (B2G)

The term B2B stands for business-to-business and refers to electronic commerce transactions between enterprises (different from transactions between enterprises and other groups, like consumers (individuals) and the government). B2B refers to commercial transactions between the responding enterprise and other enterprises (e.g. manufacturer and a wholesaler, a wholesaler and a retailer). Business-to-business electronic commerce typically takes the form of processes between trading partners and is performed in higher volumes than business-to-consumer applications (e.g. use of e-marketplaces or via the respondent's websites using login/password procedures). B2B e-marketplaces connect buyers and sellers through a hub where online transactions can be executed. The distinction of e-commerce between B2B and B2C is important because B2B transactions have advantages for enterprises, e.g. reduction of product cycle times, lowering stock levels and increasing trade volumes.

B2G stands for business-to-government and includes the electronic commercial transactions between the responding enterprise and public authorities that are conducted via a website or apps.

Question B5: *During 2023, did your enterprise have web sales to customers located in the following geographic areas?*

a) Own country

b) Other EU countries

c) Rest of the world

[Scope: enterprises which have received orders that were placed via a website or apps, i.e. B1 = Yes]

[Type: one single answer per item needed, i.e. Tick only one; binary (Yes/No); multiple items may be expected]

In this question, the respondent enterprise that had web sales (via enterprises websites or apps and/or e-commerce marketplace websites or apps used by several enterprises for trading goods or services), during 2023, is asked about the location of their customers. Three answer options are provided, namely:

a) Own country: the customer is located in the same country as the enterprise;

b) Other EU countries: the customer is located in one of the other EU countries (any of the 27 EU countries other than the country of the enterprise);

c) Rest of the world: the customer is located outside the EU27 member states.

The respondents shall tick all options that apply.

Question B6: *What was the percentage breakdown of the value of web sales in 2023 to customers located in the following geographic areas?*

(Please refer to value of web sales you reported in B2)

If you cannot provide the exact percentages an approximation will suffice.

[Scope: enterprises received orders placed via a website or apps, i.e. B1 = Yes; optional question]

[Type: numerical, percentage values that add up to 100%]

Own country	☐☐☐,☐ %
Other EU countries	☐☐☐,☐ %
Rest of the world	☐☐☐,☐ %

For respondents that had web sales (via enterprises websites or apps and/or e-commerce marketplace websites or apps used by several enterprises for trading goods or services), during 2023, a percentage breakdown of the turnover by location of customer is requested. The same three location categories as in question B5 are used, namely: a) Own country, b) Other EU countries and c) Rest of the world.

The respondent is requested to answer to this question with an approximation, if the exact percentages are not available.

In case of implementing a web questionnaire, it is recommended that B5 serves as a ‘smart filter’. B6 should only be answered if at least two of the possible responses in question B5, a), b) or c) have been answered with ‘Yes’, otherwise question B7 should be answered.

Questions on EDI-type sales

EDI-type sales cover **orders placed** by your customers via EDI-type messages, meaning:

- in an agreed or standard format suitable for automated processing;
- EDI-type order message created from the business system of the customer;

- including orders transmitted via EDI-service provider;
- including automatic system-generated demand-driven orders;
- including orders received directly into enterprise resource planning (ERP) system.

Examples of EDI: EDIFACT, XML/EDI (e.g. UBL, Rosettanet, [please add national examples])

Question B7: *During 2023, did your enterprise have EDI-type sales of goods or services?*

[Scope: enterprises with access to the internet, i.e. A1 > 0]

[Type: single answer (i.e. Tick only one); binary (Yes/No); filter question]

Question B8: *What was the value of your EDI-type sales? (please refer to the provided definition of EDI-type sales)*

[Scope: enterprises which had EDI-type sales of goods or services, i.e. B7 = Yes]

[Type: numerical, absolute or percentage values]

Please answer to a) OR b)

a) What was the value of your EDI-type sales of goods or services, in 2023? (National currency, excluding VAT)

b) What percentage of total turnover was generated by EDI-type sales of goods or services, in 2023?

If you cannot provide the exact percentage, an approximation will suffice.

For turnover, the SBS definition is to be used. See background variable X3 in this document.

Please refer to the definition of EDI in the beginning of Chapter 2.3.2.3. and the guidelines below.

Guidelines for the distinction of e-commerce, web or EDI type in specific cases

1. Distinction between EDI-type sales and web sales in a situation where both technologies are involved

Case/situation:

The responding enterprise has a website or app with sales functionalities. The customer chooses the product and the transaction is completed by pressing the Send button. The website might create an EDI-type message that is sent: to the sales department to prepare electronically the transportation documents and the goods; to the accounting department to prepare the electronic invoice; and to the department responsible for dispatching to prepare the planning of the dispatch of the goods. The same case can be replicated if the customer is not using the enterprise's website or app but the website or app of an online shop which produces the same EDI messages.

Classification/explanation:

According to the e-commerce definition[7], this is referred to as **web sales** because the order was placed via the enterprise's website or app (or a website of a third party selling on behalf of the enterprise – online store) even if the enterprise received it as an EDI-type message. The reporting enterprise should avoid double counting the sales.

2. Selling credit online via apps, e.g. on mobile phones.

This concerns:

a) **Reporting enterprises whose principle economic activity is selling credit**: the fact that they sell credit over the internet should be broadly considered as e-commerce. It should be considered as a commercial transaction, similar to selling vouchers (with/without specifying product) and it would be initially registered as liability towards other enterprises (e.g. retailers, wholesalers, service providers). The 'service fee' should be registered as turnover when the final client ultimately uses the credit/voucher and orders the products. It may be that, for taxation purposes, there is a 'timing issue' for registering the 'service fee' before the final client uses the credit/voucher.

b) **Reporting enterprises that sell credit/vouchers and do not refer to any of their specific products (e.g. gift cards with a certain preloaded amount)**: From the accounting perspective, selling credit over the internet is not e-commerce (no specific product is ordered), so it should be registered as advance payment and not as turnover. It should be registered as commercial transaction and turnover only when the customer orders specific products via the internet (e-commerce) or otherwise (brick-and-mortar business).

c) **Reporting enterprises that sell credit/vouchers and implicitly refer to single/specific products (e.g. set of tickets for the cinema)**: In principle this should be the same as above. However, in some countries, in accordance with national tax law, it may be registered as commercial transaction and turnover because the specific product and its respective price is specified on the credit/voucher.

In the context of the survey, the most important issue is to avoid double counting (when selling credit and when receiving orders) and to enable enterprises to respond as accurately as possible, depending on how e-commerce has been implemented into their accounting system.

3. Clarification on sales over an extranet: web sales or EDI-sales

Enterprises (respondents) may receive orders placed over an extranet and these usually concern business-to-business transactions. It may not always be sufficiently clear for respondents whether the orders received over an extranet should be considered as web sales or EDI-type sales. If it is unclear, the level of automation for placing the order should be considered and clarifications may be required from the respondents. In fact, respondents should know how their customers place orders besides just EDI-type messages.

The OECD definition of e-commerce is based on **the way the order is placed**. If the order is placed via forms/web forms via extranet, it is clearly web sales, regardless of the fact that an exchange of EDI-type messages follows as explained in case 1) above. The issue to be clarified with the enterprise (respondent) should be exactly on the operation of their partner when placing

[7] The type of e-commerce transaction is defined by the method of making the order. This approach should mitigate the interpretation problems where both types, EDI and web, are used in the process. An example is a situation where an order is made by the customer through a web application but the information is transmitted to the seller as an EDI-message. Here the type of selling application is, however, web; EDI is only a business application to transmit information.

the order.

Only the description of the actual operation for placing the order would provide the necessary information for web or EDI-type sales and, in certain cases, the NSI must take a decision. However, it is suggested that 'web sales' are completely excluded before declaring the sales as 'EDI-sales'.

4. Frame agreements

Goods or services that are agreed to be sold in a frame agreement, but are ordered later via web or EDI-type orders are also included in e-commerce sales. This means a situation where enterprises reach an agreement for an agreed number/amount/volume/limit of products – not necessarily online – for an agreed time frame, but the products are ordered electronically later when they are needed (it could be multiple smaller orders summing up to the frame agreement). In this case, the initial sale (frame agreement) should not be included as e-commerce, but the order of the products via web or EDI message should be (and only this turnover generated by the orders for the specific reference year).

5. Examples of special cases

Examples on what IS e-commerce:

a) e-commerce website or app that offers the possibility to fill in a web form (online) hence the order is placed using a website or app.

The is the usual case of an enterprise having web sales.

b) e-commerce website or app that offers the possibility to fill in a web form (online), hence the order is placed using a website or app and the responding enterprise enters the website and retrieves online the order in any format.

This refers to the use of 'methods specifically designed for the purpose of receiving orders' over a website even though the selling enterprise (responding enterprise) retrieves (downloads) the order on its own initiative.

Examples on what is NOT e-commerce:

The following cases are not considered as e-commerce web sales because the web form or the PDF order form is used as a Word document, text document or an email. The website does not provide the means to 'place the order' but in principle to 'construct the order' that is eventually placed differently (email, other electronic message via the website or post). In these cases, the orders are not made necessarily on web pages, i.e. they can be made by printing a PDF document which is then 'reintroduced' into a computer system and, because of this break, there is no e-commerce because there is a human intervention in the process, and a 'break' in the automation.

a) e-commerce website or app that offers the possibility to fill in a **web form** (online). The form is 'printed' in PDF and sent by the customer to the responding enterprise as an **email attachment**.

b) e-commerce website or app that offers the possibility to fill in a **web form** (online). The form is 'printed' in PDF and sent by the customer to the responding enterprise **by post**.

c) e-commerce website or app that offers the possibility to fill in a PDF order form (not necessarily online). The form is 'saved' in **PDF**, completed and sent by the customer to the responding enterprise as an **email attachment**.

d) e-commerce website or app that offers the possibility to fill in a PDF order form (not necessarily online). The form is 'saved' in **PDF**, completed and sent by the customer to the responding enterprise via the same **website as an attachment to a message**.

e) e-commerce website that offers the possibility to fill in a PDF order form (not necessarily online). The form is 'saved' in **PDF**, completed and sent by the customer to the responding enterprise **by post**.

2.3.2.4. Module C: ICT specialists and skills

Question C1: *Does your enterprise employ ICT specialists?*

ICT specialists are employees for whom ICT is the main job. For example, to develop, operate or maintain ICT systems or applications.

[Scope: all enterprises]

[Type: single answer (i.e. Tick only one); binary (Yes/No)]

The purpose of this question is to measure if enterprises employ ICT specialists, and hence indirectly the incidence of ICT specialist skills. Skills of ICT specialists refer to skills for developing, operating, maintaining ICT systems or applications.

Additionally, ICT specialists have the relevant skills to specify, design, install, support, manage, evaluate or perform research activities. Use of the term ICT specialists: In general, 'specialists' are considered professionals in an ICT field e.g. database specialists, ICT support specialists, etc. Alternatively, the term 'ICT professionals' could be used. It would strictly correspond to the international standard classification of occupations ISCO 08 (25 Information and communications technology professionals, broken down to 251 Software and applications developers and analysts, and 252 Database and network professionals).

However, the scope needs to be wider than the strict definition of ISCO 08 'ICT professionals'. Therefore, in the context of the survey the clarification 'ICT is their main job' is important. In particular because respondents would not have the specific classification in mind when answering the questions. The term 'ICT specialists' is used and includes the following ISCO 08 codes and groups (occupations)[8]:

133 Information and communications technology service managers

2152 Electronics engineers

2153 Telecommunications engineers

2166 Graphic and multimedia designers 2356 Information technology trainers

2434 Information and communications technology sales professionals

25 Information and communications technology professionals

251 Software and applications developers and analysts

252 Database and network professionals

35 Information and communications technicians

[8] For more information on ISCO 08 codes and groups: International Standard Classification of Occupations (ISCO) - ILOSTAT.

351 Information and communications technology operations and user support technicians

352 Telecommunications and broadcasting technicians

7422 Information and communications technology installers and servicers

In the questions on ICT specialists and skills (C1, C2, C3, C4 and C5), reference is made to persons employed by the responding enterprise, excluding those who provide their services as ICT specialists to the responding enterprise on behalf of other enterprise(s), and are either employed by the other enterprise(s) or are self-employed (outsourcing).

Question C2: *Did your enterprise provide any type of training to develop ICT related skills of the persons employed during 2023?*

[Scope: all enterprises]

[Type: single answer per item (i.e. Tick only one); binary (Yes/No); multiple items may be expected]

a) Training for ICT specialists

Tick No if your enterprise didn't employ ICT specialists during 2023.

b) Training for other persons employed

The purpose of this question is to identify enterprises that provide professional ICT training to their persons employed either internally or using external trainers. Training in reply option a) refers exclusively to ICT specialist skills (if the enterprise didn't employ ICT specialists during 2023 respondents should tick 'No'). Reply option b) refers to ICT professional training aiming at developing or upgrading ICT related skills of other persons employed than ICT specialists (skills for the using specific work-related applications (including specialised software tools) or generic software tools).

The intention of the question is to measure the provision of training to develop ICT related skills which does not mean that employees and self-employed persons actually attended the training.

ICT skills refer to skills related for instance to tasks such as: management of online marketing or e-commerce; management of enterprise's social media profile; programming languages; design or management of websites or apps; management of databases or data analysis; maintenance of computer networks, servers, etc.; IT-security or privacy management; use or design of specific software applications; management of telecommunications systems and networks etc. The upper mentioned list of ICT skills refers merely to some prominent examples and is not exhaustive; other ICT related skills depending on the functions and the needs of the enterprise could be included.

Question C3: *Did your enterprise recruit or try to recruit ICT specialists during 2023?*

[Scope: all enterprises]

[Type: single answer (i.e. Tick only one); binary (Yes/No); filter question]

The aim of the current and the next question is to identify mismatches or shortages[9] of ICT specialists in the labour market, in case where the enterprise recruited or tried to recruit personnel for jobs requiring specific ICT specialist skills. If C3 is 'No' respondents should continue with question C6.

Question C4: *During 2023, did your enterprise have vacancies for ICT specialists that were difficult to fill?*

[Scope: enterprises that recruited or tried to recruit ICT specialists during 2023, i.e. C3 = Yes]

[Type: single answer (i.e. Tick only one); binary (Yes/No)]

The aim of this question is to identify mismatches or shortages in the labour market concerning ICT specialist skills when the response to question C3 is 'Yes'. Question C4 refers only to difficulties due to external factors, e.g. general lack of applicants, lack of applicants with skills or experience relevant for the post. The question does not refer to situations when the post could not have been filled due to some internal organisational difficulties during the recruitment.

Question C5: *Did your enterprise have any of the following difficulties to recruit ICT specialists during 2023?*

[Scope: enterprises that recruited or tried to recruit ICT specialists during 2023 and those vacancies were hard to fill, i.e. C3 = Yes and C4 = Yes; optional question]

[Type: single answer per item (i.e. Tick only one); binary (Yes/No); multiple items may be expected]

For enterprises that have had difficulties filling vacancies for jobs requiring ICT specialist skills, the following reasons are collected:

a) Lack of applications

Enterprises tried to recruit personnel with ICT specialist skills and had difficulties because there were not enough candidates/ applicants responding to the vacancy announcement. The difference between a) and b) is that the former refers to quantity (no one applied) and the latter refers rather to quality of applications (there were people applying, but the qualifications related to their education/training were not appropriate).

b) Applicants' lack of relevant ICT related qualifications from education and/or training

Enterprises tried to recruit personnel with ICT specialist skills and had difficulties to find candidates with adequate knowledge in the specific ICT field that the enterprise required (qualifications of candidates were not appropriate for the enterprise). One important element in this item is that it refers to formal education. The difference between a) and b) is that the former refers to quantity (not having people applying) and the latter refers to quality (there were people applying, but the qualifications were not appropriate). This item does not include the work experience as part of the

[9] Mismatch between the job seekers' competencies and those required or expected by the employers. Shortages indicate an insufficient number of skilled people in the labour market due to not enough job seekers, too low pay rates offered for them, low unemployment.

skill of the candidates.

c) Applicants' lack of relevant work experience

Enterprises had difficulties in recruiting a person for a job vacancy because there were not enough candidates with the required work experience needed for the vacant position.

d) Applicants' salary expectations too high

Enterprises found difficulties in recruiting personnel with ICT specialist skills, because the remuneration expected by the candidates was more than what the enterprise was willing to offer for the specific vacant position.

As the list does not include all reasons in having difficulties when recruiting ICT specialists, it is possible that none of the items a) to d) is ticked 'Yes'.

Question C6: *Who performed your enterprise's ICT functions in 2023? (e.g. maintenance of ICT infrastructure; support for office software; development or support of business management software/systems and/or web solutions; security and data protection)*

[Scope: all enterprises]

[Type: single answer per item (i.e. Tick only one); binary (Yes/No); multiple items may be expected]

a) own employees (incl. those employed in parent or affiliate enterprises)

b) external suppliers

The aim of this question is to obtain information who performed ICT functions in the enterprise.

Note that 'external supplier'[10] refers to any other enterprise that performed the ICT functions like the examples mentioned in brackets under the question. Affiliates and parent enterprises are not considered as external suppliers.

ICT functions refer to:

- Maintenance of ICT infrastructure – software components that are necessary to operate the hardware (e.g. operating system software, drivers for peripherals, network software), as well as updating and upgrading system related software. Maintenance functions also include the repair of broken equipment, and the extension of existing infrastructure in terms of hardware and subsequently of the necessary software.
- Providing support for office software – software dedicated to producing documents, presentations, worksheets, graphs, charts, etc. Support in the context of office automation software might for example refer to installation of office software, training on the effective use of the software, granting access to users for databases.
- Development of business management software/systems – e.g. ERP — enterprise resource planning used to manage resources by sharing information among different functional areas

[10] This would be consistent with the global value chain methodology - Business Statistics Framework for Global Value Chains - 2024 edition (europa.eu).

such as accounting, planning, production, marketing; CRM software application for managing information about customers; human resources information management. Development includes providing tailor-made, customised solutions for the responding enterprises as well as customisation of purchased, packaged, off-the-shelf software (e.g. SAP) or software that is provided as a service over the cloud. In addition, all types of maintenance (corrective, adaptive, perfective and preventive) of software solutions due to business evolution — e.g. correction of bugs, updating and extension of software functionalities, customisation and adjustment is part of development. The development/purchase of custom-made (ad hoc solutions) 'business management software/systems' is also in the scope as 'developed by external supplier'.

- Support for business management software/systems – e.g. ERP, CRM, HR, databases. Support refers to providing advice for solving problems, for improving the effectiveness of the use of the solutions, training, etc.
- Development of web solutions – e.g. development of own enterprise's website, apps, e-commerce solutions, etc. Ad hoc programming or introducing links to 'social media on an enterprise's website should be considered as development, extending the existing functionalities of the website.
- Development of web applications is a process which may include: defining goals and audience scope, technology selection, developing of technical architecture and structure, third party vendors analysis and selection (e.g. SSL certificate, payment gateway) designing layout and interface, quality, security and usability testing.

  There are two types of web applications: the browser based and the client-based web applications. The former are web applications accessible over a webpage (html and Javascript) and run within the web browser. The latter are installed on a device (e.g. smartphone), run without going through a browser but they use web protocols. Having such apps an enterprise may for example manage its database, use it to scale and automate business growth for sales, marketing, operations (business automation). Using such apps allows a customer to place an order or buy credit (e-commerce) for example.
- Support for web solutions – e.g. support of own enterprise's websites, apps, e-commerce solutions, excluding hosting your enterprise's website. Support for 'web solutions' includes assistance to update e.g. lists of products, prices and other information of the enterprise, but not extending functionalities of the web solution. Providing support in the context of e-commerce marketplaces could be considered in the cases of training, provision of instructions and guidelines, etc. Hosting an enterprise's website (i.e. provision of servers, computers or computer space, and software tools) or providing the infrastructure by an e-commerce platform is not considered to be on its own merits 'support for web solutions'. ICT security and data protection – e.g. security testing, training on security, resolving ICT security incidents, excluding upgrades of pre-packaged software.

Note: Due to the different observation periods, the case where C1 is ticked "No" and C6a is ticked "Yes" or C1 is ticked "Yes" and C6a is ticked "No" should not be treated as an error without verification.

In cases where no one carries out the mentioned functions, the enterprise should tick "No" to both questions. Answering "Yes" to both questions is possible, in case that the ICT functions are carried out by both own employees and by external suppliers.

2.3.2.5. Module D: ICT Security

[Scope: enterprises with access to the internet, i.e. if A1>0]

ICT security means measures, controls and procedures applied on enterprise's ICT systems to ensure integrity, authenticity, availability and confidentiality of enterprise's data and systems. Security risks, measures implemented and incidents experienced can be classified by the following conceptual approaches:

- Confidentiality, Integrity, and Availability (CIA) Triad with the attributes
  - **Confidentiality** (limited access, restrictions of access to information to prevent the disclosure of information to unauthorized individuals or systems).
  - **Integrity** (the assurance of the accuracy and consistency of systems and data). Integrity is a pre-requisite for ensuring confidentiality. Without it, encryption is useless.
  - **Availability** (making sure that the computing systems, the security controls, and the communication channels are functioning correct) and newly added non-repudiation or authenticity[11].
- "3 A" basic security tools: Authentication, Authorisation, Auditing.
- Maturity scale: enterprises which are conscious of security risks; enterprises which have implemented some security measures; enterprises which ensure regular maintenance and updates of security policy and measures.

Question D1: *Does your enterprise apply any of the following ICT security measures on its ICT systems?*

[Scope: enterprises with access to the internet, i.e. if A1>0]

[Type: single answer per item, i.e. Tick only one; binary (Yes/No); multiple items may be expected]

The intention of the questions is to find out what security measures enterprises apply on their own ICT systems in order to ensure integrity, authenticity, availability and confidentiality of data and information systems. This includes also security measures on operational technologies, meaning the use of IT (hardware and software) to monitor and control physical processes, devices and infrastructure (including purchased and provided by external suppliers).

The answer options reflect the information system-related security facilities: identification, authorisation and authentication.

Identification is the ability to identify uniquely a user of a system or an application that is running in the system. It is the process of associating a user with something that has occurred on a server, on a network, or with some other resource. This information is almost always logged. Usually, it is a username or some type of very unique identifier assigned to that particular function.

Authentication is the process that determines whether a client really is who they say they are.

[11] Information security

Authentication can be done with the help of passwords (authentication by knowledge), or with additional devices, such as smart cards, hardware tokens or identity cards (authentication by ownership). The last possibility would be authentication by characteristics, i.e. using biometrical authentication, such as fingerprints scanner or retina patterns. A strong identification is defined by at least a combination of two authentication methods, e.g. passwords and smart cards. The authentication methods can be classified in: a) Static passwords (These do not changed unless they expire or user changes them); b) One-time password (OTP) such as Personal Identification Numbers (PINs) delivered through SMS texts or otherwise c) Digital certificates (e.g. x.509 and such); d) Biometric credential.

Authorization is the process that determines what an authenticated client can and cannot do on a network. In general, identification and authentication of users are used in the context of authorisation.

a) Authentication via strong password (e.g. minimum length, use of numbers and special characters, changed periodically)

The strong password may have one or several requirements concerning for instance a minimum length (specified minimum number of characters), characters to be a mix of uppercase, lowercase alphanumeric and special characters, forcing users to change passwords periodically, passwords to be transmitted and stored in encrypted form. The described conditions follow the ISO norm 9594-1.

Note: In scope is the usage of the authentication via strong password for authentication to any ICT system or software of the enterprise, including remote access (e.g. the enterprise's network, business software, any other applications).

b) Authentication via biometric methods used to access the enterprise's ICT system (e.g. authentication based on fingerprints, voice, face)

Authentication via biometric methods involves using some part of a person's physical or behavioural characteristics to authenticate them. The biometric authentication can be done based on for instance a fingerprint, an iris scan, a retina scan, voice recognition, face recognition or some other physical characteristic. A single characteristic or multiple characteristics could be used, depending on the infrastructure and the level of security desired. With biometric authentication, the physical characteristic being examined is usually mapped to a username. This username is used to make decisions after the person has been authenticated. In some cases, the user must enter the username when attempting to authenticate; in others, a lookup is done on the biometric sample in order to determine the username.

Biometric authentication is performed by doing a comparison of the physical aspect a person presents for authentication against a copy that has been stored. For example, one would place their finger on a fingerprint scanner for comparison against the stored sample. If the fingerprint matches the stored sample, then the authentication is considered to be successful.[12]

Note: Usage of biometric methods for purpose not related with ICT systems and hardware protection (e.g. access for enterprises premises not related to IT) is not in the scope.

[12] Biometric Authentication - an overview | ScienceDirect Topics

In scope is the usage of the authentication via biometric methods to access the enterprise's ICT system or software, including remote access (e.g. the enterprise's network, business software, any other applications).

c) Authentication based on a combination of at least two authentication mechanisms (i.e. combination of e.g. user-defined password, one-time password (OTP), code generated via a security token or received via a smartphone, biometric method (e.g. based on fingerprints, voice, face))

This answer option covers authentication methods where identification is defined by a combination of at least two authentication methods.

Note: In scope is the usage of authentication based on a combination of at least two authentication mechanisms to access any ICT system or software of the enterprise, including remote access (e.g. network, business software, any other applications).

d) Encryption of data, documents or e-mails

Encryption is the process of encoding messages or information in a way that only authorized persons can access it. Encryption does not of itself prevent interference but denies the message content to the interceptor. In an encryption scheme, the intended information or message, referred to as plaintext, is encrypted using an encryption algorithm, generating cipher text that can be read only after decryption. An authorized recipient can easily decrypt the message with the key provided by the originator to recipients, but not to unauthorized interceptors. The purpose of encryption is to ensure that only somebody who is authorized to access data (e.g. a text message or a file), will be able to read it, using the decryption key. Somebody who is not authorized can be excluded, because he or she does not have the required key, without which it is impossible to read the encrypted information.

e) Data backup to a separate location (including backup to the cloud)

Offsite data backup is part of the off-site data protection strategy of sending critical data from the main site to another location by means of removable storage media, e.g. magnetic type, external hard-disks, or electronically via remote backup services. Offsite data protection is normally part of a contingency plan that describes disaster recovery actions in case of security incidents. Included is the backup of data in the cloud (enterprises purchasing cloud computing services).

f) Network access control (management of user rights in enterprise's network)

Access control to enterprise network enables access and enforces security policy based on the state of the computer and the identity of the user.

Network access control (NAC), also called network admission control, is a method of bolstering the security of a proprietary network by restricting the availability of network resources to endpoint devices that comply with a defined security policy[13]. For example, if a connecting system

[13] Explanation | CyberSecurity Training | www.cybertraining365.com

doesn't have the standard corporate anti-virus package, the user should get a different access control policy than if everything is installed and all the signatures are up-to-date[14].

Note: User rights outside of the enterprise's IT system are out of scope.

g) VPN (Virtual Private Network extends a private network across a public network to enable secure exchange of data over public network)

A virtual private network (VPN) extends a private network across a public network, such as the Internet. It enables users to send and receive data across shared or public networks as if their computing devices were directly connected to the private network. Applications running across the VPN may therefore benefit from the functionality, security, and management of the private network. VPNs may allow employees to securely access a corporate intranet while located outside the office. They are used to securely connect geographically separated offices of an organization, creating one cohesive network.

h) ICT security monitoring system used to detect suspicious activity (e.g. intrusion detection or prevention system that monitors users' or devices' behaviour, network traffic)

Please exclude antivirus software and default firewall solution included in the operating system of personal computers and routers.

The answer option covers ICT security monitoring system used to detect suspicious activity, except antivirus software and default firewall solution included in the operating system of personal computers and routers. The answer option covers using e.g. ICT intrusion detection methods such as Next Generation Intrusion Prevention System (NGIPS), Next Generation Firewall (NGFW), or Intrusion detection system (IDS).

Example of the **Next Generation Intrusion Prevention System (NGIPS)** can be the NSFOCUS Next Generation Intrusion Prevention System provides advanced multi-stage AI analysis for detection and mitigation of tomorrow and beyond unknown and zero-day advanced persistent threats (APTs). NSFOCUS goes beyond signature and behaviour-based detection, using cutting edge Intelligent Detection advanced intelligence heuristics learning technology for network and application threat detection. NGIPS also combines AI with state-of-the-art threat intelligence to detect malicious sites and botnets. An optional virtual sandboxing capability can be added to the NGIPS system using the NSFOCUS Threat Analysis System. The TAS uses multiple innovative detection engines to identify known and zero-day APTs, including IP reputation engines, anti-virus engines, static and dynamic analysis engines and virtual sandbox execution mimicking live hardware environments.[15]

A **next-generation firewall (NGFW)** is a part of the third generation of firewall technology, combining a traditional firewall with other network device filtering functions, such as an application firewall using in-line deep packet inspection (DPI), an intrusion prevention system (IPS). Other techniques might also be employed, such as TLS/SSL encrypted traffic inspection, website filtering, QoS/bandwidth management, antivirus inspection and third-party identity management integration (i.e. LDAP, RADIUS, Active Directory). NGFWs include the typical functions of traditional

[14] What is NAC anyway? | Network World
[15] Next Generation Intrusion Prevention (NGIPS) - NSFOCUS, Inc., a global network and cyber security leader, protects enterprises and carriers from advanced cyber attacks. (nsfocusglobal.com)

firewalls such as packet filtering, network- and port-address translation (NAT), stateful inspection, and virtual private network (VPN) support. The goal of next-generation firewalls is to include more layers of the OSI model, improving filtering of network traffic that is dependent on the packet contents.

NGFWs perform deeper inspection compared to stateful inspection performed by the first- and second-generation firewalls. NGFWs use a more thorough inspection style, checking packet payloads and matching signatures for harmful activities such as exploitable attacks and malware.[16]

An **intrusion detection system (IDS; also intrusion protection system or IPS)** is a device or software application that monitors a network or systems for malicious activity or policy violations. Any intrusion activity or violation is typically reported either to an administrator or collected centrally using a security information and event management (SIEM) system. A SIEM system combines outputs from multiple sources and uses alarm filtering techniques to distinguish malicious activity from false alarms.

IDS types range in scope from single computers to large networks. The most common classifications are network intrusion detection systems (NIDS) and host-based intrusion detection systems (HIDS). A system that monitors important operating system files is an example of an HIDS, while a system that analyses incoming network traffic is an example of a NIDS. It is also possible to classify IDS by detection approach. The most well-known variants are signature-based detection (recognizing bad patterns, such as malware) and anomaly-based detection (detecting deviations from a model of “good” traffic, which often relies on machine learning). Another common variant is reputation-based detection (recognizing the potential threat according to the reputation scores). Some IDS products have the ability to respond to detected intrusions. Systems with response capabilities are typically referred to as an intrusion prevention system. Intrusion detection systems can also serve specific purposes by augmenting them with custom tools, such as using a honeypot to attract and characterize malicious traffic.[17]

i) Maintaining log files that enable analysis after ICT security incidents

Information system application activities or user activities can be logged and used for analysis in case of security incidents in order to take appropriate actions to prevent these kinds of incidents in future or to quantify occurred damage.

Note: Activities relate to computer activities.

j) ICT risk assessment, i.e. periodical assessment of probability and consequences of ICT security incidents

Traditional risk assessment includes general ICT-related issues such as accidental outages, hardware failures, and uptime. There are basically three risk management components:

- Evaluation and assessment, to identify assets and evaluate their properties and characteristics.
- Risk assessment, to discover threats and vulnerabilities that pose risk to assets.

[16] Next-generation firewall - Wikipedia
[17] Intrusion detection system - Wikipedia

- Risk mitigation, to address risk by transferring, eliminating or accepting it. ICT risk assessments are performed to allow enterprises to assess, identify and modify their overall ICT security.

k) ICT security tests (e.g. performing penetration tests, testing security alert system, review of security measures, testing of backup systems)

The intention of ICT security tests it to test established ICT security measures. Penetration tests for example can help determine whether a system is vulnerable to attack, if the defences were sufficient, and which defences (if any) the test defeated.

Question D2: *Does your enterprise make persons employed aware of their obligations in ICT security related issues in the following ways?*

[Scope: enterprises with access to the internet, i.e. if A1>0]

[Type: single answer per item, i.e. Tick only one; binary (Yes/No); multiple items may be expected]

This question should provide information on methods applied by enterprises to raise awareness on ICT security issues among their persons employed. The reply options distinguish between obligatory and non-obligatory methods of awareness raising. In addition, there is a distinction between legally binding and non-binding commitment. Enterprises might apply different levels of obligations, e.g. between staff working in IT departments and other staff. In this case, the options should refer to all staff.

a) Voluntary training or internally available information (e.g. information on the intranet)

This option covers information offers that can be followed or retrieved by persons employed voluntarily, e.g. ICT security information on the intranet of an enterprise or as information leaflets. Information can be also provided by enterprise ICT department.

b) Compulsory training courses or viewing compulsory material

This option covers compulsory trainings or presentations on the information security policy, precautions or principles. The difference between training and compulsory material (which can be a presentation) lies in the degree of interactivity or active involvement of the trainees. The option includes eLearning techniques with control of participation. Positive replies would indicate a higher level of awareness of the enterprise compared to training on staff initiative.

c) By contract (e.g. contract of employment)

This option focuses on the legal aspects of information security policy. Persons employed could be informed and bound to the information security policy by contract or letter of appointment. Usually, this would be accompanied by complementary actions such as handover of an information document or participation in training activities. The purpose of the formulation of this item is to point to the contract of employment as primary source of making staff aware via contracts. At the same time, other types of contracts, which e.g. might specifically aim at ICT security obligations, should not be excluded from the replies of the enterprises.

Question D3: *Does your enterprise have document(s) on measures, practices or procedures on ICT security?*

(Documents on ICT security and confidentiality of data cover employee training in ICT use, ICT security measures, the evaluation of ICT security measures, plans for updating ICT security documents, etc.)

[Scope: enterprises with access to the internet, i.e. if A1>0]

[Type: one single answer needed, i.e. Tick only one; binary (Yes/No); filter question]

ICT security document(s) refer to document(s) describing how an enterprise plans to maintain security and confidentiality of data. Documents can describe how employees will be trained in the safe usage of ICT, which security measures will be implemented and carried out, procedures for evaluating the effectiveness of ICT security measures, plans for updating the ICT security document(s), etc. Enterprises can have one or more documents regarding ICT security. The measures, practices or procedures should be enforced in the enterprise. The document(s) can be in a paper or electronic form.

The existence of document(s) on measures, practices or procedures on ICT security (an ICT security policy) in an enterprise means that the enterprise is aware of the importance of its ICT and the risks related to them. The enterprise has made the effort to formulate a document(s) by identifying the systems, their risks and how to deal with these risks. Focus is on mentioned measures, practices or procedures on ICT security that are actually applied. A proxy for such kind of document(s) is the term “policy formally defined”, i.e. policy that is documented and adopted by the enterprise. Normally, such document(s) on ICT security contains the objectives and aims of measures, practices or procedures. The document should contain or make reference to definitions of the relevant terms related to ICT security. In addition, it should contain documentation or make reference to documentation of the enterprise information systems. A main component would be the assessment of security risks in terms of likelihood of occurrence of incidents and their possible impact on the operations of the enterprise. Ideally, the document should distinguish between different groups of actors and their responsibilities and functions in relation to information system operations and incident handling. Finally, the document(s) should describe security controls and measures as well as contingency planning in case of ICT security incidents.

The degree of documentation might depend on the size of the enterprise and on the results of an ICT security risk assessment. In addition, the document(s) on ICT security might not cover all the above mentioned aspects. It may be possible that enterprises (e.g. SMEs) don’t employ ICT specialists and thus outsource various ICT functions. These external service providers may have such documents or ‘formally defined ICT security policy’ that addresses different risks and which goal is to protect ICT infrastructure of their business partners.

Concerning “outsourcing” (e.g. in the case of using cloud computing services): an enterprise should report having document(s) on ICT security when a) it is the ICT security policy, formally defined in document(s) on ICT security by the enterprise, or b) the enterprise has a service provision contract or a formal agreement that foresees issues related to the enterprise’s ICT security concerning the risks to be addressed and the standards to be followed. In the latter - case (b) - the ICT security policy of the external supplier of ICT services (e.g. provider of cloud computing services, subcontractor for ICT functions related to maintenance, development, or support) is directly applied in order to protect the enterprise against ICT security risks according to certain standards. In these cases, enterprises that benefit from the security policy of their service provider should answer with ‘Yes’ to D3 and the following question.

The existence of such documents in an enterprise means that the enterprise is aware of the

importance of its ICT and the risks related to them. It is assumed that the existence of documents on ICT security and the frequency of reviewing these documents is positively correlated to the readiness of the enterprises to report ICT security incidents.

Question D4: *When was the document(s) on measures, practices or procedures on ICT security defined or most recently reviewed?*

(Documents on ICT security and confidentiality of data cover employee training in ICT use, ICT security measures, the evaluation of ICT security measures, plans for updating ICT security documents, etc.)

[Scope: enterprises which have document(s) on measures, practices or procedures on ICT security, i.e. D3 = Yes]

[Type: one single answer needed, i.e. Tick only one; optional question]

This question should provide information on the latest period in time that the document(s) on ICT security was reviewed/defined according to the following responses:

a) within the last 12 months,

b) more than 12 months and up to 24 months ago and

c) more than 24 months ago.

Question D5: *During 2023, did your enterprise experience any ICT related security incident leading to the following consequences?*

[Scope: enterprises with access to the internet, i.e. if A1>0]

[Type: single answer per item, i.e. Tick only one; binary (Yes/No); multiple items may be expected]

This question concerns information communication system security related incidents. Enterprises might be reluctant to report on these types of incidents as this is likely to be considered as sensitive information. Therefore it would be important to include a strong statement on data confidentiality when submitting this questionnaire to the enterprise. Moreover, users consider statistics on security related incidents as very important information for policy purposes. The number of questions on this sensitive issue is kept to the absolute minimum and no quantitative information is asked. The options are built around the general elements of information security: availability, integrity and confidentiality.

a) Unavailability of ICT services due to hardware or software failures

This answer option covers cases when the ICT services of the enterprise are unavailable due to failure of the hardware or software.

b) Unavailability of ICT services due to attack from outside, e.g. ransomware attacks, Denial of Service attacks

The distinction of this answer option from item a) is the cause of the unavailability of the ICT services, which in this instance is caused by malicious attacks from outside.

Attacks from outside the enterprise can lead to unavailability of services. A typical example would be a distributed denial of service (DDoS) attack that attempts to make an information system resource unavailable to its intended users. A common method for DDoS attacks involves saturating the target information system with external communication requests so that it cannot respond adequately to legitimate requests.

c) Destruction or corruption of data due to hardware or software failures

This option covers issues of data integrity related to unintended incidents caused by hardware or software failures. Unintended incidents could be crashes of servers or hard disks due to hardware failures or crashes of servers due to software failures, e.g. erroneous updates.

d) Destruction or corruption of data due to infection of malicious software or unauthorised intrusion

This answer option covers issues of data integrity related to attacks with a malicious purpose. In addition, this option comprises as well failures of data integrity due to unauthorised access (internal or external intrusion) to the ICT system of an enterprise. An intrusion is an attempt to bypass security controls on an information system. Means of intrusion can be eavesdropping, viruses, worms, Trojan horses, logic or time bombs, brute force attacks, etc.

e) Disclosure of confidential data due to intrusion, pharming, phishing attack, intentional actions by own employees

This answer option refers only to cases of disclosure of confidential data due to malicious actions.

The data can be information on persons, staff or clients, intellectual property in terms of trade secrets, information or other confidential information such as e.g. business numbers or information on managerial decisions. A possible way for getting access to confidential data could be brute force attacks for disclosing passwords or phishing and pharming techniques.

Phishing is a criminally fraudulent attempt to acquire sensitive information such as usernames, passwords and credit card details by masquerading as a trustworthy entity in an electronic communication. The term “pharming” connotes an attack to redirect the traffic of a website to another, bogus website in order to acquire sensitive information.

Intellectual property is defined as “intangible property that is the result of creativity”. These assets can be artistic or commercial. Intellectual property can be protected by law. Types of legal protection include copyrights, patents and trademarks. Under intellectual property law, owners are granted certain exclusive rights to intangible assets. In this case, intellectual property is accessible by third parties but only the owner may exploit the assets. In case of unauthorised exploitation, the right holder can initiate legal actions.

Trade secrets are intellectual properties that are not legally protected. “A trade secret is a formula, practice, process, design, instrument, pattern, or compilation of information which is not generally known or reasonably ascertainable, by which a business can obtain an economic advantage over competitors or customers. In some jurisdictions, such secrets are referred to as ‘confidential information’ or ‘classified information’”. The option only refers to intellectual property in terms of trade secrets as they are not protected by law and their disclosure would have negative effects on the economic performance of an enterprise.

f) Disclosure of confidential data due to unintentional actions by own employees

This answer option refers only to cases of disclosure of confidential data was caused by unintentional actions of enterprise's own employees.

Note: 'No' to all items is possible in cases where none of the mentioned consequences were experienced due to any ICT related security incident.

2.3.2.6. Module E: Artificial Intelligence

[Scope: enterprises with access to the internet, i.e. A1 > 0]

Artificial intelligence refers to systems that use technologies, such **as text mining, computer vision, speech recognition, natural language generation, machine learning, and deep learning**, to gather and/or use data to predict, recommend or decide, with varying levels of autonomy, the best action to achieve specific goals.

Artificial intelligence systems **can be purely software-based**, e.g.:

- chatbots and business virtual assistants based on natural language processing;
- face recognition systems based on computer vision or speech recognition systems;
- machine translation software;
- data analysis based on machine learning;

or **embedded in devices**, e.g.:

- autonomous robots for warehouse automation or production assembly works;
- autonomous drones for production surveillance or parcel handling.

Question E1: *Does your enterprise use any of the following artificial intelligence technologies?*

Scope: enterprises with access to the internet, i.e. if A1>0]

[Type: single answer per item (i.e. Tick only one); binary (Yes/No); multiple items may be expected; filter question]

The aim of this question is to provide information on the use of artificial intelligence technologies used by enterprises, e.g. text mining, speech recognition, natural language generation. The answer options **a) to g)** may not be an exhaustive list of artificial intelligence technologies but constitute the most prevalent AI technologies that enterprises use nowadays.

The question is about use. So, if an enterprise creates AI solutions for others to use, i.e. 'for sale', then it should not answer 'Yes' to E1. However, if an enterprise creates AI solutions and uses them for their business, then it should answer 'Yes' to the relevant answer options in E1.

The responding enterprise is asked to choose among the following AI technologies; multiple answers are allowed:

a) AI technologies performing analysis of written language (e.g. text mining)

This answer option refers to the use of AI technologies performing analysis of written language, also referred to as text mining.

Text mining, also known as text analysis, is the process of transforming unstructured text data into meaningful and actionable information, identifying facts, relationships and assertions that would otherwise remain buried in the mass of unstructured or low-structured textual big data. Once extracted, this information is converted into a structured form that can be further analysed, or presented directly using clustered HTML tables, mind maps, charts, etc. Text mining employs a variety of methodologies to process the text, one of the most important of these being Natural Language Processing (NLP)[18].

Natural language processing – a subfield of computer science, information engineering and artificial intelligence concerned with the interactions between computers and human (natural) languages, in particular how to program computers to process and analyse large amounts of natural language data.

Examples for the usage of technologies performing analysis of written language (text mining):

- Social media: Text mining software packages are available for analysing social media applications to monitor and analyse the online plain text from internet news, blogs, email, etc. Text mining tools help to identify and analyse number of posts, likes and followers on the social media. This kind of analysis show people's reaction to different posts, news and how they spread around. It shows the behaviour of people belonging to specific age group or communities having similarity and variation in views about the same post.

- Business intelligence: Text mining plays a significant role in business intelligence that helps organizations and enterprises analyse their customers and competitors in order to take better decisions. It provides a deeper insight about business and gives information how to improve the customer satisfaction and gain competitive advantages. By bringing text mining into the equation, business intelligence (BI) tools are able to leverage unstructured data in addition to structured data and expand the data sets which model uses to generate business insight. The text mining tools like IBM text analytics, Rapid miner, GATE help take decisions about the organization that generate alerts about good and bad performance, market changeover that help to take remedial actions. It also helps in telecommunication industry, business and commerce applications and customer chain management system.[19] BI can uses AI or not. However, only BI using AI is in the scope.

- Risk Management: No matter the industry, insufficient risk analysis is often a leading cause of failure. Adoption of risk management software based on text mining technology can dramatically increase the ability to mitigate risk. This provides enterprises with easy access to thousands of sources (and petabytes) of text documents, enabling them to link together information and access the right information at the right time.

- Knowledge Management: When managing large volumes of text documents, ability to find important information quickly is essential. Text mining solutions can help enterprises locate

[18] NIST Cloud Computing Security Reference Architecture
[19] Text Mining Techniques Applications and Issues

information quickly and accurately, enabling them to unlock deeper insights and deliver products to the market faster.

- Cybercrime Prevention: The anonymous nature of the internet and the many communication features operated through it contribute to the increased risk of internet-based crimes. Text mining is making cybercrime prevention easier for enterprises by establishing more context around the intelligence they are being fed. This enables them to pinpoint real threats and limit the number of false positives created by keywords taken out of context.

- Enhanced Customer Service: Text mining and natural language processing have proven immensely helpful for customer care teams. They are commonly adopted to improve the customer experience by leveraging valuable information sources such as surveys, trouble tickets and customer call notes to improve the quality, effectiveness and speed of problem resolution. Text analysis is also central to chatbots as it captures the information necessary to provide a rapid, automated response to the customer. As a result, enterprises can dramatically reduce their reliance on call centre operators to solve customer problems.

- Contextual Advertising: Digital advertising is a moderately new and growing field of application for text analytics. Compared to the traditional cookie-based approach, contextual advertising analyzes the text on a webpage to understand the content on a deeper level. For example, someone reading an article about the top new book releases may be served an ad for Kindle due to its relevance. This is not only beneficial in terms of serving more accurate and targeted advertisements, but it completely preserves user privacy.

- Spam Filtering: Email is an effective, fast and reasonably cheap way to communicate, but it comes with a dark side: spam. Today, spam is a major issue for internet service providers, increasing their costs for service management and hardware software updating. For users, spam is both an entry point for viruses and a detriment to productivity. Text mining techniques can be implemented to improve the effectiveness of statistical filtering methods by leveraging established prior knowledge. This not only makes for more efficient email management, but it improves the user experience considerably.[20]

b) AI technologies converting spoken language into machine-readable format (speech recognition)

This answer option refers to the use of AI technologies converting spoken language into machine-readable format, also referred to as speech recognition.

Speech recognition is an interdisciplinary subfield of computer science and computational linguistics that develops methodologies and technologies that enable the recognition and translation of spoken language into text by computers. It is also known as automatic speech recognition (ASR), computer speech recognition or speech to text (STT). It incorporates knowledge and research in the computer science, linguistics and computer engineering fields.

Natural language processing – is a subfield of computer science, information engineering and artificial intelligence concerned with the interactions between computers and human (natural)

[20] Text Mining Examples to Leverage Right Now

languages, in particular how to program computers to process and analyse large amounts of natural language data.

Although the science of speech recognition dates many decades back it has benefited substantially by AI. AI based speech recognition employs the latest machine learning advancements. These are high-learnability neural network algorithms and rich lexicons that enable fine recognition accuracy.

Examples for the usage of technologies converting spoken language into machine-readable format (speech recognition):

- Speech recognition is used in call centre, where it handles incoming customer calls.
- Digital assistants use speech recognition: Google Voice, Amazon Alexa, Microsoft Cortana, and Apple's Siri.

c) AI technologies generating written or spoken language (natural language generation, speech synthesis)

This answer option refers to the use of AI technologies generating written or spoken language, also referred to as natural language generation.

Natural language generation – is the ability of a computer program to convert structured data into natural language representation. It can be used to produce long documents that summarize or explain the contents of computer databases, for example generating technical manuals, generating product descriptions for a large e-commerce site, making news reports (automated journalism) or summarizing medical records. It can also be used to generate short blurbs of text in interactive conversations (a chatbot) which might even be read out loud by a text-to-speech system (artificial production of human speech).

Examples of the usage of Technologies generating written or spoken language (natural language generation):

- Chatbot: A chatbot is an AI software that can simulate a conversation (or a chat) with a user in natural language through messaging applications, websites, mobile apps or through the telephone. Formulating responses to questions in natural language is one of the most typical examples of Natural Language Processing applied in various enterprises' end-use applications.

d) AI technologies identifying objects or persons based on images or videos (image recognition, image processing)

This answer option refers to the use of AI technologies identifying objects based on images, also referred to as image recognition or image processing. Image processing and image recognition are examples of computer vision application.

Computer vision[21] is an interdisciplinary scientific field that deals with how computers can gain high-level understanding from digital images or videos. From the perspective of engineering, it seeks to understand and automate tasks that the human visual system can do. Computer vision tasks include methods for acquiring, processing, analysing and understanding digital images, and extraction of high-dimensional data from the real world in order to produce numerical or symbolic information, e.g. in the forms of decisions. The scientific discipline of computer vision is concerned with the theory behind artificial systems that extract information from images. The image data can take many forms, such as video sequences, views from multiple cameras, multi-dimensional data from a 3D scanner or medical scanning device. Sub-domains of computer vision include scene reconstruction, event detection, video tracking, object recognition, face recognition, motion estimation, and image restoration.

Examples of the usage of Technologies identifying objects or persons based on images (image recognition, image processing):

- User identification and authentication via biometric methods implemented by the enterprise (e.g. based on fingerprints, faces).
- Drones equipped with image recognition capabilities can provide vision-based automatic monitoring, inspection and control of the assets located in remote areas.
- In manufacturing – inspecting production lines, evaluating critical points on a regular basis within the premises. Monitoring the quality of final products to reduce defects. Assessing the condition of workers can help manufacturing industries to have a complete control of different activities in the systems. Examples of usage are: predictive maintenance, defect detection, safety (manufacturers are using computer vision to protect their employees by providing the safest working environment they can), assembling products and components (computer vision can assist in assembling products accurately and reducing the amount of time it takes to complete a product build), reading barcodes (computer vision can read barcodes to quickly and easily track components and packages at all stages of development, through a manufacturing facility to departure and beyond. Many manufacturers have begun incorporating barcode into their systems to better route products through the proper assembly lines).[22]
- Autonomous vehicles with image recognition can identify activity on the road and take necessary actions. Mini robots can help logistics industries to locate and transfer objects from one place to another. It also maintains the database of the product movement history to prevent the product from being misplaced or stolen.

e) Machine learning (e.g. deep learning) for data analysis

This answer option refers to the use of AI technologies based on machine learning for data analysis, for instance deep learning.

[21] Computer visio
[22] Computer vision use cases for manufacturing

Machine learning involves ‘training’ a computer model to better perform an automated task, e.g. data analytics. Machine learning uses algorithms whose performance improves as they are exposed to more data over time.

Deep learning is a subset of machine learning in which multi-layered neural networks learn from vast amounts of data.

Neural networks (Artificial neural networks (ANN) or connectionist systems) are computing systems vaguely inspired by the biological neural networks. The neural network itself is not an algorithm, but rather a framework for many different machine learning algorithms to work together and process complex data inputs. Such systems "learn" to perform tasks by considering examples, generally without being programmed with any task-specific rules.

Although neural networks are not explicitly mentioned in this answer option, they are in scope of this item.

Examples of the usage of Machine learning (e.g. deep learning) for data analysis:

- Recommender systems based on machine learning – they are utilized in a variety of areas and are most commonly recognized as playlist generators for video and music services like Netflix, YouTube and Spotify, product recommenders for services such as Amazon, or content recommenders for social media platforms such as Facebook and Twitter.

- Dynamic pricing also referred to as surge pricing, demand pricing, or time-based pricing is a pricing strategy in which businesses set flexible prices for products or services based on current market demands. Enterprises are able to change prices based on algorithms that take into account competitor pricing, supply and demand, and other external factors in the market

- Cyber threat detection – machine learning has become a vital technology for cybersecurity. Machine learning pre-emptively stamps out cyber threats and bolsters security infrastructure through pattern detection, real-time cybercrime mapping and thorough penetration testing. Microsoft uses its own cybersecurity platform, Windows Defender Advanced Threat Protection (ATP), for preventative protection, breach detection, automated investigation and response. Windows Defender ATP IS built into Windows 10 devices, automatically updates and employs cloud AI and multiple levels of machine learning algorithms to spot threats.

f) AI technologies automating different workflows or assisting in decision-making (e.g. AI-based software robotic process automation)

This answer option refers to the use of AI technologies automating different workflows or assisting in decision making, also referred to as Artificial Intelligence based software robotic process automation.

AI based software robotic process automation is process automation based on AI technologies. Robotic process automation (RPA) is a fast-emerging process automation approach that uses software robots to replicate human tasks. After recording a process workflow, a virtual bot mimics

the actions performed by humans in the application's graphical user interface and automates their execution[23].

While process automation is not a new term and could be done without AI, i.e. rules-based automation, this answer option refers only to the cases where AI is used in order to boost the benefits of traditional process automation. So as RPA gets paired with AI disciplines such as natural language processing or computer vision, the possibilities for effective automation grow considerably.

Converging AI with RPA enables businesses to automate more complex, end-to-end processes than ever before, and integrate predictive modelling and insights into these processes (which could for instance even identify new processes to automate[24] to help humans work smarter and faster.[25]

Examples of the usage of technologies automating different workflows or assisting in decision-making (AI-based software robotic process automation):

Automated bots are excellent at carrying out repetitive jobs much faster and with much more precision than human counterparts. A software bot is a programme designed to automate tasks. Typically, these tasks are simple, repetitive and routine. So, a software bot can perform them quicker and more efficiently than a human could. Software bots can take several different forms. For example, one of the most well-known types of software bot today is a chatbot. Other types of bot include web crawler bots and rule-based automation bots.

RPA that is used to work in **conjunction with people** by automating repetitive processes (attended automation) is not within the scope of the question.

g) AI technologies enabling physical movement of machines via autonomous decisions based on observation of surroundings (autonomous robots, self-driving vehicles, autonomous drones)

This answer option refers to the use of AI technologies enabling physical movement of machines via autonomous decisions based on observation of surroundings, such as autonomous robots, self-driving vehicles, autonomous drones.

Only systems that have hardware parts (machine-like) and use artificial intelligence to learn and perform their tasks should be included in this answer option. Pure software robots or robots that automate the production process through a repetitive action without any use of AI technologies are excluded from the scope.

Examples of the usage of technologies enabling physical movement of machines via autonomous decisions based on observation of surroundings (autonomous robots, self-driving vehicles, autonomous drones):

This answer may refer for instance to robots that use machine learning in order to learn how to better perform a task, or drones that are able to choose the best route for example for parcel

[23] Process Mining and Robotic Process Automation: A Perfect Match.pdf
[24] Robotic Process Automation
[25] Robotic Process Automation vs. AI

delivery based on machine learning, or a self-driving vehicle that uses a combination of machine learning and computer vision for driving safely.

Question E2: *Does your enterprise use artificial intelligence software or systems for any of the following purposes?*

[Scope: enterprises that use AI technologies, i.e. E1a) = yes or E1b) = yes or E1c) = yes or E1d) = yes or E1e) = yes or E1f) = yes or E1g) = yes]

[Type: single answer per item (i.e. Tick only one); binary (Yes/No); multiple items may be expected]

Question **E2** is relevant for enterprises that use one of the AI technologies listed in question E1, i.e. enterprises that ticked yes in at least one of the answer options in question E1. The question aims at identifying the purpose of the use of AI technologies in the enterprises. More specifically, the enterprises are asked to reply whether they use AI technologies for the specific purposes by business function, namely marketing or sales, production process, organisation of business administration processes, for management of enterprises, for logistics, for ICT security, for human resources management or recruiting.

Note: As the list of purposes for using AI technologies is not exhaustive, it may occur that each answer category in E2 is ticked 'No'.

a) Use of AI for marketing or sales

some of the examples may be:

- **customer profiling, price optimisation, personalised marketing offers, market analysis based on machine learning**
- **chatbots based on natural language processing for customer support**
- **autonomous robots for order processing**

This answer option is related to the use of AI technologies for marketing or sales purposes.

They may refer to chatbots that are able to analyse written language based on natural language processing and provide real-time customer support, e.g. by answering to customer questions and needs as a real person (preprogrammed chatbots that are not able to learn and only respond to specific predefined requests should be excluded from the scope). Other examples that can be included in this category are software programmes that can perform customer profiling, or recommender systems that can propose personalised marketing offers and the knowledge they have acquired about the enterprises' target market based on neural networks (e.g. through online advertisement or even phone notifications when the customer approaches a specific shop) or artificial intelligence software programmes that are able to perform market analysis in order to guide and steer the marketing decisions of the enterprises.

b) Use of AI for production or service processes

some of examples may be:

- **predictive maintenance or process optimisation based on machine learning**
- **tools to classify products or find defects in products based on computer vision**
- **autonomous drones for production surveillance, security or inspection tasks**
- **assembly works performed by autonomous robots**

This answer option is related to the use of AI technologies for production or service purposes.

AI can boost the automation of the production line by integrating industrial robots into the workflow and teaching them to perform labour-intensive, dangerous or mundane tasks, thereby improving productivity while maintaining quality and safety. Examples include autonomous robots that can perform assembly works or autonomous drones that perform inspection and surveillance tasks thanks to computer vision and machine learning, or systems that are able to classify products or find defects thanks to computer vision, or to ensure predictive maintenance for assessing the state of machinery and/or prevention of equipment failure thanks to algorithms based on machine learning and neural networks.

Predictive maintenance uses condition-monitoring equipment to evaluate an asset's performance in real time. A key element in this process is the internet of things (IoT). IoT allows for different assets and systems to connect, work together, and share, analyse and action data. IoT relies on predictive maintenance sensors to capture information, make sense of it and identify any areas that need attention.

Some examples of using predictive maintenance and predictive maintenance sensors include vibration analysis, oil analysis, thermal imaging, and equipment observation.

c) Use of AI for organisation of business administration processes or management

some of the examples may be:

- **business virtual assistants based on machine learning and/or natural language processing, e.g. for document drafting**
- **data analysis or strategic decision making based on machine learning, e.g. risk assessment, based on machine learning**
- **planning or business forecasting based on machine learning**
- **human resources management based on machine learning or natural language processing, e.g. candidates pre-selection screening, employee profiling or performance analysis**

This answer option is related to the use of AI technologies for any business administration processes or management of enterprises, including human resources management.

They may refer to business virtual assistants that are able to perform secretarial and assistance tasks based on machine learning and natural language processing and generation. They also may refer to voice-to-text conversion systems which are based on speech recognition and can produce written records (e.g. meeting minutes). Other systems may include automated planning or scheduling based on machine learning and any other intelligent automation of workflows within the enterprises that is based on AI software-based process automation.

They may refer to business software or systems based on AI algorithms (e.g. neural networks) that are able to analyse data and help board members to take investment and other corporate decisions in the boards of enterprises. The business software or system may also be able to facilitate the distribution of information to shareholders via AI software robotic process automation or even ensure representation and voting by shareholders in general meetings of the enterprise (through natural language generation). Other examples include intelligent systems that are used for sales and/or business forecasting or risk assessment based on machine learning.

The human resources management based on machine learning or natural language processing may refer to systems that are able to do candidate preselection screening and automation of recruiting based on text mining and machine learning, or to automate the recruiting and/or onboarding process based on AI software robotics process automation. Onboarding is the process of integrating a new employee into the organisation and its culture. Tactics used in this process include formal meetings, lectures, videos, printed materials, or computer-based orientations to introduce newcomers to their new jobs and organisations.

Enterprises may use machine learning methods to predict employee absences from work. Employees are one of the enterprise's most important resources. Their absence negatively affects productivity. Analyses may uncover reasons of absences that can be addressed by the enterprise e.g. improve the mental health of staff, invest in safety practices to prevent accidents at work, and improve workplace morale.

Other examples include personalised training offers (online course and digital classroom, best time frame for new courses and schedules to fit personal preferences) based on AI-powered staff profiling or human resources virtual assistants (HR bots), and electronic employee helpdesks that are able to handle staff requests based on natural language processing and generation. Moreover, they can use machine learning for employee performance analysis, and computer vision for sentiment and group analysis.

d) Use of AI for logistics

some of the examples may be:

- **autonomous robots for pick-and-pack solutions in warehouses for parcel shipping, tracing, distribution and sorting**
- **route optimisation based on machine learning**

This answer option is related to the use of AI technologies for logistics.

Logistics is the process of planning and executing the efficient transportation and storage of goods from the point of origin to the point of consumption. The goal of logistics is to meet customer requirements in a timely, cost-effective manner.

Examples of usage include AI that refers to autonomous robots for pick-and-pack solutions in warehouses, or for parcel handling (shipping, tracing, distribution and sorting), or parcel delivery and route optimisation that increase safety and efficiency, based on computer vision and machine learning. Other examples may include computer vision inventory management and execution. AI-powered technology is already able to extract characteristics of products such as price tags, shelf condition, and brand to provide real-time warehouse inventory management.

Visual inspection powered by AI is identifying damage, classifying the damage type, and determining the appropriate corrective action faster than ever before.

Robotic process automation – the combination of robotic process automation and AI can automate and streamline routine business tasks to cut corners significantly (collecting and processing data files; shipment scheduling and tracking; order processing and sending confirmation emails; capturing, researching, and closing out loads).[26]

e) Use of AI for ICT security

some of the examples may be:

- **face recognition based on computer vision for authentication of ICT users**
- **detection and prevention of cyberattacks based on machine learning**

This answer option is related to the use of AI technologies for the enterprise's ICT security.

They may refer to biometric authentication systems (fingerprints, face, iris, voice) based on computer vision and/or natural language processing. Other examples may include intelligent AI-powered antivirus software, spam filter applications (SpamAssassin), fraud detection, botnet detection, hacking incident forecasting or cyberattack detection and prevention systems (network intrusion detection and prevention) that protect the enterprise networks and systems from intrusion and other ICT security incidents.

Ordinary antivirus or spam filter applications that do not use any artificial intelligence technologies are out of the scope of this question.

f) Use of AI for accounting, controlling or finance management

some of the examples may be:

- **machine learning to analyse data that helps to make financial decisions**
- **invoice processing based on machine learning**
- **machine learning or natural language processing for bookkeeping documents**

This answer option is related to the use of AI technologies for accounting, controlling or finance management.

They may refer to intelligent systems that are used for accounting and bookkeeping services (e.g. for validation transaction activities, automatic suggestions or completion of accounting codes), and for invoice processing (e.g. to read invoices, extract the needed data, code the invoices, track outstanding invoices and automate the follow-up collection process). They may also refer to machine learning algorithms that analyse data to identify potential fraud issues and flag them for review to avoid loss of revenue. Another example may be analysis of large volumes of

[26] Artificial Intelligence in Logistics: Exploring Applications and Opportunities

transactions to detect hidden errors or trends. In procurement, invoicing or billings, the AI can detect potential issues with future purchases, such as late payments and/or deliveries. AI-enabled systems can support auditing, and compliance with corporate, state, and federal regulations, by monitoring the pertinent documents and raising alerts where necessary.

Machine learning or natural language processing for bookkeeping documents refers to the usage of AI to automate and simplify bookkeeping tasks, e.g. AI can quickly and accurately record transactions in ledgers, and automate processes by extracting data from invoices, receipts, and other documents using optical character recognition. The example can also be translated as usage of machine learning or natural language processing for bookkeeping tasks.

g) Use of AI for research and development (R&D) or innovation activity (excluding research on AI)

some of the examples may be:

- **analysis of data for conducting research, solving research problems, developing a new or significantly improved product/service based on machine learning**

This answer option is related to the use of artificial intelligence technologies for research and development (R&D) or innovation activity. AI technologies that are not used for the R&D and innovation activity, but which are used to be embedded in the enterprise's new or improved product or service, are not within the scope of this answer option, nor is R&D activity on AI.

They may refer to machine learning to analyse data to obtain useful insights, categorise, predict, and make evidence-based decisions for solving research problems or developing new or significantly improved products/services. AI might be used in the planning or implementation.

R&D[27] activity is creative and systematic work undertaken in order to increase the stock of knowledge - including knowledge of humankind, culture and society - and to devise new applications of available knowledge. For an activity to be an R&D activity, it must satisfy five core criteria (at least in principle), namely the activity must be novel, creative, uncertain, systematic and transferable. The term R&D covers three types of activity: basic research, applied research and experimental development.

An **innovation**[28] is a new or improved product or process (or combination thereof) that differs significantly from the unit's previous products or processes and which has been made available to potential users (product) or brought into use by the unit (process).

The minimum requirement for an innovation to occur is that the product or process (marketing or organisational) method must be new or significantly improved in the enterprise. This includes not only products, processes and methods that enterprises are the first to develop but also those that have been adopted from other enterprises or organisations.

[27] OECD (2015), Frascati Manual 2015: Guidelines for Collecting and Reporting Data on Research and Experimental Development, The Measurement of Scientific, Technological and Innovation Activities, OECD Publishing, Paris.
[28] OECD/Eurostat (2018), Oslo Manual 2018: Guidelines for Collecting, Reporting and Using Data on Innovation, 4th Edition, The Measurement of Scientific, Technological and Innovation Activities, OECD Publishing, Paris/Eurostat, Luxembourg.

Question E3: *How did your enterprise acquire the artificial intelligence software or systems that it uses?*

[Scope: enterprises that use AI technologies, i.e. E1a) = yes or E1b) = yes or E1c) = yes or E1d) = yes or E1e) = yes or E1f) = yes or E1g) = yes; optional question]

[Type: single answer per item (i.e. Tick only one); binary (Yes/No); multiple items may be expected]

Question **E3** is optional and should be answered by enterprises that use one of the AI technologies listed in question E1, i.e. enterprises that ticked 'yes' in at least one of the answer options in question E1. The question aims at identifying how the enterprises acquired the AI technologies that they are using. Five ways of acquisition, with **varying levels of involvement on the part of the enterprise**, have been identified; namely the AI technologies were developed by own employees (including those employed in the parent or affiliate enterprises); commercial software or systems were modified by own employees (including those employed in the parent or affiliate enterprises); open-source software or systems were modified by own employees (including those employed in the parent or affiliate enterprises); commercial software or systems ready to use were purchased (including examples where it was already incorporated in a purchased item or system); and external providers were contracted to develop or modify AI technologies.

a) They were developed by own employees (including those employed in the parent or affiliate enterprise)

This answer option covers the cases where the enterprise uses AI software or a system, as listed in question E1, that was fully developed by its own employees (including those employed in parent or affiliate enterprises as affiliates and parent enterprises are not considered as external suppliers according to the current practices in global value chain statistics).

For instance, the enterprise's employees developed from scratch a machine learning algorithm that makes sales or business forecasting.

b) Commercial software or systems were modified by own employees (including those employed in parent or affiliate enterprise)

This answer option covers the cases where the enterprise purchased a commercial AI software or system as listed in question E1. The commercial software or system was then modified by its own employees (including those employed in parent or affiliate enterprises) before being used by the enterprise (e.g. the software was adapted to the enterprise's need).

For instance, the enterprise bought a commercial AI software that does customer profiling and the enterprise's employees further developed and customised the software for their own market needs and/or adapted it to better integrate into their own systems.

c) Open-source software or systems were modified by own employees (including those

employed in parent or affiliate enterprise)

This answer option covers the cases where the enterprise acquired an open-source AI software or system as listed in question E1. The open-source software or system was modified by its own employees (including those employed in parent or affiliate enterprises) before being used by the enterprise (e.g. the software was adapted to the enterprise's need).

For instance, the enterprise acquired an open-source AI software that does customer profiling and the enterprise's employees further developed and customised the software for their own market needs and/or adapted it to better integrate into their own systems.

d) Commercial software or systems ready to use were purchased (including examples where it was already incorporated in a purchased item or system)

This answer option covers the cases where the enterprise purchased a commercial AI software or system as listed in question E1. The commercial software or system was used directly by the enterprise without any further development. This answer option includes cases where the AI technology is only part of a bigger system.

For instance, the respondent enterprise bought a ready-to-use software for detection and prevention of cyberattacks based on machine learning. The software was directly installed on the enterprise's network without any further customisation.

e) External providers were contracted to develop or modify them

This answer option covers the cases where the enterprise contracted external service providers to develop for the respondent enterprise a customised AI software or system as listed in question E1.

For instance, the respondent enterprise contracted an external provider to provide them with a chatbot based on natural language processing specially developed for their customer support. The external provider may have developed the chatbot entirely from scratch or modified an already existing chatbot to meet the customer enterprise's needs. The ready-made systems that are not customised are outside the scope of this answer option and fall under answer options d).

Note that 'external supplier' refers to any other enterprise that developed the AI software or system. Affiliates and parent enterprises are not considered as external suppliers and are therefore covered under answer option a). This complies with the current practices in global value chain statistics.

According to the definitions in the international sourcing survey, there are four types of sourcing based on 'location' and 'control'. Accordingly, outsourcing (i.e. external suppliers) includes sourcing to an enterprise that is not affiliated, otherwise it is considered as insourcing (i.e. own staff), hence the proposed clarification would be consistent with global value chains.

Note: If one of the questions in E1 was ticked 'yes', then the enterprise had to acquire or purchase AI software or systems from somewhere. In cases where one of the E1 questions is ticked 'yes' and all of the questions in E3 are ticked 'no', it should be considered, if feasible, to contact the enterprise and verify their response (e.g. information obtained, and from whom they obtained or purchased the software or system).

Question E4: *Does your enterprise process data on individuals using AI technologies?*

Some of examples of data on individuals (e.g. employees, job applicants or customers) are: sex, age, racial or ethnic origin, disability, religion or belief, sexual orientation, facial images, record of purchases, occupation or address.

[Scope: enterprises that use AI technologies, i.e. E1a) = yes or E1b) = yes or E1c) = yes or E1d) = yes or E1e) = yes or E1f) = yes or E1g) = yes; optional question]

[Type: single answer (i.e. Tick only one); binary (Yes/No)]

The intention of the question is to find out if an enterprise processes data on individuals (e.g. employees, job applicants or customers) using AI technologies. Some examples of data on individuals are data on sex, age, racial or ethnic origin, disability, religion or belief, sexual orientation, facial images, purchase records, occupation or address.

An enterprise might process data on individuals using AI technologies when, for example, doing customer profiling or pre-selection screening of job candidates, or when analysing employee performance.

Question E5. *Does your enterprise have any measures to check the results generated by AI technologies for possible biases towards individuals based on sex, age, racial or ethnic origin, disability, religion or belief, sexual orientation?*

Some of examples of measures to check for possible bias:

- **analysing the results of various machine learning models**
- **examining the dataset that was used to train the machine learning model**
- **data augmentation (which involves techniques to artificially generate additional data points from existing data, i.e. synthetic data)**

[Scope: enterprises that have any measures to check the results generated by AI technologies for possible biases towards individuals based on sex, age, racial or ethnic origin, disability, religion or belief, sexual orientation, i.e. E4 = yes; optional question]

[Type: single answer (i.e. Tick only one); binary (Yes/No)]

AI has many benefits, such as improving healthcare, education and decision-making, increasing the efficiency of production, boosting security, and improving customer service. However, it also has risks (e.g. non-transparent decision-making, gender-based or other kinds of discrimination). Bias refers to the inherent risks of any societal or economic activity. AI bias can affect and discriminate against people, if left unchecked by the social control mechanisms that govern human behaviour. Given the potential risks of using AI, it is necessary to monitor awareness of them and take measures to ensure that the subsequent use of AI systems does not lead to biases.

The aim is to find out if an enterprise has any measures in place to check the results generated by AI technologies for possible biases towards individuals based on sex, age, racial or ethnic origin, disability, religion or belief, or sexual orientation.

AI bias is ‘an anomaly in the output of machine learning algorithms, due to prejudiced assumptions

made during the algorithm development process or prejudices in the training data'[29]. AI systems contain biases due to:

- unconscious errors in thinking that affect individuals' judgments and decisions and
- a lack of complete data (incomplete data may not be representative and may therefore include bias.).

An AI system can be as good as the quality of its input data, so it is important to minimise biases in AI. To avoid results generated by AI technologies that are biased towards individuals based on sex, age, racial or ethnic origin, disability, religion or belief, or sexual orientation, an enterprise can use different measures. Some measures to check for possible biases are:

- analysing the results of various machine learning models;
- examining the dataset used to train machine learning models (to see if it is representative and large enough to prevent common biases such as sampling biases;
- monitoring models over time for biases;
- augmenting data (using techniques to artificially generate additional data points from existing data, i.e. synthetic data);
- diversifying the organisation (diversification in the AI community makes it easier to identify biases, with people who first notice biases generally being users from the community affected by them).[30]

Question E6: *Has your enterprise ever considered using any of the artificial intelligence technologies listed in question E1?*

[Scope: enterprises that had not used any of the AI technologies listed in question E1, i.e. E1a) = no and E1b) = no and E1c) = no and E1d) = no and E1e) = no and E1f) = no and E1g) = no; optional question]

[Type: single answer (i.e. Tick only one); binary (Yes/No)]

Question **E6** is asked to those enterprises that have not used any of the AI technologies listed in question E1, answer options a) to g).

The question is aiming to filter out those enterprises that have considered using AI technologies but have not yet used them. This question is a filter for the following question **E7**.

Question E7: *What are the reasons for your enterprise not to use any of the Artificial Intelligence technologies listed in question E1?*

[Scope: enterprises that had not used any of the AI technologies listed in question E1, but

[29] Bias in AI: What it is, Types, Examples & 6 Ways to Fix it in 2023 (aimultiple.com)
[30] See footnote 27.

have considered using them, i.e. E6 = Yes; optional question]

[Type: single answer per item (i.e. Tick only one); binary (Yes/ No); multiple items may be expected]

Question **E7** is aiming to collect information on the possible reasons (e.g. associated costs, lack of expertise, ethical concerns) why an enterprise has not used any of the AI technologies listed in question E1, although they have considered it (they have replied yes to question E6).

Multiple reasons for not using AI technologies may be valid for one enterprise.

a) The costs seemed too high

This answer option includes cases where the enterprise estimated that the costs related to using the AI technologies listed in question E1 were higher than the expected benefits. Therefore, the enterprise decided finally not to use them. The costs related to AI technologies use may include: costs of acquiring the AI systems; staff costs (costs of hiring a specialist, training, etc.); hardware/software costs due to incompatibility with existing equipment/software/systems; costs for purchasing training data for the AI systems; or other unexpected costs due to uncertainty factors e.g. due to legal consequences or violation of data privacy.

b) There is a lack of relevant expertise in the enterprise

This item covers cases where enterprises have decided not to use any AI because they did not have enough relevant expertise in the enterprise, namely sufficient human resources, knowledge or skills, e.g. the required specialists are insufficiently available within the enterprise, or it is difficult to hire them. AI is an advanced and complex ICT technology that may require specialised knowledge to be managed and maintained even when acquired as a ready-to-use system. Therefore, some enterprises may be reluctant to use any AI systems because they do not have, or find it difficult to employ, specialised personnel with the required expertise.

c) Incompatibility with existing equipment, software or systems

This item covers cases where enterprises have decided not to use any AI technologies because of incompatibility with existing equipment, software or systems. The introduction of AI technologies in the enterprise may not necessarily be compatible with the existing systems in use in the enterprise. It may therefore bring a certain uncertainty and require a resource investment in terms of the integration of the new system into the existing systems. An enterprise might therefore choose not to use AI technology because this might require a complete change in the technical infrastructure that the enterprise is already using, which could bring about additional acquisition costs, need for specialised personnel, need for additional training for the enterprise's employees, or even the introduction of a new system for customers or suppliers of the enterprise.

d) Difficulties with availability or quality of the necessary data

This answer option covers cases where the responding enterprise has not used any AI technologies due to difficulties with the availability or quality of the necessary data from within or from outside the responding enterprise. Data are essential for AI systems to be trained and operate efficiently. Difficulties with the availability or quality of the necessary data refers to cases

where there is no available data within the enterprise or it is difficult and costly to access them from external sources, or the available data is not detailed enough or do not include the necessary information to enable the training of the AI systems for the specific purpose of the enterprise (e.g. the enterprise may not be able to gain access to data regarding consumer behaviour in their customer segment for training an AI software for marketing or sales).

e) Concerns regarding violation of data protection and privacy

This answer option refers to cases where the respondent enterprise has not used any AI technologies due to concerns regarding violation of data protection and privacy.

One of the key elements of trustworthy AI technologies is the privacy and data governance, along with other requirements such as technical robustness, transparency, and diversity. AI systems must guarantee privacy and data protection throughout a system's entire life cycle. This includes the information initially provided by the user, as well as the information generated about the user over the course of their interaction with the system (e.g. outputs that the AI system generated for specific users or how users responded to particular recommendations). Digital records of human behaviour may allow AI systems to infer not only individual preferences, but also their sexual orientation, age, gender, and religious or political views. To allow individuals to trust the data gathering process, it must be ensured that data collected about them will not be used to unlawfully or unfairly discriminate against them[31].

f) Lack of clarity regarding the legal consequences (e.g. liability in the case of damage caused by the use of artificial intelligence)

This answer option refers to cases where the respondent enterprise has not used any AI technologies due to lack of clarity regarding the legal consequences.

Lawful AI is particularly important for the European Commission in the context of trustworthy AI[32]. AI systems do not operate in a lawless world. Several of the legally binding rules at European, national and international level already apply or are relevant to the development, deployment and use of AI systems today. Legal sources include, but are not limited to: EU primary law (the Treaties of the European Union and its Charter of Fundamental Rights); EU secondary law (such as the General Data Protection Regulation, the Product Liability Directive, the Regulation on the Free Flow of Non-Personal Data, anti-discrimination Directives, consumer law, and Safety and Health at Work Directives); the UN Human Rights treaties and the Council of Europe conventions (such as the European Convention on Human Rights); and numerous EU Member State laws. Besides generally applicable rules, various domain-specific rules exist that apply to particular AI applications (such as the Medical Device Regulation in the healthcare sector).

The law provides both positive and negative obligations, which means that it should not only be interpreted with reference to what cannot be done, but also with reference to what should be

[31] Ethics guidelines for trustworthy AI
[32] See a footnote 29.

done and what may be done. The law not only prohibits certain actions but also enables others. In this regard, it can be noted that the EU Charter contains articles on the 'freedom to conduct a business' and the 'freedom of the arts and sciences', alongside articles addressing areas to ensure AI's trustworthiness, such as for instance data protection and non-discrimination.

g) Ethical considerations

This answer option refers to cases where the respondent enterprise has not used any AI technologies because of ethical considerations. This is a particularly important topic for the EU (as well as internationally), as it tries to regulate the use of AI in order to tackle the associated risks. The EU is considered a front runner with regard to establishing a framework on ethical rules for AI. In this context, the European Commission has issued ethical guidelines[33] for trustworthy AI that includes the ethical dimension, i.e. ensuring adherence to ethical principles and values of respect for human autonomy, prevention of harm, fairness and explicability. Trustworthy AI should be: (i) lawful - respecting all applicable laws and regulations; (ii) ethical - respecting ethical principles and values; and (iii) robust - both from a technical and social perspective since, even with good intentions, AI systems can cause unintentional harm[34].

Achieving trustworthy AI requires not only compliance with the law, which is but one of its three components. Laws are not always up to speed with technological developments, can at times be out of step with ethical norms, or may simply not be well suited to addressing certain issues. For AI systems to be trustworthy, they should therefore also be ethical, ensuring alignment with ethical norms.

h) Artificial intelligence technologies are not useful for the enterprise

This answer option refers to enterprises that have not used any AI technologies because they were not considered useful for the enterprise, for instance not needed because of the activity of the enterprise.

2.3.2.7. Module X: Background information

The background variables have several purposes. Firstly, they are used for breakdowns. That is the case for the 'Main economic activity of the enterprise' and 'Average number of employees and self-employed persons (persons employed)'.

Secondly, they are needed to weight the percentages of turnover from e-commerce. The background variable 'Total turnover' is used for that purpose. The variable 'Average number of employees and self-employed persons (persons employed)' is similarly used to weight the percentage of persons employed using computers, the percentage of persons employed using computers with access to the internet, etc. The number of employees and self-employed persons

[33] See a footnote 29.
[34] Ethics guidelines for trustworthy AI | Shaping Europe's digital future (europa.eu)

(persons employed) is also used to weight the qualitative variables.

Thirdly, the background variables are used in the sampling design. Namely, 'Main economic activity' and 'Average number of employees and self-employed persons (persons employed)' are used to stratify the sample.

The background variables described so far may be collected through the ICT survey questionnaire or obtained from alternative sources. The alternative sources are mainly the registers and one main business survey, usually used to produce the structural business statistics. It is very important that the background information is at least consistent with the structural business statistics.

Question X1: *Main economic activity of the enterprise*

[Scope: all enterprises]

[Type: categorical]

The main (or principal) economic activity is identified as the activity that contributes most to the total value added of the enterprise. The principal activity so identified does not necessarily account for 50% or more of the enterprise's total value added. The classification of principal activity is determined by reference to NACE, first at the highest level of classification and then at more detailed levels ('top-down' method).

The nomenclature NACE Rev. 2 is available on: https://ec.europa.eu/eurostat/web/nace

The main economic activity of the enterprise should be classified by NACE Rev. 2 at its highest level of detail (4 digits). Nevertheless, only the following level of detail which is used in the breakdown is strictly necessary.

NACE Rev. 2 groupings		
Aggregates for calculation of **national** NACE Rev. 2 aggregates		
1	10 - 33 + 35 - 39 + 41 - 43 + 45 - 47 + 49 - 53 + 55 - 56 + 58 - 63 + 68 - 75 + 77 - 82 + 95.1	
2	10 - 33	Manufacturing
3	10 - 18	Manufacture of products based on food, beverages, tobacco, textile, leather, wood, pulp and paper; publishing and printing

4	19 - 23	Manufacture of coke, refined petroleum products, chemical products, basic pharmaceutical products, rubber and plastics, other non-metallic mineral products
5	24 - 25	Manufacture of basic metals and fabricated metal products excluding machines and equipment
6	26 - 33	Manufacture of computers, electric and optical products, electrical equipment, machinery and equipment n.e.c., motor vehicles, other transport equipment, furniture, other manufacturing, repair and installation of machinery and equipment
7	35 - 39	Production and distribution of electricity, gas, steam and air conditioning; water supply, sewerage, waste management and remediation activities
8	41 - 43	Construction
9	45 - 47	Wholesale and retail trade; repair of motor vehicles and motorcycles
10	47	Retail trade
11	49 - 53	Transport and storage
12	55	Accommodation
13	55 - 56	Accommodation and food service activities
14	58 - 63	Information and communication
15	68	Real estate activities
16	69 - 75	Professional, scientific and technical activities
17	77 - 82	Administrative and support service activities
	26.1 - 26.4 + 26.8 + 46.5 + 58.2 + 61 + 62.01 + 62.02 + 62.03 + 62.09 + 63.1 + 95.1	Manufacture of electronic components and boards, consumer electronics, magnetic and optical media; wholesale of information and communication equipment; software publishing; telecommunications; computer programming, consultancy and facilities management activities, other information technology and computer service activities; data processing, hosting and related activities, web

		portals; repair of computers and communication equipment or ICT sector

Aggregates for calculation of **European** NACE Rev. 2 aggregates		
3a	10 - 12	Manufacture of beverages, food and tobacco products
3b	13 - 15	Manufacture of textiles, wearing apparel, leather and related products
3c	16 - 18	Manufacture of wood and products of wood and cork, except furniture; articles of straw and plaiting materials; paper and paper products; printing and reproduction of recorded media
4a	19	Manufacture of coke and refined petroleum product
4b	20	Manufacture of chemicals and chemical products
4c	21	Manufacture of basic pharmaceutical products and pharmaceutical preparations
4d	22 - 23	Manufacture of rubber and plastic products; other non-metallic mineral products
6a	26	Manufacture of computer, electronic and optical products
6b	27	Manufacture of electrical equipment, machinery and equipment n.e.c.
6c	28	Manufacture of machinery and equipment n.e.c.
6d	29 - 30	Manufacture of motor vehicles, trailers and semi-trailers; other transport equipment
6e	31 - 33	Manufacture of furniture; other manufacturing; repair and installation of machinery and equipment
7a	35	Electricity, gas, steam and air conditioning supply
7b	36 - 39	Water collection, treatment and supply; sewerage; waste collection, treatment and disposal activities; materials recovery; remediation activities and other waste management services
9a	45	Wholesale and retail trade and repair of motor vehicles and motorcycles

Aggregates for calculation of **European** NACE Rev. 2 aggregates		
9b	46	Wholesale trade, except of motor vehicles and motorcycles
14a	58 - 60	Publishing activities; motion picture, video and television programme production, sound recording and music publishing activities; programming and broadcasting activities
14b	61	Telecommunications
14c	62 - 63	Computer programming, consultancy and related activities; information service activities
16a	69 - 71	Legal and accounting activities; activities of head offices; management consultancy activities; architectural and engineering activities; technical testing and analysis
16b	72	Scientific research and development
16c	73 - 75	Advertising and market research; other professional, scientific and technical activities; veterinary activities
17a	77 - 78 + 80-82	Activities for: rental and leasing; employment; security and investigation; services to buildings and landscape; office administrative, office support and other business support
17b	79	Travel agency, tour operator and other reservation service and related activities
18a	95.1	Repair of computers and communication equipment

The NACE Rev. 2 categories are grouped together for dissemination purposes into several aggregates organised in five hierarchal levels. At the first level, there are two categories distinguishing 'Manufacturing, Energy and Construction' and 'Non-financial services'. At the second level, activities are grouped at the Section level of NACE, making 11 categories. The content of these groupings is described above.

X2: *Average number of employees and self-employed persons (persons employed), during 2023*

[Scope: all enterprises]

[Type: categorical]

With the introduction of the Regulation (EU) 2019/2152, the variable 'persons employed' is replaced by the variable 'employees and self-employed persons'. This change in the denomination of the variable does not imply any change in the scope. The two variables

represent exactly the same concept. For the sake of user friendliness, the term ‘employees and self-employed persons’ is only used in the introductory part of the questionnaire and in Module X; in the rest of the questionnaire, the term ‘persons employed’ is used.

For the purpose of general harmonisation of enterprise ICT usage statistics and the more general field of business statistics, the concept of employees and self-employed persons used here is taken from Regulation (EU) 2020/1197 of 30 July 2020 implementing Regulation 2020/2152 on European Business Statistics (p. 92, Variable 120101: Number of employees and self-employed persons).

The **number of employees and self-employed persons** is the sum of the **number of employees** and **number of self-employed persons**.

The **number of employees** represents the average number of persons who were, at some time during the reference period, employees of the statistical unit.

Explanatory note:

While the employment relationship, which qualifies the parties (into employee and employer), is defined in specific legislation or contracts, the term ‘employee’ usually means a person hired by the statistical unit to provide services to it on a regular basis, in exchange for benefits, and where the services provided are not part of an independent business. For the sake of clarity, apprentices, if hired under such conditions, are considered employees.

The average should be calculated as the arithmetic mean of the number of employees over the shortest time periods of equal length fitting into the reference period, for which regular observations are practicable (e.g. daily, weekly, monthly, quarterly).

The **number of self-employed persons** is the average number of persons who were at some time during the reference period the sole owners or joint owners of the statistical unit in which they work. Family workers and outworkers whose income is a function of the value of the outputs of the statistical unit are also included.

Note: In order to check the comparability of data, it is necessary to indicate whether voluntary workers have been included under this heading or not.

The number of employees and self-employed persons should be measured as the yearly average during the previous calendar year. For comparability reasons, the number of employees and self-employed persons should not be confused with the number of employees (that excludes unpaid workers) or the number of employees in FTE units.

The average number of employees and self-employed persons is coded into five size categories, of which three are compulsory and the other two are optional.

Size categories (according to the number of employees and self-employed persons)		
Compulsory		
1	10 to 49	Small enterprises
2	50 to 249	Medium enterprises

	3	250 or more	Large enterprises
Optional			
	4	Less than 2	Small micro-enterprises
	5	2 to 9	Big micro-enterprises

Question X3: *Total turnover (in value terms, excluding VAT), for 2023*

[Scope: all enterprises]

[Type: numerical]

This background variable is needed to weight the percentage of turnover resulting from orders received via computer networks (value of web sales and value of EDI-type sales in Module B: e-Commerce sales).

For the purpose of general harmonisation of enterprise ICT usage statistics and the more general field of business statistics, the concept of turnover used here is taken from Regulation (EU) 2020/1197 of 30 July 2020 (p. 104, Variable 140301: Net turnover) implementing Regulation (EU) 2019/2152:

For all activities except for NACE 64, 65 and some activities of NACE 66 net turnover consists of all income arising during the reference period in the course of ordinary activities of the statistical unit, and is presented net of all price reductions, discounts and rebates granted by it.

Income is defined as increases in economic benefits during the reference period in the form of inflows or enhancements of assets or decreases of liabilities that result in increases in equity, other than those relating to contributions from equity participants.

The inflows referred to are arising from contracts with customers and are realised through the satisfaction by the statistical unit of performance obligations as provided for in said contracts. Usually, a performance obligation is represented by the sale (transfer) of goods or the rendering of services, however, the gross inflows can also contain revenues obtained as a yield on the use by others of the statistical unit's assets.

Excluded from net turnover are:

- all taxes, duties or levies linked directly to revenue;
- any amounts collected on behalf of any principal, if the statistical unit is acting as an agent in its relationship with said principal;
- all income not arising in the course of ordinary activities of the statistical unit. Usually, these types of income are classified as 'Other (operating) income', 'Financial income', 'Extra-ordinary income' or under a similar heading, depending on the respective set of generally accepted accounting standards used to prepare the financial statements.

Infra-annual statistics may not be able to take into account aspects such as annual price reductions, subsidies, rebates and discounts.

3

Data transmission

Once the data have been collected by the NSIs, they must be transmitted to Eurostat. This chapter provides guidance on how to compute aggregates and transmit the data.

3.1. General description

Data files on ICT usage and e-commerce in enterprises transmitted to Eurostat include statistical data elements laid down in the Implementing Act. The Implementing Act lists 65 mandatory variables and 30 optional variables.

Metadata and quality reports complement those data.

3.2. Codification, indicators and breakdown aggregates

NSIs have to codify microdata, compute aggregates and break them down into different combinations depending on the enterprises' activity, sector and size (number of employees and self-employed persons). Those steps are necessary prerequisites for transmitting data to Eurostat.

3.2.1. Codification of microdata

The following table provides guidance on how NSIs can codify microdata gathered using the model questionnaire in order to develop the indicators that need to be sent to Eurostat.

The use of the following codification is not mandatory but recommended as the scope of the indicators to be transmitted to Eurostat are based on that codification.

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		Module A: Access and use of the internet	
		Access to the Internet	
		A1: How many persons employed have access to the internet for business purposes?	All enterprises
		(including fixed line, fixed wireless and mobile telephone network connection)	
EMPIUSEVAL	Nnnnn	Absolute number	
	Blank	No answer	
EMPIUSEPCT	Nnn	Percentage (0-100, round to the nearest integer)	
	Blank	No answer	
		Derived value	
EMPIUSE	Nnnnn	Reconstructed absolute number:	
	Blank	IF EMPIUSEVAL<> Blank THEN EMPIUSEVAL	
		ELSEIF EMPIUSEPCT<>Blank AND EMP <> Blank	
		THEN EMPIUSEPCT * EMP / 100	
		ELSE Blank	
		Use of a fixed line connection to the internet for business purposes	
		A2: Does your enterprise use any type of fixed connection to the internet? (e.g. ADSL, SDSL, VDSL, fiber optics technology (FTTP), cable technology, fixed wireless)	Enterprises where EMPIUSE > 0
FIXBB	1	Yes	
	0	No	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
		A3: What is the maximum contracted download speed of the fastest fixed internet connection of your enterprise?	Enterprises where FIXBB=1
ISPDF1	1	a) less than 30 Mbit/s	
	2	b) at least 30 but less than 100 Mbit/s	
	3	c) at least 100 but less than 500 Mbit/s	
	4	d) at least 500 but less than 1 Gbit/s	
	5	e) at least 1 Gbit/s	
	Blank	No answer	
	9	Not applicable (FIXBB=Blank or FIXBB<>1)	
		A4: Is the speed of your fixed line connection(s) to the internet usually sufficient for the actual needs of the enterprise?	Enterprises where FIXBB=1
ISPDFOK	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (FIXBB=Blank or FIXBB<>1)	
		Use of a mobile connection to the internet for business purposes	
		A5: How many persons employed use a portable device provided by the enterprise, that allows connection to the internet via mobile telephone networks, for business purposes?	Enterprises where EMPIUSE > 0

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		(e.g. portable computers, or other portable devices such as smartphones)	
EMPMD2VAL	Nnnnn	Absolute number (use as many digits as necessary)	
(optional)	Blank	No answer or option not included	
	-1	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
EMPMD2PCT	Nnn	Percentage (0-100, round to the nearest integer)	
(optional)	Blank	No answer or option not included	
	-1	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
		Derived value	
EMPMD2	Nnnnn	Reconstructed absolute number:	
(optional)	Blank	IF EMPIUSE=Blank or EMPIUSE=0 THEN -1	
	-1	ELSEIF EMPMD2VAL<> Blank THEN EMPMD2VAL	
		ELSEIF EMPMD2PCT<>Blank AND EMP <> Blank	
		THEN EMPMD2PCT * EMP / 100	
		ELSE Blank	
		Remote access and meetings via the internet	
		A6: Do any of the persons employed have remote access to the following? (via computers or portable devices such as smartphones)	Enterprises where EMPIUSE > 0
RA_M		a) E-mail system of the enterprise	
	1	Yes	
	0	No	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
RA_D		b) Documents of the enterprise (e.g. files, spreadsheets, presentations, charts, photos)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
RA_S		c) Business applications or software of the enterprise (e.g. access to accounting, sales, orders, CRM) *Please exclude applications used for internal communication, e.g. Skype, MS Teams, Yammer.*	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
		A7: Does your enterprise conduct remote meetings (via e.g. Skype, Zoom, MS Teams, WebEx)?	Enterprises where EMPIUSE > 0
RM	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		Other use of the internet	
ADS (optional)		**A8: Does your enterprise pay to advertise on the internet?** (e.g. adverts on search engines, on social media, on other websites or apps)	Enterprises where EMPIUSE > 0
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
(optional)		**A9: Does your enterprise pay to advertise on the internet using any of the following targeted advertising methods?**	Enterprises where ADS=1
ADS_KW		a) Based on content or keywords searched by internet users	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (ADS=Blank or ADS<>1)	
ADS_TRK		b) Based on the tracking of internet users' past activities or profile	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (ADS=Blank or ADS<>1)	
ADS_LOC		c) Based on the geolocation of internet users	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (ADS=Blank or ADS<>1)	
ADS_OTH		d) Any other method of targeted advertising on the internet not specified above	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (ADS=Blank or ADS<>1)	
		Module B: e-Commerce sales	
		Web sales of goods or services	
		B1: During 2023, did your enterprise have web sales of goods or services via:	Enterprises where EMPIUSE > 0
AWS_COWN		a) your enterprise's websites or apps? (including extranets)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
AWS_CMP		b) e-commerce marketplace websites or apps used by several enterprises for trading goods or services? (e.g. e-Bookers, Booking, hotels.com, eBay, Amazon, Amazon Business, Alibaba, Rakuten, TimoCom)	
	1	Yes	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
		B2: What was the value of your web sales?	
AWSVALVAL	Nnnnn	a) What was the value of your web sales of goods or services, in 2023?	Enterprises where AWS_COWN =1 or AWS_CMP=1
	Blank	No answer	
	-1	Not applicable ((AWS_COWN=Blank or AWS_COWN<>1) and (AWS_CMP=Blank or AWS_CMP<>1))	
AWSVALPCT	Nnn	b) What percentage of the turnover was generated by web sales of goods or services, in 2023? (0-100, please provide at least one decimal)	
	Blank	No answer	
	-1	Not applicable ((AWS_COWN=Blank or AWS_COWN<>1) and (AWS_CMP=Blank or AWS_CMP<>1))	
		Derived values	
AWSVAL	Nnnnn	Reconstructed absolute number:	
	Blank	IF (AWS_COWN=Blank or AWS_COWN<>1) and (AWS_CMP=Blank or AWS_CMP<>1) THEN -1	
	-1	ELSEIF AWSVALVAL <> Blank THEN AWSVALVAL	
		ELSEIF AWSVALPCT <>Blank AND TOVT <> Blank	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		THEN AWSVALPCT * TOVT / 100	
		ELSE Blank	
		B3: What was the percentage breakdown of the value of the web sales in 2023 for the following:	
AWS_COWNPCT		a) via your enterprise's websites or apps? (including extranets)	Enterprises where AWS_COWN =1 and AWS_CMP=1
	Nnn	Percentage (0-100)	
	Blank	No answer	
	-1	Not applicable ((AWS_COWN=Blank or AWS_COWN<>1) or (AWS_CMP=Blank or AWS_CMP<>1))	
AWS_CMPPCT		b) via e-commerce marketplace websites or apps used by several enterprises for trading goods or services? (e.g. e-Bookers, Booking, hotels.com, eBay, Amazon, Amazon Business, Alibaba, Rakuten, TimoCom)	
	Nnn	Percentage (0-100)	
	Blank	No answer	
	-1	Not applicable ((AWS_COWN=Blank or AWS_COWN<>1) or (AWS_CMP=Blank or AWS_CMP<>1))	
		Derived values	
		The following two variables reconstruct the turnover resulting from the orders received via a website or apps (in monetary terms, excluding VAT) in 2023, via the respective channels (enterprise's website or e-commerce marketplace), based on the replies to questions B1 and	Enterprises where AWS_COWN =1 or AWS_CMP=1

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		B3. The formulae below indicate how they could be calculated:	
AWSVAL_COWN	Nnnnn	Reconstructed absolute number:	
	Blank	IF AWS_COWN=1 THEN	
	-1	IF AWS_CMP=Blank or AWS_CMP=0	
		THEN AWSVAL	
		ELSE	
		IF AWS_COWNPCT=Blank	
		THEN Blank	
		ELSE AWSVAL * AWS_COWNPCT / 100	
		ELSEIF AWS_COWN=0 THEN 0	
		ELSEIF AWS_COWN=Blank THEN Blank	
		ELSE -1	
AWSVAL_CMP	Nnnnn	Reconstructed absolute number:	
	Blank	IF AWS_CMP=1 THEN	
	-1	IF AWS_COWN=Blank or AWS_COWN=0	
		THEN AWSVAL	
		ELSE	
		IF AWS_CMPPCT=Blank	
		THEN Blank	
		ELSE AWSVAL * AWS_CMPPCT / 100	
		ELSEIF AWS_CMP=0 THEN 0	
		ELSEIF AWS_CMP=Blank THEN Blank	
		ELSE -1	
		B4: What was the percentage breakdown of the value of the web sales in 2023 by type of customer:	Enterprises where AWS_COWN

Variable name	Code	Description	Filter/ Remarks (Standard codification)
			=1 or AWS_CMP=1
AWSVALCPCT		a) Sales to private consumers (B2C)	
	Nnn	Percentage (0-100)	
	Blank	No answer	
	-1	Not applicable ((AWS_COWN=Blank or AWS_COWN<>1) and (AWS_CMP=Blank or AWS_CMP<>1))	
		b) Sales to other enterprises (B2B) and Sales to public sector (B2G)	
AWSVALBGPCT	Nnn	Percentage (0-100)	
	Blank	No answer	
	-1	Not applicable ((AWS_COWN=Blank or AWS_COWN<>1) and (AWS_CMP=Blank or AWS_CMP<>1))	
		Derived values	
AWSVALC	Nnnnn	Reconstructed absolute number:	
	Blank	IF (AWS_COWN=Blank or AWS_COWN<>1) and (AWS_CMP=Blank or AWS_CMP<>1) THEN -1	
	-1	ELSEIF AWSVALCPCT<>Blank AND AWSVAL <> Blank	
		THEN AWSVALCPCT * AWSVAL / 100	
		ELSE Blank	
AWSVALBG	Nnnnn	Reconstructed absolute number:	
	Blank	IF (AWS_COWN=Blank or AWS_COWN<>1) and (AWS_CMP=Blank or AWS_CMP<>1) THEN -1	
	-1	ELSEIF AWSVALBGPCT<>Blank AND AWSVAL <> Blank	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		THEN AWSVALBGPCT * AWSVAL / 100	
		ELSE Blank	
		B5: During 2023, did your enterprise have web sales to customers located in the following geographic areas?	Enterprises where AWS_COWN =1 or AWS_CMP=1
AWSHM		a) Own country	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable ((AWS_COWN=Blank or AWS_COWN<>1) and (AWS_CMP=Blank or AWS_CMP<>1))	
AWSEU		b) Other EU countries	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable ((AWS_COWN=Blank or AWS_COWN<>1) and (AWS_CMP=Blank or AWS_CMP<>1))	
AWSWW		c) Rest of the world	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable ((AWS_COWN=Blank or AWS_COWN<>1) and (AWS_CMP=Blank or AWS_CMP<>1))	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
(optional)		**B6: What was the percentage breakdown of the value of web sales in 2023 to customers located in the following geographic areas?** (Please refer to value of web sales you reported in B2)	
AWSHM_PCT		a) Own country	Enterprises where at least 2 of AWSHM, AWSEU and AWSWW are 1
	Nnn	Percentage (0-100)	
	Blank	No answer	
	-1	Not applicable (at most 1 of AWSHM, AWSEU and AWSWW is 1, or AWSHM, AWSEU and AWSWW are blank or not applicable)	
AWSEU_PCT		b) Other EU countries	
	Nnn	Percentage (0-100)	
	Blank	No answer	
	-1	Not applicable (at most 1 of AWSHM, AWSEU and AWSWW is 1, or AWSHM, AWSEU and AWSWW are blank or not applicable)	
AWSWW_PCT		c) Rest of the world	
	Nnn	Percentage (0-100)	
	Blank	No answer	
	-1	Not applicable (at most 1 of AWSHM, AWSEU and AWSWW is 1, or AWSHM, AWSEU and AWSWW are blank or not applicable)	
		Derived values	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		The following three variables reconstruct the turnover resulting from the orders received via a website or apps (in monetary terms, excluding VAT) in 2023, broken down by the customer location (home country, other EU country, other non-EU country), based on the replies to questions B5 and B6. The formulae below indicate how they could be calculated:	
AWSVAL_HM	Nnnnn	Reconstructed absolute number:	Enterprises where AWS_COWN =1 or AWS_CMP=1
	Blank	IF AWSHM=1 THEN	
	-1	IF (AWSEU=Blank or AWSEU=0) and (AWSWW=Blank or AWSWW=0)	
		THEN AWSVAL	
		ELSE	
		IF AWSHM_PCT=Blank	
		THEN Blank	
		ELSE AWSVAL * AWSHM_PCT / 100	
		ELSEIF AWSHM=0 THEN 0	
		ELSEIF AWSHM=Blank THEN Blank	
		ELSE -1	
AWSVAL_EU	Nnnnn	Reconstructed absolute number:	
	Blank	IF AWSEU=1 THEN	
	-1	IF (AWSHM=Blank or AWSHM=0) and (AWSWW=Blank or AWSWW=0)	
		THEN AWSVAL	
		ELSE	
		IF AWSEU_PCT=Blank	
		THEN Blank	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		ELSE AWSVAL * AWSEU_PCT / 100	
		ELSEIF AWSEU=0 THEN 0	
		ELSEIF AWSEU=Blank THEN Blank	
		ELSE -1	
AWSVAL_WW	Nnnnn	Reconstructed absolute number:	
	Blank	IF AWSWW=1 THEN	
	-1	IF (AWSHM=Blank or AWSHM=0) and (AWSEU=Blank or AWSEU=0)	
		THEN AWSVAL	
		ELSE	
		IF AWSWW_PCT=Blank	
		THEN Blank	
		ELSE AWSVAL * AWSWW_PCT / 100	
		ELSEIF AWSWW=0 THEN 0	
		ELSEIF AWSWW=Blank THEN Blank	
		ELSE -1	
		EDI-type sales	
		B7: During 2023, did your enterprise have EDI-type sales of goods or services?	Enterprises where EMPIUSE > 0
AXSELL	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		B8: What was the value of your EDI-type sales?	Enterprises where AXSELL=1
AXSVALVAL	Nnnnn	a) What was the value of your EDI-type sales of goods or services, in 2023?	
	Blank	No answer	
	-1	Not applicable (AXSELL=Blank or AXSELL<>1)	
AXSVALPCT	Nnn	b) What percentage of total turnover was generated by EDI-type sales of goods or services, in 2023?	
	Blank	No answer	
	-1	Not applicable (AXSELL=Blank or AXSELL<>1)	
		Derived value	
AXSVAL	Nnnnn	Reconstructed absolute number:	
	Blank	IF AXSELL<>1 THEN -1	
	-1	ELSEIF AXSVALVAL <> Blank THEN AXSVALVAL	
		ELSEIF AXSVALPCT <>Blank AND TOVT <> Blank	
		THEN AXSVALPCT * TOVT / 100	
		ELSE Blank	
		Module C: ICT specialists and skills	
		C1: Does your enterprise employ ICT specialists? ICT specialists are employees for whom ICT is the main job. For example, to develop, operate or maintain ICT systems or applications.	All enterprises
ITSP2	1	Yes	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	0	No	
	Blank	No answer	
		C2: Did your enterprise provide any type of training to develop ICT related skills of the persons employed, during 2023?	All enterprises
		a) Training for ICT specialists Tick No if your enterprise didn't employ any ICT specialists during 2023	
ITSPT2	1	Yes	
	0	No	
	Blank	No answer	
		b) Training for other persons employed	
ITUST2	1	Yes	
	0	No	
	Blank	No answer	
		C3: Did your enterprise recruit or try to recruit ICT specialists, during 2023?	All enterprises
ITSPRCR2	1	Yes	
	0	No	
	Blank	No answer	
		C4: During 2023, did your enterprise have vacancies for ICT specialists that were difficult to fill?	Enterprises where ITSPRCR2=1
ITSPVAC2	1	Yes	
	0	No	
	Blank	No answer	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	9	Not applicable (ITSPRCR2=Blank or ITSPRCR2<>1)	
(optional)		**C5: Did your enterprise have any of the following difficulties to recruit ICT specialists during 2023?**	Enterprises where ITSPVAC2=1
		a) Lack of applications	
ITSPDLA	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (ITSPVAC2=Blank or ITSPVAC2<>1)	
		b) Applicants' lack of relevant ICT related qualifications from education and/or training	
ITSPDLET	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (ITSPVAC2=Blank or ITSPVAC2<>1)	
		c) Applicants' lack of relevant work experience	
ITSPDLWE	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (ITSPVAC2=Blank or ITSPVAC2<>1)	
		d) Applicants' salary expectations too high	
ITSPDSAL	1	Yes	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	0	No	
	Blank	No answer	
	9	Not applicable (ITSPVAC2=Blank or ITSPVAC2<>1)	
		C6: Who performed your enterprise's ICT functions in 2023? (e.g. maintenance of ICT infrastructure; support for office software; development or support of business management software/systems and/or web solutions; security and data protection)	All enterprises
		a) own employees (incl. those employed in parent or affiliate enterprises)	
IT_OWN	1	Yes	
	0	No	
	Blank	No answer	
		b) external suppliers	
IT_EXT	1	Yes	
	0	No	
	Blank	No answer	
		Module D: ICT security	
		D1: Does your enterprise use any of the following ICT security measures?	Enterprises where EMPIUSE > 0
SECMSPSW		a) Authentication via strong password (e.g. minimum length, use of numbers and special characters, changed periodically)	
	1	Yes	
	0	No	
	Blank	No answer	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECMUIBM		b) Authentication via biometric methods used to access the enterprise's ICT system (e.g. authentication based on fingerprints, voice, face)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECMDUO		c) Authentication based on a combination of at least two authentication mechanisms (i.e. combination of e.g. user-defined password, one-time password (OTP), code generated via a security token or received via a smartphone, biometric method (e.g. based on fingerprints, voice, face))	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECMDENC		d) Encryption of data, documents or e-mails	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
SECMOSBU		e) Data backup to a separate location (including backup to the cloud)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECMNAC		f) Network access control (management of user rights in the enterprise's network)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECMVPN		g) VPN (Virtual Private Network extends a private network across a public network to enable secure exchange of data over public network)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECMSMS1		h) ICT security monitoring system used to detect suspicious activity (e.g. intrusion detection or prevention systems that monitors users' or devices' behaviour, network traffic) Please exclude antivirus software and default firewall solution included in the	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		operating system of personal computers and routers.	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECMLOG		i) Maintaining log files that enable analysis after ICT security incidents	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECMRASS		j) ICT risk assessment, i.e. periodical assessment of probability and consequences of ICT security incidents	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECMTST		k) ICT security tests (e.g. performing penetration tests, testing security alert system, review of security measures, testing of backup systems)	
	1	Yes	
	0	No	
	Blank	No answer	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECM_CNT		Derived value: ICT security measures	Enterprises where EMPIUSE > 0
	0-11	Give one point for each of the following 11 conditions, if true:	
		SECMSPSW=1, SECMUIBM=1, SECMDUO=1, SECMDENC=1, SECMOSBU=1, SECMNAC=1, SECMVPN=1, SECMSMS=1, SECMLOG=1, SECMRASS=1, SECMTST=1	
	-1	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
		D2: Does your enterprise make persons employed aware of their obligations in ICT security related issues in the following ways?	Enterprises where EMPIUSE > 0
SECAWVTGI		a) Voluntary training or internally available information (e.g. information on the intranet)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECAWCTP		b) Compulsory training courses or viewing compulsory material	
	1	Yes	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECAWCONT		c) By contract (e.g. contract of employment)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECPOL2		**D3: Does your enterprise have document(s) on measures, practices or procedures on ICT security?** *Documents on ICT security and confidentiality of data cover employee training in ICT use, ICT security measures, the evaluation of ICT security measures, plans for updating ICT security documents, etc.*	Enterprises where EMPIUSE > 0
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SECPREV		**D4: When were your enterprise's document(s) on measures, practices or procedures on ICT security, defined or most recently reviewed?** *Documents on ICT security and confidentiality of data cover employee training in ICT use, ICT security measures, the evaluation of ICT security*	Enterprises where SECPOL2=1

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		measures, plans for updating ICT security documents, etc.	
(optional)	1	a) within the last 12 months	
	2	b) more than 12 months and up to 24 months ago	
	3	c) more than 24 months ago	
	Blank	No answer	
	9	Not applicable (SECPOL2=Blank or SECPOL2<>1)	
		D5: During 2023, did your enterprise experience any ICT related security incident leading to the following consequences?	Enterprises where EMPIUSE > 0
SEC2IUSVF		a) Unavailability of ICT services due to hardware or software failures	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SEC2IUSVA		b) Unavailability of ICT services due to attack from outside, e.g. ransomware attacks, Denial of Service attacks	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SEC2IDCDF		c) Destruction or corruption of data due to hardware or software failures	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SEC2IDCDA		d) Destruction or corruption of data due to infection of malicious software or unauthorised intrusion	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SEC2ICNFA		e) Disclosure of confidential data due to intrusion, pharming, phishing attack, intentional actions by own employees	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
SEC2ICNFF		f) Disclosure of confidential data due to unintentional actions by own employees	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		Module E: Artificial Intelligence	
		E1: Does your enterprise use any of the following Artificial Intelligence (AI) technologies?	Enterprises where EMPIUSE > 0
AI_TTM		a) AI technologies performing analysis of written language (e.g. text mining)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
AI_TSR		b) AI Technologies converting spoken language into machine-readable format (speech recognition)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
AI_TNLG		c) AI Technologies generating written or spoken language (natural language generation, speech synthesis)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
AI_TIR		d) AI Technologies identifying objects or persons based on images or videos (image recognition, image processing)	
	1	Yes	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
AI_TML		e) Machine learning (e.g. deep learning) for data analysis	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
AI_TPA		f) AI Technologies automating different workflows or assisting in decision making (e.g. AI based software robotic process automation)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	
AI_TAR		g) AI Technologies enabling physical movement of machines via autonomous decisions based on observation of surroundings (autonomous robots, self-driving vehicles, autonomous drones)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		E2: Does your enterprise use Artificial Intelligence software or systems for any of the following purposes?	Enterprises where AI_TTM=1 or AI_TSR=1 or AI_TNLG=1 or AI_TIR=1 or AI_TML=1 or AI_TPA=1 or AI_TAR=1
AI_PMS		a) Use of AI for marketing or sales	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable ((AI_TTM=Blank or AI_TTM<>1) and (AI_TSR=Blank or AI_TSR<>1) and (AI_TNLG=Blank or AI_TNLG<>1) and (AI_TIR=Blank or AI_TIR<>1) and (AI_TML=Blank or AI_TML<>1) and (AI_TPA=Blank or AI_TPA<>1) and (AI_TAR=Blank or AI_TAR<>1))	
AI_PPP		b) Use of AI for production or service processes	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable ((AI_TTM=Blank or AI_TTM<>1) and (AI_TSR=Blank or AI_TSR<>1) and (AI_TNLG=Blank or AI_TNLG<>1) and (AI_TIR=Blank or AI_TIR<>1) and (AI_TML=Blank or AI_TML<>1) and (AI_TPA=Blank or AI_TPA<>1) and (AI_TAR=Blank or AI_TAR<>1))	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
AI_PBAM		c) Use of AI for organisation of business administration processes or management	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable ((AI_TTM=Blank or AI_TTM<>1) and (AI_TSR=Blank or AI_TSR<>1) and (AI_TNLG=Blank or AI_TNLG<>1) and (AI_TIR=Blank or AI_TIR<>1) and (AI_TML=Blank or AI_TML<>1) and (AI_TPA=Blank or AI_TPA<>1) and (AI_TAR=Blank or AI_TAR<>1))	
AI_PLOG		d) Use of AI for logistics	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable ((AI_TTM=Blank or AI_TTM<>1) and (AI_TSR=Blank or AI_TSR<>1) and (AI_TNLG=Blank or AI_TNLG<>1) and (AI_TIR=Blank or AI_TIR<>1) and (AI_TML=Blank or AI_TML<>1) and (AI_TPA=Blank or AI_TPA<>1) and (AI_TAR=Blank or AI_TAR<>1))	
AI_PITS		e) Use of AI for ICT security	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable ((AI_TTM=Blank or AI_TTM<>1) and (AI_TSR=Blank or AI_TSR<>1) and (AI_TNLG=Blank or AI_TNLG<>1) and (AI_TIR=Blank or AI_TIR<>1) and (AI_TML=Blank or	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		AI_TML<>1) and (AI_TPA=Blank or AI_TPA<>1) and (AI_TAR=Blank or AI_TAR<>1))	
AI_PFIN		f) Use of AI for accounting, controlling or finance management	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable ((AI_TTM=Blank or AI_TTM<>1) and (AI_TSR=Blank or AI_TSR<>1) and (AI_TNLG=Blank or AI_TNLG<>1) and (AI_TIR=Blank or AI_TIR<>1) and (AI_TML=Blank or AI_TML<>1) and (AI_TPA=Blank or AI_TPA<>1) and (AI_TAR=Blank or AI_TAR<>1))	
AI_PRDI		g) Use of AI for research and development (R&D) or innovation activity (excluding research on AI)	
	1	Yes	
	0	No	
	Blank	No answer	
	9	Not applicable ((AI_TTM=Blank or AI_TTM<>1) and (AI_TSR=Blank or AI_TSR<>1) and (AI_TNLG=Blank or AI_TNLG<>1) and (AI_TIR=Blank or AI_TIR<>1) and (AI_TML=Blank or AI_TML<>1) and (AI_TPA=Blank or AI_TPA<>1) and (AI_TAR=Blank or AI_TAR<>1))	
(optional)		**E3: How did your enterprise acquire the Artificial Intelligence software or systems that it uses?**	Enterprises where AI_TTM=1 or AI_TSR=1 or

Variable name	Code	Description	Filter/ Remarks (Standard codification)
			AI_TNLG=1 or AI_TIR=1 or AI_TML=1 or AI_TPA=1 or AI_TAR=1
AI_ADOWN		a) They were developed by own employees (including those employed in parent or affiliate enterprise)	
	1	Yes	
	0	No	
	Blank	No answer or option not included	
	9	Not applicable ((AI_TTM=Blank or AI_TTM<>1) and (AI_TSR=Blank or AI_TSR<>1) and (AI_TNLG=Blank or AI_TNLG<>1) and (AI_TIR=Blank or AI_TIR<>1) and (AI_TML=Blank or AI_TML<>1) and (AI_TPA=Blank or AI_TPA<>1) and (AI_TAR=Blank or AI_TAR<>1))	
AI_AMOWN		b) Commercial software or systems were modified by own employees (including those employed in parent or affiliate enterprise)	
	1	Yes	
	0	No	
	Blank	No answer or option not included	
	9	Not applicable ((AI_TTM=Blank or AI_TTM<>1) and (AI_TSR=Blank or AI_TSR<>1) and (AI_TNLG=Blank or AI_TNLG<>1) and (AI_TIR=Blank or AI_TIR<>1) and (AI_TML=Blank or AI_TML<>1) and (AI_TPA=Blank or AI_TPA<>1) and (AI_TAR=Blank or AI_TAR<>1))	
AI_AOS		c) Open-source software or systems were modified by own employees (including	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		those employed in parent or affiliate enterprise)	
	1	Yes	
	0	No	
	Blank	No answer or option not included	
	9	Not applicable ((AI_TTM=Blank or AI_TTM<>1) and (AI_TSR=Blank or AI_TSR<>1) and (AI_TNLG=Blank or AI_TNLG<>1) and (AI_TIR=Blank or AI_TIR<>1) and (AI_TML=Blank or AI_TML<>1) and (AI_TPA=Blank or AI_TPA<>1) and (AI_TAR=Blank or AI_TAR<>1))	
AI_ARDY		d) Commercial software or systems ready to use were purchased (including examples where it was already incorporated in a purchased item or system)	
	1	Yes	
	0	No	
	Blank	No answer or option not included	
	9	Not applicable ((AI_TTM=Blank or AI_TTM<>1) and (AI_TSR=Blank or AI_TSR<>1) and (AI_TNLG=Blank or AI_TNLG<>1) and (AI_TIR=Blank or AI_TIR<>1) and (AI_TML=Blank or AI_TML<>1) and (AI_TPA=Blank or AI_TPA<>1) and (AI_TAR=Blank or AI_TAR<>1))	
AI_AEXT		e) External providers were contracted to develop or modify them	
	1	Yes	
	0	No	
	Blank	No answer or option not included	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	9	Not applicable ((AI_TTM=Blank or AI_TTM<>1) and (AI_TSR=Blank or AI_TSR<>1) and (AI_TNLG=Blank or AI_TNLG<>1) and (AI_TIR=Blank or AI_TIR<>1) and (AI_TML=Blank or AI_TML<>1) and (AI_TPA=Blank or AI_TPA<>1) and (AI_TAR=Blank or AI_TAR<>1))	
AI_PDI (optional)		**E4: Does your enterprise process data on individuals using AI technologies?**	Enterprises where AI_TTM=1 or AI_TSR=1 or AI_TNLG=1 or AI_TIR=1 or AI_TML=1 or AI_TPA=1 or AI_TAR=1
	1	Yes	
	0	No	
	Blank	No answer or option not included	
	9	Not applicable ((AI_TTM=Blank or AI_TTM<>1) and (AI_TSR=Blank or AI_TSR<>1) and (AI_TNLG=Blank or AI_TNLG<>1) and (AI_TIR=Blank or AI_TIR<>1) and (AI_TML=Blank or AI_TML<>1) and (AI_TPA=Blank or AI_TPA<>1) and (AI_TAR=Blank or AI_TAR<>1))	
AI_BIAS (optional)		**E5: Does your enterprise have any measures to check the results generated by AI technologies for possible biases towards individuals based on sex, age, racial or ethnic origin, disability, religion or belief, sexual orientation?**	Enterprises where AI_PDI=1
	1	Yes	
	0	No	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	Blank	No answer or option not included	
	9	Not applicable (AI_PDI=Blank or AI_PDI<>1)	
AI_EC (optional)		**E6: Has your enterprise ever considered using any of the Artificial Intelligence technologies listed in question E1?**	Enterprises where (AI_TTM=Blank or AI_TTM=0) and (AI_TSR=Blank or AI_TSR=0) and (AI_TNLG=Blank or AI_TNLG=0) and (AI_TIR=Blank or AI_TIR=0) and (AI_TML=Blank or AI_TML=0) and (AI_TPA=Blank or AI_TPA=0) and (AI_TAR=Blank or AI_TAR=0)
	1	Yes	
	0	No	
	Blank	No answer or option not included	
	9	Not applicable (AI_TTM<>0 or AI_TSR<>0 or AI_TNLG<>0 or AI_TIR<>0 or AI_TML<>0 or AI_TPA<>0 or AI_TAR<>0)	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
(optional)		**E7: What are the reasons for your enterprise not to use any of the Artificial Intelligence technologies listed in question E1?**	Enterprises where AI_EC=1
AI_BCST		a) The costs seem too high	
	1	Yes	
	0	No	
	Blank	No answer or option not included	
	9	Not applicable (AI_EC=Blank or AI_EC<>1)	
AI_BLE		b) There is a lack of relevant expertise in the enterprise	
	1	Yes	
	0	No	
	Blank	No answer or option not included	
	9	Not applicable (AI_EC=Blank or AI_EC<>1)	
AI_BINC		c) Incompatibility with existing equipment, software or systems	
	1	Yes	
	0	No	
	Blank	No answer or option not included	
	9	Not applicable (AI_EC=Blank or AI_EC<>1)	
AI_BDDT		d) Difficulties with availability or quality of the necessary data	
	1	Yes	
	0	No	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	Blank	No answer or option not included	
	9	Not applicable (AI_EC=Blank or AI_EC<>1)	
AI_BCDP		e) Concerns regarding violation of data protection and privacy	
	1	Yes	
	0	No	
	Blank	No answer or option not included	
	9	Not applicable (AI_EC=Blank or AI_EC<>1)	
AI_BLEG		f) Lack of clarity about the legal consequences (e.g. liability in case of damage caused by the use of Artificial Intelligence)	
	1	Yes	
	0	No	
	Blank	No answer or option not included	
	9	Not applicable (AI_EC=Blank or AI_EC<>1)	
AI_BEC		g) Ethical considerations	
	1	Yes	
	0	No	
	Blank	No answer or option not included	
	9	Not applicable (AI_EC=Blank or AI_EC<>1)	
AI_BNU		h) Artificial Intelligence technologies are not useful for the enterprise	
	1	Yes	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
	0	No	
	Blank	No answer or option not included	
	9	Not applicable (AI_EC=Blank or AI_EC<>1)	
		Module X: Background Information	
		X1: Main economic activity of the enterprise, during 2023 (Nace rev 2.0)	All enterprises
NACE2	xx.xx	Nace code of the enterprise, 4 digits	
		X2: Average number of persons employed, during 2023	All enterprises
EMPL	Nnnnnn	Number of employed persons (use as many digits as necessary)	
		X3: Total turnover (in monetary terms, excluding VAT), for 2023	All enterprises
TOVT	Nnnnnn	Total turnover (use as many digits as necessary)	
		Y1: Enterprise ID	All enterprises
ENT_ID	XxNnnn nnn	Unique id of the enterprise (2 letters for country code, then 7 digits)	
		Y2: Enterprise Weight	All enterprises
ENT_WGHT	Nnnnnn. nnnnnn	Grossing up factor of the enterprise (6 digits, decimal point, 6 digits)	
DI4_INDEX	0-12	**Derived value: Digital Intensity index, v4**	
		Give one point for each of the following 12 conditions, if true:	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		EMPIUSE<>Blank and EMPL<>Blank and EMPIUSE > 0.50 * EMPL	
		ITSP2=1	
		ISPDF1=2 or ISPDF1=3 or ISPDF1=4 or ISPDF1=5	
		RM=1	
		SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1	
		ITSPT2=1 or ITUST2=1	
		SECM_CNT between 3 and 11	
		SECPOL2=1	
		RA_M=1 or RA_D=1 or RA_S=1	
		AI_TTM=1 or AI_TSR=1 or AI_TNLG=1 or AI_TIR=1 or AI_TML=1 or AI_TPA=1 or AI_TAR=1	
		(AWSVAL<>Blank and AWSVAL>=TOVT*0.01) or (AXSVAL<>Blank and AXSVAL>=TOVT*0.01) or (AWSVAL<>Blank and AWSVAL<>-1 and AXSVAL<>Blank and AXSVAL<>-1 and (AWSVAL+AXSVAL)>=TOVT*0.01)	
		TOVT<>Blank and AWSVAL<>Blank and AWSVALC<>Blank AND AWSVALC>0 AND (AWSVAL>0.01 * TOVT) and AWSVALC>0.1 * AWSVAL	
AI_CNT	0-7	**Derived value: Artificial intelligence technologies counter** Give one point for each of the following 7 conditions, if true:	
		AI_TTM=1	
		AI_TSR=1	
		AI_TNLG=1	
		AI_TIR=1	

Variable name	Code	Description	Filter/ Remarks (Standard codification)
		AI_TML=1	
		AI_TPA=1	
		AI_TAR=1	
	9	Not applicable (EMPIUSE = Blank or EMPIUSE = 0)	

3.2.2. Digital intensity index

The last variable of the table above is the digital intensity index (codified as DI3_INDEX). It measures the use of different digital technologies at enterprise level. The digital intensity index score of an enterprise is determined by how many of the selected digital technologies it uses and is between 0 and 12.

The index is used to calculate certain aggregates such as the number of enterprises that use ICT security measures according to their digital intensity index.

The table below provides guidance on how to calculate it.

DI4_INDEX	0-12	Derived value: Digital Intensity index	Description (references to MQ 2024 see section 3)
		Give one point for each of the following 12 conditions, if true:	
		EMPIUSE<>Blank and EMPL<>Blank and EMPIUSE > 0.50 * EMPL	Enterprises where more than 50% of the persons employed had access to the internet for business purposes
		ITSP2=1	Employ ICT specialists
		ISPDF1=2 or ISPDF1=3 or ISPDF1=4 or ISPDF1=5	The maximum contracted download speed of the fastest fixed line internet connection is at least 30 Mb/s
		RM=1	Enterprises which conducted remote meetings (via e.g. Skype, Zoom, MS Teams, WebEx, etc.)
		SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1	Enterprises make persons employed aware of their obligations in ICT security related issues

DI4_INDEX	0-12	Derived value: Digital Intensity index	Description (references to MQ 2024 see section 3)
		ITSPT2=1 or ITUST2=1	Any type of training provided to develop ICT related skills of the persons employed, during 2023
		SECM_CNT between 3 and 11	Use at least 3 ICT security measures
		SECPOL2=1	Enterprise with document(s) on measures, practices or procedures on ICT security
		RA_M=1 or RA_D=1 or RA_S=1	Any of the persons employed having remote access to any of the following: e-mail, documents, business apps
		AI_TTM=1 or AI_TSR=1 or AI_TNLG=1 or AI_TIR=1 or AI_TML=1 or AI_TPA=1 or AI_TAR=1	Use any AI technology
		(AWSVAL<>Blank and AWSVAL>=TOVT*0.01) or (AXSVAL<>Blank and AXSVAL>=TOVT*0.01) or (AWSVAL<>Blank and AWSVAL<>-1 and AXSVAL<>Blank and AXSVAL<>-1 and (AWSVAL+AXSVAL)>=TOVT*0.01)	Used any computer networks for sales (at least 1%)
		TOVT<>Blank and AWSVAL<>Blank and AWSVALC<>Blank and AWSVALC>0 AND (AWSVAL>0.01 * TOVT) and AWSVALC>0.1 * AWSVAL	Enterprises where web sales are more than 1% of the total turnover and B2C web sales more than 10% of the web sales

3.2.3. Indicator coding and scope

Indicators have to be computed by NSIs using the microdata gathered. The indicators are aggregates of microdata. The table below presents, among other indicators (column Variable codes), their description and the questions of the model questionnaire they are derived from. The indicators' scope can be computed using the codification of microdata presented above in Section 3.2.1.

Some indicators computed by Eurostat are also presented in the table for information.

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
back	ENT_SAMPLE	Number of enterprises in final (net) sample - non-raised figures	Count (ENT_ID) unraised	FALSE	2023
back	ENT	Number of enterprises in population (surveyed size and industry groups corresponding to raised figures)	Count (ENT_ID)	FALSE	2023
back	EMPL	Average number of employed persons in population (surveyed size and industry groups corresponding to raised figures)	∑ (EMPL)	FALSE	2023
back	TOVT	Total turnover in population, in value terms, excluding VAT (surveyed size and industry groups corresponding to raised figures)	∑ (TOVT)	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from A1	E_IUSE	Enterprises where persons employed have access to the internet for business purposes	Count (ENT_ID) where EMPIUSE<> Blank and EMPIUSE>0	FALSE	2023
Derived from A1	E_IUSE_GT 10	Enterprises where more than 10% of the persons employed have access to the internet for business purposes	Count (ENT_ID) where EMPIUSE<> Blank and EMPL<>Bla nk and EMPIUSE > 0.10 * EMPL	FALSE	2023
Derived from A1	E_IUSE_GT 50	Enterprises where more than 50% of the persons employed have access to the internet for business purposes	Count (ENT_ID) where EMPIUSE<> Blank and EMPL<>Bla nk and EMPIUSE > 0.50 * EMPL	FALSE	2023
Derived from A1	E_IUSE_GE 10A	Enterprises where at least 10 persons employed have access to the internet for business purposes	Count (ENT_ID) where EMPIUSE<> Blank and EMPIUSE >= 10	FALSE	2023
A2	E_FIXBB	Use any type of fixed connection	Count (ENT_ID) where FIXBB=1	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		to the internet			
A2	E_FIXBBX	Don't use any type of fixed connection to the internet	Count (ENT_ID) where FIXBB=0	FALSE	2023
A2	E_FIXBBZ	Don't know if they use any type of fixed connection to the internet	Count (ENT_ID) where FIXBB=Blank	TRUE	2023
A3a	E_ISPDF_LT30	The maximum contracted download speed of the fastest fixed internet connection is less than 30 Mb/s	Count (ENT_ID) where ISPDF1=1	FALSE	2023
A3b	E_ISPDF1_30_100	The maximum contracted download speed of the fastest fixed internet connection is at least 30 Mb/s but less than 100 Mb/s	Count (ENT_ID) where ISPDF1=2	FALSE	2023
A3c	E_ISPDF_100_500	The maximum contracted download speed of the fastest fixed	Count (ENT_ID) where ISPDF1=3	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		internet connection is at least 100 Mb/s but less than 500 Mb/s			
A3d	E_ISPDF_500_1G	The maximum contracted download speed of the fastest fixed internet connection is at least 500 Mb/s but less than 1 Gb/s	Count (ENT_ID) where ISPDF1=4	FALSE	2023
A3e	E_ISPDF_GE1G	The maximum contracted download speed of the fastest fixed internet connection is at least 1 Gb/s	Count (ENT_ID) where ISPDF1=5	FALSE	2023
Derived from A3b to A3e	E_ISPDF1_GE30	The maximum contracted download speed of the fastest fixed internet connection is at least 30 Mb/s	Count (ENT_ID) where ISPDF1=2 or ISPDF1=3 or ISPDF1=4 or ISPDF1=5	FALSE	2023
Derived from A3c to A3e	E_ISPDF1_GE100	The maximum contracted download	Count (ENT_ID) where ISPDF1=3	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		speed of the fastest fixed internet connection is at least 100 Mb/s	or ISPDF1=4 or ISPDF1=5		
A4	E_ISPDF1OK	The speed of the fixed line connection(s) to the internet is sufficient for the actual needs of the enterprise	Count (ENT_ID) where ISPDFOK=1	FALSE	2020
A4	E_ISPDF1OKX	The speed of the fixed line connection(s) to the internet is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where ISPDFOK=0	FALSE	2020
Derived from A3b to A3e and A4	E_ISPDF1OK_GE30	The speed of the fixed line connection(s) to the internet (at least 30 Mb/s) is sufficient for the actual needs of the enterprise	Count (ENT_ID) where (ISPDF1=2 or ISPDF1=3 or ISPDF1=4 or ISPDF1=5) and ISPDFOK=1	FALSE	2020
Derived from A3a and A4	E_ISPDFOKX_LT30	The speed of the fixed line connection(s	Count (ENT_ID) where ISPDF1=1	FALSE	2020

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		) to the internet (less than 30Mb/s) is not sufficient for the actual needs of the enterprise	and ISPDFOK=0		
Derived from A3b and A4	E_ISPDF1OKX_30_100	The speed of the fixed line connection(s) to the internet ([3-100[Mb/s) is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where ISPDF1=2 and ISPDFOK=0	FALSE	2020
Derived from A3c and A4	E_ISPDFOKX_100_500	The speed of the fixed line connection(s) to the internet ([100-500[Mb/s) is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where ISPDF1=3 and ISPDFOK=0	FALSE	2020
Derived from A3d and A4	E_ISPDFOKX_500_1G	The speed of the fixed line connection(s) to the internet ([500-1000[Mb/s) is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where ISPDF1=4 and ISPDFOK=0	FALSE	2020

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from A3e and A4	E_ISPDFOKX_GE1G	The speed of the fixed line connection(s) to the internet (at least 1 Gb/s) is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where ISPDF1=5 and ISPDFOK=0	FALSE	2020
Derived from A3b to A3e and A4	E_ISPDF1OKX_GE30	The speed of the fixed line connection(s) to the internet (at least 30 Mb/s) is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where (ISPDF1=2 or ISPDF1=3 or ISPDF1=4 or ISPDF1=5) and ISPDFOK=0	FALSE	2020
Derived from A3c to A3e and A4	E_ISPDF1OKX_GE100	The speed of the fixed line connection(s) to the internet (at least 100 Mb/s) is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where (ISPDF1=3 or ISPDF1=4 or ISPDF1=5) and ISPDFOK=0	FALSE	2020
Derived from A5	E_EMPMD2_GT0	Enterprises where some employed persons use a portable device provided by	Count (ENT_ID) where EMPMD2<> Blank and EMPMD2>0	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		the enterprise, that allows internet connection via mobile telephone networks, for business purposes			
Derived from A5	E_EMPMD2_GT10	Enterprises where more than 10% of the employed persons use a portable device provided by the enterprise, that allows internet connection via mobile telephone networks, for business purposes	Count (ENT_ID) where EMPMD2<> Blank and EMPL<>Blank and (EMPMD2> EMPL * 0.1)	TRUE	2022
Derived from A5	E_EMPMD2_GT20	Enterprises where more than 20% of the employed persons use a portable device provided by the enterprise, that allows internet connection via mobile	Count (ENT_ID) where EMPMD2<> Blank and EMPL<>Blank and (EMPMD2> EMPL * 0.2)	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		telephone networks, for business purposes			
Derived from A5	E_EMPMD2_GT50	Enterprises where more than 50% of the employed persons use a portable device provided by the enterprise, that allows internet connection via mobile telephone networks, for business purposes	Count (ENT_ID) where EMPMD2<> Blank and EMPL<>Blank and (EMPMD2> EMPL * 0.5)	TRUE	2022
Derived from A5	E_PMD	Provide portable devices that allow a mobile connection to the internet using mobile telephone networks, for business purposes	Count (ENT_ID) where EMPMD2>0	TRUE	2022
Derived from A5	E_PMDX	Does not provide portable devices that allow a mobile connection	Count (ENT_ID) where EMPMD2=0 or EMPMD2=Blank	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		to the internet using mobile telephone networks, for business purposes			
Derived from A2 and A5	E_BROAD3	Connect to the Internet via fixed or mobile broadband (as of 2018)	Count (ENT_ID) where FIXBB=1 or EMPMD2>0	TRUE	2022
Derived from A2 and A5	E_BROAD3X	Connect to the Internet but not via fixed or mobile broadband (as of 2018)	Count (ENT_ID) where (FIXBB=0 or FIXBB=Blank) and (EMPMD2=0 or EMPMD2=Blank)	TRUE	2022
A6a	E_RA_M	Enterprises with employed persons with remote access to the email system of the enterprise	Count (ENT_ID) where RA_M=1	FALSE	2022
A6a	E_RA_MX	Enterprises with employed persons with no remote access to the email system of	Count (ENT_ID) where RA_M=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		the enterprise			
A6b	E_RA_D	Enterprises with employed persons with remote access to the documents of the enterprise (e.g. files, spreadsheets, presentations, charts, photos)	Count (ENT_ID) where RA_D=1	FALSE	2022
A6b	E_RA_DX	Enterprises with employed persons with no remote access to the documents of the enterprise (e.g. files, spreadsheets, presentations, charts, photos)	Count (ENT_ID) where RA_D=0	FALSE	2022
A6c	E_RA_S	Enterprises with employed persons with remote access to the business applications or software	Count (ENT_ID) where RA_S=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		of the enterprise (e.g. access to accounting, sales, orders, CRM)			
A6c	E_RA_SX	Enterprises with employed persons with no remote access to the business applications or software of the enterprise (e.g. access to accounting, sales, orders, CRM)	Count (ENT_ID) where RA_S=0	FALSE	2022
Derived A6a to A6c	E_RA	Enterprises with employed persons with remote access to the email system or documents, business applications or software of the enterprise	Count (ENT_ID) where RA_M=1 or RA_D=1 or RA_S=1	FALSE	2022
Derived A6a to A6c	E_RA_ALL	Enterprises with employed persons with	Count (ENT_ID) where RA_M=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		remote access to the email system and documents and business applications or software of the enterprise	and RA_D=1 and RA_S=1		
Derived A6b to A6c	E_RA_DS	Enterprises with employed persons with remote access to the documents, business applications or software of the enterprise	Count (ENT_ID) where RA_D=1 or RA_S=1	FALSE	2022
A7	E_RM	Enterprises which conducted remote meetings (via e.g. Skype, Zoom, MS Teams, WebEx, etc.)	Count (ENT_ID) where RM=1	FALSE	2022
A7	E_RMX	Enterprises which did not conduct remote meetings (via e.g. Skype, Zoom, MS	Count (ENT_ID) where RM=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		Teams, WebEx, etc.)			
A8	E_ADS	Pay to advertise on the Internet	Count (ENT_ID) where ADS=1	TRUE	2023
A8	E_ADSX	Don't pay to advertise on the Internet	Count (ENT_ID) where ADS=0	TRUE	2023
A9a	E_ADS_KW	Pay to advertise on the internet, based on content or keywords searched by internet users	Count (ENT_ID) where ADS_KW=1	TRUE	2023
A9a	E_ADS_KWX	Pay to advertise on the internet, but not based on content or keywords searched by internet users	Count (ENT_ID) where ADS_KW=0	TRUE	2023
A9b	E_ADS_TRK	Pay to advertise on the internet, based on the tracking of internet users' past activities or profile	Count (ENT_ID) where ADS_TRK=1	TRUE	2023
A9b	E_ADS_TRKX	Pay to advertise on the internet,	Count (ENT_ID)	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		but not based on the tracking of internet users' past activities or profile	where ADS_TRK=0		
A9c	E_ADS_LOC	Pay to advertise on the internet, based on the geolocation of internet users	Count (ENT_ID) where ADS_LOC= 1	TRUE	2023
A9c	E_ADS_LOCX	Pay to advertise on the internet, but not based on the geolocation of internet users	Count (ENT_ID) where ADS_LOC= 0	TRUE	2023
A9d	E_ADS_OTH	Pay to advertise on the internet, based on any other method of target advertising	Count (ENT_ID) where ADS_OTH= 1	TRUE	2023
A9d	E_ADS_OTHX	Pay to advertise on the internet, but not based on any other method of target advertising	Count (ENT_ID) where ADS_OTH= 0	TRUE	2023
A9a to A9c	E_ADS3	Pay to advertise on the internet,	Count (ENT_ID) where	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		based on content, keywords, internet users' past activities or profile or the geolocation	ADS_KW=1 or ADS_TRK=1 or ADS_LOC= 1		
B1a	E_AWS_COWN	Enterprises which sold via their own websites or apps	Count (ENT_ID) where AWS_COWN=1	FALSE	2023
B1a	E_AWS_COWNX	Enterprises which did not sell via their own websites or apps	Count (ENT_ID) where AWS_COWN=0	FALSE	2023
B1b	E_AWS_CMP	Enterprises which sold via an e-commerce marketplace	Count (ENT_ID) where AWS_CMP= 1	FALSE	2023
B1b	E_AWS_CMPX	Enterprises which did not sell via an e-commerce marketplace	Count (ENT_ID) where AWS_CMP= 0	FALSE	2023
Derived from B1a and B1b	E_AWS_CBOTH	Enterprises which sold via their own websites or apps and via an e-commerce marketplace	Count (ENT_ID) where AWS_COWN=1 and AWS_CMP= 1	FALSE	2023
Derived from B1a and B1b	E_AWSELL	Have received orders via	Count (ENT_ID) where	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		websites or apps (web sales)	(AWS_COWN=1 or AWS_CMP=1)		
Derived from B1a and B1b	E_AWSELLX	Have not received orders via websites or apps	Count (ENT_ID) where (AWS_COWN=0 and AWS_CMP=0)	FALSE	2023
Derived from B1a and B1b	E_AWS_CMPQ	Enterprises which sold via an e-commerce marketplace but not via their own websites or apps	Count (ENT_ID) where AWS_COWN=0 and AWS_CMP=1	FALSE	NEW derived variable
B2	E_AWSVALS	Sales via websites or apps, excluding VAT (<1% of turnover)	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL < TOVT * 1%	FALSE	2023
B2	E_AWSVALB	Sales via websites or apps, excluding VAT (>= 1% of turnover)	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL >= TOVT * 1%	FALSE	2023
B2	E_AWSVAL	Total sales via websites or apps, excluding VAT	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
B2	E_WSEL0	Total sales via websites or apps equal to or above 0	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>=0	FALSE	2023
B2	E_WSEL1	Total sales via websites or apps equal to or above 1% of turnover	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>= TOVT * 1%	FALSE	2023
B2	E_WSEL2	Total sales via websites or apps equal to or above 2% of turnover	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>= TOVT * 2%	FALSE	2023
B2	E_WSEL5	Total sales via websites or apps equal to or above 5% of turnover	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>= TOVT * 5%	FALSE	2023
B2	E_WSEL10	Total sales via websites or apps equal to or above 10% of turnover	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
			AWSVAL>= TOVT * 10%		
B2	E_WSEL25	Total sales via websites or apps equal to or above 25% of turnover	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>= TOVT * 25%	FALSE	2023
B2	E_WSEL50	Total sales via websites or apps equal to or above 50% of turnover	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>= TOVT * 50%	FALSE	2023
B2	E_WSELZ	Don't know the % of sales via websites or apps	Count (ENT_ID) where AWSVAL=Blank	TRUE	2023
Derived from B1 and B2	E_AWSVAL_CMPQ	Web sales by enterprises which only sell through an e-commerce marketplace	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1 and AWS_COWN=0 and AWS_CMP= 1	FALSE	NEW derived variable
B3a	E_AWSVAL_COWN	Sales via their own websites or apps	∑(AWSVAL_COWN) where AWSVAL_COWN <> Blank and	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
			AWSVAL_COWN <> -1		
B3b	E_AWSVAL_CMP	Sales via an e-commerce marketplace	∑(AWSVAL_CMP) where AWSVAL_CMP <> Blank and AWSVAL_CMP <> -1	FALSE	2023
Derived from B1b, B2 and B3b	E_AWS_CMP_GE20	Enterprises which sold via an e-commerce marketplace for at least 20% of the web sales	Count (ENT_ID) where AWS_CMP= 1 and (AWSVAL_CMP >= AWSVAL * 0.20)	FALSE	2023
Derived from B1b, B2 and B3b	E_AWS_CMP_GE50	Enterprises which sold via an e-commerce marketplace for at least 50% of the web sales	Count (ENT_ID) where AWS_CMP= 1 and (AWSVAL_CMP >= AWSVAL * 0.50)	FALSE	2023
Derived from B1b, B2 and B3b	T_AWS_CMP_GE20	Turnover of the enterprises where web sales via an e-commerce marketplace were at least 20% of the web sales	∑ (TOVT) where AWS_CMP= 1 and (AWSVAL_CMP >= AWSVAL * 0.20)	FALSE	2023
Derived from B1b, B2 and B3b	T_AWS_CMP_GE50	Turnover of the enterprises where web sales via an	∑ (TOVT) where AWS_CMP= 1 and (AWSVAL_C	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		e-commerce marketplace were at least 50% of the web sales	MP >= AWSVAL * 0.50)		
B4a	E_AWSVAL_B2C	Sales via websites or apps (B2C)	∑ (AWSVALC) where AWSVALC<>Blank and AWSVALC<>-1	FALSE	2023
B4b	E_AWSVAL_B2BG	Sales via websites or apps (B2B) and (B2G)	∑ (AWSVALBG) where AWSVALBG<>Blank and AWSVALBG<>-1	FALSE	2023
Derived from B4a	E_AWS_B2C	Enterprises which sold via websites or apps - B2C	Count (ENT_ID) where AWSVALC<>Blank and AWSVALC>0	FALSE	2023
Derived from B4b	E_AWS_B2BG	Enterprises which sold via websites or apps - B2B and B2G	Count (ENT_ID) where AWSVALBG<>Blank and AWSVALBG>0	FALSE	2023
Derived from B1b and B4a	E_AWS_B2C_CMP	Enterprises which sold via websites or apps - B2C and via an e-commerce marketplace	Count (ENT_ID) where (AWSVALC<>Blank and AWSVALC>0) and AWS_CMP=1	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from B2 and B4a	E_AWSVAL_B2C_GE10WS	Enterprises where B2C sales via websites or apps were 10% or more of the total web sales	Count (ENT_ID) where AWSVALC<>Blank AND AWSVALC>0 and AWSVAL<>Blank and AWSVAL<>-1 and AWSVALC>=0.1 * AWSVAL	FALSE	2023
Derived from B2 and B4a	E_AWSVAL_B2C_GE5WS	Enterprises where B2C sales via websites or apps were 5% or more of the total web sales	Count (ENT_ID) where AWSVALC<>Blank AND AWSVALC>0 and AWSVAL<>Blank and AWSVAL<>-1 and AWSVALC>=0.05 * AWSVAL	FALSE	2023
Derived from B2 and B4a	E_AWS_B2C_GT1WS	Enterprises where B2C sales via websites or apps were more than 1% of the total web sales	Count (ENT_ID) where AWSVALC<>Blank AND AWSVALC>0 and AWSVAL<>Blank and AWSVAL<>-1 and AWSVALC>0.01 * AWSVAL	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from B2 and B4a	E_AWSVAL_GT1_B2C_GT10WS	B2C web sales of enterprises where web sales > 1% of total turnover and B2C web sales > 10% of the web sales	∑ (AWSVALC) where TOVT<>Blank and AWSVAL<>Blank and AWSVALC<>Blank AND (AWSVAL>0.01 * TOVT) and AWSVALC>0.1 * AWSVAL	FALSE	2023
Derived from B2 and B4a	E_AWS_GT1_B2C_GT10WS	Enterprises where web sales were more than 1% of the total turnover and B2C web sales more than 10% of the web sales	Count (ENT_ID) where TOVT<>Blank and AWSVAL<>Blank and AWSVALC<>Blank AND AWSVALC>0 AND (AWSVAL>0.01 * TOVT) and AWSVALC>0.1 * AWSVAL	FALSE	2023
Derived from B2 and B4a	T_AWS_GT1_B2C_GT10WS	Turnover of the enterprises where web sales were more than 1% of the total turnover and B2C web sales more than 10% of	∑ (TOVT) where TOVT<>Blank and AWSVAL<>Blank and AWSVALC<>Blank AND AWSVALC>0 AND (AWSVAL>0.01 * TOVT)	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		the web sales	and AWSVALC> 0.1 * AWSVAL		
Derived from B1b, B2 and B4a	E_AWSVAL_B2C_GE10WS_CMP	Enterprises where B2C sales via websites or apps were 10% or more of the total web sales and which sold via an e-commerce marketplace	Count (ENT_ID) where AWSVALC<>Blank AND AWSVALC> 0 and AWSVAL<> Blank and AWSVAL<>-1 and AWSVALC>=0.1 * AWSVAL and AWS_CMP=1	FALSE	2023
B5a	E_AWSHM	Received orders placed via a website or apps from customers in own country	Count (ENT_ID) where AWSHM=1	FALSE	2022
B5b	E_AWSEU	Received orders placed via a website or apps from customers in other EU countries	Count (ENT_ID) where AWSEU=1	FALSE	2022
B5c	E_AWSWW	Received orders placed via a website or apps from customers in	Count (ENT_ID) where AWSWW=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		the rest of the world			
Derived from B5b and B5c	E_AWSFOR	Received orders placed via a website or apps from customers in foreign countries (EU or rest of the world)	Count (ENT_ID) where AWSEU=1 or AWSWW=1	FALSE	2022
Derived from B3a and B5a	E_AWSVAL_HM_COWN	Sales via their own websites or apps from customers in the own country	∑(AWSVAL_COWN) where AWSVAL_COWN <> Blank and AWSVAL_COWN <> -1 and AWSHM=1	FALSE	2022
Derived from B3b and B5a	E_AWSVAL_HM_CMP	Sales via an e-commerce marketplace from customers in the own country	∑(AWSVAL_CMP) where AWSVAL_CMP <> Blank and AWSVAL_CMP <> -1 and AWSHM=1	FALSE	2022
Derived from B3a and B5b	E_AWSVAL_EU_COWN	Sales via their own websites or apps from customers in other EU countries	∑(AWSVAL_COWN) where AWSVAL_COWN <> Blank and AWSVAL_COWN <> -1 and AWSEU=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from B3b and B5b	E_AWSVAL_EU_CMP	Sales via an e-commerce marketplace from customers in other EU countries	∑(AWSVAL_CMP) where AWSVAL_CMP <> Blank and AWSVAL_CMP <> -1 and AWSEU=1	FALSE	2022
Derived from B3a and B5c	E_AWSVAL_WW_COWN	Sales via their own websites or apps from customers located in the rest of the world	∑(AWSVAL_COWN) where AWSVAL_COWN <> Blank and AWSVAL_COWN <> -1 and AWSWW=1	FALSE	2022
Derived from B3b and B5c	E_AWSVAL_WW_CMP	Sales via an e-commerce marketplace from customers located in the rest of the world	∑(AWSVAL_CMP) where AWSVAL_CMP <> Blank and AWSVAL_CMP <> -1 and AWSWW=1	FALSE	2022
Derived from B1, B2 and B5a	E_AWSVAL_HM_CMPQ	Web sales by enterprises which only sell through an e-commerce marketplace from customers in the own country	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1 and AWSHM=1 and AWS_COWN=0 and AWS_CMP=1	FALSE	NEW derived variable
Derived from B1, B2 and B5b	E_AWSVAL_EU_CMPQ	Web sales by enterprises	∑ (AWSVAL) where AWSVAL <>	FALSE	NEW derived variable

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		which only sell through an e-commerce marketplace from customers in other EU countries	Blank and AWSVAL <> -1 and AWSEU=1 and AWS_COWN=0 and AWS_CMP=1		
Derived from B1, B2 and B5c	E_AWSVAL_WW_CMPQ	Web sales by enterprises which only sell through an e-commerce marketplace from customers located in the rest of the world	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1 and AWSWW=1 and AWS_COWN=0 and AWS_CMP=1	FALSE	NEW derived variable
B6a	E_AWSVAL_HM	Sales via a website or apps - from customers located in the own country	∑(AWSVAL_HM) where AWSVAL_HM <> Blank and AWSVAL_HM <> -1	TRUE	2022
B6b	E_AWSVAL_EU	Sales via a website or apps - from customers located in other EU countries	∑(AWSVAL_EU) where AWSVAL_EU <> Blank and AWSVAL_EU <> -1	TRUE	2022
B6c	E_AWSVAL_WW	Sales via a website or apps - from customers located in	∑(AWSVAL_WW) where AWSVAL_WW <> Blank and	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		the rest of the world	AWSVAL_WW <> -1		
Derived from B2, B5b, B5c, B6b and B6c	E_AWSFOR_GE20	Received orders placed via a website or apps from customers in foreign countries, with an export turnover at least 20% of the web sales	Count (ENT_ID) where (AWSEU=1 or AWSWW=1) and (AWSVAL_EU + AWSVAL_WW >= 0.20 * AWSVAL)	TRUE	2022
Derived from B2, B5b, B5c, B6b and B6c	E_AWSFOR_GE50	Received orders placed via a website or apps from customers in foreign countries, with an export turnover at least 50% of the web sales	Count (ENT_ID) where (AWSEU=1 or AWSWW=1) and (AWSVAL_EU + AWSVAL_WW >= 0.50 * AWSVAL)	TRUE	2022
Derived from B1, B2, B5b, B5c, B6b and B6c	E_AWSFOR_GE20_CMP	Sold via an e-commerce marketplace and received orders placed via a website or apps from customers in foreign countries, with an export	Count (ENT_ID) where AWS_CMP= 1 and (AWSEU=1 or AWSWW=1) and (AWSVAL_EU + AWSVAL_W	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		turnover at least 20% of the web sales	W >= 0.20 * AWSVAL)		
Derived from B1, B2, B5b, B5c, B6b and B6c	E_AWSFOR_GE50_CMP	Sold via an e-commerce marketplace and received orders placed via a website or apps from customers in foreign countries, with an export turnover at least 50% of the web sales	Count (ENT_ID) where AWS_CMP= 1 and (AWSEU=1 or AWSWW=1) and (AWSVAL_EU + AWSVAL_WW >= 0.50 * AWSVAL)	TRUE	2022
B7	E_AXSELL	Have received orders via EDI-type messages	Count (ENT_ID) where AXSELL=1	FALSE	2023
B7	E_AXSELLX	Have not received orders via EDI-type messages	Count (ENT_ID) where AXSELL=0	FALSE	2023
B7	E_AXSELLZ	Do not know if have received orders via EDI-type messages	Count (ENT_ID) where AXSELL=Blank	TRUE	2023
Derived from B1 and B7	E_AESELL	Have received orders via computer networks	Count (ENT_ID) where (AWS_COWN=1 or	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
			AWS_CMP= 1) or AXSELL=1		
Derived from B2 and B7	E_WSEL1Q	Enterprises with web sales via websites or apps equal to or above 1% of turnover, and no orders received via EDI-type messages	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>= TOVT * 1% and (AXSELL=0 or AXSELL=Blank)	FALSE	2023
B8	E_AXSVALS	Total sales via EDI type messages, excluding VAT(<1% of turnover)	∑ (AXSVAL) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL<TOVT * 1%	FALSE	2023
B8	E_AXSVALB	Total sales via EDI type messages, excluding VAT(>=1% of turnover)	∑ (AXSVAL) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=TOVT * 1%	FALSE	2023
B8	E_AXSVAL	Total sales via EDI type messages, excluding VAT	∑ (AXSVAL]) where AXSVAL <> Blank and AXSVAL <> -1	FALSE	2023
B8	E_XSEL0	Enterprises with sales via EDI-type	Count (ENT_ID) where	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		messages equal to or above 0	AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=0		
B8	E_XSEL1	Enterprises with sales via EDI-type messages equal to or above 1% of turnover	Count (ENT_ID) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=TOVT * 1%	FALSE	2023
B8	E_XSEL2	Enterprises with sales via EDI-type messages equal to or above 2% of turnover	Count (ENT_ID) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=TOVT * 2%	FALSE	2023
B8	E_XSEL5	Enterprises with sales via EDI-type messages equal to or above 5% of turnover	Count (ENT_ID) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=TOVT * 5%	FALSE	2023
B8	E_XSEL10	Enterprises with sales via EDI-type messages equal to or above 10% of turnover	Count (ENT_ID) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=TOVT * 10%	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
B8	E_XSEL25	Enterprises with sales via EDI-type messages equal to or above 25% of turnover	Count (ENT_ID) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=TOVT * 25%	FALSE	2023
B8	E_XSEL50	Enterprises with sales via EDI-type messages equal to or above 50% of turnover	Count (ENT_ID) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=TOVT * 50%	FALSE	2023
B8	E_XSELZ	Don't know the % of sales via EDI-type messages	Count (ENT_ID) where AXSVAL=Blank	TRUE	2023
Derived from B2 and B8	E_ESELL	Enterprises which used any computer networks for sales (at least 1%)	Count (ENT_ID) where (AWSVAL<> Blank and AWSVAL>= TOVT*1%) or (AXSVAL<> Blank and AXSVAL>=TOVT*1%) or (AWSVAL<> Blank and AWSVAL<>-1 and AXSVAL<>Blank and AXSVAL<>-1 and	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
			(AWSVAL+AXSVAL)>=TOVT*1%)		
Derived from B2 and B8	E_ETURN	Total electronic sales, excluding VAT	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1 + ∑ (AXSVAL) where AXSVAL <> Blank and AXSVAL <> -1	FALSE	2023
Derived from B2, B4a and B8	E_AWSVAL_B2C_GE10EC	Enterprises where B2C sales via websites or apps were 10% or more of the e-commerce turnover	Count (ENT_ID) where AWSVALC<>Blank AND AWSVALC> 0 and AWSVAL<> Blank and AXSVAL<>Blank and ((AXSVAL = -1 and AWSVALC>=0.1 * AWSVAL) or (AXSVAL <> -1 and AWSVALC>=0.1 * (AWSVAL+AXSVAL)))	FALSE	2023
C1	E_ITSP2	Employ ICT specialists	Count(ENT_ID) where ITSP2=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
C1	E_ITSP2X	Don't employ ICT specialists	Count(ENT_ID) where ITSP2=0	FALSE	2022
C2a	E_ITSPT2	Have provided training to develop ICT skills of personnel: for ICT specialists	Count(ENT_ID) where ITSPT2=1	FALSE	2022
C2a	E_ITSPT2X	Have not provided training to develop ICT skills of personnel: for ICT specialists	Count(ENT_ID) where ITSPT2=0	FALSE	2022
C2b	E_ITUST2	Have provided training to develop ICT skills of personnel: for other persons employed	Count(ENT_ID) where ITUST2=1	FALSE	2022
C2b	E_ITUST2X	Have not provided training to develop ICT skills of personnel: for other persons employed	Count(ENT_ID) where ITUST2=0	FALSE	2022
Derived from C2a and C2b	E_ITT2	Have provided training to develop ICT	Count(ENT_ID) where ITSPT2=1 or ITUST2=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		skills of personnel			
Derived from C1 and C2a	E_ITSP2_ITT2	Employ ICT specialists and have provided training to develop ICT skills of personnel	Count(ENT_ID) where ITSP2=1 and (ITSPT2=1 or ITUST2=1)	FALSE	2022
Derived from C1 and C2b	E_ITSP2X_ITT2	Don't employ ICT specialists, but have provided training to develop ICT skills of personnel	Count(ENT_ID) where ITSP2=0 and ITUST2=1	FALSE	2022
C3	E_ITSPRCR2	Have recruited/tried to recruit ICT specialists	Count(ENT_ID) where ITSPRCR2=1	FALSE	2022
C3	E_ITSPRCR2X	Have not recruited/tried to recruit ICT specialists	Count(ENT_ID) where ITSPRCR2=0	FALSE	2022
Derived from C1 and C3	E_ITSP2_OR_RCR2	Employ ICT specialists or have recruited/tried to recruit ICT specialists	Count(ENT_ID) where ITSP2=1 or ITSPRCR2=1	FALSE	2022
C4	E_ITSPVAC2	Had hard-to-fill vacancies for ICT specialists	Count(ENT_ID) where ITSPVAC2=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
C4	E_ITSPVAC2X	Had no hard-to-fill vacancies for ICT specialists	Count(ENT_ID) where ITSPVAC2=0	FALSE	2022
Derived from C1 and C4	E_ITSP2_VAC2	Employ ICT specialists and had hard-to-fill vacancies for ICT specialists	Count(ENT_ID) where ITSP2=1 and ITSPVAC2=1	FALSE	2022
C5a	E_ITSPDLA	Enterprises had the following difficulty to recruit ICT specialists: Lack of applications	Count(ENT_ID) where ITSPDLA=1	TRUE	2022
C5a	E_ITSPDLAX	Enterprises did not have the following difficulty to recruit ICT specialists: Lack of applications	Count(ENT_ID) where ITSPDLA=0	TRUE	2022
C5b	E_ITSPDLET	Enterprises had the following difficulty to recruit ICT specialists: Applicants' lack of relevant ICT qualifications from education and/or training	Count(ENT_ID) where ITSPDLET=1	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
C5b	E_ITSPDLETX	Enterprises did not have the following difficulty to recruit ICT specialists: Applicants' lack of relevant ICT qualifications from education and/or training	Count(ENT_ID) where ITSPDLET=0	TRUE	2022
C5c	E_ITSPDLWE	Enterprises had the following difficulty to recruit ICT specialists: Applicants' lack of relevant work experience	Count(ENT_ID) where ITSPDLWE=1	TRUE	2022
C5c	E_ITSPDLWEX	Enterprises did not have the following difficulty to recruit ICT specialists: Applicants' lack of relevant work experience	Count(ENT_ID) where ITSPDLWE=0	TRUE	2022
C5d	E_ITSPDSAL	Enterprises had the following difficulty to recruit ICT specialists: Applicants'	Count(ENT_ID) where ITSPDSAL=1	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		salary expectation too high			
C5d	E_ITSPDSALX	Enterprises did not have the following difficulty to recruit ICT specialists: Applicants' salary expectation too high	Count(ENT_ID) where ITSPDSAL=0	TRUE	2022
C6a	E_IT_OWN	ICT functions were performed by own employees (incl. those employed in parent or affiliate enterprises)	Count(ENT_ID) where IT_OWN=1	FALSE	2022
C6a	E_IT_OWNX	ICT functions were not performed by own employees (incl. those employed in parent or affiliate enterprises)	Count(ENT_ID) where IT_OWN=0	FALSE	2022
C6b	E_IT_EXT	ICT functions were performed by external suppliers	Count(ENT_ID) where IT_EXT=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
C6b	E_IT_EXTX	ICT functions were not performed by external suppliers	Count(ENT_ID) where IT_EXT=0	FALSE	2022
Derived from C6a and C6b	E_IT_OWNQ	ICT functions were entirely performed by own employees (incl. those employed in parent or affiliate enterprises)	Count(ENT_ID) where IT_OWN=1 and IT_EXT=0	FALSE	2022
Derived from C6a and C6b	E_IT_EXTQ	ICT functions were entirely performed by external suppliers	Count(ENT_ID) where IT_OWN=0 and IT_EXT=1	FALSE	2022
Derived from C1 and C6a	E_IT_OWN_ITSP2	Employ ICT specialists and ICT functions were performed by own employees (incl. those employed in parent or affiliate enterprises)	Count(ENT_ID) where IT_OWN=1 AND ITSP2=1	FALSE	2022
Derived from C1 and C6b	E_IT_EXT_ITSP2	Employ ICT specialists and ICT functions were performed	Count(ENT_ID) where IT_EXT=1 AND ITSP2=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		by external suppliers			
Derived from C2a and C2b, C6b	E_IT_EXT_ITT2	Provided training to develop ICT skills of personnel and ICT functions were performed by external suppliers	Count(ENT_ID) where IT_EXT=1 AND (ITSPT2=1 or ITUST2=1)	FALSE	2022
D1a	E_SECMSPSW	ICT security measure used: strong password authentication, i.e. minimum length of 8 mixed characters, periodical change	Count(ENT_ID) where SECMSPSW=1	FALSE	2022
D1a	E_SECMSPSWX	ICT security measure not used: strong password authentication, i.e. minimum length of 8 mixed characters, periodical change	Count(ENT_ID) where SECMSPSW=0	FALSE	2022
D1b	E_SECMUIBM	ICT security measure used: user identification and	Count(ENT_ID) where SECMUIBM=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		authentication via biometric methods implemented by the enterprise (e.g. based on fingerprints, voice, faces)			
D1b	E_SECMUIBMX	ICT security measure not used: user identification and authentication via biometric methods implemented by the enterprise (e.g. based on fingerprints, voice, faces)	Count(ENT_ID) where SECMUIBM=0	FALSE	2022
D1c	E_SECMDUO	ICT security measure used: at least two authentication mechanisms (i.e. combination of e.g. user-defined password, one-time password (OTP), code generated via a	Count(ENT_ID) where SECMDUO=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		security token or received via a smartphone, biometric method faces)			
D1c	E_SECMDUOX	ICT security measure not used: at least two authentication mechanisms (i.e. combination of e.g. user-defined password, one-time password (OTP), code generated via a security token or received via a smartphone, biometric method (e.g. based on fingerprints,	Count(ENT_ID) where SECMDUO=0	FALSE	2022
D1d	E_SECMDENC	ICT security measure used: encryption techniques for data, documents or e-mails	Count(ENT_ID) where SECMDENC=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
D1d	E_SECMDENCX	ICT security measure not used: encryption techniques for data, documents or e-mails	Count(ENT_ID) where SECMDENC=0	FALSE	2022
D1e	E_SECMOSBU	ICT security measure used: data backup to a separate location (including backup to the cloud)	Count(ENT_ID) where SECMOSBU=1	FALSE	2022
D1e	E_SECMOSBUX	ICT security measure not used: data backup to a separate location (including backup to the cloud)	Count(ENT_ID) where SECMOSBU=0	FALSE	2022
D1f	E_SECMNAC	ICT security measure used: network access control (management of access by devices and users to the enterprise's network)	Count(ENT_ID) where SECMNAC=1	FALSE	2022
D1f	E_SECMNACX	ICT security measure not used: network	Count(ENT_ID) where SECMNAC=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		access control (management of access by devices and users to the enterprise's network)			
D1g	E_SECMVPN	ICT security measure used: VPN (Virtual Private Network extends a private network across a public network to enable secure exchange of data over public network)	Count(ENT_ID) where SECMVPN=1	FALSE	2022
D1g	E_SECMVPNX	ICT security measure not used: VPN (Virtual Private Network extends a private network across a public network to enable secure exchange of data over	Count(ENT_ID) where SECMVPN=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		public network)			
D1h	E_SECMSMS1	ICT security measure used: monitoring system that allows to detect suspicious activity in the ICT systems and alerts the enterprises about it, other than standalone anti-virus software	Count(ENT_ID) where SECMSMS1=1	FALSE	NEW
D1h	E_SECMSMS1X	ICT security measure not used: monitoring system that allows to detect suspicious activity in the ICT systems and alerts the enterprises about it, other than standalone anti-virus software	Count(ENT_ID) where SECMSMS1=0	FALSE	NEW
D1i	E_SECMLOG	ICT security measure used: maintaining log files for	Count(ENT_ID) where SECMLOG=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		analysis after security incidents			
D1i	E_SECMLOGX	ICT security measure not used: maintaining log files for analysis after security incidents	Count(ENT_ID) where SECMLOG=0	FALSE	2022
D1j	E_SECMRASS	ICT security measure used: ICT risk assessment, i.e. periodically assessment of probability and consequences of ICT security incidents	Count(ENT_ID) where SECMRASS=1	FALSE	2022
D1j	E_SECMRASSX	ICT security measure not used: ICT risk assessment, i.e. periodically assessment of probability and consequences of ICT security incidents	Count(ENT_ID) where SECMRASS=0	FALSE	2022
D1k	E_SECMTST	ICT security measure used: ICT	Count(ENT_ID) where	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		security tests (e.g. performing penetration tests, testing security alert system, review of security measures, testing of backup systems)	SECMTST= 1		
D1k	E_SECMTSTX	ICT security measure not used: ICT security tests (e.g. performing penetration tests, testing security alert system, review of security measures, testing of backup systems)	Count(ENT_ID) where SECMTST= 0	FALSE	2022
Derived from D1a to D1k	E_SECMGE1	Use any ICT security measure	Count(ENT_ID) where SECM_CNT >=1	FALSE	2022
Derived from D1a to D1k	E_SECMGE3	Use at least 3 ICT security measures	Count(ENT_ID) where SECM_CNT >=3	FALSE	2022
Derived from D1a to D1k	E_SECMGE5	Use at least 5 ICT security measures	Count(ENT_ID) where SECM_CNT >=5	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from D1a to D1k	E_SECMGE7	Use at least 7 ICT security measures	Count(ENT_ID) where SECM_CNT >=7	FALSE	2022
Derived from D1a to D1k	E_SECMALL	Use all ICT security measures	Count(ENT_ID) where SECM_CNT =11	FALSE	2022
D2a	E_SECAWVTGI	Make persons employed aware of their obligations in ICT security related issues by voluntary training or internally available information (e.g. information on the intranet)	Count(ENT_ID) where SECAWVTGI=1	FALSE	2022
D2a	E_SECAWVTGIX	Don't make persons employed aware of their obligations in ICT security related issues by voluntary training or internally available information (e.g.	Count(ENT_ID) where SECAWVTGI=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		information on the intranet)			
D2b	E_SECAWCTP	Make persons employed aware of their obligations in ICT security related issues by compulsory training courses or viewing compulsory material	Count(ENT_ID) where SECAWCTP=1	FALSE	2022
D2b	E_SECAWCTPX	Don't make persons employed aware of their obligations in ICT security related issues by compulsory training courses or viewing compulsory material	Count(ENT_ID) where SECAWCTP=0	FALSE	2022
D2c	E_SECAWCONT	Make persons employed aware of their obligations in ICT security	Count(ENT_ID) where SECAWCONT=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		related issues by contract (e.g. contract of employment)			
D2c	E_SECAWCONTX	Don't make persons employed aware of their obligations in ICT security related issues by contract (e.g. contract of employment)	Count(ENT_ID) where SECAWCONT=0	FALSE	2022
Derived from D2a to D2c	E_SECAWANY	Make persons employed aware of their obligations in ICT security related issues	Count(ENT_ID) where SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1	FALSE	2022
Derived from D2a to D2c	E_SECAWNONE	Don't make persons employed aware of their obligations in ICT security related issues	Count(ENT_ID) where SECAWVTGI=0 and SECAWCTP=0 and SECAWCONT=0	FALSE	2022
Derived from C2a and D2a	E_SECAWVTGI_ITSPT2	Make persons	Count(ENT_ID) where	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		employed aware of their obligations in ICT security related issues by voluntary training or internally available information and have provided training to develop ICT skills for ICT specialists	SECAWVTGI=1 and ITSPT2=1		
Derived from C2b and D2a	E_SECAWVTGI_ITUST2	Make persons employed aware of their obligations in ICT security related issues by voluntary training or internally available information and have provided training to develop ICT skills for other persons employed	Count(ENT_ID) where SECAWVTGI=1 and ITUST2=1	FALSE	2022
Derived from C2a and D2b	E_SECAWCTP_ITSPT2	Make persons	Count(ENT_ID) where	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		employed aware of their obligations in ICT security related issues by compulsory training courses or viewing compulsory material and have provided training to develop ICT skills for ICT specialists	SECAWCTP=1 and ITSPT2=1		
Derived from C2b and D2b	E_SECAWCTP_ITUST2	Make persons employed aware of their obligations in ICT security related issues by compulsory training courses or viewing compulsory material and have provided training to develop ICT skills for other persons employed	Count(ENT_ID) where SECAWCTP=1 and ITUST2=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from C2a and D2c	E_SECAWCONT_ITSPT2	Make persons employed aware of their obligations in ICT security related issues by contract and have provided training to develop ICT skills for ICT specialists	Count(ENT_ID) where SECAWCONT=1 and ITSPT2=1	FALSE	2022
Derived from C2b and D2c	E_SECAWCONT_ITUST2	Make persons employed aware of their obligations in ICT security related issues by contract and have provided training to develop ICT skills for other persons employed	Count(ENT_ID) where SECAWCONT=1 and ITUST2=1	FALSE	2022
Derived from C2a and D2	E_SECAWANY_ITSPT2	Make persons employed aware of their obligations in ICT security	Count(ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCO	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		related issues and have provided training to develop ICT skills for ICT specialists	NT=1) and ITSPT2=1		
Derived from C2b and D2	E_SECAWANY_ITUST2	Make persons employed aware of their obligations in ICT security related issues and have provided training to develop ICT skills for other persons employed	Count(ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1) and ITUST2=1	FALSE	2022
D3	E_SECPOL2	Have document(s) on measures, practices or procedures on ICT security	Count(ENT_ID) where SECPOL2=1	FALSE	2022
D3	E_SECPOL2X	Have no document(s) on measures, practices or procedures on ICT security	Count(ENT_ID) where SECPOL2=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from C1 and D3	E_SECPOL2_ITSP2	Employ ICT specialists and have document(s) on measures, practices or procedures on ICT security	Count(ENT_ID) where ITSP2=1 and SECPOL2=1	FALSE	2022
Derived from D2a to D2c and D3	E_SECAWANY_POL2	Make persons employed aware of their obligations in ICT security related issues and have document(s) on measures, practices or procedures on ICT security	Count(ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1) and SECPOL2=1	FALSE	2022
D4a	E_SECPREV_CY	The document(s) on measures, practices or procedures were defined or most recently reviewed: within the last 12 months	Count(ENT_ID) where SECPREV=1	TRUE	2022
D4b	E_SECPREV_1_2	The document(s)	Count(ENT_ID) where	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		on measures, practices or procedures were defined or most recently reviewed: more than 12 months and up to 24 months ago	SECPREV=2		
D4c	E_SECPREV_MT2	The document(s) on measures, practices or procedures were defined or most recently reviewed: more than 24 months ago	Count(ENT_ID) where SECPREV=3	TRUE	2022
Derived from D4a and D4b	E_SECPREV_LE2	The document(s) on measures, practices or procedures were defined or most recently reviewed: within the last 24 months	Count(ENT_ID) where SECPREV=1 or SECPREV=2	TRUE	2022
D5a	E_SEC2IUSVF	Have experienced at least once problems due to ICT	Count(ENT_ID) where SEC2IUSVF=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		security incident: unavailability of ICT services due to hardware or software failures			
D5a	E_SEC2IUSVFX	Have not experienced at least once problems due to ICT security incident: unavailability of ICT services due to hardware or software failures	Count(ENT_ID) where SEC2IUSVF=0	FALSE	2022
D5b	E_SEC2IUSVA	Have experienced at least once problems due to ICT security incident: unavailability of ICT services (e.g. Ransomware attacks, Denial of Service attacks)	Count(ENT_ID) where SEC2IUSVA=1	FALSE	2022
D5b	E_SEC2IUSVAX	Have not experienced at least once problems due to ICT security	Count(ENT_ID) where SEC2IUSVA=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		incident: unavailability of ICT services (e.g. Ransomware attacks, Denial of Service attacks)			
Derived from D5a and D5b	E_SEC2IUSV	Have experienced at least once problems due to ICT security incident: unavailability of ICT services (e.g. Denial of Service attacks, ransomware attacks, hardware or software failures - excluding mechanical failure, theft)	Count(ENT_ID) where SEC2IUSVF =1 or SEC2IUSVA =1	FALSE	2022
D5c	E_SEC2IDCDF	Have experienced at least once problems due to ICT security incident: destruction or corruption of data due to hardware or software failures	Count(ENT_ID) where SEC2IDCDF =1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
D5c	E_SEC2IDCDFX	Have not experienced at least once problems due to ICT security incident: destruction or corruption of data due to hardware or software failures	Count(ENT_ID) where SEC2IDCDF =0	FALSE	2022
D5d	E_SEC2IDCDA	Have experienced at least once problems due to ICT security incident: destruction or corruption of data due to infection of malicious software or unauthorised intrusion	Count(ENT_ID) where SEC2IDCDA =1	FALSE	2022
D5d	E_SEC2IDCDAX	Have not experienced at least once problems due to ICT security incident: destruction or corruption of data due to infection of malicious software or unauthorised intrusion	Count(ENT_ID) where SEC2IDCDA =0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from D5c and D5d	E_SEC2IDCD	Have experienced at least once problems due to ICT security incident: destruction or corruption of data (e.g. due to infection of malicious software or unauthorised intrusion, hardware or software failures)	Count(ENT_ID) where SEC2IDCDF =1 or SEC2IDCDA =1	FALSE	2022
D5e	E_SEC2ICNFA	Have experienced at least once problems due to ICT security incident: disclosure of confidential data due to intrusion, pharming, phishing attack, intentional actions by own employees	Count(ENT_ID) where SEC2ICNFA =1	FALSE	2022
D5e	E_SEC2ICNFAX	Have not experienced at least once problems due to ICT security incident:	Count(ENT_ID) where SEC2ICNFA =0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		disclosure of confidential data due to intrusion, pharming, phishing attack, intentional actions by own employees			
D5f	E_SEC2ICNFF	Have experienced at least once problems due to ICT security incident: disclosure of confidential data due to unintentional actions by own employees	Count(ENT_ID) where SEC2ICNFF=1	FALSE	2022
D5f	E_SEC2ICNFFX	Have not experienced at least once problems due to ICT security incident: disclosure of confidential data due to unintentional actions by own employees	Count(ENT_ID) where SEC2ICNFF=0	FALSE	2022
Derived from D5e and D5f	E_SEC2ICNF	Have experienced at least once problems	Count(ENT_ID) where SEC2ICNFA=1 or	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		due to ICT security incident: disclosure of confidential data (e.g. due to intrusion, pharming, phishing attack, actions by own employees (intentionally or unintentionally))	SEC2ICNFF=1		
Derived from D5a to D5f	E_SEC2IANY	Have experienced at least once any problem due to ICT security incident: unavailibility of ICT services, destruction or corruption of data, disclosure of confidential data (for any reason)	Count(ENT_ID) where SEC2IUSVF=1 or SEC2IUSVA=1 or SEC2IDCDF=1 or SEC2IDCDA=1 or SEC2ICNFA=1 or SEC2ICNFF=1	FALSE	2022
Derived from D5a to D5f	E_SEC2IGE3	Have experienced at least once 3 or more problems due to ICT security incident: unavailibility	Count(ENT_ID) where SEC2IUSVF+SEC2IUSVA+SEC2IDCDF+SEC2IDCDA+SEC2ICNFA+SEC2ICNFF	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		of ICT services, destruction or corruption of data, disclosure of confidential data (for any reason)	between 3 and 6		
E1a	E_AI_TTM	Enterprises use AI technologies performing analysis of written language	Count(ENT_ID) where AI_TTM=1	FALSE	2023
E1a	E_AI_TTMX	Enterprises do not use AI technologies performing analysis of written language	Count(ENT_ID) where AI_TTM=0	FALSE	2023
E1b	E_AI_TSR	Enterprises use AI technologies converting spoken language into machine-readable format	Count(ENT_ID) where AI_TSR=1	FALSE	2023
E1b	E_AI_TSRX	Enterprises do not use AI technologies converting spoken language into	Count(ENT_ID) where AI_TSR=0	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		machine-readable format			
E1c	E_AI_TNLG	Enterprises use AI technologies generating written or spoken language	Count(ENT_ID) where AI_TNLG=1	FALSE	2023
E1c	E_AI_TNLGX	Enterprises do not use AI technologies generating written or spoken language	Count(ENT_ID) where AI_TNLG=0	FALSE	2023
E1d	E_AI_TIR	Enterprises use AI technologies identifying objects or persons based on images	Count(ENT_ID) where AI_TIR=1	FALSE	2023
E1d	E_AI_TIRX	Enterprises do not use AI technologies identifying objects or persons based on images	Count(ENT_ID) where AI_TIR=0	FALSE	2023
E1e	E_AI_TML	Enterprises use machine learning (e.g. deep learning) for data analysis	Count(ENT_ID) where AI_TML=1	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E1e	E_AI_TMLX	Enterprises do not use machine learning (e.g. deep learning) for data analysis	Count(ENT_ID) where AI_TML=0	FALSE	2023
E1f	E_AI_TPA	Enterprises use AI technologies automating different workflows or assisting in decision making	Count(ENT_ID) where AI_TPA=1	FALSE	2023
E1f	E_AI_TPAX	Enterprises do not use AI technologies automating different workflows or assisting in decision making	Count(ENT_ID) where AI_TPA=0	FALSE	2023
E1g	E_AI_TAR	Enterprises use AI technologies enabling physical movement of machines via autonomous decisions based on observation of surroundings	Count(ENT_ID) where AI_TAR=1	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E1g	E_AI_TARX	Enterprises do not use AI technologies enabling physical movement of machines via autonomous decisions based on observation of surroundings	Count(ENT_ID) where AI_TAR=0	FALSE	2023
Derived from E1	E_AI_TANY	Enterprises use at least 1 of the listed artificial intelligence technologies : (AI_TTM, AI_TSR, AI_TNLG, AI_TIR, AI_TML, AI_TPA, AI_TAR)	Count (ENT_ID) where (AI_TTM+AI_TSR+AI_TNLG+AI_TIR+AI_TML+AI_TPA+AI_TAR) between 1 and 7	FALSE	2023
Derived from E1	E_AI_TX	Enterprises doesn't use artificial intelligence technologies .	Count (ENT_ID) where AI_TTM=0 and AI_TSR=0 and AI_TNLG=0 and AI_TIR=0 and AI_TML=0 and AI_TPA=0	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
			and AI_TAR=0		
Derived from E1	E_AI_TGE2	Enterprises use at least 2 of the listed artificial intelligence technologies : (AI_TTM, AI_TSR, AI_TNLG, AI_TIR, AI_TML, AI_TPA, AI_TAR)	Count (ENT_ID) where (AI_TTM+AI_TSR+AI_TNLG+AI_TIR+AI_TML+AI_TPA+AI_TAR) between 2 and 7	FALSE	2023
Derived from E1	E_AI_TGE3	Enterprises use at least 3 of the listed artificial intelligence technologies : (AI_TTM, AI_TSR, AI_TNLG, AI_TIR, AI_TML, AI_TPA, AI_TAR)	Count (ENT_ID) where (AI_TTM+AI_TSR+AI_TNLG+AI_TIR+AI_TML+AI_TPA+AI_TAR) between 3 and 7	FALSE	2023
E2a	E_AI_PMS	Enterprises use AI technologies for marketing or sales	Count(ENT_ID) where AI_PMS=1	FALSE	2023
E2a	E_AI_PMSX	Enterprises do not use AI technologies for	Count(ENT_ID) where AI_PMS=0	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		marketing or sales			
E2b	E_AI_PPP	Enterprises use AI technologies for production or service processes	Count(ENT_ID) where AI_PPP=1	FALSE	2023
E2b	E_AI_PPPX	Enterprises do not use AI technologies for production or service processes	Count(ENT_ID) where AI_PPP=0	FALSE	2023
E2c	E_AI_PBAM	Enterprises use AI technologies for organisation of business administration processes or management	Count(ENT_ID) where AI_PBAM=1	FALSE	2023
E2c	E_AI_PBAMX	Enterprises do not use AI technologies for organisation of business administration processes or management	Count(ENT_ID) where AI_PBAM=0	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E2e	E_AI_PLOG	Enterprises use AI technologies for logistics	Count(ENT_ID) where AI_PLOG=1	FALSE	2023
E2e	E_AI_PLOGX	Enterprises do not use AI technologies for logistics	Count(ENT_ID) where AI_PLOG=0	FALSE	2023
E2f	E_AI_PITS	Enterprises use AI technologies for ICT security	Count(ENT_ID) where AI_PITS=1	FALSE	2023
E2f	E_AI_PITSX	Enterprises do not use AI technologies for ICT security	Count(ENT_ID) where AI_PITS=0	FALSE	2023
E2g	E_AI_PFIN	Enterprises use AI technologies for accounting, controlling or finance managemen t	Count(ENT_ID) where AI_PFIN=1	FALSE	2023
E2g	E_AI_PFINX	Enterprises do not use AI technologies for accounting, controlling or finance managemen t	Count(ENT_ID) where AI_PFIN=0	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E2h	E_AI_PRDI	Enterprises use AI technologies for research and development (R&D) or innovation activity	Count(ENT_ID) where AI_PRDI=1	FALSE	2023
E2h	E_AI_PRDIX	Enterprises do not use AI technologies for research and development (R&D) or innovation activity	Count(ENT_ID) where AI_PRDI=0	FALSE	2023
Derived from E2	E_AI_P1ANY	Enterprises use artificial intelligence for at least 1 of the listed purposes: (AI_PMS, AI_PPP, AI_PBAM, AI_PLOG, AI_PITS, AI_PFIN, AI_PRDI)	Count (ENT_ID) where (AI_PMS+AI_PPP+AI_PBAM+AI_PLOG+AI_PITS+AI_PFIN+AI_PRDI) between 1 and 7	FALSE	2023
Derived from E2	E_AI_P1GE2	Enterprises use artificial intelligence for at least 2 of the listed purposes: (AI_PMS, AI_PPP, AI_PBAM, AI_PLOG, AI_PITS,	Count (ENT_ID) where (AI_PMS+AI_PPP+AI_PBAM+AI_PLOG+AI_PITS+AI_PFIN+AI_PRDI) between 2 and 7	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		AI_PFIN, AI_PRDI)			
Derived from E2	E_AI_P1GE3	Enterprises use artificial intelligence for at least 3 of the listed purposes: (AI_PMS, AI_PPP, AI_PBAM, AI_PLOG, AI_PITS, AI_PFIN, AI_PRDI)	Count (ENT_ID) where (AI_PMS+AI_PPP+AI_PBAM+AI_PLOG+AI_PITS+AI_PFIN+AI_PRDI) between 3 and 7	FALSE	2023
E3a	E_AI_ADOWN	The enterprises' AI technologies were developed by own employees	Count(ENT_ID) where AI_ADOWN=1	TRUE	2023
E3a	E_AI_ADOWNX	The enterprises' AI technologies were not developed by own employees	Count(ENT_ID) where AI_ADOWN=0	TRUE	2023
E3b	E_AI_AMOWN	The enterprises' AI technologies were commercial software or systems modified by own employees	Count(ENT_ID) where AI_AMOWN=1	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E3b	E_AI_AMOWNX	The enterprises' AI technologies were not commercial software or systems modified by own employees	Count(ENT_ID) where AI_AMOWN=0	TRUE	2023
E3c	E_AI_AOS	The enterprises' AI technologies were open-source software or systems modified by own employees	Count(ENT_ID) where AI_AOS=1	TRUE	2023
E3c	E_AI_AOSX	The enterprises' AI technologies were not open-source software or systems modified by own employees	Count(ENT_ID) where AI_AOS=0	TRUE	2023
E3d	E_AI_ARDY	The enterprises' AI technologies were purchased commercial software or systems ready to use	Count(ENT_ID) where AI_ARDY=1	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E3d	E_AI_ARDYX	The enterprises' AI technologies were not purchased commercial software or systems ready to use	Count(ENT_ID) where AI_ARDY=0	TRUE	2023
E3e	E_AI_AEXT	The enterprises' AI technologies were developed or modified by external providers	Count(ENT_ID) where AI_AEXT=1	TRUE	2023
E3e	E_AI_AEXTX	The enterprises' AI technologies were not developed or modified by external providers	Count(ENT_ID) where AI_AEXT=0	TRUE	2023
E4	E_AI_PDI	Enterprises which process data on individuals using AI technologies	Count(ENT_ID) where AI_PDI=1	TRUE	NEW
E4	E_AI_PDIX	Enterprises which do not process data on individuals using AI technologies	Count(ENT_ID) where AI_PDI=0	TRUE	NEW

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E5	E_AI_BIAS	Enterprises which have measures to check the results generated by AI technologies for possible biases towards individuals	Count(ENT_ID) where AI_BIAS=1	TRUE	NEW
E5	E_AI_BIASX	Enterprises which do not have measures to check the results generated by AI technologies for possible biases towards individuals	Count(ENT_ID) where AI_BIAS=0	TRUE	NEW
Derived from E5, E3a	E_AI_BIAS_ADOWN	Enterprises which have measures to check the results generated by AI for possible biases towards individuals, and their AI technologies were developed by own employees	Count(ENT_ID) where AI_BIAS=1 and AI_ADOWN =1	TRUE	NEW

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from E5, E3b	E_AI_BIAS_AMOWN	Enterprises which have measures to check the results generated by AI for possible biases towards individuals, and their AI technologies were commercial software or systems modified by own employees	Count(ENT_ID) where AI_BIAS=1 and AI_AMOWN =1	TRUE	NEW
Derived from E5, E3c	E_AI_BIAS_AOS	Enterprises which have measures to check the results generated by AI for possible biases towards individuals, and their AI technologies were open-source software or systems modified by own employees	Count(ENT_ID) where AI_BIAS=1 and AI_AOS=1	TRUE	NEW
Derived from E5, E3d	E_AI_BIAS_ARDY	Enterprises which have measures to check the	Count(ENT_ID) where AI_BIAS=1	TRUE	NEW

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		results generated by AI for possible biases towards individuals, and their AI technologies were purchased commercial software or systems ready to use	and AI_ARDY=1		
Derived from E5, E3e	E_AI_BIAS_AEXT	Enterprises which have measures to check the results generated by AI for possible biases towards individuals, and their AI technologies were developed or modified by external providers	Count(ENT_ID) where AI_BIAS=1 and AI_AEXT=1	TRUE	NEW
E6	E_AI_EC	Enterprises have ever considered to use AI technologies	Count(ENT_ID) where AI_EC=1	TRUE	2023
E6	E_AI_ECX	Enterprises have never considered to use AI technologies	Count(ENT_ID) where AI_EC=0	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E7a	E_AI_BCST	Enterprises do not use AI technologies, because the costs seem too high	Count(ENT_ID) where AI_BCST=1	TRUE	2023
E7a	E_AI_BCSTX	Enterprises do not use AI technologies, but not because the costs seem too high	Count(ENT_ID) where AI_BCST=0	TRUE	2023
E7b	E_AI_BLE	Enterprises do not use AI technologies, because of a lack of relevant expertise	Count(ENT_ID) where AI_BLE=1	TRUE	2023
E7b	E_AI_BLEX	Enterprises do not use AI technologies, but not because of a lack of relevant expertise	Count(ENT_ID) where AI_BLE=0	TRUE	2023
E7c	E_AI_BINC	Enterprises do not use AI technologies, because of incompatibility with existing equipment,	Count(ENT_ID) where AI_BINC=1	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		software or systems			
E7c	E_AI_BINCX	Enterprises do not use AI technologies, but not because of incompatibility with existing equipment, software or systems	Count(ENT_ID) where AI_BINC=0	TRUE	2023
E7d	E_AI_BDDT	Enterprises do not use AI technologies, because of difficulties with availability or quality of the necessary data	Count(ENT_ID) where AI_BDDT=1	TRUE	2023
E7d	E_AI_BDDTX	Enterprises do not use AI technologies, but not because of difficulties with availability or quality of the necessary data	Count(ENT_ID) where AI_BDDT=0	TRUE	2023
E7e	E_AI_BCDP	Enterprises do not use AI technologies, because od	Count(ENT_ID) where AI_BCDP=1	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		concerns regarding violation of data protection and privacy			
E7e	E_AI_BCDPX	Enterprises do not use AI technologies, but not because od concerns regarding violation of data protection and privacy	Count(ENT_ID) where AI_BCDP=0	TRUE	2023
E7f	E_AI_BLEG	Enterprises do not use AI technologies, because of a lack of clarity about the legal consequences	Count(ENT_ID) where AI_BLEG=1	TRUE	2023
E7f	E_AI_BLEGX	Enterprises do not use AI technologies, but not because of a lack of clarity about the legal consequences	Count(ENT_ID) where AI_BLEG=0	TRUE	2023
E7g	E_AI_BEC	Enterprises do not use AI	Count(ENT_ID) where AI_BEC=1	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		technologies, because of ethical considerations			
E7g	E_AI_BECX	Enterprises do not use AI technologies, but not because of ethical considerations	Count(ENT_ID) where AI_BEC=0	TRUE	2023
E7h	E_AI_BNU	Enterprises do not use AI technologies, because artificial Intelligence technologies are not useful for the enterprise	Count(ENT_ID) where AI_BNU=1	TRUE	2023
E7h	E_AI_BNUX	Enterprises do not use AI technologies, but not because artificial Intelligence technologies are not useful for the enterprise	Count(ENT_ID) where AI_BNU=0	TRUE	2023
A1	P_IUSE	Persons employed have access to the internet for	∑ (EMPIUSE) where EMPIUSE<> Blank and	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		business purposes	EMPIUSE<> -1		
Derived from A1 (Eurostat computed) 1	P_IUSEX	Persons employed who do not have access to the internet for business purposes	∑ (EMPL) - ∑ (EMPIUSE) where EMPIUSE<> Blank and EMPIUSE<> -1	FALSE	2023
Derived from B1	P_AWSELL	Persons employed by enterprises which have received orders via websites or apps	∑ (EMPL) where (AWS_COWN=1 or AWS_CMP= 1)	FALSE	2023
Derived from B7	P_AXSELL	Persons employed by enterprises which have received orders via EDI-type messages	∑ (EMPL) where AXSELL=1	FALSE	2023
Derived from B1 and B7	P_AESELL	Persons employed by enterprises which have received orders via computer networks	∑ (EMPL) where (AWS_COWN=1 or AWS_CMP= 1) or AXSELL=1	FALSE	2023
Derived, see section 5 and Annex 3	E_DI4_VLO	Enterprises with very low digital intensity index	Count (ENT_ID) where DI4_INDEX>=0 and DI4_INDEX<=3	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived, see section 5 and Annex 3	E_DI4_LO	Enterprises with low digital intensity index	Count (ENT_ID) where DI4_INDEX>=4 and DI4_INDEX<=6	FALSE	2022
Derived, see section 5 and Annex 3	E_DI4_HI	Enterprises with high digital intensity index	Count (ENT_ID) where DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2022
Derived, see section 5 and Annex 3	E_DI4_VHI	Enterprises with very high digital intensity index	Count (ENT_ID) where DI4_INDEX>=10 and DI4_INDEX<=12	FALSE	2022
Derived, see section 5 and Annex 3 (Eurostat computed) 2	E_DI4_GELO	Enterprises with at least low (basic) digital intensity index	[Count(ENT_ID) where DI4_INDEX>=4 and DI4_INDEX<=6] + [Count (ENT_ID) where DI4_INDEX>=7 and DI4_INDEX<=9] + [Count (ENT_ID) where DI4_INDEX>=10 and DI4_INDEX<=12]	TRUE	2022
Derived, see section 5 and E1	E_DI4_VLO_AI_TANY	Enterprises with very low digital intensity	Count (ENT_ID) where DI4_INDEX>	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		index and use any artificial intelligence technology	=0 and DI4_INDEX<=3 and (AI_TTM=1 or AI_TSR=1 or AI_TNLG=1 or AI_TIR=1 or AI_TML=1 or AI_TPA=1 or AI_TAR=1)		
Derived, see section 5 and E1	E_DI4_LO_AI_TANY	Enterprises with low digital intensity index and use any artificial intelligence technology	Count (ENT_ID) where DI4_INDEX>=4 and DI4_INDEX<=6 and (AI_TTM=1 or AI_TSR=1 or AI_TNLG=1 or AI_TIR=1 or AI_TML=1 or AI_TPA=1 or AI_TAR=1)	FALSE	2024
Derived, see section 5 and E1	E_DI4_HI_AI_TANY	Enterprises with high digital intensity index and use any artificial intelligence technology	Count (ENT_ID) where DI4_INDEX>=7 and DI4_INDEX<=9 and (AI_TTM=1 or AI_TSR=1 or AI_TNLG=1 or AI_TIR=1 or AI_TML=1 or AI_TPA=1 or AI_TAR=1)	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived, see section 5 and E1	E_DI4_VHI_AI_TANY	Enterprises with very high digital intensity index that use any artificial intelligence technology	Count (ENT_ID) where DI4_INDEX>=10 and DI4_INDEX<=12 and (AI_TTM=1 or AI_TSR=1 or AI_TNLG=1 or AI_TIR=1 or AI_TML=1 or AI_TPA=1 or AI_TAR=1)	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VLO_SECMGE1	Enterprises with very low digital intensity index, which use any ICT security measure	Count (ENT_ID) where SECM_CNT>=1 and DI4_INDEX>=0 and DI4_INDEX<=3	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_LO_SECMGE1	Enterprises with low digital intensity index, which use any ICT security measure	Count (ENT_ID) where SECM_CNT>=1 and DI4_INDEX>=4 and DI4_INDEX<=6	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_HI_SECMGE1	Enterprises with high digital intensity index, which use any ICT security measure	Count (ENT_ID) where SECM_CNT>=1 and DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived, see section 5 and Annex 3	E_DI4_VHI_SECMGE1	Enterprises with very high digital intensity index, which use any ICT security measure	Count (ENT_ID) where SECM_CNT >=1 and DI4_INDEX>=10 and DI4_INDEX<=12	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VLO_SECMGE3	Enterprises with very low digital intensity index, which use 3 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=3 and DI4_INDEX>=0 and DI4_INDEX<=3	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_LO_SECMGE3	Enterprises with low digital intensity index, which use 3 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=3 and DI4_INDEX>=4 and DI4_INDEX<=6	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_HI_SECMGE3	Enterprises with high digital intensity index, which use 3 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=3 and DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VHI_SECMGE3	Enterprises with very high digital intensity index, which use 3 or more ICT	Count (ENT_ID) where SECM_CNT >=3 and DI4_INDEX>=10 and	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		security measures	DI4_INDEX<=12		
Derived, see section 5 and Annex 3	E_DI4_VLO_SECMGE5	Enterprises with very low digital intensity index, which use 5 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=5 and DI4_INDEX>=0 and DI4_INDEX<=3	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_LO_SECMGE5	Enterprises with low digital intensity index, which use 5 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=5 and DI4_INDEX>=4 and DI4_INDEX<=6	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_HI_SECMGE5	Enterprises with high digital intensity index, which use 5 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=5 and DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VHI_SECMGE5	Enterprises with very high digital intensity index, which use 5 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=5 and DI4_INDEX>=10 and DI4_INDEX<=12	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VLO_SECMGE7	Enterprises with very low digital intensity	Count (ENT_ID) where SECM_CNT	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		index, which use 7 or more ICT security measures	>=7 and DI4_INDEX>=0 and DI4_INDEX<=3		
Derived, see section 5 and Annex 3	E_DI4_LO_SECMGE7	Enterprises with low digital intensity index, which use 7 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=7 and DI4_INDEX>=4 and DI4_INDEX<=6	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_HI_SECMGE7	Enterprises with high digital intensity index, which use 7 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=7 and DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VHI_SECMGE7	Enterprises with very high digital intensity index, which use 7 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=7 and DI4_INDEX>=10 and DI4_INDEX<=12	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VLO_SECAWANY	Enterprises with very low digital intensity index, which make persons employed aware of their	Count (ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1) and DI4_INDEX>	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		obligations in ICT security related issues	=0 and DI4_INDEX<=3		
Derived, see section 5 and Annex 3	E_DI4_LO_SECAWANY	Enterprises with low digital intensity index, which make persons employed aware of their obligations in ICT security related issues	Count (ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1) and DI4_INDEX>=4 and DI4_INDEX<=6	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_HI_SECAWANY	Enterprises with high digital intensity index, which make persons employed aware of their obligations in ICT security related issues	Count (ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1) and DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VHI_SECAWANY	Enterprises with very high digital intensity index, which make persons employed	Count (ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCO	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		aware of their obligations in ICT security related issues	NT=1) and DI4_INDEX>=10 and DI4_INDEX<=12		
Derived, see section 5 and Annex 3	E_DI4_VLO_SECPOL2	Enterprises with very low digital intensity index, in which have document(s) on measures, practices or procedures on ICT security	Count (ENT_ID) where SECPOL2=1 and DI4_INDEX>=0 and DI4_INDEX<=3	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_LO_SECPOL2	Enterprises with low digital intensity index, in which have document(s) on measures, practices or procedures on ICT security	Count (ENT_ID) where SECPOL2=1 and DI4_INDEX>=4 and DI4_INDEX<=6	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_HI_SECPOL2	Enterprises with high digital intensity index, in which have document(s) on measures, practices or	Count (ENT_ID) where SECPOL2=1 and DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		procedures on ICT security			
Derived, see section 5 and Annex 3	E_DI4_VHI_SECPOL2	Enterprises with very high digital intensity index, in which have document(s) on measures, practices or procedures on ICT security	Count (ENT_ID) where SECPOL2=1 and DI4_INDEX>=10 and DI4_INDEX<=12	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VLO_SECPREV_CY	Enterprises with very low digital intensity index, in which the document(s) on measures, practices or procedures were defined or most recently reviewed within the last 12 months	Count (ENT_ID) where SECPREV=1 and DI4_INDEX>=0 and DI4_INDEX<=3	TRUE	2024
Derived, see section 5 and Annex 3	E_DI4_LO_SECPREV_CY	Enterprises with low digital intensity index, in which the document(s) on measures,	Count (ENT_ID) where SECPREV=1 and DI4_INDEX>=4 and DI4_INDEX<=6	TRUE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		practices or procedures were defined or most recently reviewed within the last 12 months			
Derived, see section 5 and Annex 3	E_DI4_HI_SECPREV_CY	Enterprises with high digital intensity index, in which the document(s) on measures, practices or procedures were defined or most recently reviewed within the last 12 months	Count (ENT_ID) where SECPREV= 1 and DI4_INDEX>=7 and DI4_INDEX<=9	TRUE	2024
Derived, see section 5 and Annex 3	E_DI4_VHI_SECPREV_CY	Enterprises with very high digital intensity index, in which the document(s) on measures, practices or procedures were defined or most recently reviewed within the	Count (ENT_ID) where SECPREV= 1 and DI4_INDEX>=10 and DI4_INDEX<=12	TRUE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		last 12 months			
back	ENT_SAMPLE	Number of enterprises in final (net) sample - non-raised figures	Count (ENT_ID) unraised	FALSE	2023
back	ENT	Number of enterprises in population (surveyed size and industry groups corresponding to raised figures)	Count (ENT_ID)	FALSE	2023
back	EMPL	Average number of employed persons in population (surveyed size and industry groups corresponding to raised figures)	∑ (EMPL)	FALSE	2023
back	TOVT	Total turnover in population, in value terms, excluding VAT (surveyed size and industry groups correspondin	∑ (TOVT)	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		g to raised figures)			
Derived from A1	E_IUSE	Enterprises where persons employed have access to the internet for business purposes	Count (ENT_ID) where EMPIUSE<> Blank and EMPIUSE>0	FALSE	2023
Derived from A1	E_IUSE_GT 10	Enterprises where more than 10% of the persons employed have access to the internet for business purposes	Count (ENT_ID) where EMPIUSE<> Blank and EMPL<>Bla nk and EMPIUSE > 0.10 * EMPL	FALSE	2023
Derived from A1	E_IUSE_GT 50	Enterprises where more than 50% of the persons employed have access to the internet for business purposes	Count (ENT_ID) where EMPIUSE<> Blank and EMPL<>Bla nk and EMPIUSE > 0.50 * EMPL	FALSE	2023
Derived from A1	E_IUSE_GE 10A	Enterprises where at least 10 persons employed have access to the internet for business purposes	Count (ENT_ID) where EMPIUSE<> Blank and EMPIUSE >= 10	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
A2	E_FIXBB	Use any type of fixed connection to the internet	Count (ENT_ID) where FIXBB=1	FALSE	2023
A2	E_FIXBBX	Don't use any type of fixed connection to the internet	Count (ENT_ID) where FIXBB=0	FALSE	2023
A2	E_FIXBBZ	Don't know if they use any type of fixed connection to the internet	Count (ENT_ID) where FIXBB=Blank	TRUE	2023
A3a	E_ISPDF_LT30	The maximum contracted download speed of the fastest fixed internet connection is less than 30 Mb/s	Count (ENT_ID) where ISPDF1=1	FALSE	2023
A3b	E_ISPDF1_30_100	The maximum contracted download speed of the fastest fixed internet connection is at least 30 Mb/s but less than 100 Mb/s	Count (ENT_ID) where ISPDF1=2	FALSE	2023
A3c	E_ISPDF_100_500	The maximum contracted	Count (ENT_ID)	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		download speed of the fastest fixed internet connection is at least 100 Mb/s but less than 500 Mb/s	where ISPDF1=3		
A3d	E_ISPDF_500_1G	The maximum contracted download speed of the fastest fixed internet connection is at least 500 Mb/s but less than 1 Gb/s	Count (ENT_ID) where ISPDF1=4	FALSE	2023
A3e	E_ISPDF_GE1G	The maximum contracted download speed of the fastest fixed internet connection is at least 1 Gb/s	Count (ENT_ID) where ISPDF1=5	FALSE	2023
Derived from A3b to A3e	E_ISPDF1_GE30	The maximum contracted download speed of the fastest fixed internet connection is at least 30 Mb/s	Count (ENT_ID) where ISPDF1=2 or ISPDF1=3 or ISPDF1=4 or ISPDF1=5	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from A3c to A3e	E_ISPDF1_GE100	The maximum contracted download speed of the fastest fixed internet connection is at least 100 Mb/s	Count (ENT_ID) where ISPDF1=3 or ISPDF1=4 or ISPDF1=5	FALSE	2023
A4	E_ISPDF1OK	The speed of the fixed line connection(s) to the internet is sufficient for the actual needs of the enterprise	Count (ENT_ID) where ISPDFOK=1	FALSE	2020
A4	E_ISPDF1OKX	The speed of the fixed line connection(s) to the internet is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where ISPDFOK=0	FALSE	2020
Derived from A3b to A3e and A4	E_ISPDF1OK_GE30	The speed of the fixed line connection(s) to the internet (at least 30 Mb/s) is sufficient for the actual needs of the enterprise	Count (ENT_ID) where (ISPDF1=2 or ISPDF1=3 or ISPDF1=4 or ISPDF1=5) and ISPDFOK=1	FALSE	2020

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from A3a and A4	E_ISPDFOKX_LT30	The speed of the fixed line connection(s) to the internet (less than 30Mb/s) is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where ISPDF1=1 and ISPDFOK=0	FALSE	2020
Derived from A3b and A4	E_ISPDF1OKX_30_100	The speed of the fixed line connection(s) to the internet ([3-100[Mb/s) is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where ISPDF1=2 and ISPDFOK=0	FALSE	2020
Derived from A3c and A4	E_ISPDFOKX_100_500	The speed of the fixed line connection(s) to the internet ([100-500[Mb/s) is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where ISPDF1=3 and ISPDFOK=0	FALSE	2020
Derived from A3d and A4	E_ISPDFOKX_500_1G	The speed of the fixed line connection(s) to the internet ([500-1000[Mb/s) is not	Count (ENT_ID) where ISPDF1=4 and ISPDFOK=0	FALSE	2020

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		sufficient for the actual needs of the enterprise			
Derived from A3e and A4	E_ISPDFOKX_GE1G	The speed of the fixed line connection(s) to the internet (at least 1 Gb/s) is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where ISPDF1=5 and ISPDFOK=0	FALSE	2020
Derived from A3b to A3e and A4	E_ISPDF1OKX_GE30	The speed of the fixed line connection(s) to the internet (at least 30 Mb/s) is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where (ISPDF1=2 or ISPDF1=3 or ISPDF1=4 or ISPDF1=5) and ISPDFOK=0	FALSE	2020
Derived from A3c to A3e and A4	E_ISPDF1OKX_GE100	The speed of the fixed line connection(s) to the internet (at least 100 Mb/s) is not sufficient for the actual needs of the enterprise	Count (ENT_ID) where (ISPDF1=3 or ISPDF1=4 or ISPDF1=5) and ISPDFOK=0	FALSE	2020
Derived from A5	E_EMPMD2_GT0	Enterprises where some	Count (ENT_ID)	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		employed persons use a portable device provided by the enterprise, that allows internet connection via mobile telephone networks, for business purposes	where EMPMD2<> Blank and EMPMD2>0		
Derived from A5	E_EMPMD2_GT10	Enterprises where more than 10% of the employed persons use a portable device provided by the enterprise, that allows internet connection via mobile telephone networks, for business purposes	Count (ENT_ID) where EMPMD2<> Blank and EMPL<>Blank and (EMPMD2> EMPL * 0.1)	TRUE	2022
Derived from A5	E_EMPMD2_GT20	Enterprises where more than 20% of the employed persons use a portable device provided by the	Count (ENT_ID) where EMPMD2<> Blank and EMPL<>Blank and (EMPMD2> EMPL * 0.2)	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		enterprise, that allows internet connection via mobile telephone networks, for business purposes			
Derived from A5	E_EMPMD2_GT50	Enterprises where more than 50% of the employed persons use a portable device provided by the enterprise, that allows internet connection via mobile telephone networks, for business purposes	Count (ENT_ID) where EMPMD2<> Blank and EMPL<>Blank and (EMPMD2> EMPL * 0.5)	TRUE	2022
Derived from A5	E_PMD	Provide portable devices that allow a mobile connection to the internet using mobile telephone networks, for business purposes	Count (ENT_ID) where EMPMD2>0	TRUE	2022
Derived from A5	E_PMDX	Does not provide	Count (ENT_ID)	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		portable devices that allow a mobile connection to the internet using mobile telephone networks, for business purposes	where EMPMD2=0 or EMPMD2=Blank		
Derived from A2 and A5	E_BROAD3	Connect to the Internet via fixed or mobile broadband (as of 2018)	Count (ENT_ID) where FIXBB=1 or EMPMD2>0	TRUE	2022
Derived from A2 and A5	E_BROAD3X	Connect to the Internet but not via fixed or mobile broadband (as of 2018)	Count (ENT_ID) where (FIXBB=0 or FIXBB=Blank) and (EMPMD2=0 or EMPMD2=Blank)	TRUE	2022
A6a	E_RA_M	Enterprises with employed persons with remote access to the email system of the enterprise	Count (ENT_ID) where RA_M=1	FALSE	2022
A6a	E_RA_MX	Enterprises with employed persons with	Count (ENT_ID) where RA_M=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		no remote access to the email system of the enterprise			
A6b	E_RA_D	Enterprises with employed persons with remote access to the documents of the enterprise (e.g. files, spreadsheets, presentations, charts, photos)	Count (ENT_ID) where RA_D=1	FALSE	2022
A6b	E_RA_DX	Enterprises with employed persons with no remote access to the documents of the enterprise (e.g. files, spreadsheets, presentations, charts, photos)	Count (ENT_ID) where RA_D=0	FALSE	2022
A6c	E_RA_S	Enterprises with employed persons with remote	Count (ENT_ID) where RA_S=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		access to the business applications or software of the enterprise (e.g. access to accounting, sales, orders, CRM)			
A6c	E_RA_SX	Enterprises with employed persons with no remote access to the business applications or software of the enterprise (e.g. access to accounting, sales, orders, CRM)	Count (ENT_ID) where RA_S=0	FALSE	2022
Derived A6a to A6c	E_RA	Enterprises with employed persons with remote access to the email system or documents, business applications or software of the enterprise	Count (ENT_ID) where RA_M=1 or RA_D=1 or RA_S=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived A6a to A6c	E_RA_ALL	Enterprises with employed persons with remote access to the email system and documents and business applications or software of the enterprise	Count (ENT_ID) where RA_M=1 and RA_D=1 and RA_S=1	FALSE	2022
Derived A6b to A6c	E_RA_DS	Enterprises with employed persons with remote access to the documents, business applications or software of the enterprise	Count (ENT_ID) where RA_D=1 or RA_S=1	FALSE	2022
A7	E_RM	Enterprises which conducted remote meetings (via e.g. Skype, Zoom, MS Teams, WebEx, etc.)	Count (ENT_ID) where RM=1	FALSE	2022
A7	E_RMX	Enterprises which did not conduct remote meetings	Count (ENT_ID) where RM=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		(via e.g. Skype, Zoom, MS Teams, WebEx, etc.)			
A8	E_ADS	Pay to advertise on the Internet	Count (ENT_ID) where ADS=1	TRUE	2023
A8	E_ADSX	Don't pay to advertise on the Internet	Count (ENT_ID) where ADS=0	TRUE	2023
A9a	E_ADS_KW	Pay to advertise on the internet, based on content or keywords searched by internet users	Count (ENT_ID) where ADS_KW=1	TRUE	2023
A9a	E_ADS_KWX	Pay to advertise on the internet, but not based on content or keywords searched by internet users	Count (ENT_ID) where ADS_KW=0	TRUE	2023
A9b	E_ADS_TRK	Pay to advertise on the internet, based on the tracking of internet users' past activities or profile	Count (ENT_ID) where ADS_TRK=1	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
A9b	E_ADS_TRKX	Pay to advertise on the internet, but not based on the tracking of internet users' past activities or profile	Count (ENT_ID) where ADS_TRK=0	TRUE	2023
A9c	E_ADS_LOC	Pay to advertise on the internet, based on the geolocation of internet users	Count (ENT_ID) where ADS_LOC=1	TRUE	2023
A9c	E_ADS_LOCX	Pay to advertise on the internet, but not based on the geolocation of internet users	Count (ENT_ID) where ADS_LOC=0	TRUE	2023
A9d	E_ADS_OTH	Pay to advertise on the internet, based on any other method of target advertising	Count (ENT_ID) where ADS_OTH=1	TRUE	2023
A9d	E_ADS_OTHX	Pay to advertise on the internet, but not based on any other method of target advertising	Count (ENT_ID) where ADS_OTH=0	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
A9a to A9c	E_ADS3	Pay to advertise on the internet, based on content, keywords, internet users' past activities or profile or the geolocation	Count (ENT_ID) where ADS_KW=1 or ADS_TRK=1 or ADS_LOC= 1	TRUE	2023
B1a	E_AWS_CO WN	Enterprises which sold via their own websites or apps	Count (ENT_ID) where AWS_COW N=1	FALSE	2023
B1a	E_AWS_CO WNX	Enterprises which did not sell via their own websites or apps	Count (ENT_ID) where AWS_COW N=0	FALSE	2023
B1b	E_AWS_CM P	Enterprises which sold via an e-commerce marketplace	Count (ENT_ID) where AWS_CMP= 1	FALSE	2023
B1b	E_AWS_CM PX	Enterprises which did not sell via an e-commerce marketplace	Count (ENT_ID) where AWS_CMP= 0	FALSE	2023
Derived from B1a and B1b	E_AWS_CB OTH	Enterprises which sold via their own websites or apps and via an e-commerce marketplace	Count (ENT_ID) where AWS_COW N=1 and AWS_CMP= 1	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from B1a and B1b	E_AWSELL	Have received orders via websites or apps (web sales)	Count (ENT_ID) where (AWS_COWN=1 or AWS_CMP= 1)	FALSE	2023
Derived from B1a and B1b	E_AWSELLX	Have not received orders via websites or apps	Count (ENT_ID) where (AWS_COWN=0 and AWS_CMP= 0)	FALSE	2023
Derived from B1a and B1b	E_AWS_CMPQ	Enterprises which sold via an e-commerce marketplace but not via their own websites or apps	Count (ENT_ID) where AWS_COWN=0 and AWS_CMP= 1	FALSE	NEW derived variable
B2	E_AWSVALS	Sales via websites or apps, excluding VAT (<1% of turnover)	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL < TOVT * 1%	FALSE	2023
B2	E_AWSVALB	Sales via websites or apps, excluding VAT (>= 1% of turnover)	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL >= TOVT * 1%	FALSE	2023
B2	E_AWSVAL	Total sales via websites or apps,	∑ (AWSVAL) where AWSVAL <>	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		excluding VAT	Blank and AWSVAL <> -1		
B2	E_WSEL0	Total sales via websites or apps equal to or above 0	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>=0	FALSE	2023
B2	E_WSEL1	Total sales via websites or apps equal to or above 1% of turnover	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>= TOVT * 1%	FALSE	2023
B2	E_WSEL2	Total sales via websites or apps equal to or above 2% of turnover	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>= TOVT * 2%	FALSE	2023
B2	E_WSEL5	Total sales via websites or apps equal to or above 5% of turnover	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>= TOVT * 5%	FALSE	2023
B2	E_WSEL10	Total sales via websites or apps equal to or	Count (ENT_ID) where AWSVAL <>	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		above 10% of turnover	Blank and AWSVAL <> -1 and AWSVAL>= TOVT * 10%		
B2	E_WSEL25	Total sales via websites or apps equal to or above 25% of turnover	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>= TOVT * 25%	FALSE	2023
B2	E_WSEL50	Total sales via websites or apps equal to or above 50% of turnover	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>= TOVT * 50%	FALSE	2023
B2	E_WSELZ	Don't know the % of sales via websites or apps	Count (ENT_ID) where AWSVAL=Blank	TRUE	2023
Derived from B1 and B2	E_AWSVAL_CMPQ	Web sales by enterprises which only sell through an e-commerce marketplace	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1 and AWS_COWN=0 and AWS_CMP= 1	FALSE	NEW derived variable
B3a	E_AWSVAL_COWN	Sales via their own websites or apps	∑(AWSVAL_COWN) where AWSVAL_C	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
			OWN <> Blank and AWSVAL_C OWN <> -1		
B3b	E_AWSVAL _CMP	Sales via an e-commerce marketplace	∑(AWSVAL_ CMP) where AWSVAL_C MP <> Blank and AWSVAL_C MP <> -1	FALSE	2023
Derived from B1b, B2 and B3b	E_AWS_CM P_GE20	Enterprises which sold via an e-commerce marketplace for at least 20% of the web sales	Count (ENT_ID) where AWS_CMP= 1 and (AWSVAL_C MP >= AWSVAL * 0.20)	FALSE	2023
Derived from B1b, B2 and B3b	E_AWS_CM P_GE50	Enterprises which sold via an e-commerce marketplace for at least 50% of the web sales	Count (ENT_ID) where AWS_CMP= 1 and (AWSVAL_C MP >= AWSVAL * 0.50)	FALSE	2023
Derived from B1b, B2 and B3b	T_AWS_CM P_GE20	Turnover of the enterprises where web sales via an e-commerce marketplace were at least 20% of the web sales	∑ (TOVT) where AWS_CMP= 1 and (AWSVAL_C MP >= AWSVAL * 0.20)	FALSE	2023
Derived from B1b, B2 and B3b	T_AWS_CM P_GE50	Turnover of the enterprises	∑ (TOVT) where AWS_CMP=	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		where web sales via an e-commerce marketplace were at least 50% of the web sales	1 and (AWSVAL_CMP >= AWSVAL * 0.50)		
B4a	E_AWSVAL_B2C	Sales via websites or apps (B2C)	∑ (AWSVALC) where AWSVALC<>Blank and AWSVALC<>-1	FALSE	2023
B4b	E_AWSVAL_B2BG	Sales via websites or apps (B2B) and (B2G)	∑ (AWSVALBG) where AWSVALBG <>Blank and AWSVALBG <>-1	FALSE	2023
Derived from B4a	E_AWS_B2C	Enterprises which sold via websites or apps - B2C	Count (ENT_ID) where AWSVALC<>Blank and AWSVALC>0	FALSE	2023
Derived from B4b	E_AWS_B2BG	Enterprises which sold via websites or apps - B2B and B2G	Count (ENT_ID) where AWSVALBG <>Blank and AWSVALBG >0	FALSE	2023
Derived from B1b and B4a	E_AWS_B2C_CMP	Enterprises which sold via websites or apps - B2C and via an e-	Count (ENT_ID) where (AWSVALC<>Blank and AWSVALC>0) and	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		commerce marketplace	AWS_CMP= 1		
Derived from B2 and B4a	E_AWSVAL _B2C_GE10 WS	Enterprises where B2C sales via websites or apps were 10% or more of the total web sales	Count (ENT_ID) where AWSVALC< >Blank AND AWSVALC> 0 and AWSVAL<> Blank and AWSVAL<>- 1 and AWSVALC> =0.1 * AWSVAL	FALSE	2023
Derived from B2 and B4a	E_AWSVAL _B2C_GE5 WS	Enterprises where B2C sales via websites or apps were 5% or more of the total web sales	Count (ENT_ID) where AWSVALC< >Blank AND AWSVALC> 0 and AWSVAL<> Blank and AWSVAL<>- 1 and AWSVALC> =0.05 * AWSVAL	FALSE	2023
Derived from B2 and B4a	E_AWS_B2 C_GT1WS	Enterprises where B2C sales via websites or apps were more than 1% of the total web sales	Count (ENT_ID) where AWSVALC< >Blank AND AWSVALC> 0 and AWSVAL<> Blank and AWSVAL<>- 1 and AWSVALC>	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
			0.01 * AWSVAL		
Derived from B2 and B4a	E_AWSVAL_GT1_B2C_GT10WS	B2C web sales of enterprises where web sales > 1% of total turnover and B2C web sales > 10% of the web sales	∑ (AWSVALC) where TOVT<>Blank and AWSVAL<>Blank and AWSVALC<>Blank AND (AWSVAL>0.01 * TOVT) and AWSVALC>0.1 * AWSVAL	FALSE	2023
Derived from B2 and B4a	E_AWS_GT1_B2C_GT10WS	Enterprises where web sales were more than 1% of the total turnover and B2C web sales more than 10% of the web sales	Count (ENT_ID) where TOVT<>Blank and AWSVAL<>Blank and AWSVALC<>Blank AND AWSVALC>0 AND (AWSVAL>0.01 * TOVT) and AWSVALC>0.1 * AWSVAL	FALSE	2023
Derived from B2 and B4a	T_AWS_GT1_B2C_GT10WS	Turnover of the enterprises where web sales were more than 1% of the total turnover and	∑ (TOVT) where TOVT<>Blank and AWSVAL<>Blank and AWSVALC<>Blank AND AWSVALC>	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		B2C web sales more than 10% of the web sales	0 AND (AWSVAL>0 .01 * TOVT) and AWSVALC> 0.1 * AWSVAL		
Derived from B1b, B2 and B4a	E_AWSVAL _B2C_GE10 WS_CMP	Enterprises where B2C sales via websites or apps were 10% or more of the total web sales and which sold via an e-commerce marketplace	Count (ENT_ID) where AWSVALC< >Blank AND AWSVALC> 0 and AWSVAL<> Blank and AWSVAL<>-1 and AWSVALC> =0.1 * AWSVAL and AWS_CMP= 1	FALSE	2023
B5a	E_AWSHM	Received orders placed via a website or apps from customers in own country	Count (ENT_ID) where AWSHM=1	FALSE	2022
B5b	E_AWSEU	Received orders placed via a website or apps from customers in other EU countries	Count (ENT_ID) where AWSEU=1	FALSE	2022
B5c	E_AWSWW	Received orders placed via a	Count (ENT_ID)	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		website or apps from customers in the rest of the world	where AWSWW=1		
Derived from B5b and B5c	E_AWSFOR	Received orders placed via a website or apps from customers in foreign countries (EU or rest of the world)	Count (ENT_ID) where AWSEU=1 or AWSWW=1	FALSE	2022
Derived from B3a and B5a	E_AWSVAL_HM_COWN	Sales via their own websites or apps from customers in the own country	∑(AWSVAL_COWN) where AWSVAL_COWN <> Blank and AWSVAL_COWN <> -1 and AWSHM=1	FALSE	2022
Derived from B3b and B5a	E_AWSVAL_HM_CMP	Sales via an e-commerce marketplace from customers in the own country	∑(AWSVAL_CMP) where AWSVAL_CMP <> Blank and AWSVAL_CMP <> -1 and AWSHM=1	FALSE	2022
Derived from B3a and B5b	E_AWSVAL_EU_COWN	Sales via their own websites or apps from customers in other EU countries	∑(AWSVAL_COWN) where AWSVAL_COWN <> Blank and AWSVAL_COWN <> -1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
			and AWSEU=1		
Derived from B3b and B5b	E_AWSVAL_EU_CMP	Sales via an e-commerce marketplace from customers in other EU countries	∑(AWSVAL_CMP) where AWSVAL_CMP <> Blank and AWSVAL_CMP <> -1 and AWSEU=1	FALSE	2022
Derived from B3a and B5c	E_AWSVAL_WW_COWN	Sales via their own websites or apps from customers located in the rest of the world	∑(AWSVAL_COWN) where AWSVAL_COWN <> Blank and AWSVAL_COWN <> -1 and AWSWW=1	FALSE	2022
Derived from B3b and B5c	E_AWSVAL_WW_CMP	Sales via an e-commerce marketplace from customers located in the rest of the world	∑(AWSVAL_CMP) where AWSVAL_CMP <> Blank and AWSVAL_CMP <> -1 and AWSWW=1	FALSE	2022
Derived from B1, B2 and B5a	E_AWSVAL_HM_CMPQ	Web sales by enterprises which only sell through an e-commerce marketplace from customers in the own country	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1 and AWSHM=1 and AWS_COWN=0 and AWS_CMP=1	FALSE	NEW derived variable

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from B1, B2 and B5b	E_AWSVAL_EU_CMPQ	Web sales by enterprises which only sell through an e-commerce marketplace from customers in other EU countries	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1 and AWSEU=1 and AWS_COWN=0 and AWS_CMP=1	FALSE	NEW derived variable
Derived from B1, B2 and B5c	E_AWSVAL_WW_CMPQ	Web sales by enterprises which only sell through an e-commerce marketplace from customers located in the rest of the world	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1 and AWSWW=1 and AWS_COWN=0 and AWS_CMP=1	FALSE	NEW derived variable
B6a	E_AWSVAL_HM	Sales via a website or apps - from customers located in the own country	∑(AWSVAL_HM) where AWSVAL_HM <> Blank and AWSVAL_HM <> -1	TRUE	2022
B6b	E_AWSVAL_EU	Sales via a website or apps - from customers located in other EU countries	∑(AWSVAL_EU) where AWSVAL_EU <> Blank and AWSVAL_EU <> -1	TRUE	2022
B6c	E_AWSVAL_WW	Sales via a website or apps - from customers	∑(AWSVAL_WW) where AWSVAL_WW <> Blank	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		located in the rest of the world	and AWSVAL_WW <> -1		
Derived from B2, B5b, B5c, B6b and B6c	E_AWSFOR_GE20	Received orders placed via a website or apps from customers in foreign countries, with an export turnover at least 20% of the web sales	Count (ENT_ID) where (AWSEU=1 or AWSWW=1) and (AWSVAL_EU + AWSVAL_WW >= 0.20 * AWSVAL)	TRUE	2022
Derived from B2, B5b, B5c, B6b and B6c	E_AWSFOR_GE50	Received orders placed via a website or apps from customers in foreign countries, with an export turnover at least 50% of the web sales	Count (ENT_ID) where (AWSEU=1 or AWSWW=1) and (AWSVAL_EU + AWSVAL_WW >= 0.50 * AWSVAL)	TRUE	2022
Derived from B1, B2, B5b, B5c, B6b and B6c	E_AWSFOR_GE20_CMP	Sold via an e-commerce marketplace and received orders placed via a website or apps from customers in foreign countries, with an	Count (ENT_ID) where AWS_CMP= 1 and (AWSEU=1 or AWSWW=1) and (AWSVAL_EU + AWSVAL_W	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		export turnover at least 20% of the web sales	W >= 0.20 * AWSVAL)		
Derived from B1, B2, B5b, B5c, B6b and B6c	E_AWSFOR_GE50_CMP	Sold via an e-commerce marketplace and received orders placed via a website or apps from customers in foreign countries, with an export turnover at least 50% of the web sales	Count (ENT_ID) where AWS_CMP= 1 and (AWSEU=1 or AWSWW=1) and (AWSVAL_EU + AWSVAL_WW >= 0.50 * AWSVAL)	TRUE	2022
B7	E_AXSELL	Have received orders via EDI-type messages	Count (ENT_ID) where AXSELL=1	FALSE	2023
B7	E_AXSELLX	Have not received orders via EDI-type messages	Count (ENT_ID) where AXSELL=0	FALSE	2023
B7	E_AXSELLZ	Do not know if have received orders via EDI-type messages	Count (ENT_ID) where AXSELL=Blank	TRUE	2023
Derived from B1 and B7	E_AESELL	Have received orders via	Count (ENT_ID) where (AWS_COW	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		computer networks	N=1 or AWS_CMP= 1) or AXSELL=1		
Derived from B2 and B7	E_WSEL1Q	Enterprises with web sales via websites or apps equal to or above 1% of turnover, and no orders received via EDI-type messages	Count (ENT_ID) where AWSVAL <> Blank and AWSVAL <> -1 and AWSVAL>= TOVT * 1% and (AXSELL=0 or AXSELL=Bl ank)	FALSE	2023
B8	E_AXSVALS	Total sales via EDI type messages, excluding VAT(<1% of turnover)	∑ (AXSVAL) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL<TO VT * 1%	FALSE	2023
B8	E_AXSVALB	Total sales via EDI type messages, excluding VAT(>=1% of turnover)	∑ (AXSVAL) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=T OVT * 1%	FALSE	2023
B8	E_AXSVAL	Total sales via EDI type messages, excluding VAT	∑ (AXSVAL]) where AXSVAL <> Blank and AXSVAL <> -1	FALSE	2023
B8	E_XSEL0	Enterprises with sales	Count (ENT_ID)	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		via EDI-type messages equal to or above 0	where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=0		
B8	E_XSEL1	Enterprises with sales via EDI-type messages equal to or above 1% of turnover	Count (ENT_ID) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=TOVT * 1%	FALSE	2023
B8	E_XSEL2	Enterprises with sales via EDI-type messages equal to or above 2% of turnover	Count (ENT_ID) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=TOVT * 2%	FALSE	2023
B8	E_XSEL5	Enterprises with sales via EDI-type messages equal to or above 5% of turnover	Count (ENT_ID) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=TOVT * 5%	FALSE	2023
B8	E_XSEL10	Enterprises with sales via EDI-type messages equal to or above 10% of turnover	Count (ENT_ID) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=TOVT * 10%	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
B8	E_XSEL25	Enterprises with sales via EDI-type messages equal to or above 25% of turnover	Count (ENT_ID) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=TOVT * 25%	FALSE	2023
B8	E_XSEL50	Enterprises with sales via EDI-type messages equal to or above 50% of turnover	Count (ENT_ID) where AXSVAL <> Blank and AXSVAL <> -1 and AXSVAL>=TOVT * 50%	FALSE	2023
B8	E_XSELZ	Don't know the % of sales via EDI-type messages	Count (ENT_ID) where AXSVAL=Blank	TRUE	2023
Derived from B2 and B8	E_ESELL	Enterprises which used any computer networks for sales (at least 1%)	Count (ENT_ID) where (AWSVAL<> Blank and AWSVAL>= TOVT*1%) or (AXSVAL<> Blank and AXSVAL>=TOVT*1%) or (AWSVAL<> Blank and AWSVAL<>-1 and AXSVAL<>Blank and AXSVAL<>-1 and	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
			(AWSVAL+AXSVAL)>=TOVT*1%)		
Derived from B2 and B8	E_ETURN	Total electronic sales, excluding VAT	∑ (AWSVAL) where AWSVAL <> Blank and AWSVAL <> -1 + ∑ (AXSVAL) where AXSVAL <> Blank and AXSVAL <> -1	FALSE	2023
Derived from B2, B4a and B8	E_AWSVAL_B2C_GE10EC	Enterprises where B2C sales via websites or apps were 10% or more of the e-commerce turnover	Count (ENT_ID) where AWSVALC<>Blank AND AWSVALC> 0 and AWSVAL<> Blank and AXSVAL<>Blank and ((AXSVAL = -1 and AWSVALC>=0.1 * AWSVAL) or (AXSVAL <> -1 and AWSVALC>=0.1 * (AWSVAL+AXSVAL)))	FALSE	2023
C1	E_ITSP2	Employ ICT specialists	Count(ENT_ID) where ITSP2=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
C1	E_ITSP2X	Don't employ ICT specialists	Count(ENT_ID) where ITSP2=0	FALSE	2022
C2a	E_ITSPT2	Have provided training to develop ICT skills of personnel: for ICT specialists	Count(ENT_ID) where ITSPT2=1	FALSE	2022
C2a	E_ITSPT2X	Have not provided training to develop ICT skills of personnel: for ICT specialists	Count(ENT_ID) where ITSPT2=0	FALSE	2022
C2b	E_ITUST2	Have provided training to develop ICT skills of personnel: for other persons employed	Count(ENT_ID) where ITUST2=1	FALSE	2022
C2b	E_ITUST2X	Have not provided training to develop ICT skills of personnel: for other persons employed	Count(ENT_ID) where ITUST2=0	FALSE	2022
Derived from C2a and C2b	E_ITT2	Have provided training to develop ICT	Count(ENT_ID) where ITSPT2=1 or ITUST2=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		skills of personnel			
Derived from C1 and C2a	E_ITSP2_ITT2	Employ ICT specialists and have provided training to develop ICT skills of personnel	Count(ENT_ID) where ITSP2=1 and (ITSPT2=1 or ITUST2=1)	FALSE	2022
Derived from C1 and C2b	E_ITSP2X_ITT2	Don't employ ICT specialists, but have provided training to develop ICT skills of personnel	Count(ENT_ID) where ITSP2=0 and ITUST2=1	FALSE	2022
C3	E_ITSPRCR2	Have recruited/tried to recruit ICT specialists	Count(ENT_ID) where ITSPRCR2=1	FALSE	2022
C3	E_ITSPRCR2X	Have not recruited/tried to recruit ICT specialists	Count(ENT_ID) where ITSPRCR2=0	FALSE	2022
Derived from C1 and C3	E_ITSP2_OR_RCR2	Employ ICT specialists or have recruited/tried to recruit ICT specialists	Count(ENT_ID) where ITSP2=1 or ITSPRCR2=1	FALSE	2022
C4	E_ITSPVAC2	Had hard-to-fill vacancies for ICT specialists	Count(ENT_ID) where ITSPVAC2=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
C4	E_ITSPVAC2X	Had no hard-to-fill vacancies for ICT specialists	Count(ENT_ID) where ITSPVAC2=0	FALSE	2022
Derived from C1 and C4	E_ITSP2_VAC2	Employ ICT specialists and had hard-to-fill vacancies for ICT specialists	Count(ENT_ID) where ITSP2=1 and ITSPVAC2=1	FALSE	2022
C5a	E_ITSPDLA	Enterprises had the following difficulty to recruit ICT specialists: Lack of applications	Count(ENT_ID) where ITSPDLA=1	TRUE	2022
C5a	E_ITSPDLAX	Enterprises did not have the following difficulty to recruit ICT specialists: Lack of applications	Count(ENT_ID) where ITSPDLA=0	TRUE	2022
C5b	E_ITSPDLET	Enterprises had the following difficulty to recruit ICT specialists: Applicants' lack of relevant ICT qualifications from education and/or training	Count(ENT_ID) where ITSPDLET=1	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
C5b	E_ITSPDLETX	Enterprises did not have the following difficulty to recruit ICT specialists: Applicants' lack of relevant ICT qualifications from education and/or training	Count(ENT_ID) where ITSPDLET=0	TRUE	2022
C5c	E_ITSPDLWE	Enterprises had the following difficulty to recruit ICT specialists: Applicants' lack of relevant work experience	Count(ENT_ID) where ITSPDLWE=1	TRUE	2022
C5c	E_ITSPDLWEX	Enterprises did not have the following difficulty to recruit ICT specialists: Applicants' lack of relevant work experience	Count(ENT_ID) where ITSPDLWE=0	TRUE	2022
C5d	E_ITSPDSAL	Enterprises had the following difficulty to recruit ICT specialists: Applicants'	Count(ENT_ID) where ITSPDSAL=1	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		salary expectation too high			
C5d	E_ITSPDSALX	Enterprises did not have the following difficulty to recruit ICT specialists: Applicants' salary expectation too high	Count(ENT_ID) where ITSPDSAL=0	TRUE	2022
C6a	E_IT_OWN	ICT functions were performed by own employees (incl. those employed in parent or affiliate enterprises)	Count(ENT_ID) where IT_OWN=1	FALSE	2022
C6a	E_IT_OWNX	ICT functions were not performed by own employees (incl. those employed in parent or affiliate enterprises)	Count(ENT_ID) where IT_OWN=0	FALSE	2022
C6b	E_IT_EXT	ICT functions were performed by external suppliers	Count(ENT_ID) where IT_EXT=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
C6b	E_IT_EXTX	ICT functions were not performed by external suppliers	Count(ENT_ID) where IT_EXT=0	FALSE	2022
Derived from C6a and C6b	E_IT_OWNQ	ICT functions were entirely performed by own employees (incl. those employed in parent or affiliate enterprises)	Count(ENT_ID) where IT_OWN=1 and IT_EXT=0	FALSE	2022
Derived from C6a and C6b	E_IT_EXTQ	ICT functions were entirely performed by external suppliers	Count(ENT_ID) where IT_OWN=0 and IT_EXT=1	FALSE	2022
Derived from C1 and C6a	E_IT_OWN_ITSP2	Employ ICT specialists and ICT functions were performed by own employees (incl. those employed in parent or affiliate enterprises)	Count(ENT_ID) where IT_OWN=1 AND ITSP2=1	FALSE	2022
Derived from C1 and C6b	E_IT_EXT_ITSP2	Employ ICT specialists and ICT functions were performed	Count(ENT_ID) where IT_EXT=1 AND ITSP2=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		by external suppliers			
Derived from C2a and C2b, C6b	E_IT_EXT_ITT2	Provided training to develop ICT skills of personnel and ICT functions were performed by external suppliers	Count(ENT_ID) where IT_EXT=1 AND (ITSPT2=1 or ITUST2=1)	FALSE	2022
D1a	E_SECMSPSW	ICT security measure used: strong password authentication, i.e. minimum length of 8 mixed characters, periodical change	Count(ENT_ID) where SECMSPSW=1	FALSE	2022
D1a	E_SECMSPSWX	ICT security measure not used: strong password authentication, i.e. minimum length of 8 mixed characters, periodical change	Count(ENT_ID) where SECMSPSW=0	FALSE	2022
D1b	E_SECMUIBM	ICT security measure used: user identification and	Count(ENT_ID) where SECMUIBM=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		authentication via biometric methods implemented by the enterprise (e.g. based on fingerprints, voice, faces)			
D1b	E_SECMUIBMX	ICT security measure not used: user identification and authentication via biometric methods implemented by the enterprise (e.g. based on fingerprints, voice, faces)	Count(ENT_ID) where SECMUIBM=0	FALSE	2022
D1c	E_SECMDUO	ICT security measure used: at least two authentication mechanisms (i.e. combination of e.g. user-defined password, one-time password (OTP), code generated via a	Count(ENT_ID) where SECMDUO=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		security token or received via a smartphone, biometric method faces)			
D1c	E_SECMDUOX	ICT security measure not used: at least two authentication mechanisms (i.e. combination of e.g. user-defined password, one-time password (OTP), code generated via a security token or received via a smartphone, biometric method (e.g. based on fingerprints,	Count(ENT_ID) where SECMDUO=0	FALSE	2022
D1d	E_SECMDENC	ICT security measure used: encryption techniques for data, documents or e-mails	Count(ENT_ID) where SECMDENC=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
D1d	E_SECMDENCX	ICT security measure not used: encryption techniques for data, documents or e-mails	Count(ENT_ID) where SECMDENC=0	FALSE	2022
D1e	E_SECMOSBU	ICT security measure used: data backup to a separate location (including backup to the cloud)	Count(ENT_ID) where SECMOSBU=1	FALSE	2022
D1e	E_SECMOSBUX	ICT security measure not used: data backup to a separate location (including backup to the cloud)	Count(ENT_ID) where SECMOSBU=0	FALSE	2022
D1f	E_SECMNAC	ICT security measure used: network access control (management of access by devices and users to the enterprise's network)	Count(ENT_ID) where SECMNAC=1	FALSE	2022
D1f	E_SECMNACX	ICT security measure not used: network	Count(ENT_ID) where SECMNAC=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		access control (management of access by devices and users to the enterprise's network)			
D1g	E_SECMVPN	ICT security measure used: VPN (Virtual Private Network extends a private network across a public network to enable secure exchange of data over public network)	Count(ENT_ID) where SECMVPN=1	FALSE	2022
D1g	E_SECMVPNX	ICT security measure not used: VPN (Virtual Private Network extends a private network across a public network to enable secure exchange of data over	Count(ENT_ID) where SECMVPN=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		public network)			
D1h	E_SECMSMS1	ICT security measure used: monitoring system that allows to detect suspicious activity in the ICT systems and alerts the enterprises about it, other than standalone anti-virus software	Count(ENT_ID) where SECMSMS1=1	FALSE	NEW
D1h	E_SECMSMS1X	ICT security measure not used: monitoring system that allows to detect suspicious activity in the ICT systems and alerts the enterprises about it, other than standalone anti-virus software	Count(ENT_ID) where SECMSMS1=0	FALSE	NEW
D1i	E_SECMLOG	ICT security measure used: maintaining log files for	Count(ENT_ID) where SECMLOG=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		analysis after security incidents			
D1i	E_SECMLOGX	ICT security measure not used: maintaining log files for analysis after security incidents	Count(ENT_ID) where SECMLOG=0	FALSE	2022
D1j	E_SECMRASS	ICT security measure used: ICT risk assessment, i.e. periodically assessment of probability and consequences of ICT security incidents	Count(ENT_ID) where SECMRASS=1	FALSE	2022
D1j	E_SECMRASSX	ICT security measure not used: ICT risk assessment, i.e. periodically assessment of probability and consequences of ICT security incidents	Count(ENT_ID) where SECMRASS=0	FALSE	2022
D1k	E_SECMTST	ICT security measure used: ICT	Count(ENT_ID) where	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		security tests (e.g. performing penetration tests, testing security alert system, review of security measures, testing of backup systems)	SECMTST=1		
D1k	E_SECMTSTX	ICT security measure not used: ICT security tests (e.g. performing penetration tests, testing security alert system, review of security measures, testing of backup systems)	Count(ENT_ID) where SECMTST=0	FALSE	2022
Derived from D1a to D1k	E_SECMGE1	Use any ICT security measure	Count(ENT_ID) where SECM_CNT>=1	FALSE	2022
Derived from D1a to D1k	E_SECMGE3	Use at least 3 ICT security measures	Count(ENT_ID) where SECM_CNT>=3	FALSE	2022
Derived from D1a to D1k	E_SECMGE5	Use at least 5 ICT security measures	Count(ENT_ID) where SECM_CNT>=5	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from D1a to D1k	E_SECMGE7	Use at least 7 ICT security measures	Count(ENT_ID) where SECM_CNT >=7	FALSE	2022
Derived from D1a to D1k	E_SECMALL	Use all ICT security measures	Count(ENT_ID) where SECM_CNT =11	FALSE	2022
D2a	E_SECAWVTGI	Make persons employed aware of their obligations in ICT security related issues by voluntary training or internally available information (e.g. information on the intranet)	Count(ENT_ID) where SECAWVTGI=1	FALSE	2022
D2a	E_SECAWVTGIX	Don't make persons employed aware of their obligations in ICT security related issues by voluntary training or internally available information (e.g.	Count(ENT_ID) where SECAWVTGI=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		information on the intranet)			
D2b	E_SECAWCTP	Make persons employed aware of their obligations in ICT security related issues by compulsory training courses or viewing compulsory material	Count(ENT_ID) where SECAWCTP=1	FALSE	2022
D2b	E_SECAWCTPX	Don't make persons employed aware of their obligations in ICT security related issues by compulsory training courses or viewing compulsory material	Count(ENT_ID) where SECAWCTP=0	FALSE	2022
D2c	E_SECAWCONT	Make persons employed aware of their obligations in ICT security	Count(ENT_ID) where SECAWCONT=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		related issues by contract (e.g. contract of employment)			
D2c	E_SECAWCONTX	Don't make persons employed aware of their obligations in ICT security related issues by contract (e.g. contract of employment)	Count(ENT_ID) where SECAWCONT=0	FALSE	2022
Derived from D2a to D2c	E_SECAWANY	Make persons employed aware of their obligations in ICT security related issues	Count(ENT_ID) where SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1	FALSE	2022
Derived from D2a to D2c	E_SECAWNONE	Don't make persons employed aware of their obligations in ICT security related issues	Count(ENT_ID) where SECAWVTGI=0 and SECAWCTP=0 and SECAWCONT=0	FALSE	2022
Derived from C2a and D2a	E_SECAWVTGI_ITSPT2	Make persons	Count(ENT_ID) where	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		employed aware of their obligations in ICT security related issues by voluntary training or internally available information and have provided training to develop ICT skills for ICT specialists	SECAWVTGI=1 and ITSPT2=1		
Derived from C2b and D2a	E_SECAWVTGI_ITUST2	Make persons employed aware of their obligations in ICT security related issues by voluntary training or internally available information and have provided training to develop ICT skills for other persons employed	Count(ENT_ID) where SECAWVTGI=1 and ITUST2=1	FALSE	2022
Derived from C2a and D2b	E_SECAWCTP_ITSPT2	Make persons	Count(ENT_ID) where	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		employed aware of their obligations in ICT security related issues by compulsory training courses or viewing compulsory material and have provided training to develop ICT skills for ICT specialists	SECAWCTP =1 and ITSPT2=1		
Derived from C2b and D2b	E_SECAWCTP_ITUST2	Make persons employed aware of their obligations in ICT security related issues by compulsory training courses or viewing compulsory material and have provided training to develop ICT skills for other persons employed	Count(ENT_ID) where SECAWCTP =1 and ITUST2=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from C2a and D2c	E_SECAWCONT_ITSPT2	Make persons employed aware of their obligations in ICT security related issues by contract and have provided training to develop ICT skills for ICT specialists	Count(ENT_ID) where SECAWCONT=1 and ITSPT2=1	FALSE	2022
Derived from C2b and D2c	E_SECAWCONT_ITUST2	Make persons employed aware of their obligations in ICT security related issues by contract and have provided training to develop ICT skills for other persons employed	Count(ENT_ID) where SECAWCONT=1 and ITUST2=1	FALSE	2022
Derived from C2a and D2	E_SECAWANY_ITSPT2	Make persons employed aware of their obligations in ICT security	Count(ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCO	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		related issues and have provided training to develop ICT skills for ICT specialists	NT=1) and ITSPT2=1		
Derived from C2b and D2	E_SECAWARENY_ITUST2	Make persons employed aware of their obligations in ICT security related issues and have provided training to develop ICT skills for other persons employed	Count(ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1) and ITUST2=1	FALSE	2022
D3	E_SECPOL2	Have document(s) on measures, practices or procedures on ICT security	Count(ENT_ID) where SECPOL2=1	FALSE	2022
D3	E_SECPOL2X	Have no document(s) on measures, practices or procedures on ICT security	Count(ENT_ID) where SECPOL2=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from C1 and D3	E_SECPOL2_ITSP2	Employ ICT specialists and have document(s) on measures, practices or procedures on ICT security	Count(ENT_ID) where ITSP2=1 and SECPOL2=1	FALSE	2022
Derived from D2a to D2c and D3	E_SECAWANY_POL2	Make persons employed aware of their obligations in ICT security related issues and have document(s) on measures, practices or procedures on ICT security	Count(ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1) and SECPOL2=1	FALSE	2022
D4a	E_SECPREV_CY	The document(s) on measures, practices or procedures were defined or most recently reviewed: within the last 12 months	Count(ENT_ID) where SECPREV=1	TRUE	2022
D4b	E_SECPREV_1_2	The document(s)	Count(ENT_ID) where	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		on measures, practices or procedures were defined or most recently reviewed: more than 12 months and up to 24 months ago	SECPREV=2		
D4c	E_SECPREV_MT2	The document(s) on measures, practices or procedures were defined or most recently reviewed: more than 24 months ago	Count(ENT_ID) where SECPREV=3	TRUE	2022
Derived from D4a and D4b	E_SECPREV_LE2	The document(s) on measures, practices or procedures were defined or most recently reviewed: within the last 24 months	Count(ENT_ID) where SECPREV=1 or SECPREV=2	TRUE	2022
D5a	E_SEC2IUSVF	Have experienced at least once problems due to ICT	Count(ENT_ID) where SEC2IUSVF=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		security incident: unavailability of ICT services due to hardware or software failures			
D5a	E_SEC2IUSVFX	Have not experienced at least once problems due to ICT security incident: unavailability of ICT services due to hardware or software failures	Count(ENT_ID) where SEC2IUSVF=0	FALSE	2022
D5b	E_SEC2IUSVA	Have experienced at least once problems due to ICT security incident: unavailability of ICT services (e.g. Ransomware attacks, Denial of Service attacks)	Count(ENT_ID) where SEC2IUSVA=1	FALSE	2022
D5b	E_SEC2IUSVAX	Have not experienced at least once problems due to ICT security	Count(ENT_ID) where SEC2IUSVA=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		incident: unavailability of ICT services (e.g. Ransomware attacks, Denial of Service attacks)			
Derived from D5a and D5b	E_SEC2IUSV	Have experienced at least once problems due to ICT security incident: unavailability of ICT services (e.g. Denial of Service attacks, ransomware attacks, hardware or software failures - excluding mechanical failure, theft)	Count(ENT_ID) where SEC2IUSVF=1 or SEC2IUSVA=1	FALSE	2022
D5c	E_SEC2IDCDF	Have experienced at least once problems due to ICT security incident: destruction or corruption of data due to hardware or software failures	Count(ENT_ID) where SEC2IDCDF=1	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
D5c	E_SEC2IDCDFX	Have not experienced at least once problems due to ICT security incident: destruction or corruption of data due to hardware or software failures	Count(ENT_ID) where SEC2IDCDF=0	FALSE	2022
D5d	E_SEC2IDCDA	Have experienced at least once problems due to ICT security incident: destruction or corruption of data due to infection of malicious software or unauthorised intrusion	Count(ENT_ID) where SEC2IDCDA=1	FALSE	2022
D5d	E_SEC2IDCDAX	Have not experienced at least once problems due to ICT security incident: destruction or corruption of data due to infection of malicious software or unauthorised intrusion	Count(ENT_ID) where SEC2IDCDA=0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from D5c and D5d	E_SEC2IDCD	Have experienced at least once problems due to ICT security incident: destruction or corruption of data (e.g. due to infection of malicious software or unauthorised intrusion, hardware or software failures)	Count(ENT_ID) where SEC2IDCDF =1 or SEC2IDCDA =1	FALSE	2022
D5e	E_SEC2ICNFA	Have experienced at least once problems due to ICT security incident: disclosure of confidential data due to intrusion, pharming, phishing attack, intentional actions by own employees	Count(ENT_ID) where SEC2ICNFA =1	FALSE	2022
D5e	E_SEC2ICNFAX	Have not experienced at least once problems due to ICT security incident:	Count(ENT_ID) where SEC2ICNFA =0	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		disclosure of confidential data due to intrusion, pharming, phishing attack, intentional actions by own employees			
D5f	E_SEC2ICNFF	Have experienced at least once problems due to ICT security incident: disclosure of confidential data due to unintentional actions by own employees	Count(ENT_ID) where SEC2ICNFF=1	FALSE	2022
D5f	E_SEC2ICNFFX	Have not experienced at least once problems due to ICT security incident: disclosure of confidential data due to unintentional actions by own employees	Count(ENT_ID) where SEC2ICNFF=0	FALSE	2022
Derived from D5e and D5f	E_SEC2ICNF	Have experienced at least once problems	Count(ENT_ID) where SEC2ICNFA=1 or	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		due to ICT security incident: disclosure of confidential data (e.g. due to intrusion, pharming, phishing attack, actions by own employees (intentionally or unintentionally))	SEC2ICNFF=1		
Derived from D5a to D5f	E_SEC2IANY	Have experienced at least once any problem due to ICT security incident: unavailibility of ICT services, destruction or corruption of data, disclosure of confidential data (for any reason)	Count(ENT_ID) where SEC2IUSVF=1 or SEC2IUSVA=1 or SEC2IDCDF=1 or SEC2IDCDA=1 or SEC2ICNFA=1 or SEC2ICNFF=1	FALSE	2022
Derived from D5a to D5f	E_SEC2IGE3	Have experienced at least once 3 or more problems due to ICT security incident: unavailibility	Count(ENT_ID) where SEC2IUSVF+SEC2IUSVA+SEC2IDCDF+SEC2IDCDA+SEC2ICNFA+SEC2ICNFF	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		of ICT services, destruction or corruption of data, disclosure of confidential data (for any reason)	between 3 and 6		
E1a	E_AI_TTM	Enterprises use AI technologies performing analysis of written language	Count(ENT_ID) where AI_TTM=1	FALSE	2023
E1a	E_AI_TTMX	Enterprises do not use AI technologies performing analysis of written language	Count(ENT_ID) where AI_TTM=0	FALSE	2023
E1b	E_AI_TSR	Enterprises use AI technologies converting spoken language into machine-readable format	Count(ENT_ID) where AI_TSR=1	FALSE	2023
E1b	E_AI_TSRX	Enterprises do not use AI technologies converting spoken language into	Count(ENT_ID) where AI_TSR=0	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		machine-readable format			
E1c	E_AI_TNLG	Enterprises use AI technologies generating written or spoken language	Count(ENT_ID) where AI_TNLG=1	FALSE	2023
E1c	E_AI_TNLGX	Enterprises do not use AI technologies generating written or spoken language	Count(ENT_ID) where AI_TNLG=0	FALSE	2023
E1d	E_AI_TIR	Enterprises use AI technologies identifying objects or persons based on images	Count(ENT_ID) where AI_TIR=1	FALSE	2023
E1d	E_AI_TIRX	Enterprises do not use AI technologies identifying objects or persons based on images	Count(ENT_ID) where AI_TIR=0	FALSE	2023
E1e	E_AI_TML	Enterprises use machine learning (e.g. deep learning) for data analysis	Count(ENT_ID) where AI_TML=1	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E1e	E_AI_TMLX	Enterprises do not use machine learning (e.g. deep learning) for data analysis	Count(ENT_ID) where AI_TML=0	FALSE	2023
E1f	E_AI_TPA	Enterprises use AI technologies automating different workflows or assisting in decision making	Count(ENT_ID) where AI_TPA=1	FALSE	2023
E1f	E_AI_TPAX	Enterprises do not use AI technologies automating different workflows or assisting in decision making	Count(ENT_ID) where AI_TPA=0	FALSE	2023
E1g	E_AI_TAR	Enterprises use AI technologies enabling physical movement of machines via autonomous decisions based on observation of surroundings	Count(ENT_ID) where AI_TAR=1	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E1g	E_AI_TARX	Enterprises do not use AI technologies enabling physical movement of machines via autonomous decisions based on observation of surroundings	Count(ENT_ID) where AI_TAR=0	FALSE	2023
Derived from E1	E_AI_TANY	Enterprises use at least 1 of the listed artificial intelligence technologies: (AI_TTM, AI_TSR, AI_TNLG, AI_TIR, AI_TML, AI_TPA, AI_TAR)	Count (ENT_ID) where (AI_TTM+AI_TSR+AI_TNLG+AI_TIR+AI_TML+AI_TPA+AI_TAR) between 1 and 7	FALSE	2023
Derived from E1	E_AI_TX	Enterprises doesn't use artificial intelligence technologies.	Count (ENT_ID) where AI_TTM=0 and AI_TSR=0 and AI_TNLG=0 and AI_TIR=0 and AI_TML=0 and AI_TPA=0	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
			and AI_TAR=0		
Derived from E1	E_AI_TGE2	Enterprises use at least 2 of the listed artificial intelligence technologies : (AI_TTM, AI_TSR, AI_TNLG, AI_TIR, AI_TML, AI_TPA, AI_TAR)	Count (ENT_ID) where (AI_TTM+AI_TSR+AI_TNLG+AI_TIR+AI_TML+AI_TPA+AI_TAR) between 2 and 7	FALSE	2023
Derived from E1	E_AI_TGE3	Enterprises use at least 3 of the listed artificial intelligence technologies : (AI_TTM, AI_TSR, AI_TNLG, AI_TIR, AI_TML, AI_TPA, AI_TAR)	Count (ENT_ID) where (AI_TTM+AI_TSR+AI_TNLG+AI_TIR+AI_TML+AI_TPA+AI_TAR) between 3 and 7	FALSE	2023
E2a	E_AI_PMS	Enterprises use AI technologies for marketing or sales	Count(ENT_ID) where AI_PMS=1	FALSE	2023
E2a	E_AI_PMSX	Enterprises do not use AI technologies for	Count(ENT_ID) where AI_PMS=0	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		marketing or sales			
E2b	E_AI_PPP	Enterprises use AI technologies for production or service processes	Count(ENT_ID) where AI_PPP=1	FALSE	2023
E2b	E_AI_PPPX	Enterprises do not use AI technologies for production or service processes	Count(ENT_ID) where AI_PPP=0	FALSE	2023
E2c	E_AI_PBAM	Enterprises use AI technologies for organisation of business administration processes or management	Count(ENT_ID) where AI_PBAM=1	FALSE	2023
E2c	E_AI_PBAMX	Enterprises do not use AI technologies for organisation of business administration processes or management	Count(ENT_ID) where AI_PBAM=0	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E2e	E_AI_PLOG	Enterprises use AI technologies for logistics	Count(ENT_ID) where AI_PLOG=1	FALSE	2023
E2e	E_AI_PLOGX	Enterprises do not use AI technologies for logistics	Count(ENT_ID) where AI_PLOG=0	FALSE	2023
E2f	E_AI_PITS	Enterprises use AI technologies for ICT security	Count(ENT_ID) where AI_PITS=1	FALSE	2023
E2f	E_AI_PITSX	Enterprises do not use AI technologies for ICT security	Count(ENT_ID) where AI_PITS=0	FALSE	2023
E2g	E_AI_PFIN	Enterprises use AI technologies for accounting, controlling or finance managemen t	Count(ENT_ID) where AI_PFIN=1	FALSE	2023
E2g	E_AI_PFINX	Enterprises do not use AI technologies for accounting, controlling or finance managemen t	Count(ENT_ID) where AI_PFIN=0	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E2h	E_AI_PRDI	Enterprises use AI technologies for research and development (R&D) or innovation activity	Count(ENT_ID) where AI_PRDI=1	FALSE	2023
E2h	E_AI_PRDIX	Enterprises do not use AI technologies for research and development (R&D) or innovation activity	Count(ENT_ID) where AI_PRDI=0	FALSE	2023
Derived from E2	E_AI_P1ANY	Enterprises use artificial intelligence for at least 1 of the listed purposes: (AI_PMS, AI_PPP, AI_PBAM, AI_PLOG, AI_PITS, AI_PFIN, AI_PRDI)	Count (ENT_ID) where (AI_PMS+AI_PPP+AI_PBAM+AI_PLOG+AI_PITS+AI_PFIN+AI_PRDI) between 1 and 7	FALSE	2023
Derived from E2	E_AI_P1GE2	Enterprises use artificial intelligence for at least 2 of the listed purposes: (AI_PMS, AI_PPP, AI_PBAM, AI_PLOG, AI_PITS,	Count (ENT_ID) where (AI_PMS+AI_PPP+AI_PBAM+AI_PLOG+AI_PITS+AI_PFIN+AI_PRDI) between 2 and 7	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		AI_PFIN, AI_PRDI)			
Derived from E2	E_AI_P1GE3	Enterprises use artificial intelligence for at least 3 of the listed purposes: (AI_PMS, AI_PPP, AI_PBAM, AI_PLOG, AI_PITS, AI_PFIN, AI_PRDI)	Count (ENT_ID) where (AI_PMS+AI_PPP+AI_PBAM+AI_PLOG+AI_PITS+AI_PFIN+AI_PRDI) between 3 and 7	FALSE	2023
E3a	E_AI_ADOWN	The enterprises' AI technologies were developed by own employees	Count(ENT_ID) where AI_ADOWN=1	TRUE	2023
E3a	E_AI_ADOWNX	The enterprises' AI technologies were not developed by own employees	Count(ENT_ID) where AI_ADOWN=0	TRUE	2023
E3b	E_AI_AMOWN	The enterprises' AI technologies were commercial software or systems modified by own employees	Count(ENT_ID) where AI_AMOWN=1	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E3b	E_AI_AMOWNX	The enterprises' AI technologies were not commercial software or systems modified by own employees	Count(ENT_ID) where AI_AMOWN=0	TRUE	2023
E3c	E_AI_AOS	The enterprises' AI technologies were open-source software or systems modified by own employees	Count(ENT_ID) where AI_AOS=1	TRUE	2023
E3c	E_AI_AOSX	The enterprises' AI technologies were not open-source software or systems modified by own employees	Count(ENT_ID) where AI_AOS=0	TRUE	2023
E3d	E_AI_ARDY	The enterprises' AI technologies were purchased commercial software or systems ready to use	Count(ENT_ID) where AI_ARDY=1	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E3d	E_AI_ARDYX	The enterprises' AI technologies were not purchased commercial software or systems ready to use	Count(ENT_ID) where AI_ARDY=0	TRUE	2023
E3e	E_AI_AEXT	The enterprises' AI technologies were developed or modified by external providers	Count(ENT_ID) where AI_AEXT=1	TRUE	2023
E3e	E_AI_AEXTX	The enterprises' AI technologies were not developed or modified by external providers	Count(ENT_ID) where AI_AEXT=0	TRUE	2023
E4	E_AI_PDI	Enterprises which process data on individuals using AI technologies	Count(ENT_ID) where AI_PDI=1	TRUE	NEW
E4	E_AI_PDIX	Enterprises which do not process data on individuals using AI technologies	Count(ENT_ID) where AI_PDI=0	TRUE	NEW

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E5	E_AI_BIAS	Enterprises which have measures to check the results generated by AI technologies for possible biases towards individuals	Count(ENT_ID) where AI_BIAS=1	TRUE	NEW
E5	E_AI_BIASX	Enterprises which do not have measures to check the results generated by AI technologies for possible biases towards individuals	Count(ENT_ID) where AI_BIAS=0	TRUE	NEW
Derived from E5, E3a	E_AI_BIAS_ADOWN	Enterprises which have measures to check the results generated by AI for possible biases towards individuals, and their AI technologies were developed by own employees	Count(ENT_ID) where AI_BIAS=1 and AI_ADOWN =1	TRUE	NEW

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from E5, E3b	E_AI_BIAS_AMOWN	Enterprises which have measures to check the results generated by AI for possible biases towards individuals, and their AI technologies were commercial software or systems modified by own employees	Count(ENT_ID) where AI_BIAS=1 and AI_AMOWN =1	TRUE	NEW
Derived from E5, E3c	E_AI_BIAS_AOS	Enterprises which have measures to check the results generated by AI for possible biases towards individuals, and their AI technologies were open-source software or systems modified by own employees	Count(ENT_ID) where AI_BIAS=1 and AI_AOS=1	TRUE	NEW
Derived from E5, E3d	E_AI_BIAS_ARDY	Enterprises which have measures to check the	Count(ENT_ID) where AI_BIAS=1	TRUE	NEW

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		results generated by AI for possible biases towards individuals, and their AI technologies were purchased commercial software or systems ready to use	and AI_ARDY=1		
Derived from E5, E3e	E_AI_BIAS_AEXT	Enterprises which have measures to check the results generated by AI for possible biases towards individuals, and their AI technologies were developed or modified by external providers	Count(ENT_ID) where AI_BIAS=1 and AI_AEXT=1	TRUE	NEW
E6	E_AI_EC	Enterprises have ever considered to use AI technologies	Count(ENT_ID) where AI_EC=1	TRUE	2023
E6	E_AI_ECX	Enterprises have never considered to use AI technologies	Count(ENT_ID) where AI_EC=0	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
E7a	E_AI_BCST	Enterprises do not use AI technologies, because the costs seem too high	Count(ENT_ID) where AI_BCST=1	TRUE	2023
E7a	E_AI_BCSTX	Enterprises do not use AI technologies, but not because the costs seem too high	Count(ENT_ID) where AI_BCST=0	TRUE	2023
E7b	E_AI_BLE	Enterprises do not use AI technologies, because of a lack of relevant expertise	Count(ENT_ID) where AI_BLE=1	TRUE	2023
E7b	E_AI_BLEX	Enterprises do not use AI technologies, but not because of a lack of relevant expertise	Count(ENT_ID) where AI_BLE=0	TRUE	2023
E7c	E_AI_BINC	Enterprises do not use AI technologies, because of incompatibility with existing equipment,	Count(ENT_ID) where AI_BINC=1	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		software or systems			
E7c	E_AI_BINCX	Enterprises do not use AI technologies, but not because of incompatibility with existing equipment, software or systems	Count(ENT_ID) where AI_BINC=0	TRUE	2023
E7d	E_AI_BDDT	Enterprises do not use AI technologies, because of difficulties with availability or quality of the necessary data	Count(ENT_ID) where AI_BDDT=1	TRUE	2023
E7d	E_AI_BDDTX	Enterprises do not use AI technologies, but not because of difficulties with availability or quality of the necessary data	Count(ENT_ID) where AI_BDDT=0	TRUE	2023
E7e	E_AI_BCDP	Enterprises do not use AI technologies, because od	Count(ENT_ID) where AI_BCDP=1	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		concerns regarding violation of data protection and privacy			
E7e	E_AI_BCDPX	Enterprises do not use AI technologies, but not because od concerns regarding violation of data protection and privacy	Count(ENT_ID) where AI_BCDP=0	TRUE	2023
E7f	E_AI_BLEG	Enterprises do not use AI technologies, because of a lack of clarity about the legal consequences	Count(ENT_ID) where AI_BLEG=1	TRUE	2023
E7f	E_AI_BLEGX	Enterprises do not use AI technologies, but not because of a lack of clarity about the legal consequences	Count(ENT_ID) where AI_BLEG=0	TRUE	2023
E7g	E_AI_BEC	Enterprises do not use AI	Count(ENT_ID) where AI_BEC=1	TRUE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		technologies, because of ethical considerations			
E7g	E_AI_BECX	Enterprises do not use AI technologies, but not because of ethical considerations	Count(ENT_ID) where AI_BEC=0	TRUE	2023
E7h	E_AI_BNU	Enterprises do not use AI technologies, because artificial Intelligence technologies are not useful for the enterprise	Count(ENT_ID) where AI_BNU=1	TRUE	2023
E7h	E_AI_BNUX	Enterprises do not use AI technologies, but not because artificial Intelligence technologies are not useful for the enterprise	Count(ENT_ID) where AI_BNU=0	TRUE	2023
A1	P_IUSE	Persons employed have access to the internet for	∑ (EMPIUSE) where EMPIUSE<> Blank and	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		business purposes	EMPIUSE<> -1		
Derived from A1 (Eurostat computed)[1]	P_IUSEX	Persons employed who do not have access to the internet for business purposes	∑ (EMPL) - ∑ (EMPIUSE) where EMPIUSE<> Blank and EMPIUSE<> -1	TRUE	2023
Derived from A5	P_EMPMD2	Persons employed who use a portable device provided by the enterprise that allows internet connection via mobile telephone networks, for business purposes	∑ (EMPMD2) where EMPMD2<> Blank and EMPMD2<>-1	TRUE	2022
Derived from A5	P_PMD	Persons employed by enterprises which provide portable devices that allow a mobile connection to the internet using mobile telephone networks, for business purposes	∑ (EMPL) where EMPMD2>0	TRUE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from A5	P_PMDX	Persons employed by enterprises which do not provide portable devices that allow a mobile connection to the internet using mobile telephone networks, for business use	∑ (EMPL) where EMPMD2=0 or EMPMD2=Blank	TRUE	2022
Derived from A5	P_EMPMD2X	Persons employed who don't use a portable device provided by the enterprise that allows internet connection via mobile telephone networks, for business purposes	∑ (EMPL) where EMPMD2>0 - ∑ (EMPMD2) where EMPMD2<> Blank and EMPMD2>0	TRUE	2022
Derived from B1	P_AWSELL	Persons employed by enterprises which have received orders via websites or apps	∑ (EMPL) where (AWS_COWN=1 or AWS_CMP=1)	FALSE	2023

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived from B7	P_AXSELL	Persons employed by enterprises which have received orders via EDI-type messages	∑ (EMPL) where AXSELL=1	FALSE	2023
Derived from B1 and B7	P_AESELL	Persons employed by enterprises which have received orders via computer networks	∑ (EMPL) where (AWS_COWN=1 or AWS_CMP=1) or AXSELL=1	FALSE	2023
Derived, see section 5 and Annex 3	E_DI4_VLO	Enterprises with very low digital intensity index	Count (ENT_ID) where DI4_INDEX>=0 and DI4_INDEX<=3	FALSE	2022
Derived, see section 5 and Annex 3	E_DI4_LO	Enterprises with low digital intensity index	Count (ENT_ID) where DI4_INDEX>=4 and DI4_INDEX<=6	FALSE	2022
Derived, see section 5 and Annex 3	E_DI4_HI	Enterprises with high digital intensity index	Count (ENT_ID) where DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2022
Derived, see section 5 and Annex 3	E_DI4_VHI	Enterprises with very high digital intensity index	Count (ENT_ID) where DI4_INDEX>=10 and	FALSE	2022

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
			DI4_INDEX<=12		
Derived, see section 5 and Annex 3 (Eurostat computed)[2]	E_DI4_GELO	Enterprises with at least low (basic) digital intensity index	[Count(ENT_ID) where DI4_INDEX>=4 and DI4_INDEX<=6] + [Count (ENT_ID) where DI4_INDEX>=7 and DI4_INDEX<=9] + [Count (ENT_ID) where DI4_INDEX>=10 and DI4_INDEX<=12]	TRUE	2022
Derived, see section 5 and E1	E_DI4_VLO_AI_TANY	Enterprises with very low digital intensity index and use any artificial intelligence technology	Count (ENT_ID) where DI4_INDEX>=0 and DI4_INDEX<=3 and (AI_TTM=1 or AI_TSR=1 or AI_TNLG=1 or AI_TIR=1 or AI_TML=1 or AI_TPA=1 or AI_TAR=1)	FALSE	2024
Derived, see section 5 and E1	E_DI4_LO_AI_TANY	Enterprises with low digital intensity index and use any artificial	Count (ENT_ID) where DI4_INDEX>=4 and DI4_INDEX<=6 and	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		intelligence technology	(AI_TTM=1 or AI_TSR=1 or AI_TNLG=1 or AI_TIR=1 or AI_TML=1 or AI_TPA=1 or AI_TAR=1)		
Derived, see section 5 and E1	E_DI4_HI_AI_TANY	Enterprises with high digital intensity index and use any artificial intelligence technology	Count (ENT_ID) where DI4_INDEX>=7 and DI4_INDEX<=9 and (AI_TTM=1 or AI_TSR=1 or AI_TNLG=1 or AI_TIR=1 or AI_TML=1 or AI_TPA=1 or AI_TAR=1)	FALSE	2024
Derived, see section 5 and E1	E_DI4_VHI_AI_TANY	Enterprises with very high digital intensity index that use any artificial intelligence technology	Count (ENT_ID) where DI4_INDEX>=10 and DI4_INDEX<=12 and (AI_TTM=1 or AI_TSR=1 or AI_TNLG=1 or AI_TIR=1 or AI_TML=1 or AI_TPA=1 or AI_TAR=1)	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived, see section 5 and Annex 3	E_DI4_VLO_SECMGE1	Enterprises with very low digital intensity index, which use any ICT security measure	Count (ENT_ID) where SECM_CNT >=1 and DI4_INDEX>=0 and DI4_INDEX<=3	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_LO_SECMGE1	Enterprises with low digital intensity index, which use any ICT security measure	Count (ENT_ID) where SECM_CNT >=1 and DI4_INDEX>=4 and DI4_INDEX<=6	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_HI_SECMGE1	Enterprises with high digital intensity index, which use any ICT security measure	Count (ENT_ID) where SECM_CNT >=1 and DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VHI_SECMGE1	Enterprises with very high digital intensity index, which use any ICT security measure	Count (ENT_ID) where SECM_CNT >=1 and DI4_INDEX>=10 and DI4_INDEX<=12	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VLO_SECMGE3	Enterprises with very low digital intensity index, which use 3 or more ICT	Count (ENT_ID) where SECM_CNT >=3 and DI4_INDEX>=0 and	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		security measures	DI4_INDEX<=3		
Derived, see section 5 and Annex 3	E_DI4_LO_SECMGE3	Enterprises with low digital intensity index, which use 3 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=3 and DI4_INDEX>=4 and DI4_INDEX<=6	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_HI_SECMGE3	Enterprises with high digital intensity index, which use 3 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=3 and DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VHI_SECMGE3	Enterprises with very high digital intensity index, which use 3 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=3 and DI4_INDEX>=10 and DI4_INDEX<=12	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VLO_SECMGE5	Enterprises with very low digital intensity index, which use 5 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=5 and DI4_INDEX>=0 and DI4_INDEX<=3	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_LO_SECMGE5	Enterprises with low digital intensity	Count (ENT_ID) where SECM_CNT	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		index, which use 5 or more ICT security measures	>=5 and DI4_INDEX>=4 and DI4_INDEX<=6		
Derived, see section 5 and Annex 3	E_DI4_HI_SECMGE5	Enterprises with high digital intensity index, which use 5 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=5 and DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VHI_SECMGE5	Enterprises with very high digital intensity index, which use 5 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=5 and DI4_INDEX>=10 and DI4_INDEX<=12	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VLO_SECMGE7	Enterprises with very low digital intensity index, which use 7 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=7 and DI4_INDEX>=0 and DI4_INDEX<=3	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_LO_SECMGE7	Enterprises with low digital intensity index, which use 7 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=7 and DI4_INDEX>=4 and DI4_INDEX<=6	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
Derived, see section 5 and Annex 3	E_DI4_HI_SECMGE7	Enterprises with high digital intensity index, which use 7 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=7 and DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VHI_SECMGE7	Enterprises with very high digital intensity index, which use 7 or more ICT security measures	Count (ENT_ID) where SECM_CNT >=7 and DI4_INDEX>=10 and DI4_INDEX<=12	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VLO_SECAWANY	Enterprises with very low digital intensity index, which make persons employed aware of their obligations in ICT security related issues	Count (ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1) and DI4_INDEX>=0 and DI4_INDEX<=3	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_LO_SECAWANY	Enterprises with low digital intensity index, which make persons employed aware of their	Count (ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1) and DI4_INDEX>	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		obligations in ICT security related issues	=4 and DI4_INDEX<=6		
Derived, see section 5 and Annex 3	E_DI4_HI_SECAWANY	Enterprises with high digital intensity index, which make persons employed aware of their obligations in ICT security related issues	Count (ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1) and DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VHI_SECAWANY	Enterprises with very high digital intensity index, which make persons employed aware of their obligations in ICT security related issues	Count (ENT_ID) where (SECAWVTGI=1 or SECAWCTP=1 or SECAWCONT=1) and DI4_INDEX>=10 and DI4_INDEX<=12	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VLO_SECPOL2	Enterprises with very low digital intensity index, in which have document(s) on	Count (ENT_ID) where SECPOL2=1 and DI4_INDEX>=0 and	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		measures, practices or procedures on ICT security	DI4_INDEX<=3		
Derived, see section 5 and Annex 3	E_DI4_LO_SECPOL2	Enterprises with low digital intensity index, in which have document(s) on measures, practices or procedures on ICT security	Count (ENT_ID) where SECPOL2=1 and DI4_INDEX>=4 and DI4_INDEX<=6	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_HI_SECPOL2	Enterprises with high digital intensity index, in which have document(s) on measures, practices or procedures on ICT security	Count (ENT_ID) where SECPOL2=1 and DI4_INDEX>=7 and DI4_INDEX<=9	FALSE	2024
Derived, see section 5 and Annex 3	E_DI4_VHI_SECPOL2	Enterprises with very high digital intensity index, in which have document(s) on measures, practices or procedures	Count (ENT_ID) where SECPOL2=1 and DI4_INDEX>=10 and DI4_INDEX<=12	FALSE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		on ICT security			
Derived, see section 5 and Annex 3	E_DI4_VLO_SECPREV_CY	Enterprises with very low digital intensity index, in which the document(s) on measures, practices or procedures were defined or most recently reviewed within the last 12 months	Count (ENT_ID) where SECPREV= 1 and DI4_INDEX>=0 and DI4_INDEX<=3	TRUE	2024
Derived, see section 5 and Annex 3	E_DI4_LO_SECPREV_CY	Enterprises with low digital intensity index, in which the document(s) on measures, practices or procedures were defined or most recently reviewed within the last 12 months	Count (ENT_ID) where SECPREV= 1 and DI4_INDEX>=4 and DI4_INDEX<=6	TRUE	2024
Derived, see section 5 and Annex 3	E_DI4_HI_SECPREV_CY	Enterprises with high digital intensity index, in	Count (ENT_ID) where SECPREV= 1 and	TRUE	2024

Model questionnaire 2024	Variable codes (CAPITAL LETTERS)	Description	Scope	Optional	Notes / Source (if not previous year)
		which the document(s) on measures, practices or procedures were defined or most recently reviewed within the last 12 months	DI4_INDEX>=7 and DI4_INDEX<=9		
Derived, see section 5 and Annex 3	E_DI4_VHI_SECPREV_CY	Enterprises with very high digital intensity index, in which the document(s) on measures, practices or procedures were defined or most recently reviewed within the last 12 months	Count (ENT_ID) where SECPREV=1 and DI4_INDEX>=10 and DI4_INDEX<=12	TRUE	2024

3.2.4. Activity and employee number codes

Once the indicators have been computed, they must be broken down by economic activity, sector and size using the codes below.

It is mandatory to transmit the data according to the breakdowns in Sections 3.2.4.1 to 3.2.4.3. However, the breakdowns in Sections 3.2.4.4. to 3.2.4.8. are optional.

Some breakdowns computed by Eurostat are also presented in the tables below for information.

3.2.4.1. Economic activity breakdown

The economic activities to be covered are the following:

NACE Rev. 2 category	Description
Section C	Manufacturing
Section D, E	Electricity, gas and steam, water supply, sewerage and waste management
Section F	Construction
Section G	Wholesale and retail trade; repair of motor vehicles and motorcycles
Section H	Transportation and storage
Section I	Accommodation and food service activities
Section J	Information and communication
Section L	Real estate activities
Section M	Professional, scientific and technical activities
Section N	Administrative and support activities
Group 95.1	Repair of computers and communication equipment

The following activity and number_employees codes should be applied to all indicators.

For the possible calculation of **national** aggregates the economic activities are to be subdivided into the following sections using the given codes for **enterprises with 10 or more employees and self-employed persons**:

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
C10T18	EGE10	10 or more employees and self-employed persons in NACE section C10T18
C19T23	EGE10	10 or more employees and self-employed persons in NACE section C19T23
C24_25	EGE10	10 or more employees and self-employed persons in NACE section C24_25
C26T33	EGE10	10 or more employees and self-employed persons in NACE section C26T33

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
C10T33	EGE10	[C10T18]+[C19T23]+[C24_25]+[C26T33]
D35TE39	EGE10	10 or more employees and self-employed persons in NACE section D35_E39
C10TE39 *Eurostat computed*	EGE10	[C10T18]+[C19T23]+[C24_25]+[C26T33]+[D35TE39]
F41T43	EGE10	10 or more employees and self-employed persons in NACE section F41T43
G45T47	EGE10	10 or more employees and self-employed persons in NACE section G45T47
G47	EGE10	10 or more employees and self-employed persons in NACE section G47
H49T53	EGE10	10 or more employees and self-employed persons in NACE section H49T53
I55	EGE10	10 or more employees and self-employed persons in NACE section I55
I55_56	EGE10	10 or more employees and self-employed persons in NACE section I55_56
J58T63	EGE10	10 or more employees and self-employed persons in NACE section J58T63
L68	EGE10	10 or more employees and self-employed persons in NACE section L68
M69T75	EGE10	10 or more employees and self-employed persons in NACE section M69T75
N77T82	EGE10	10 or more employees and self-employed persons in NACE section N77T82 (including N79)
ICT_T	EGE10	10 or more employees and self-employed persons in NACE groups 26.1-26.4, 26.8, 46.5, 58.2, 61, 62, 63.1, 95.1

For the possible calculation of **European** aggregates the economic activities are to be subdivided into the following sections using the given codes for **enterprises with 10 or more employees and self-employed persons.**

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
C10T12	EGE10	10 or more employees and self-employed persons in NACE section C10T12

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
C13T15	EGE10	10 or more employees and self-employed persons in NACE section C13T15
C16T18	EGE10	10 or more employees and self-employed persons in NACE section C16T18
C19	EGE10	10 or more employees and self-employed persons in NACE section C19
C20	EGE10	10 or more employees and self-employed persons in NACE section C20
C21	EGE10	10 or more employees and self-employed persons in NACE section C21
C22_23	EGE10	10 or more employees and self-employed persons in NACE section C22_23
C26	EGE10	10 or more employees and self-employed persons in NACE section C26
C27	EGE10	10 or more employees and self-employed persons in NACE section C27
C28	EGE10	10 or more employees and self-employed persons in NACE section C28
C27_28 *Eurostat computed*	EGE10	10 or more employees and self-employed persons in NACE section C27_28
C29_30	EGE10	10 or more employees and self-employed persons in NACE section C29_30
C31T33	EGE10	10 or more employees and self-employed persons in NACE section C31T33
D35	EGE10	10 or more employees and self-employed persons in NACE section D35
E36T39	EGE10	10 or more employees and self-employed persons in NACE section E36T39
G45	EGE10	10 or more employees and self-employed persons in NACE section G45
G46	EGE10	10 or more employees and self-employed persons in NACE section G46
J58T60	EGE10	10 or more employees and self-employed persons in NACE section J58T60
J61	EGE10	10 or more employees and self-employed persons in NACE section J61
J62_63	EGE10	10 or more employees and self-employed persons in NACE section J62_63

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
M69T71	EGE10	10 or more employees and self-employed persons in NACE section M69T71
M72	EGE10	10 or more employees and self-employed persons in NACE section M72
M73T75	EGE10	10 or more employees and self-employed persons in NACE section M73T75
N77T82X79	EGE10	10 or more employees and self-employed persons in NACE section N77T82_X79
N79	EGE10	10 or more employees and self-employed persons in NACE section N79
S951	EGE10	10 or more employees and self-employed persons in NACE group S951

3.2.4.2. Enterprise size class

These activity and number_employees codes should be applied to all variables.

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
C10TS951XK	EGE10	10 or more employees and self-employed persons in NACE sections [C10TE39]+[F41T43]+[G45T47]+[H49T53]+[I55_56]+[J58T63]+[L68]+ [M69T75]+[N77T82]+[S951]
C10TS951XK	E10T49	10 - 49 employees and self-employed persons (small enterprises) in given NACE sections
C10TS951XK	E50T249	50 - 249 employees and self-employed persons (medium enterprises) in given NACE sections
C10TS951XK	EGE250	250 or more employees and self-employed persons (large enterprises) in given NACE sections
C10TS951XK *Eurostat computed*	E10T249	C10TS951XK [E10T49] + C10TS951XK [E50T249]

3.2.4.3. Sector enterprise size class

These activities and number_employees codes should be applied to all variables:

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
C10TF43	EGE10	10 or more employees and self-employed persons in NACE sections C to F
G45TS951XK	EGE10	10 or more employees and self-employed persons in NACE sections G to N, S951, except K
C10TF43	E10T49	10 – 49 employees and self-employed persons (small enterprises) in NACE sections C to F
G45TS951XK	E10T49	10 – 49 employees and self-employed persons (small enterprises) in NACE sections G to N, S951, except K
C10TF43	E50T249	50 – 249 employees and self-employed persons (medium enterprises) in NACE sections C to F
G45TS951XK	E50T249	50 – 249 employees and self-employed persons (medium enterprises) in NACE sections G to N, S951, except K
C10TF43	EGE250	250 or more employees and self-employed persons (large enterprises) in NACE sections C to F
G45TS951XK	EGE250	250 or more employees and self-employed persons (large enterprises) in NACE sections G to N, S951, except K
C10TF43 *Eurostat computed*	E10T249	10 – 249 employees and self-employed persons (small and medium enterprises) in NACE sections C to F C10TF43 [E10T49] + C10TF43 [E50T249]
G45TS951XK *Eurostat computed*	E10T249	10 – 249 employees and self-employed persons (small and medium enterprises) in NACE sections G to N, S951, except K G45TS951XK [E10T49]+ G45TS951XK [E50T249]

3.2.4.4. Economic activity breakdowns for micro-enterprises

The results of the statistics compiled for microenterprises are to be broken down in accordance with the NACE Rev. 2. sections, divisions and group for the following size class breakdowns: 0-9 employees and self-employed persons, 2-9 and 0-1 employees and self-employed persons, or only 2-9 employees and self-employed persons.

The economic activities are to be subdivided into the following sections using the codes for enterprises with 0-9 employees and self-employed persons.

These activities and number_employees codes should be used for all variables.

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
C10TS951XK	E0T9	0-9 employees and self-employed persons in NACE sections 10-33, 35-39, 41-43, 45-47, 49-53, 55-56, 58-63, 68-75, 77-82, 95.1
C10T18	E0T9	0-9 employees and self-employed persons in NACE section C10T18
C19T23	E0T9	0-9 employees and self-employed persons in NACE section C19T23
C24_25	E0T9	0-9 employees and self-employed persons in NACE section C24_25
C26T33	E0T9	0-9 employees and self-employed persons in NACE section C26T33
C10T33	E0T9	[C10T18]+[C19T23]+[C24_25]+[C26T33]
D35TE39	E0T9	0-9 employees and self-employed persons in NACE section D35TE39
C10TE39 Eurostat computed	E0T9	[C10T18]+[C19T23]+[C24_25]+[C26T33]+[D35TE39]
F41T43	E0T9	0-9 employees and self-employed persons in NACE section F41T43
G45T47	E0T9	0-9 employees and self-employed persons in NACE section G45T47
G47	E0T9	0-9 employees and self-employed persons in NACE section G47
H49T53	E0T9	0-9 employees and self-employed persons in NACE section H49T53
I55	E0T9	0-9 employees and self-employed persons in NACE section I55
I55_56	E0T9	0-9 employees and self-employed persons in NACE section I55_56
J58T63	E0T9	0-9 employees and self-employed persons in NACE section J58T63
L68	E0T9	0-9 employees and self-employed persons in NACE section L68
L68TM75	E0T9	0-9 employees and self-employed persons in NACE section L68TM75
M69T75	E0T9	0-9 employees and self-employed persons in NACE section M69T75
N77T82	E0T9	0-9 employees and self-employed persons in NACE section N77T82 (including N79)

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
S951	E0T9	0-9 employees and self-employed persons in NACE group S951
ICT_T	E0T9	0-9 employees and self-employed persons in NACE 26.1-26.4, 26.8, 46.5, 58.2, 61, 62, 63.1, 95.1

3.2.4.5. Sector enterprise size classes for enterprises with 0-9 employees and self-employed persons

These activities and number_employees codes should be used for all variables. The breakdowns for enterprises with 0-9 employees and self-employed persons are for NSIs providing the breakdown.

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
C10TF43	E0T9	0-9 employees and self-employed persons (very small enterprises) in NACE sections C to F
G45TS951XK	E0T9	0-9 employees and self-employed persons (very small enterprises) in NACE sections G to N, S951, except K

3.2.4.6. Optional sector breakdowns by size class for enterprises with 0-1 and 2-9 employees and self-employed persons

These activity and number_employees codes should be used for all variables.

The breakdowns for enterprises with 0-1 and 2-9 employees and self-employed persons are for NSIs providing the breakdown.

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
C10TS951XK	E0T1	0-1 employees and self-employed persons (micro-enterprises) in given NACE sections
C10TS951XK	E2T9	2-9 employees and self-employed persons (mini-enterprises) in given NACE sections
C10TF43	E0T1	0-1 employees and self-employed persons (micro-enterprises) in NACE sections C to F
C10TF43	E2T9	2-9 employees and self-employed persons (mini-enterprises) in NACE sections C to F

G45TS951XK	E0T1	0-1 employees and self-employed persons (micro-enterprises) in NACE sections G to N, S951, except K
G45TS951XK	E2T9	2-9 employees and self-employed persons (mini-enterprises) in NACE sections G to N, S951, except K

3.2.4.7. Optional breakdowns by economic activity for enterprises with 0-1 employees and self-employed persons

These activity and number_employees codes should be used for all variables.

The breakdowns for enterprises with 0-1 employees and self-employed persons are for NSIs providing the breakdown.

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
C10TS951XK	E0T1	0-1 employees and self-employed persons in NACE sections 10-33, 35-39, 41-43, 45-47, 49-53, 55-56, 58-63, 68-75, 77-82, 95.1
C10T18	E0T1	0-1 employees and self-employed persons in NACE section C10T18
C19T23	E0T1	0-1 employees and self-employed persons in NACE section C19T23
C24_25	E0T1	0-1 employees and self-employed persons in NACE section C24_25
C26T33	E0T1	0-1 employees and self-employed persons in NACE section C26T33
C10T33	E0T1	[C10T18]+[C19T23]+[C24_25]+[C26T33]
D35TE39	E0T1	0-1 employees and self-employed persons in NACE section D35TE39
C10TE39 Eurostat computed	E0T1	[C10T18]+[C19T23]+[C24_25]+[C26T33]+[D35TE39]
F41T43	E0T1	0-1 employees and self-employed persons in NACE section F41T43
G45T47	E0T1	0-1 employees and self-employed persons in NACE section G45T47
G47	E0T1	0-1 employees and self-employed persons in NACE section G47
H49T53	E0T1	0-1 employees and self-employed persons in NACE section H49T53
I55	E0T1	0-1 employees and self-employed persons in NACE section I55

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
I55_56	E0T1	0-1 employees and self-employed persons in NACE section I55_56
J58T63	E0T1	0-1 employees and self-employed persons in NACE section J58T63
L68	E0T1	0-1 employees and self-employed persons in NACE section L68
L68TM75	E0T1	0-1 employees and self-employed persons in NACE section L68TM75
M69T75	E0T1	0-1 employees and self-employed persons in NACE section M69T75
N77T82	E0T1	0-1 employees and self-employed persons in NACE section N77T82 (including N79)
S951	E0T1	0-1 employees and self-employed persons in NACE group S951
ICT_T	E0T1	0-1 employees and self-employed persons in NACE 26.1-26.4, 26.8, 46.5, 58.2, 61, 62, 63.1, 95.1

3.2.4.8. Optional breakdowns by economic activity for enterprises with 2-9 employees and self-employed persons

These activity and number_employees codes should be used for all variables.

The breakdowns for enterprises with 2-9 employees and self-employed persons are for NSIs providing the breakdown.

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
C10TS951XK	E2T9	2-9 employees and self-employed persons in NACE sections 10-33, 35-39, 41-43, 45-47, 49-53, 55-56, 58-63, 68-75, 77-82, 95.1
C10T18	E2T9	2-9 employees and self-employed persons in NACE section C10T18
C19T23	E2T9	2-9 employees and self-employed persons in NACE section C19T23
C24_25	E2T9	2-9 employees and self-employed persons in NACE section C24_25
C26T33	E2T9	2-9 employees and self-employed persons in NACE section C26T33
C10T33	E2T9	[C10T18]+[C19T23]+[C24_25]+[C26T33]

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
D35TE39	E2T9	2-9 employees and self-employed persons in NACE section D35TE39
C10TE39 Eurostat computed	E2T9	[C10T18]+[C19T23]+[C24_25]+[C26T33]+[D35TE39]
F41T43	E2T9	2-9 employees and self-employed persons in NACE section F41T43
G45T47	E2T9	2-9 employees and self-employed persons in NACE section G45T47
G47	E2T9	2-9 employees and self-employed persons in NACE section G47
H49T53	E2T9	2-9 employees and self-employed persons in NACE section H49T53
I55	E2T9	2-9 employees and self-employed persons in NACE section I55
I55_56	E2T9	2-9 employees and self-employed persons in NACE section I55_56
J58T63	E2T9	2-9 employees and self-employed persons in NACE section J58T63
L68	E2T9	2-9 employees and self-employed persons in NACE section L68
L68TM75	E2T9	2-9 employees and self-employed persons in NACE section L68TM75
M69T75	E2T9	2-9 employees and self-employed persons in NACE section M69T75
N77T82	E2T9	2-9 employees and self-employed persons in NACE section N77T82 (including N79)
S951	E2T9	2-9 employees and self-employed persons in NACE group S951
ICT_T	E2T9	2-9 employees and self-employed persons in NACE 26.1-26.4, 26.8, 46.5, 58.2, 61, 62, 63.1, 95.1

3.2.5. Regional data

Regional data is collected for a selection of indicators and activities. The data should refer to enterprises with 10+ employees.

Indicators for regional data.

The indicators which should be provided relate to questions A1, A3, B2, D1, F1 (first priority) as well as C1, C3 and E1 (second priority).

Enterprise size and Economic activities for regional data.

The economic activities to be covered are the following:

NACE category	Explanation
Section C	Manufacturing
Section D, E	Electricity, gas and steam, water supply, sewerage and waste management
Section F	Construction
Section G	Wholesale and retail trade; repair of motor vehicles and motorcycles
Section H	Transportation and storage
Section I	Accommodation and food service activities
Section J	Information and communication
Section L	Real estate activities
Section M, N	Professional, scientific, technical, administrative and support activities

The following activity and number_employees codes should be used:

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
C10T33	EGE10	10 or more employees and self-employed persons in NACE section C10T33
D35TE39	EGE10	10 or more employees and self-employed persons in NACE section D35TE39
C10TE39 *Eurostat computed*	EGE10	[C10T33]+[D35TE39]
F41T43	EGE10	10 or more employees and self-employed persons in NACE section F41T43
G45T47	EGE10	10 or more employees and self-employed persons in NACE section G45T47
H49T53	EGE10	10 or more employees and self-employed persons in NACE section H49T53
I55_56	EGE10	10 or more employees and self-employed persons in NACE section I55_56

ACTIVITY	NUMBER_ EMPLOYEES	Explanation
J58T63	EGE10	10 or more employees and self-employed persons in NACE section J58T63
L68TM75	EGE10	10 or more employees and self-employed persons in NACE section L68TM75
N77T82	EGE10	10 or more employees and self-employed persons in NACE section N77T82 (including N79)
C10TS951XK	EGE10	10 or more employees and self-employed persons in NACE sections [C10TE39]+[F41T43]+[G45T47]+[H49T53]+[I55_56]+ [J58T63]+[L68]+ [M69T75]+[N77T82]+[S951]
C10TF43	EGE10	10 or more employees and self-employed persons in NACE sections C to F
G45TS951XK	EGE10	10 or more employees and self-employed persons in NACE sections G to N, S951, except K

For **regional data**, the column REGIONAL in the data file in SDMX format should not contain _Z, but the respective NUTS[35] 1 and NUTS 2 region codes (see 12).

For countries providing regional data, where only one NUTS 1 region representing the national total exists, the variables and breakdowns requested for regional data should be included twice in the data set: first with _Z in the column REGIONAL and a second time with the respective NUTS 1 code in the column REGIONAL.

[35] NUTS stands for Nomenclature Of Territorial Units For Statistics.

Region codes

Region	Region Name
AT1	Ostösterreich
AT11	Burgenland (AT)
AT12	Niederösterreich
AT13	Wien
AT2	Südösterreich
AT21	Kärnten
AT22	Steiermark
AT3	Westösterreich
AT31	Oberösterreich
AT32	Salzburg
AT33	Tirol
AT34	Vorarlberg
BE1	Région de Bruxelles-Capitale/Brussels Hoofdstedelijk Gewest
BE10	Région de Bruxelles-Capitale/Brussels Hoofdstedelijk Gewest
BE2	Vlaams Gewest
BE21	Prov. Antwerpen
BE22	Prov. Limburg (BE)
BE23	Prov. Oost-Vlaanderen
BE24	Prov. Vlaams-Brabant
BE25	Prov. West-Vlaanderen
BE3	Région wallonne
BE31	Prov. Brabant wallon
BE32	Prov. Hainaut
BE33	Prov. Liège
BE34	Prov. Luxembourg (BE)
BE35	Prov. Namur

Region	Region Name
BG3	Severna i yugoiztochna Bulgaria
BG31	Severozapaden
BG32	Severen tsentralen
BG33	Severoiztochen
BG34	Yugoiztochen
BG4	Yugozapadna i yuzhna tsentralna Bulgaria
BG41	Yugozapaden
BG42	Yuzhen tsentralen
CY0	Kypros
CY00	Kypros
CZ0	Cesko
CZ01	Praha
CZ02	Střední Cechy
CZ03	Jihozápad
CZ04	Severozápad
CZ05	Severovýchod
CZ06	Jihovýchod
CZ07	Střední Morava
CZ08	Moravskoslezsko
DE1	Baden-Württemberg
DE11	Stuttgart
DE12	Karlsruhe
DE13	Freiburg
DE14	Tübingen
DE2	Bayern
DE21	Oberbayern
DE22	Niederbayern
DE23	Oberpfalz

Region	Region Name
DE24	Oberfranken
DE25	Mittelfranken
DE26	Unterfranken
DE27	Schwaben
DE3	Berlin
DE30	Berlin
DE4	Brandenburg
DE40	Brandenburg
DE5	Bremen
DE50	Bremen
DE6	Hamburg
DE60	Hamburg
DE7	Hessen
DE71	Darmstadt
DE72	Gießen
DE73	Kassel
DE8	Mecklenburg-Vorpommern
DE80	Mecklenburg-Vorpommern
DE9	Niedersachsen
DE91	Braunschweig
DE92	Hannover
DE93	Lüneburg
DE94	Weser-Ems
DEA	Nordrhein-Westfalen
DEA1	Düsseldorf
DEA2	Köln
DEA3	Münster
DEA4	Detmold
DEA5	Arnsberg

Region	Region Name
DEB	Rheinland-Pfalz
DEB1	Koblenz
DEB2	Trier
DEB3	Rheinhessen-Pfalz
DEC	Saarland
DEC0	Saarland
DED	Sachsen
DED2	Dresden
DED4	Chemnitz
DED5	Leipzig
DEE	Sachsen-Anhalt
DEE0	Sachsen-Anhalt
DEF	Schleswig-Holstein
DEF0	Schleswig-Holstein
DEG	Thüringen
DEG0	Thüringen
DK0	Danmark
DK01	Hovedstaden
DK02	Sjælland
DK03	Syddanmark
DK04	Midtjylland
DK05	Nordjylland
EE0	Eesti
EE00	Eesti
EL3	Attiki
EL30	Attiki
EL4	Nisia Aigaiou, Kriti
EL41	Voreio Aigaio
EL42	Notio Aigaio
EL43	Kriti

Region	Region Name
EL5	Voreia Ellada
EL51	Anatoliki Makedonia, Thraki
EL52	Kentriki Makedonia
EL53	Dytiki Makedonia
EL54	Ipeiros
EL6	Kentriki Ellada
EL61	Thessalia
EL62	Ionia Nisia
EL63	Dytiki Ellada
EL64	Sterea Ellada
EL65	Peloponnisos
ES1	Noroeste (ES)
ES11	Galicia
ES12	Principado de Asturias
ES13	Cantabria
ES2	Noreste (ES)
ES21	País Vasco
ES22	Comunidad Foral de Navarra
ES23	La Rioja
ES24	Aragón
ES3	Comunidad de Madrid
ES30	Comunidad de Madrid
ES4	Centro (ES)
ES41	Castilla y León
ES42	Castilla-la Mancha
ES43	Extremadura
ES5	Este (ES)
ES51	Cataluña
ES52	Comunitat Valenciana

Region	Region Name
ES53	Illes Balears
ES6	Sur (ES)
ES61	Andalucía
ES62	Región de Murcia
ES63	Ciudad de Ceuta
ES64	Ciudad de Melilla
ES7	Canarias
ES70	Canarias
FI1	Manner-Suomi
FI19	Länsi-Suomi
FI1B	Helsinki-Uusimaa
FI1C	Etelä-Suomi
FI1D	Pohjois- ja Itä-Suomi
FI2	Åland
FI20	Åland
FR1	Île de France
FR10	Île de France
FRB	Centre - Val de Loire
FRB0	Centre - Val de Loire
FRC	Bourgogne - Franche-Comté
FRC1	Bourgogne
FRC2	Franche-Comté
FRD	Normandie
FRD1	Basse-Normandie
FRD2	Haute-Normandie
FRE	Hauts-de-France
FRE1	Nord-Pas-de-Calais
FRE2	Picardie
FRF	Grand Est

Region	Region Name
FRF1	Alsace
FRF2	Champagne-Ardenne
FRF3	Lorraine
FRG	Pays-de-la-Loire
FRG0	Pays-de-la-Loire
FRH	Bretagne
FRH0	Bretagne
FRI	Nouvelle-Aquitaine
FRI1	Aquitaine
FRI2	Limousin
FRI3	Poitou-Charentes
FRJ	Occitanie
FRJ1	Languedoc-Roussillon
FRJ2	Midi-Pyrénées
FRK	Auvergne - Rhône-Alpes
FRK1	Auvergne
FRK2	Rhône-Alpes
FRL	Provence-Alpes-Côte d'Azur
FRL0	Provence-Alpes-Côte d'Azur
FRM	Corse
FRM0	Corse
FRY	RUP FR - Régions ultrapériphériques françaises
FRY1	Guadeloupe
FRY2	Martinique
FRY3	Guyane
FRY4	La Réunion
FRY5	Mayotte
HR0	Hrvatska

Region	Region Name
HR02	Panonska Hrvatska
HR03	Jadranska Hrvatska
HR05	Grad Zagreb
HR06	Sjeverna Hrvatska
HU1	Közép-Magyarország
HU11	Budapest
HU12	Pest
HU2	Dunántúl
HU21	Közép-Dunántúl
HU22	Nyugat-Dunántúl
HU23	Dél-Dunántúl
HU3	Alföld és Észak
HU31	Észak-Magyarország
HU32	Észak-Alföld
HU33	Dél-Alföld
IE0	Éire/Ireland
IE04	Northern and Western
IE05	Southern
IE06	Eastern and Midland
IS0	Ísland
IS00	Ísland
ITC	Nord-Ovest
ITC1	Piemonte
ITC2	Valle d'Aosta/Vallée d'Aoste
ITC3	Liguria
ITC4	Lombardia
ITF	Sud
ITF1	Abruzzo
ITF2	Molise

Region	Region Name
ITF3	Campania
ITF4	Puglia
ITF5	Basilicata
ITF6	Calabria
ITG	Isole
ITG1	Sicilia
ITG2	Sardegna
ITH	Nord-Est
ITH1	Provincia Autonoma di Bolzano/Bozen
ITH2	Provincia Autonoma di Trento
ITH3	Veneto
ITH4	Friuli-Venezia Giulia
ITH5	Emilia-Romagna
ITI	Centro (IT)
ITI1	Toscana
ITI2	Umbria
ITI3	Marche
ITI4	Lazio
LT0	Lietuva
LT01	Sostines regionas
LT02	Vidurio ir vakaru Lietuvos regionas
LU0	Luxembourg
LU00	Luxembourg
LV0	Latvija
LV00	Latvija
MT0	Malta
MT00	Malta
NL1	Noord-Nederland

Region	Region Name
NL11	Groningen
NL12	Friesland (NL)
NL13	Drenthe
NL2	Oost-Nederland
NL21	Overijssel
NL22	Gelderland
NL23	Flevoland
NL3	West-Nederland
NL33	Zuid-Holland
NL34	Zeeland
NL35	Utrecht
NL36	Zuid-Holland
NL4	Zuid-Nederland
NL41	Noord-Brabant
NL42	Limburg (NL)
NO0	Norge
NO02	Innlandet
NO06	Trøndelag/Tröödelage
NO07	Nord-Norge
NO08	Oslo og Viken
NO09	Agder og Sør-Østlandet
NO0A	Vestlandet
NO0B	Svalbard and Jan Mayen
PL2	Makroregion Poludniowy
PL21	Malopolskie
PL22	Slaskie
PL4	Makroregion Pólnocno-Zachodni
PL41	Wielkopolskie
PL42	Zachodniopomorskie

Region	Region Name
PL43	Lubuskie
PL5	Makroregion Poludniowo-Zachodni
PL51	Dolnoslaskie
PL52	Opolskie
PL6	Makroregion Pólnocny
PL61	Kujawsko-Pomorskie
PL62	Warminsko-Mazurskie
PL63	Pomorskie
PL7	Makroregion Centralny
PL71	Lódzkie
PL72	Swietokrzyskie
PL8	Makroregion Wschodni
PL81	Lubelskie
PL82	Podkarpackie
PL84	Podlaskie
PL9	Makroregion Województwo Mazowieckie
PL91	Warszawski stoleczny
PL92	Mazowiecki regionalny
PT1	Continente
PT11	Norte
PT15	Algarve
PT17	Área Metropolitana de Lisboa
PT18	Alentejo
PT19	Centro (PT)
PT2	Região Autónoma dos Açores (PT)
PT20	Região Autónoma dos Açores (PT)

Region	Region Name
PT3	Região Autónoma da Madeira (PT)
PT30	Região Autónoma da Madeira (PT)
RO1	Macroregiunea unu
RO11	Nord-Vest
RO12	Centru
RO2	Macroregiunea doi
RO21	Nord-Est
RO22	Sud-Est
RO3	Macroregiunea trei
RO31	Sud - Muntenia
RO32	Bucuresti - Ilfov
RO4	Macroregiunea patru
RO41	Sud-Vest Oltenia
RO42	Vest
SE1	Östra Sverige
SE11	Stockholm
SE12	Östra Mellansverige
SE2	Södra Sverige
SE21	Småland med öarna
SE22	Sydsverige
SE23	Västsverige
SE3	Norra Sverige
SE31	Norra Mellansverige
SE32	Mellersta Norrland
SE33	Övre Norrland
SI0	Slovenija
SI03	Vzhodna Slovenija
SI04	Zahodna Slovenija
SK0	Slovensko

Region	Region Name
SK01	Bratislavský kraj
SK02	Západné Slovensko

Region	Region Name
SK03	Stredné Slovensko
SK04	Východné Slovensko

3.3. Transmission format for the ICT usage and e-commerce in enterprises survey

Once microdata have been collected and aggregates computed, the data can be transmitted to Eurostat.

The purpose of this section is to describe how the ICT enterprise survey data should be compiled and sent to Eurostat. The 2024 results must be transmitted to Eurostat as a single txt (tab-delimited) file following the Statistical Data and Metadata eXchange (**SDMX**) CSV format.

Each data transmission is assumed to be **a full transmission**. New transmissions will **replace** previous transmissions.

For the content validation tool and Eurostat to be able to process the data, they must be transmitted in the format described below. Extra columns or codes will not be recognised. Comments fields should be used to indicate clearly any deviations Eurostat should be aware of – this is critical for ensuring the fullest comparability of the data for Eurostat's users. **Non-comparable data will not be published.**

The examples given in this section are laid out as they are for ease of reading only.

3.3.1. SDMX

SDMX is an international initiative aimed at standardising and modernising ('industrialising') the mechanisms and processes for exchanging statistical data and metadata among international organisations and their member countries.

SDMX is sponsored by seven international organisations including the Bank for International Settlements (BIS), the European Central Bank (ECB), Eurostat (Statistical Office of the European Union), the International Monetary Fund (IMF), the Organisation for Economic Co-operation and Development (OECD), the United Nations Statistical Division (UNSD), and the World Bank.

SDMX has three key components.

- A model — the information model — to describe data and metadata

The SDMX information model describes statistics in a standard way. It identifies objects and their relationships. A description is necessary to represent data, in order to make them meaningful. The descriptors are modelled according to whether they are dimensions (identifying and describing data), attributes (providing additional information about the data) or measures (representing the phenomenon to be measured). The structural descriptors are brought together

in the Data Structure Definition (DSD). This identifies the dimensions, attributes and measures of a data set, associating them with common code lists and concepts. The DSD provides all the information necessary to fully describe the data transmitted[36].

• A standard for automated communication called Content-oriented Guidelines

The Content-oriented Guidelines (COGs) are a set of recommendations within the scope of the SDMX standard that are designed to maximise interoperability. They are intended to be applicable to all statistical domains. The COGs focus on harmonising specific concepts and terminology that are common to a large number of statistical domains. Such harmonisation helps achieving an even more efficient exchange of comparable data and metadata, and builds on existing experience from implementation.

The COGs cover cross-domain concepts, code lists, subject-matter domains, a glossary, and implementation-specific guidelines[37].

- An IT architecture and set of tools for data and metadata exchange

To support more automated and efficient data and metadata exchange, standard tools and an IT architecture are required. In practice, this means that SDMX promotes the use of standard SDMX-compliant formats. It provides the necessary tools to support the Model, create SDMXcompliant files, store SDMX-related artefacts, map and transcode from existing databases, and validate the structure (and in future the content) of data files. It also provides the necessary architecture to connect IT systems to the SDMX world, enabling data to be shared more easily[38].

The SDMX standard thus provides essential support for statisticians, in that it: maximises the amount of information made available to users; enables the process to be automated; and allows web-service queries.

3.3.2. SDMX format for the ICT usage and e-commerce in enterprises survey

The first line contains the column headers. They must be in uppercase characters, and separated by a tab.

Column 1 - DATAFLOW

The dataflow column should contain ESTAT:INFOSOC_ ENT_A(1.0). This value should be repeated for each line of the file.

Column 2 - FREQ

The freq column should contain A, as the survey is annual.

[36] More details on the SDMX information model can be found in the Technical specifications.
[37] For more information see the recommended practices offered by the SDMX Content-Oriented Guidelines.
[38] For more information see SDMX explained - Eurostat (europa.eu).

Column 3 – TIME_PERIOD

Put the reference year of the survey, with four digits, e.g. 2024.

Column 4 – REF_AREA

Put here your country code. This code should follow the two alpha ISO code.

Codes to be used for the Member States, Candidate Countries and other participating countries are as follows: BE, BG, CZ, DK, DE, EE, IE, EL, ES, FR, HR, IT, CY, LV, LT, LU, HU, MT, NL, AT, PL, PT, RO, SI, SK, FI, SE, IS, NO, ME, MK, AL, RS, TR, BA.

Column 5 – TABLE_IDENTIFIER

Please use the following code

ENT2 – Enterprise/ecommerce ICT survey

Column 6 – EMBARGO_TIME

YYYY-MM-DD - example: 2024-10-31. This would mean that data must not be published before the 31st October 2024. Please leave empty if there is no embargo date. This value should be repeated for each line of the file.

Column 7 – INDICATOR – The variables, their codes and descriptions

Variable codes are the codes, which Eurostat uses for its database and publications. Data should be delivered to Eurostat using these codes in upper case. There is no given order necessary for the variables. If a variable is not collected, it should be removed from the dataset or the value entry should be left blank. In this case, please add a note of explanation in a separate file.

Scope of the observation variables by module

This shows the observation variables. Their exact scope has been defined based on section 3 "Codification of microdata" where the filters to be applied are given.

Column 8 – UNIT_MEASURE and column 18 – UNIT_MULT

Please use the following codes

UNIT_MEASURE	UNIT_MULT	Explanation
PS	0	Persons – use these codes when referring to employed persons (employees or self-employed persons)
PN	0	Pure Number – use these codes when referring to enterprises
EUR	6	Million Euros – use these codes to report amounts, in million euros. Use these codes if your country belongs to the Eurozone.
XDC	6	Million National Currency – use these codes to report amounts, in million national currency. Use these codes if your country does not belong to the Eurozone.

Column 9 – ACTIVITY

Please use the following codes

ACTIVITY	Explanation
C10T12	Manufacture of beverages, food and tobacco products
C10T18	Manufacture of products based on: food, beverages, tobacco, textile, leather, wood, pulp and paper; publishing and printing
C10T33	Manufacturing
C10TE39	Manufacturing, electricity, gas, steam and air conditioning; water supply, sewerage, waste management and remediation activities
C10TF43	Manufacturing, electricity, gas, steam and air conditioning supply; water supply and construction
C10TS951XK	All enterprises, without financial sector
C13T15	Manufacture of textiles, wearing apparel, leather and related products
C16T18	Manufacture of wood & products of wood & cork, except furniture; articles of straw & plaiting materials; paper & paper products; printing & reproduction of recorded media
C19	Manufacture of coke and refined petroleum products
C19T23	Manufacture of coke, refined petroleum, chemical & basic pharmaceutical products, rubber & plastics, other non-metallic mineral products
C20	Manufacture of chemicals and chemical products
C21	Manufacture of basic pharmaceutical products and pharmaceutical preparations
C22_23	Manufacture of rubber and plastics products, and other non-metallic mineral products
C24_25	Manufacture of basic metals & fabricated metal products excluding machines & equipment
C26	Manufacture of computer, electronic and optical products
C26T33	Manufacture of computers, electric & optical products, electrical equipment, machinery & equipment n.e.c., motor vehicles, oth. transport equipment, furniture, oth. manufacturing, repair & installation of machinery & equipment
C27	Manufacture of electrical equipment
C27_28	Manufacture of electrical equipment, machinery and equipment n.e.c.
C28	Manufacture of machinery and equipment n.e.c.
C29_30	Manufacture of motor vehicles, trailers and semi-trailers, other transport equipment

ACTIVITY	Explanation
C31T33	Manufacture of furniture and other manufacturing; repair and installation of machinery and equipment
D35	Electricity, gas, steam and air conditioning supply
D35TE39	Electricity, gas, steam and air conditioning; water supply, sewerage, waste management and remediation activities
E36T39	Water supply, sewerage, waste management and remediation
F41T43	Construction
G45	Trade of motor vehicles and motorcycles
G45T47	Wholesale and retail trade; repair of motor vehicles and motorcycles
G45TS951XK	Services, without financial sector
G46	Wholesale trade, except of motor vehicles and motorcycles
G47	Retail trade, except of motor vehicles and motorcycles
H49T53	Transport and storage
I55	Accommodation
I55_56	Accommodation and food service activities
ICT_T	NACE 26.1-26.4, 26.8, 46.5, 58.2, 61, 62, 63.1, 95.1
J58T60	Publishing activities; motion picture, video & television programme production, sound recording & music publishing; programming & broadcasting
J58T63	Information and communication
J61	Telecommunications
J62_63	Computer programming, consultancy and related activities, information service activities
L68	Real estate activities
L68TM75	Real estate; professional, scientific and technical service activities (only for micro-enterprises and regional data: Eurostat computed for national totals)
M69T71	Legal, accounting, management, architecture, engineering, technical testing and analysis activities
M69T75	Professional, scientific and technical service activities
M72	Scientific research and development
M73T75	Other professional, scientific and technical activities
N77T82	Administrative and support service activities

ACTIVITY	Explanation
N77T82X79	Activities for rental and leasing, employment, security & investigation , services to buildings &landscape, office administrative, office support & other business support
N79	Travel agency; tour operator reservation service and related activities
S951	Repair of computers and communication equipment

Column 10 – NUMBER_EMPLOYEES

Please use the following codes

NUMBER_EMPLOYEES	Explanation
E0T1	From 0 to 1 employee or self-employed
E0T9	From 0 to 9 employees or self-employed
E10T249	From 10 to 249 employees or self-employed
E10T49	From 10 to 49 employees or self-employed
E2T9	From 2 to 9 employees or self-employed
E50T249	From 50 to 249 employees or self-employed
EGE10	10 or more employees or self-employed
EGE250	250 or more employees or self-employed

Annex 2 provides an overview of the requested activity and number_employees codes combinations.

Column 11 - REGIONAL

The regional column should contain _Z for national totals.

For regional data the column should contain the respective region code.

Column 12 – CUST_BREAKDOWN

The cust_breakdown column should contain _T.

Column 13 – OBS_VALUE

All variables should be grossed up to the enterprise population, except for

- variables starting with P_ which should be **grossed up** to the population of employees and self-employed persons
- currency or monetary variables: **grossed up** to turnover (sales variables)
- sample values (remain non-raised figures).

For the values in this txt file **a dot (“.”) should be used as decimal point**. Thousand separators should not be used. In order to avoid small differences which occur in the crosschecking phase, values must not be rounded to integer figures but decimal figures should be reported instead.

The value can have maximum of 10 digits for the whole part and maximum of 7 decimal digits.

Any non-numerical cell entry will be interpreted as a missing value. If a variable is not collected and is not removed from the dataset, the value entry should be left blank and an explanation provided in a separate file.

Column 14 – OBS_STATUS and column 15 – OBS_ STATUS_1

These columns allow to flag the data. Each column can contain only one flag. Put the first flag, if any, in OBS_STATUS. Then put the second flag, if any, in OBS_ STATUS_1. The codes to be used are as follows:

OBS_STATUS	Explanation
A	Normal value (no flag)
B	Break in series
U	Low reliability

Column 16 – CONF_STATUS

These columns allow to define the confidentiality status. The codes to be used are as follows:

CONF_STATUS	Explanation
F	Free for publication
C	Confidential

Column 17 – COMMENT_OBS

This column allows to put a footnote. Actually, it may contain several footnotes, separated by a '@' character. A footnote that applies to many cells must be repeated for each cell. Right now, the maximum length of the combined footnotes is 255 characters.

Column 18 – UNIT_MULT

This column is a unit multiplier. Use 0 if the UNIT is PS or PN. Use 6 if the unit is EUR or XDC.

Column 19 – DECIMALS

This column is foreseen to allow to indicate how many of the OBS_VALUE decimals are significant. Right now, this column is not used. It should be set to 0.

3.3.3. Example

This is an example of a file that could be sent. The data is made up.

DATAFLOW	FREQ	TIME_PERIOD	REF_AREA	TABLE_ IDENTIFIER	EBARGO_ TIME
ESTAT:INFOSOC_ ENT_A(1.0)	A	2024	XY	ENT2	2024-10-11

ESTAT:INFOSOC_ENT_A(1.0)	A	2024	XY	ENT2	2024-10-11
ESTAT:INFOSOC_ENT_A(1.0)	A	2024	XY	ENT2	2024-10-11

INDICATOR	UNIT_MEASURE	ACTIVITY	NUMBER_EMPLOYEES	REGIONAL	CUST_BREAKDOWN
EMPL	PS	C10T12	EGE10	_Z	_T
EMPL	PS	C10T18	EGE10	_Z	_T
EMPL	PS	C10T33	EGE10	_Z	_T

OBS_VALUE	OBS_STATUS	OBS_STATUS_1	CONF_STATUS	COMMENT_OBS	UNIT_MULT	DECIMALS
1	U	A	F	Note 1	0	0
2	B	A	C	Note 1@Note 2	0	0
3	U	B	F	Note 2	0	0

3.4. Data confidentiality

Confidentiality is a fundamental principle of European statistics, as defined by Regulation (EC) 223/2009, Commission Regulation (EU) No 557/2013 and the European Statistics Code of Practice (ESCoP).

Recital 24 of Regulation (EC) 223/2009 on European statistics makes provision for the establishment of common principles and guidelines ensuring the confidentiality of data used for the production of European statistics and access to those confidential data, taking due account of technical developments and the requirements of users in a democratic society.

For that purpose, Article 20(4) of Regulation (EC) 223/2009 requires the following.

'Within their respective spheres of competence, the NSIs and other national authorities and the Commission (Eurostat) shall take all necessary regulatory, administrative, technical and organisational measures to ensure the physical and logical protection of confidential data (statistical disclosure control).'

'Confidential data' means:

'data which allow statistical units to be identified, either directly or indirectly, thereby disclosing individual information. Confidentiality aims at protecting data from unauthorised disclosure that could be prejudicial or harmful to the interest of the source or other relevant parties. To determine whether a statistical unit is identifiable, account shall be taken of all relevant means that might reasonably be used by a third party to identify the statistical unit' (Article 3(4) of Regulation (EC) 223/2009).

In practice, for Eurostat not to publish confidential data, compilers need to indicate during data transmission which data are confidential, using the c-flag.

It is recommended to use flags wisely and sparingly. Most cells should not be flagged.

Depending on the status of national confidentiality, there are several implications for how European aggregates are to be published.

Data used for the production of statistics by national and EU authorities are considered confidential if statistical units can be identified, directly or indirectly, and if information about individuals or businesses can be disclosed as a result.

- Direct identification means identification of the respondent (statistical unit) from their formal identifiers (e.g. name, address, identification number).
- Indirect identification means inferring a respondent's identity through a combination of variables or characteristics (e.g. age, gender, education).

Statistical disclosure control can be ensured through physical protection and statistical disclosure control (SDC). For further details, refer to the Eurostat website.

Confidentiality rules are based on the number of enterprises. Two criteria, relevant for confidential disclosure, are proposed at national level:

- criterion A – data refer to less than three statistical units;
- criterion B – one or a few enterprises contribute to over 85 % of the total volume of aggregated data.

Data compilers should explain the national confidentiality rules in the metadata and quality reports.

3.5. Transmission deadlines

The Implementing Act states that:

- the annual metadata report for survey year 2024 must be transmitted to Eurostat by 31 May 2024.
- the transmission deadline for the data relating to ICT usage and e-commerce in enterprises survey is 5 October of the survey year, i.e. 5 October 2024 for survey year 2024.
- the annual quality report for survey year 2024 must be transmitted to Eurostat by 5 November 2024.

3.6. Transmission channels

The transmission and delivery of data sets is managed by EDAMIS (Electronic Data Files Administration and Management Information System), adopted as the unique entry point for the transmission of data to Eurostat.

The EDAMIS portal is accessible via the following link: https://webgate.ec.europa.eu/edamis4

EDAMIS is made available through different networks: the internet and secure European

networks like TESTA (Trans European Services for Telematics between Administrations) and CCN (Common Communication Network).

Information on networks, comparison between the different transmission methods and step-by-step instructions for submitting files are provided in EDAMIS short and extensive user guides developed by Eurostat to which access is given on request.

3.7. Data revisions

Revisions are broadly defined as any change to the value of a statistic released to the public. They can occur when new observations (one additional month or quarter) become available and some past values are modified or when the current and/or some past values are modified. Data are generally revised to incorporate new, improved information. Revisions are therefore inevitable whenever statistics are produced that report promptly on economic developments despite the fact that some relevant information is still missing.

Further guidance on data revision principles can be found in the European Statistics Code of Practice.

3.8. Support of data providers

Specific support is available here: ESTAT-ICT-SURVEYS@ ec.europa.eu.

4

Data quality

Assessing data quality is a crucial step in providing users with high-quality statistics.

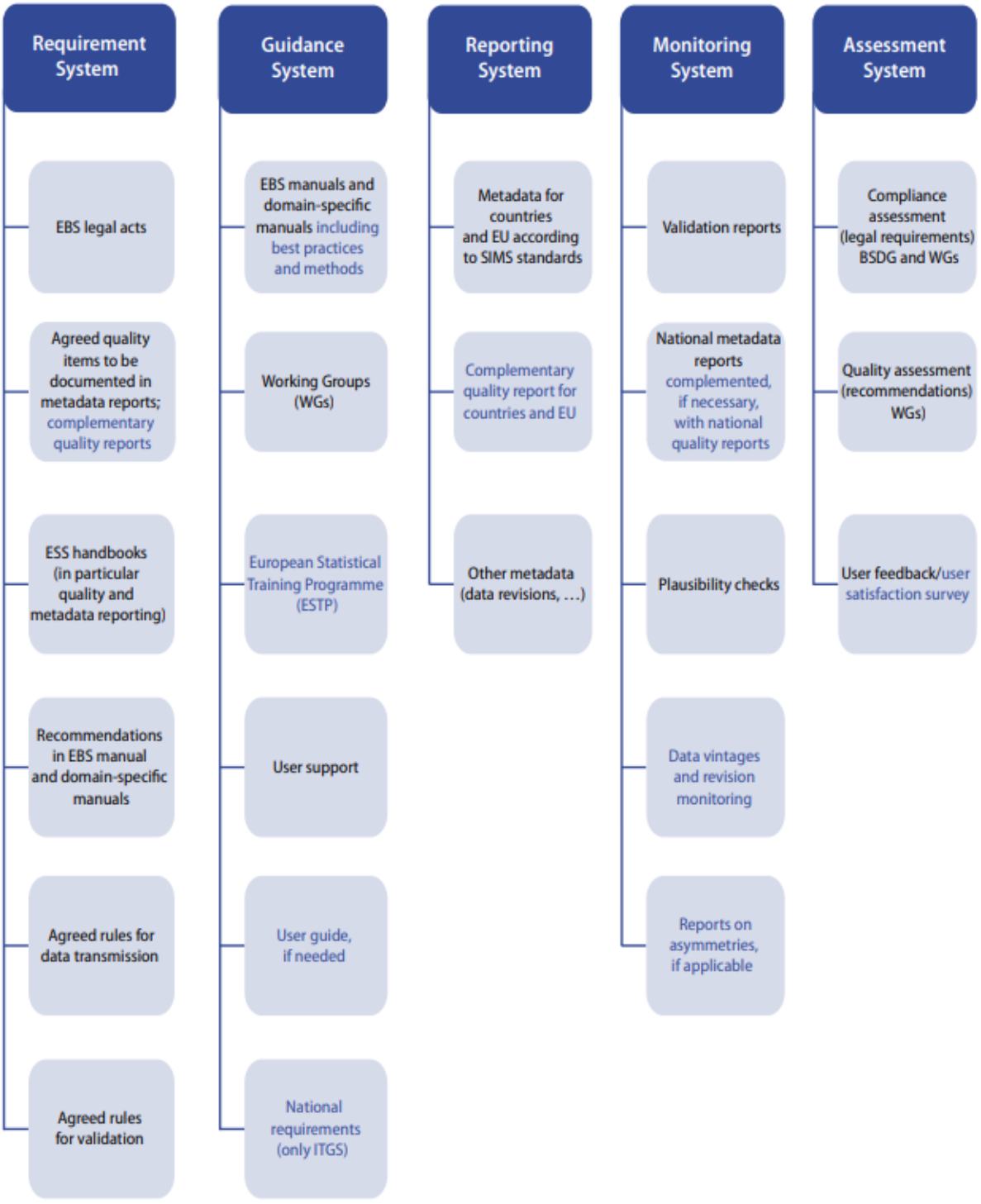
Consequently, Eurostat has a quality framework in place to assess data on ICT usage and e-commerce in enterprises. It carries out a number of checks on the data submitted to Eurostat based on validation rules. NSIs are required to send quality and metadata reports to assess the overall quality of their data.

4.1. Quality framework

4.1.1. European business statistics quality framework

Efforts were made to standardise the quality framework for business and trade statistics under the EBS Regulation. As shown below, the core EBS quality framework involves different measures that ensure high-quality statistics and allow users to understand the quality issues for the statistics concerned.

The core set of measures are written in black and ensure the quality of business and trade statistics. These should eventually be in place for all business and trade statistics. Measures written in blue are considered optional depending on the needs of the individual domains.


Requirement System
EBS legal acts
Agreed quality items to be documented in metadata reports; complementary quality reports
ESS handbooks (in particular quality and metadata reporting)
Recommendations in EBS manual and domain-specific manuals
Agreed rules for data transmission
Agreed rules for validation
Guidance System
EBS manuals and domain-specific manuals including best practices and methods
Working Groups (WGs)
European Statistical Training Programme (ESTP)
User support
User guide, if needed
National requirements (only ITGS)
Reporting System
Metadata for countries and EU according to SIMS standards
Complementary quality report for countries and EU
Other metadata (data revisions, …)
Monitoring System
Validation reports
National metadata reports complemented, if necessary, with national quality reports
Plausibility checks
Data vintages and revision monitoring
Reports on asymmetries, if applicable
Assessment System
Compliance assessment (legal requirements) BSDG and WGs
Quality assessment (recommendations) WGs)
User feedback/user satisfaction survey

4.1.2. Implementing the framework for data on ICT usage and e-commerce in enterprises

The table below explains how the EBS quality framework is applied when collecting data on ICT usage and e-commerce in enterprises.

REQUIREMENT SYSTEM **The requirements system comprises all legal requirements, standards and rules agreed at working-group level.**	
European business statistics legal acts	See Section 2.1 – Legislative background
Quality items to be documented in metadata reports	12.3.1. Data completeness – rate (rates are not requested but deviation from question / item in model questionnaire have to be provided) 14.2.1. Punctuality – delivery (Number of days between the delivery date of data and the target date on which they were scheduled for delivery)
European statistical system (ESS) handbooks, particularly the handbook quality and metadata reporting	ESS handbook for quality and metadata reports
Recommendations	See Chapter 2 – Data compilation
Rules for submitting data	See Chapter 3 – Data transmission
Rules for validating data	See Annex 2 – Validation rules
GUIDANCE SYSTEM **The guidance system provides further advice to help compilers improve and ensure the quality of data.**	
European business statistics manuals and domain-specific manuals, including best practices and methods	ESS manual
Expert groups	Information Society Statistics Working Group Information Society Statistics Task Force
European statistical training programme courses	Not applicable

User support	Questions to be addressed to ESTAT-ICT-SURVEYS@ec.europa.eu.

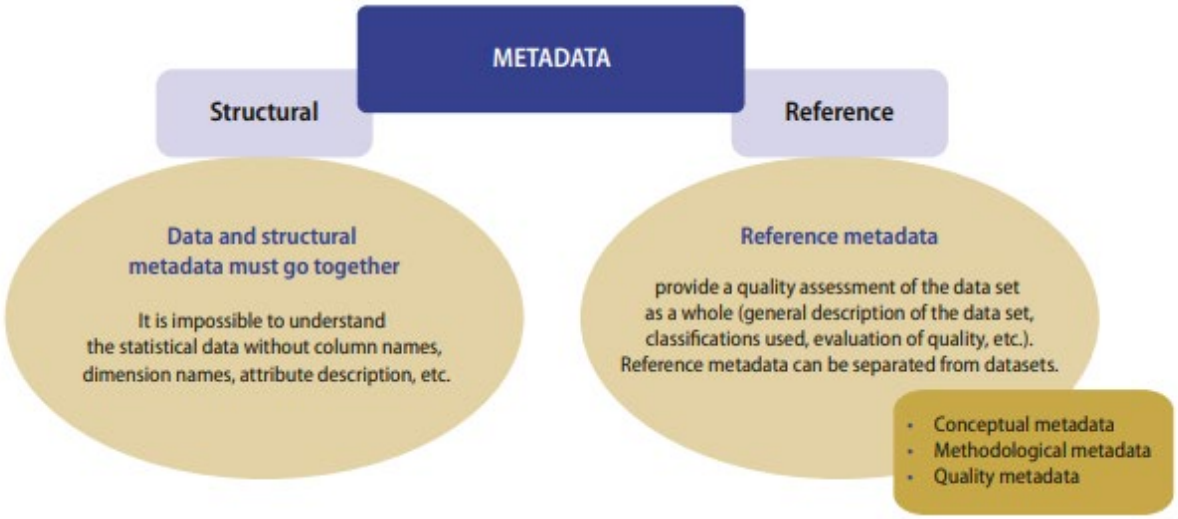
REPORTING SYSTEM **The reporting system is used by data providers to report on the quality of the European business and trade statistics sent to Eurostat.**	
Metadata for countries and EU according to the single integrated metadata structure reporting standards	Metadata are available via the following link: ESS Metadata Handler EU metadata are available via the following link: Eurostat metadata
MONITORING SYSTEM **The monitoring system comprises all measures Eurostat uses to keep track of the quality of European business and trade statistics.**	
Validation reports	Countries contacted in case of issues.
National metadata reports complemented with national quality reports if necessary	Metadata collected according to the single integrated metadata structure.
Plausibility checks	Checks for consistency within data sets, cross-country data and time series analyses.
ASSESSMENT SYSTEM **The assessment system describes measurements Eurostat uses to analyse how European statistical system countries comply with legal requirements. It also analyses the quality of national data submitted to Eurostat.**	
Compliance assessment (legal requirements)	Annual compliance assessment.
Quality assessment	Annual quality reviews before publication, recommendations issued in this manual, experience of NSIs taken into account when preparing the model questionnaire.
User feedback / user satisfaction surveys	Feedback collected from regular users and via questions sent out by user support

4.2. Metadata and quality reports

Metadata, also described as «data about data», provide information about data and are essential for their understanding. They allow users to make comparisons between data and assess the quality of data. Metadata can be expressed as text (for example descriptions), values (for example percentage rates) and codes (from controlled vocabularies such as code lists).

There are two types of metadata: structural metadata and reference metadata.

Structural metadata cover the information that is used to identify and describe the data. Some examples of types of structural metadata are: variable names; variable codes; classifications used; technical descriptions of a data set (e.g. data formats, time dimensions, etc.); and data set locations. Structural metadata need to be linked with data so that they can be searched for and identified.



Reference metadata describe the contents and quality of statistical data. According to the latest version of the *European statistical system handbook for quality and metadata reports*, metadata are subdivided into:

- conceptual metadata – these explain the concepts used;
- methodological metadata – these refer to methods used in preparing the statistical data;
- quality metadata – these refer to and explain quality dimensions of the statistical outputs.

Several tools have been developed to produce high-quality and standardised metadata within the ESS. The single integrated metadata structure (**SIMS**) was created to support quality reporting on European statistics. It provides a standard, integrated and comprehensible framework for metadata and quality reporting in the ESS. It was formed by integrating and standardising two reporting structures. In particular, the Euro-SDMX metadata structure and the ESS standard for

quality reports structure[39]. It is a template for the ESS reference metadata report structure and quality report, which contain information about quality concepts at different levels of detail.

4.2.1. Metadata and quality report on ICT usage and e-commerce in enterprises

For the ICT usage and e-commerce survey, the EBS Regulation asks reporting countries to submit quality and metadata reports to Eurostat each year. The information needed for data treatment should be included in the metadata report and all remaining information should be included in the quality report. The content of those reports is then combined into one national reference metadata report for each country.

The deadlines to submit the quality and metadata reports are laid down in the Implementing Act. The deadline to send Eurostat the metadata report is before the quality report.

The current structure of the reference metadata report is the following:

Contact: information about the organisation, contact name and details (address, email, telephone number) of the person in charge in the organisation.

Metadata update: dates when the metadata have been certified, posted and updated.

Statistical presentation: description of the data and classifications used (e.g. NACE Rev. 2); statistical concepts and definitions; coverage of the statistical domain; statistical population and statistical units data refer to; time coverage and reference area.

Unit of measure: for the survey on ICT usage and e-commerce in enterprises, the percentage of enterprises, of turnover, and of employees and self-employed persons is used.

Reference period: the reference period is usually the survey year, but for certain variables it is the year before the survey year.

Institutional mandate: information about complementary national legislation constituting the legal basis for the survey.

Confidentiality: this includes confidentiality policy – provisions concerning confidentiality in legal acts; and data treatment – rules applied to keep confidential data undisclosed.

Release policy: this covers the data release schedule/ calendar and where this calendar can be found. The release policy refers to several principles as laid down in the Regulation on European statistics and in the European statistics code of practice. For example, objectivity, impartiality, confidentiality and accessibility.

Frequency of dissemination: data on ICT usage and e-commerce in enterprises are disseminated annually. Accessibility and clarity: this refers to various formats used in the

[39] More information and a visualisation of this structure is available in the European Statistical System (ESS) handbook for quality and metadata reports — 2020 edition, p 238-241.

dissemination data on ICT usage and e-commerce in enterprises at national level. For example, news releases, publications and online databases.

Quality management: quality assurance and quality assessment describe the systems and frameworks in place to manage the quality of surveys and processes on ICT usage and e-commerce in enterprises.

Relevance: the main national users and their needs are taken into account when developing the survey on ICT usage and e-commerce in enterprises. User satisfaction is also taken into account either through a survey or other methods.

Accuracy: accuracy of data is the closeness of computations or estimates to the exact or true values that the statistics were intended to measure. It is assessed based on overall comments on accuracy and the amount of sampling and non-sampling errors. These include: coverage errors; measurement errors; non-response errors; processing errors; and model assumption errors.

Timeliness and punctuality: Timeliness refers to the time elapsed between the event or phenomenon the data describe taking place and the time at which the data becomes available. Punctuality refers to the time lag between the actual delivery of the data and the target date that they should have been delivered. For example, 5 October 2024.

Coherence and comparability: European statistics should be coherent, maintaining internal consistency over time. It should also be possible to compare statistics between regions and countries, as well as combine and make joint use of related data from different sources. Information on geographical and intertemporal comparability, and cross-domain coherence are provided in the metadata report.

Cost and burden: provides information on the cost of collecting and producing data on ICT usage and e-commerce in enterprises, and the burden on respondents.

Data revision: data revision is any change in a value of a statistic released to the public. This includes data revision policy applicable to data output and how it can be implemented practically.

Statistical processing: information about: data on ICT usage and e-commerce in enterprises source data (e.g. frame population and sampling); data collection frequency; type of data collection (e.g. paper, web, electronic); data validation; and data compilation.

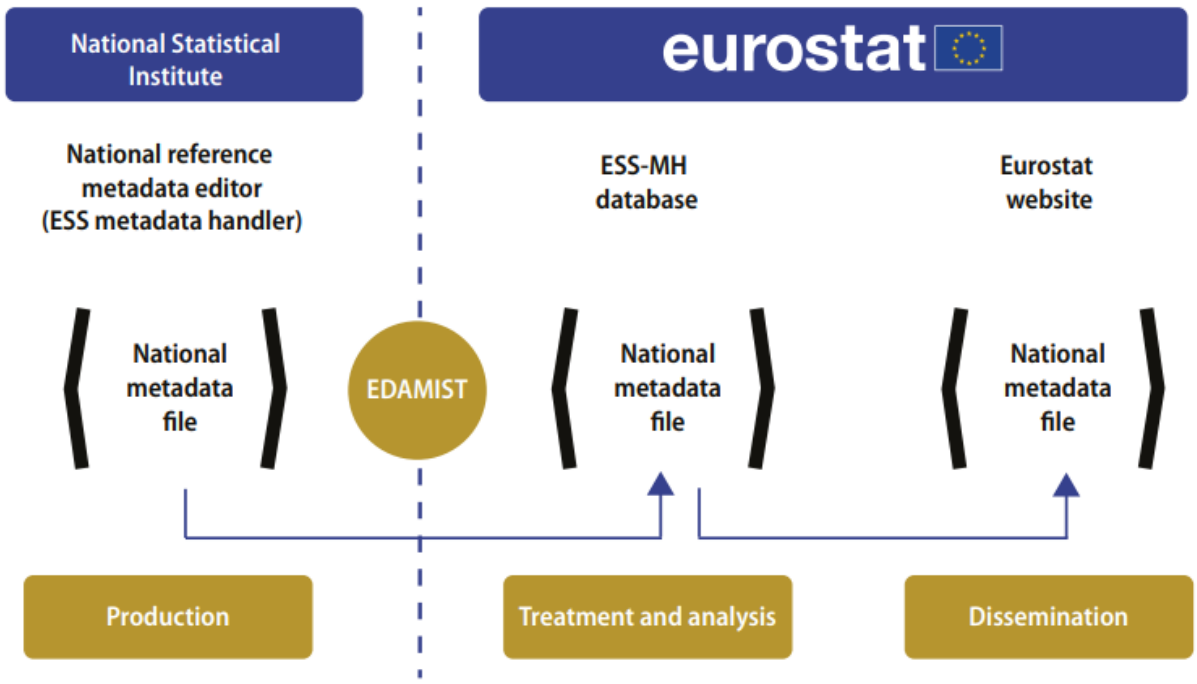
Comments: contains problems encountered, lessons learnt, any other comments, and the list of annexes.

Related metadata: not applicable for this survey.

Annexes: links to annexes.

4.2.2. European Statistical System metadata handler

Reporting countries should deliver the metadata and quality report using the MH. The ESS Metadata Handler (ESS-MH) is a web-based application that supports the production, exchange and dissemination of reference metadata in the ESS. This MH accommodates SDMX-compliant files based on SIMS. It supports standardising reference metadata and quality reports in the ESS. The diagram below presents the high-level business process to report SDMX-compliant reference metadata. It also presents use of the MH.



Eurostat and NSIs use this application to produce metadata files. The application is accessible through a password system[40].

Information about the metadata file on ICT usage and e-commerce in enterprises is provided below:

File name: INFOSOC_ETNSI_ A_CC_YEAR_0000

Domain: INFOSOC

Metadata flow: INFOSOC_ETNSI_ A

Typology: simsie - Sims structure for INFOSOC enterprises

Country: CC (country code)

Organisation – provider: name of the organisation providing the metadata file

Reference year: YEAR

Period: A0

Status: Draft; Ready for validation; Validated; Ready for publication, Published

Modified on: dd/mm/yyyy hh:mm:ss

By: username

[40] For any issues, contact: ESTAT-DATA-METADATA-SERVICES@ec.europa.eu

Published on: dd/mm/yyyy hh:mm:ss

Owner: username

The main application features of the metadata file for users are:

copy an existing file to create a new one;

recall the file to make changes;

download;

view history;

preview;

print.

After the changes have been made, the file is sent to the domain manager at Eurostat for validation. The domain manager will approve the file if all information is clear. If not, they can ask for clarifications. Reporting countries are asked to correct or add missing information. The file is then submitted again for approval.

The template of the national reference metadata report is provided in Annex 3 – metadata reporting template.

4.3. Data validation by Eurostat

Although reporting countries are responsible for the quality of the data provided, Eurostat carries out a series of checks to ensure the accuracy of data submission format and the absence of errors. The validation process is currently structured according to the validation levels classification established by the ESS.

Validation level 0: consistency with expected IT structural requirements;

Validation level 1: consistency within the data set;

Validation level 2: consistency with other data sets within the same domain and the same data source;

Validation level 3: consistency within the same domain between different data sources;

Validation level 4: consistency between separate domains in the same data provider;

Validation level 5: consistency with data of other data providers;

Only levels 0 to 3 are currently used to validate data on ICT usage and e-commerce in enterprises. Checks with the highest priority include: format checks; checks on the completeness of the file and uniqueness of records (level 0); and checks on data consistency (level 1). Failure to pass these levels will result in the file being rejected. The other checks may result in a list of warnings for which the reporting country is asked either to send revised data or to confirm the data correctness according to the type of warning spotted. A description of the different data checks is provided below. The rules applied are included in Annex 2 – validation rules.

Validation level 0: consistency with expected IT structural requirements

The first step is to check whether the file complies with the structure and format required by the

submission format presented in Section 3.3.2. The checks performed at this stage refer to the:

validity of format: data are expected to be sent in the sole SDMX-TXT format as defined in Chapter 3. The number of columns in the file should be in line with the submission format presented in Section 3.3.2.

validity of codes: these checks are performed on each dimension and attribute at record level. Their aim is to make sure that each reported code belongs to the code list related to that particular dimension or attribute.

Complying with the submission format is the highest priority. Failing to pass checks at this level will result in the file being rejected.

Validation level 1: consistency within the data set

The next step in the validation process is to analyse the content of the file. The checks are divided into the three categories below:

- **completeness of the file**: this check verifies that the number of records in the file is equal to the total number expected for this data set.
- **inter-record consistency checks**: these verify the consistency between the observation value of two or more records. These records can be linked by an equality or an inequality. The link is described in a consistency rule. Typically, the consistency of the total with the sum of details will be verified through these types of checks.
- **consistency checks on ratios**: ICT data are shared on the Eurostat website as percentages. Checks are run on these percentages to assess their plausibility by comparing them with a configurable range defined for every unit.

Validation level 2: consistency with other data sets within the same domain and the same data source

In this step, two series of checks are performed:

- variation over previous reference years: data provided for a new reference year are compared with those from previous years. The variation should fall within a pre-defined range.
- consistency checks on reference data and flags are carried out and compared with the information in the quality report provided by the NSIs.

If the variation of the data falls outside the pre-defined range when compared with previous reference years, Eurostat will contact the country concerned and ask them to clarify and confirm the accuracy of the data. Eurostat will also contact countries if there is a discrepancy in the reference data.

Validation level 3: consistency within the same domain and a different data source

Data on ICT usage and e-commerce in enterprises are compared with similar data reported by other countries. The comparison is carried out at the level of main indicators, enterprise sizes and NACE activities, and is based on the most recent detailed data submitted to Eurostat.

If there is significant inconsistency between the two sources, Eurostat will contact the reporting countries to ask them to provide clarifications and possible corrections.

5

Data dissemination

Once Eurostat has received and validated the data transferred by the NSIs, the European aggregates are calculated and the complete database with the results from the surveys is published online.

5.1. Data description

Eurostat publishes the 'ICT usage and e-commerce in enterprises' data gathered by the NSIs on the basis of the model questionnaire along with EU metadata. The results of the surveys are published as ratios (e.g. the percentage of companies with e-commerce sales that represent at least 1% of turnover).

A ratio is the result of the division of a numerator and a denominator. Numerators and denominators are aggregated variables.

For example, a ratio is computed by dividing the aggregated variable E_IUSE (enterprises where persons employed have access to the internet) by the aggregated variable ENT (total enterprises), and the result is E_IUSE with unit PC_ENT (percentage of all enterprises).

Similarly, EU ratios are computed by dividing the sum (over all Member States) of the aggregated variable E_IUSE by the sum of the aggregated variable ENT.

Given that data published in this domain are based on model questionnaires that change every year, Eurostat provides, on a dedicated webpage, a table that lists the available ratios, enterprise sizes and activities (NACE Rev. 2) per year.

5.2. Confidentiality and flags

The aggregated data received from the NSIs also contain some flags to indicate confidentiality, statistical reliability, and break in series. Then, the ratios are computed and they inherit the flags. Data flagged as unreliable are not published at national level, but are taken into account for calculating the EU aggregates. Data flagged as confidential are neither published at national level nor taken into account for EU aggregates.

5.3. Dissemination channels

ICT usage and e-commerce in enterprises data are accessible on Eurostat's website through different paths: the data navigation tree, the 'Statistics Explained' articles, the news items and the dedicated webpage.

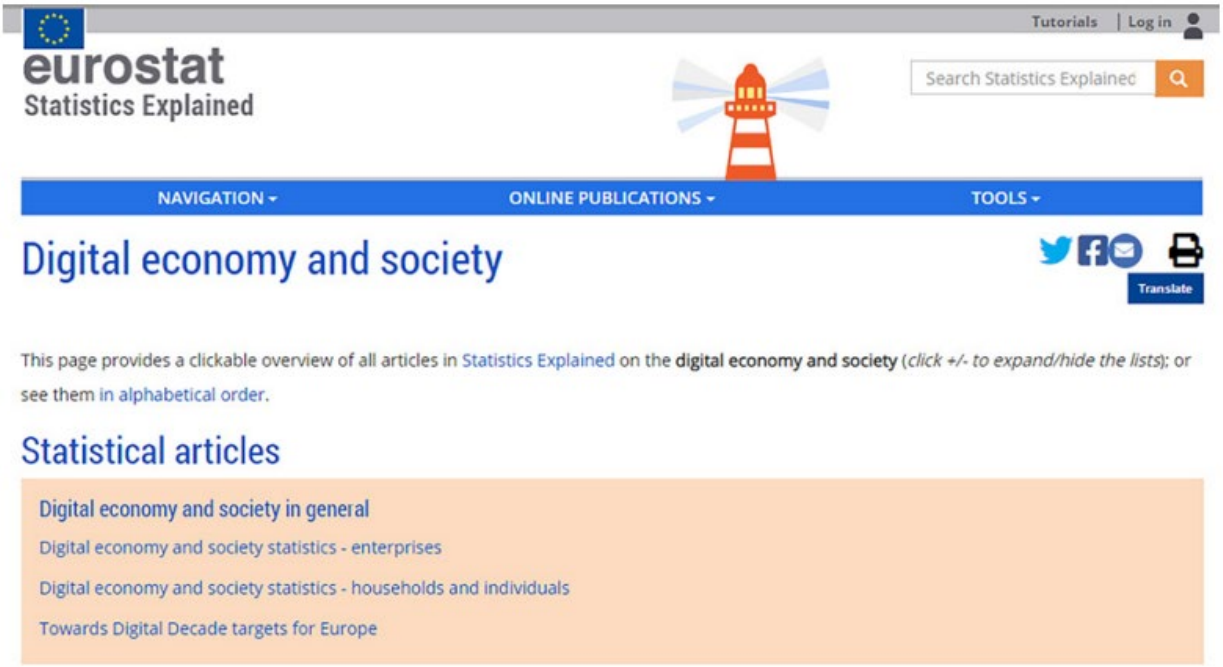
Eurostat navigation tree

ICT usage and e-commerce in enterprises data are published on Database - Eurostat (europa.eu) under the 'ICT usage in enterprises' sub-branch of the 'science, technology, digital society' heading.



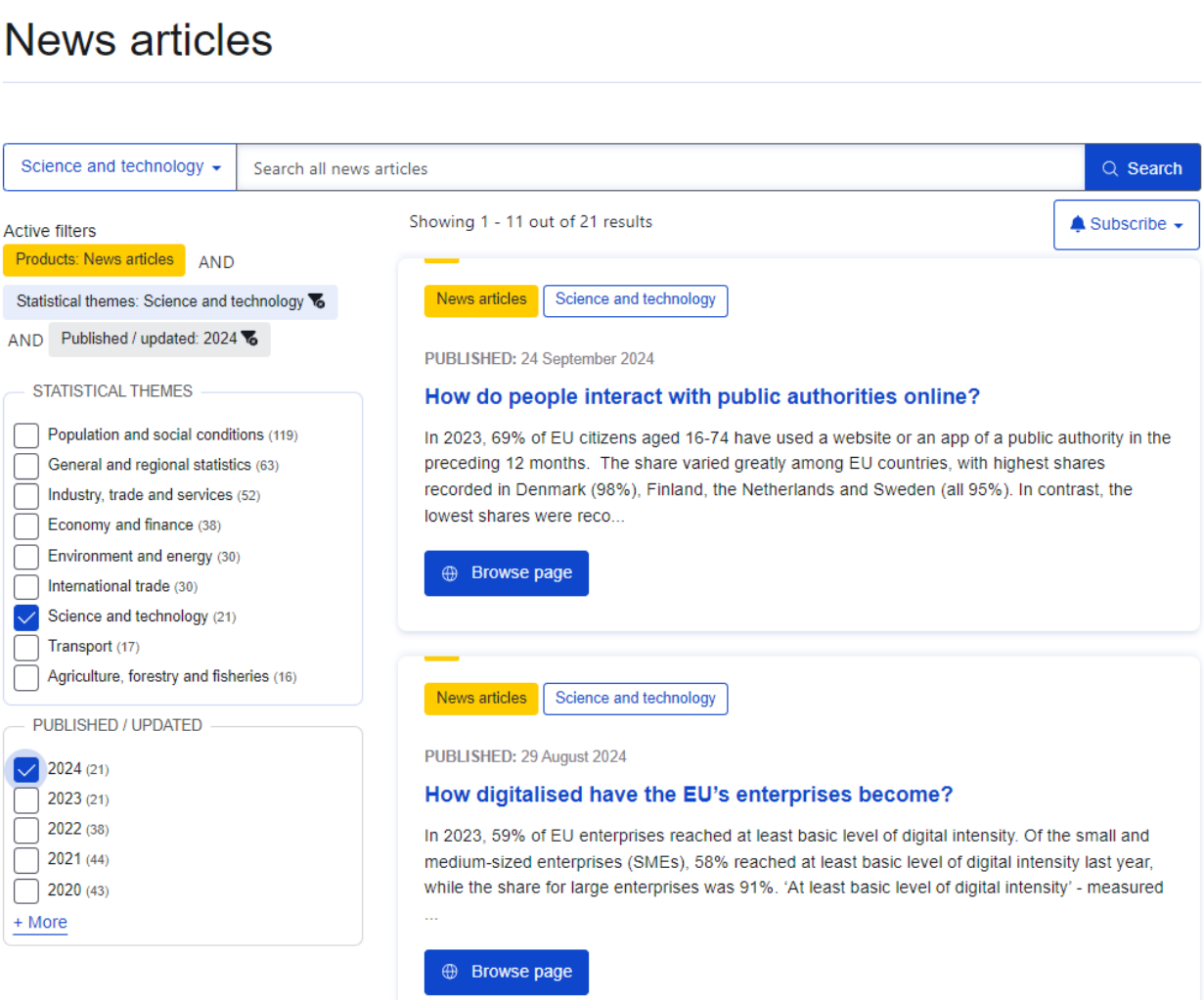
Statistics Explained

'Statistics Explained' is an official Eurostat website presenting statistical topics in an easily understandable way. The Digital economy and society page contains the links to the main statistical articles related to ICT data for enterprises, household and individuals.



News Items

News items are displayed in reverse chronological order on the News page (accessible by clicking on 'News' in the drop-down menu anywhere in the Eurostat portal and then on News articles). They provide short texts and visuals on our data, including first data releases. From the menu on the left, the theme of interest can be selected and articles on ICT in enterprises (among other things) can be viewed.



Dedicated webpage

In addition, a dedicated website on the digital economy and society provides information on the data, links to the database and a comprehensive database covering all data collected, publications, visualisations, methodological information and legislation.

The subpage on the Comprehensive database provides information on the variables and breakdowns collected over time, a description of how to use the database, model questionnaires, country-specific notes, and other related information.

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Annexes

Annex 1 – Model questionnaire

Annex 2 – Validation rules

Annex 3 – Metadata reporting template

Annex 1 – Model questionnaire

EU Survey on ICT Usage and e-commerce in Enterprises

2024

General outline of the survey

Sampling unit:	Enterprise
Scope / Target Population:	**Economic activity:** Enterprises classified in the following categories of NACE Rev. 2: - Section C – *"Manufacturing"*; - Section D, E – *"Electricity, gas, steam and air conditioning supply"* "*Water supply, sewerage, waste management and remediation activities";* - Section F – *"Construction"*; - Section G – *"Wholesale and retail trade; repair of motor vehicles and motorcycles"*; - Section H – *"Transportation and storage"*; - Section I – *"Accommodation and food service activities"*; - Section J – *"Information and communication"*; - Section L – *"Real estate activities"*; - Section M – *"Professional, scientific and technical activities"*; - Section N – *"Administrative and support service activities";* - Group 95.1 – "*Repair of computers and communication equipment*" **Enterprise size:** Enterprises with 10 or more employees and self-employed persons[41]. Optional: enterprises with number of employees and self-employed persons between 0 and 9. **Geographic scope:** Enterprises located in any part of the territory of the country.

[41] With the introduction of the Framework Regulation on European Business Statistics the variable "persons employed" is replaced by the variable "employees and self-employed persons". This change in the denomination of the variable does not imply any change in the scope. The two variables represent exactly the same concept. For the sake of user friendliness, the term "employees and self-employed persons" is only used in the introductory part of the questionnaire and in Module X, while in the rest of the questionnaire the term "persons employed" is used.

Sampling unit:	Enterprise
Reference period:	Where not specified respondents should consider as reference their current situation (survey period in 2024). Year 2023 for the value or % of sales data and where specified.
Recommended survey period:	First quarter 2024.
Questionnaire:	The layout of the national questionnaire should be defined by the country. However, countries should follow the order of the list of variables enclosed, if possible. The background information (Module X) should be placed at the end of the questionnaire. This information can be obtained in 3 different ways: from national registers, from Structural Business Statistics or collected directly with the ICT usage survey. Every effort should be made to obtain them from the most recent SBS survey. Countries can include additional questions.
Target respondent:	A decision maker with major responsibility for ICT-related issues in the enterprise (the ICT manager or a senior professional in the ICT department). In smaller enterprises, the respondent should be someone at the level of managing director or the owner. In any case the respondent should not be someone with responsibilities only in accounting.
Sample size, stratification:	The sampling design and the resulting sample size should be appropriate for obtaining accurate, reliable and representative results on the variables and items in the model questionnaire. This objective should be achieved for the overall proportions as well as for the proportions for the different breakdowns of the population defined below: NACE and size class. NACE breakdown and enterprise size class breakdown are not required to be cross-tabulated. This requirement aims at ensuring the collection of a complete dataset – without empty, confidential or unreliable cells – for these indicators – with an exception for those broken down by economic activity for the calculation of European NACE aggregates.
NACE breakdown:	**(To be applied to: all variables; enterprises with 10 or more employees and self-employed persons; whole territory of the country.)** Data should be broken down by the following NACE Rev. 2 aggregates for possible calculation of **national** NACE Rev. 2 aggregates: 1 10-33, 35-39, 41-43, 45-47, 49-53, 55-56, 58-63, 68-75, 77-82, 95.1

Sampling unit:	Enterprise

2 10-33

3 10-18

4 19-23

5 24-25

6 26-33

7 35-39

8 41-43

9 45-47

10 47

11 49-53

12 55

13 55-56

14 58-63

15 68

16 69-75

17 77-82

18 26.1-26.4, 26.8, 46.5, 58.2, 61, 62, 63.1, 95.1

Breakdowns for which national data should be provided with the purpose of possible calculation of **European** NACE aggregates.

The production and transmission of these aggregates with an accuracy that allows the release at national level is optional. The production and transmission of these aggregates with an accuracy that may not allow the release at national level (use of flag u: unreliable) but are accurate enough to be combined with other countries' aggregates to be released at European level is mandatory.)

3a 10-12

3b 13-15

3c 16-18

4a 19

4b 20

4c 21

Sampling unit:	Enterprise
	4d 22-23 6a 26 6b 27 6c 28 6d 29-30 6e 31-33 7a 35 7b 36-39 9a 45 9b 46 14a 58-60 14b 61 14c 62-63 16a 69-71 16b 72 16c 73-75 17a 77-78 + 80-82 17b 79 18a 95.1
Size class breakdown:	**(To be applied to: all variables; aggregate of all mandatory NACE aggregates [1 to 18 defined above]; whole territory of the country.)** Data should be broken down by the following size classes according to the number of employees and self-employed persons: 1 10 or more 2 10-49 (small enterprises) 3 50-249 (medium enterprises) 4 250 or more (large enterprises) **Optional**: 5 0-9 6 0-1

Sampling unit:	Enterprise
	7 2-9
Weighting of results:	Results should in general be weighted by number of enterprises. Turnover weighting should be used for sales related questions. Quantitative variables in the e-Commerce module related to sales should be weighted by total turnover. Weighting by the number of employees and self-employed persons should be applied for variables related to questions A1, A5, and for other variables e.g. % sending orders via a website or EDI-type messages., as specified in the transmission format document.
Treatment of non-response/'Do not know':	**Unit non-response**: The non-respondent units should be assumed to resemble those who have responded to the survey and be treated as non-selected units. For this, the weighting or the grossing up factors should be adjusted: the design weight Nh / nh is replaced by Nh / mh where Nh is the size of stratum h, nh is the sample size in stratum h and mh is the number of respondents in stratum h. **Item non-response**: Logical corrections should be made, when information can be deduced from other variables, and priority given to further contacts with enterprises to collect the missing information. For the categorical variables (e.g. the YES/NO questions), respondents with item non response or 'do not know' should not be imputed with values from respondents who answered the question. Numerical variables shouldn't be imputed (see also Methodological Manual).
Tabulation of results:	For the categorical variables, estimates should be made for the total number of enterprises for each response category, tabulated using the breakdowns specified above. For the quantitative variables (turnover, sales and number of employees and self-employed persons), when collected in absolute or percentage terms (and not in percentage classes), estimates should be made for the total values in absolute terms, tabulated using breakdowns as specified in the transmission format document.
Data transmission:	Results are to be sent to Eurostat following the transmission format described in a forthcoming Eurostat document.

Disclaimer: References to third-party brands, products and trademarks are for the sake of clarification and are not intended to promote the use of such products.

ICT-ENT 2024 - Model Questionnaire V1.4 – Response burden and numbering

Module	Description	Mandatory variables	Optional variables
A	**Access and use of the internet**	8	6
	Access to the internet	1	0
	Use of a fixed line connection to the internet for business purposes	3	0
	Use of a mobile connection to the internet for business purposes	0	1
	Remote access and meetings via the internet	4	0
	Other use of internet	0	5
B	**e-Commerce sales**	12	3
	Web sales of goods or services	10	3
	EDI-type sales	2	0
C	**ICT specialists and skills**	7	4
D	**ICT security**	21	1
E	**Artificial Intelligence**	14	16
	Total number of questions/responses	62	30
X	Background characteristics	3	0
	Total number of **questions/responses with background characteristics**	65	30

EU SURVEY ON ICT USAGE AND E-COMMERCE IN ENTERPRISES
2024
MODEL QUESTIONNAIRE VERSION 1.4

Module A: Access and use of the internet

(Scope: all enterprises)

A1 **How many persons employed have access to the internet for business purposes?**

(including fixed line, fixed wireless and mobile telephone network connection)

(Filter question)

(Number) _____

If you can't provide this value,

please indicate an estimate of the percentage of the total number of persons employed who have access to the internet for business purposes

␣ ␣ ␣ ␣ ,␣ **%**

If the value=0, go to C1

Use of a fixed line connection to the internet for business purposes

A2. **Does your enterprise use any type of fixed connection to the internet?** (e.g. ADSL, SDSL, VDSL, fiber optics technology (FTTP), cable technology, fixed wireless)

(Add national examples)

(Filter question)

Yes ☐

No ☐

->go to **A5** (if optional included),

else go to **A6**

A3. **What is the maximum contracted download speed of the fastest fixed internet connection of your enterprise?**

(additional categories at national level can be added, if needed)

(Tick only one)

a) less than 30 Mbit/s ☐

b) at least 30 but less than 100 Mbit/s ☐

c) at least 100 Mbit/s but less than 500 Mbit/s ☐

d) at least 500 Mbit/s but less than 1 Gbit/s ☐

e) at least 1 Gbit/s ☐

A4.	**Is the speed of your fixed connection(s) to the internet usually sufficient for the actual needs of the enterprise?**	Yes ☐ No ☐

Use of a mobile connection to the internet for business purposes

The following question refers to the usage of portable devices connecting to the internet through mobile telephone networks for business purposes. This includes the use of devices where the enterprise provides the devices and pays for the subscription and the costs of use fully or at least up to a limit.

A5.	**How many persons employed use a portable device provided by the enterprise, that allows connection to the internet via mobile telephone networks, for business purposes?** (e.g. portable computers or other portable devices such as smartphones) If you can't provide this value, **please indicate an estimate of the percentage of the total number of persons employed who use a portable device provided by the enterprise, that allows connection to the internet via mobile telephone networks, for business purposes** (e.g. portable computers, or other portable devices such as smartphones) ***– Optional***	*(Number)* _____ ␣ ␣ ␣ %

Remote access and meetings via the internet
Scope: enterprises with access to the internet, i.e. if A1>0)

A6.	**Do any of the persons employed have remote access to the following?** (via computers or portable devices such as smartphones)	**Yes**	**No**
	a) E-mail system of the enterprise	☐	☐
	b) Documents of the enterprise (e.g. files, spreadsheets, presentations, charts, photos)	☐	☐
	c) Business applications or software of the enterprise (e.g. access to accounting, sales, orders, CRM) *Please exclude applications used for internal communication, e.g. Skype, MS Teams, Yammer.*	☐	☐

A7.	**Does your enterprise conduct remote meetings (via e.g. Skype, Zoom, MS Teams, WebEx)?**	**Yes** ☐	**No** ☐ ->go to **A8** (if optional included) else go to **B1**

	Other use of the internet **– Optional**		
A8.	**Does your enterprise pay to advertise on the internet?** (e.g. adverts on search engines, on social media, on other web-sites or apps) (Filter question)	**Yes** ☐	**No** ☐ ->go to **B1**

A9.	**Does your enterprise pay to advertise on the internet using any of the following targeted advertising methods?**	**Yes**	**No**
	a) Based on content or keywords searched by internet users	☐	☐
	b) Based on the tracking of internet users' past activities or profile	☐	☐
	c) Based on the geolocation of internet users	☐	☐
	d) Any other method of targeted advertising on the internet not specified above	☐	☐

Module B: e-Commerce sales

(Scope: enterprises with access to the internet, i.e. if A1>0)

In e-commerce sales of goods or services, the order is placed via web sites, apps or EDI-type messages (EDI: Electronic Data interchange) by methods specifically designed for the purpose of receiving orders.

The payment may be done online or offline.

e-Commerce **does not include** orders written in e-mail.

Please report **web and EDI-type sales separately**. They are defined by the method of placing the order**:**

- WEB sales: the **customer** places the order on a website or through an app;
- EDI type sales: **an EDI-type order message is created from the business system of the customer.**

Web sales of goods or services

Web sales cover orders, bookings and reservations placed by your customers via

- your enterprise's **websites or apps:**
  - online store (web shop);
  - web forms;
  - extranet (web shop or web forms) ;
  - booking/reservation applications for services;
  - apps for mobile devices or computers;
- **e-commerce marketplace websites or apps** (used by several enterprises for trading goods or services).

Orders written in e-mail are **not** counted as web sales.

B1.	During 2023, did your enterprise have web sales of goods or services via:	Yes	No
	a) your enterprise's websites or apps? (including extranets)	☐	☐
	b) e-commerce marketplace websites or apps used by several enterprises for trading goods or services? **(**e.g. e-Bookers, Booking, hotels.com, eBay, Amazon, Amazon Business, Alibaba, Rakuten, TimoCom) *[Please add national examples of e-commerce marketplaces incl. government marketplaces]*	☐	☐

If both B1 a) and B1 b) = "No" then go to B7

B2. **What was the value of your web sales?**

(please refer to the provided definition of web sales)

Please answer to a) OR b)

a) What was the value of your web sales of goods or services, in 2023?	(National currency, excluding VAT) _____
OR **b) What percentage of total turnover was generated by web sales of goods or services, in 2023?** *If you cannot provide the exact percentage an approximation will suffice.*	⊔ ⊔ ⊔ , ⊔ %

	Question B3 should be answered only if both B1 a) and B1 b) = "Yes"	
B3.	**What was the percentage breakdown of the value of web sales in 2023 for the following:** (Please refer to value of web sales you reported in B2) *If you cannot provide the exact percentages an approximation will suffice.*	
	a) via your enterprise's websites or apps? (including extranets)	⊔ ⊔ ⊔ %
	b) via e-commerce marketplace websites or apps used by several enterprises for trading goods or services? **(**e.g. e-Bookers, Booking, hotels.com, eBay, Amazon, Amazon Business, Alibaba, Rakuten, TimoCom) *[Please add national examples of e-commerce marketplaces incl. government marketplaces]*	⊔ ⊔ ⊔ %
	TOTAL	**1 0 0** %

B4.	**What was the percentage breakdown of the value of web sales in 2023 by type of customer:** (Please refer to value of web sales you reported in B2) *If you cannot provide the exact percentages an approximation will suffice.*	
	a) Sales to private consumers (**B2C**)	⊔ ⊔ ⊔ %
	b) Sales to other enterprises (**B2B**) and Sales to public sector (**B2G**)	⊔ ⊔ ⊔ %
	TOTAL	**1 0 0** %

B5.	**During 2023, did your enterprise have web sales to customers located in the following geographic areas?**		
		Yes	**No**
	a) Own country	☐	☐
	b) Other EU countries	☐	☐
	c) Rest of the world	☐	☐

	The following question (B6) should only be answered if at least two of the above possible responses in question B5 a) , b) or c) are answered with "Yes", otherwise go to B7

B6.	**What was the percentage breakdown of the value of web sales in 2023 to customers located in the following geographic areas?** ***– Optional*** (Please refer to value of web sales you reported in **B2**) *If you cannot provide the exact percentages an approximation will suffice.*	
	a) Own country	⊔ ⊔ ⊔ %
	b) Other EU countries	⊔ ⊔ ⊔ %
	c) Rest of the world	⊔ ⊔ ⊔ %
	TOTAL	**1 0 0** %

EDI-type sales

EDI-type sales cover **orders placed** by customers via EDI-type messages (EDI: Electronic Data interchange) meaning:

- in an **agreed or standard format** suitable for automated processing;
- EDI-type order message created from the **business system** of the customer;
- including orders transmitted via EDI-service provider;
- including automatic system generated **demand driven orders**;
- including orders received directly into your **ERP** system.

Examples of EDI: EDIFACT, XML/EDI (e.g. UBL, Rosettanet, *[please add national examples]*).

B7.	**During 2023, did your enterprise have EDI-type sales of goods or services?** (Filter question)	**Yes** ☐	**No** ☐ -> go to **C1**

B8.	**What was the value of your EDI-type sales?** (please refer to the provided definition of EDI-type sales) Please answer to a) OR b)	
	a) What was the value of your EDI-type sales of goods or services, in 2023?	(National currency, excluding VAT)

		_____
	OR **b) What percentage of total turnover was generated by EDI-type sales of goods or services, in 2023?** *If you cannot provide the exact percentage an approximation will suffice.*	␣ ␣ ␣ , ␣ %

	Module C: ICT specialists and skills **(Scope: all enterprises)**		
C1.	**Does your enterprise employ ICT specialists?** ICT specialists are persons employed for whom **ICT is the main job.** For example, to develop, operate or maintain ICT systems or applications.	**Yes** ☐	**No** ☐
C2.	**Did your enterprise provide any type of training to develop ICT related skills of the persons employed, during 2023?**	**Yes**	**No**
	a) Training for ICT specialists *Tick "No" if your enterprise didn't employ ICT specialists during 2023.*	☐	☐
	b) Training for other persons employed	☐	☐
C3.	**Did your enterprise recruit or try to recruit ICT specialists during 2023?** (Filter question)	**Yes** ☐	**No** ☐ ->go to **C6**
C4.	**During 2023, did your enterprise have vacancies for ICT specialists that were difficult to fill?**	**Yes** ☐	**No** ☐ ->go to **C6**
C5.	**Did your enterprise have any of the following difficulties to recruit ICT specialists during 2023? – *Optional***	**Yes**	**No**
	a) Lack of applications	☐	☐
	b) Applicants' lack of relevant ICT related qualifications from education and/or training;	☐	☐
	c) Applicants' lack of relevant work experience	☐	☐
	d) Applicants' salary expectations too high	☐	☐
C6.	**Who performed your enterprise's ICT functions in 2023** (e.g. maintenance of ICT infrastructure, support for office software, development or support of business management software/systems and/or web solutions, security and data protection)**?**	**Yes**	**No**

	a) own employees (incl. those employed in parent or affiliate enterprises)	☐	☐
	b) external suppliers	☐	☐

Module D: ICT Security

(Scope: enterprises with access to the internet, i.e. if A1>0)

ICT security means measures, controls and procedures applied on enterprise's ICT systems to ensure integrity, authenticity, availability and confidentiality of enterprise's data and systems.

D1.	**Does your enterprise apply any of the following ICT security measures on its ICT systems?**	**Yes**	**No**
	a) authentication via **strong password** (e.g. minimum length, use of numbers and special characters, changed periodically)	☐	☐
	b) authentication via **biometric methods** used to access the enterprise's ICT system (e.g. authentication based on fingerprints, voice, face)	☐	☐
	c) authentication **based on** a combination of **at least two authentication mechanisms** (i.e. combination of e.g. user-defined password, one-time password (OTP), code generated via a security token or received via a smartphone, biometric method (e.g. based on fingerprints, voice, face))	☐	☐
	d) Encryption of data, documents or e-mails	☐	☐
	e) Data backup to a separate location (including backup to the cloud)	☐	☐
	f) Network access control (management of user rights in enterprise's network)	☐	☐
	g) VPN (Virtual Private Network extends a private network across a public network to enable secure exchange of data over public network)	☐	☐
	h) ICT security monitoring system used to detect suspicious activity (e.g. intrusion detection or prevention systems that monitors users' or devices' behaviour, network traffic) * *Please exclude antivirus software and default firewall solution included in the operating system of personal computers and routers.*	☐	☐
	i) Maintaining log files that enable analysis after ICT security incidents	☐	☐
	j) ICT risk assessment, i.e. periodical assessment of probability and consequences of ICT security incidents	☐	☐
	k) ICT security tests (e.g. performing penetration tests, testing security alert system, review of security measures, testing of backup systems)	☐	☐

D2.	**Does your enterprise make persons employed aware of their obligations in ICT security related issues in the following ways?**	**Yes**	**No**

	a) Voluntary training or internally available information (e.g. information on the intranet)	☐	☐
	b) Compulsory training courses or viewing compulsory material	☐	☐
	c) By contract (e.g. contract of employment)	☐	☐

D3.	**Does your enterprise have document(s) on measures, practices or procedures on ICT security?** (Filter question) (Documents on ICT security and confidentiality of data cover employee training in ICT use, ICT security measures, the evaluation of ICT security measures, plans for updating ICT security documents, etc.)	**Yes**☐	**No**☐ ->go to **D5**

D4.	**When were your enterprise's document(s) on measures, practices or procedures on ICT security, defined or most recently reviewed?** (Documents on ICT security and confidentiality of data cover employee training in ICT use, ICT security measures, the evaluation of ICT security measures, plans for updating ICT security documents, etc.) *(Tick only one)* ***– Optional***	
	*[42]a) within the last 12 months	☐
	b) more than 12 months and up to 24 months ago	☐
	c) more than 24 months ago	☐

D5.	**During 2023, did your enterprise experience any ICT related security incident leading to the following consequences?**	**Yes**	**No**
	a) Unavailability of ICT services due to hardware or software failures	☐	☐
	b) Unavailability of ICT services due to attack from outside, e.g. ransomware attacks, Denial of Service attacks	☐	☐
	c) Destruction or corruption of data due to hardware or software failures	☐	☐
	d) Destruction or corruption of data due to infection of malicious software or unauthorised intrusion	☐	☐
	e) Disclosure of confidential data due to intrusion, pharming, phishing attack, intentional actions by own employees	☐	☐
	f) Disclosure of confidential data due to unintentional actions by own employees	☐	☐

[42] Formulation in the national questionnaires should correspond to: a) <= 12 months; b) > 12 months and <= 24 months; and c) more than 24 months

Module E: Artificial Intelligence

(Scope: enterprises with access to the internet, i.e. if A1>0)

Artificial intelligence refers to systems that use technologies such as: **text mining, computer vision, speech recognition, natural language generation, machine learning, deep learning** to gather and/or use data to predict, recommend or decide, with varying levels of autonomy, the best action to achieve specific goals.

Artificial intelligence systems **can be purely software based**, e.g.:

- chatbots and business virtual assistants based on natural language processing;
- face recognition systems based on computer vision or speech recognition systems;
- machine translation software;
- data analysis based on machine learning;

or **embedded in devices**, e.g.:

- autonomous robots for warehouse automation or production assembly works;
- autonomous drones for production surveillance or parcel handling.

E1.	**Does your enterprise use any of the following Artificial Intelligence (AI) technologies?**	**Yes**	**No**
	a) AI technologies performing analysis of written language (e.g. text mining)	☐	☐
	b) AI Technologies converting spoken language into machine-readable format (speech recognition)	☐	☐
	c) AI Technologies generating written or spoken language (natural language generation, speech synthesis)	☐	☐
	d) AI Technologies identifying objects or persons based on images or videos (image recognition, image processing)	☐	☐
	e) Machine learning (e.g. deep learning) for data analysis	☐	☐
	f) AI Technologies automating different workflows or assisting in decision making (e.g. **AI based** software robotic process automation)	☐	☐
	g) AI Technologies enabling physical movement of machines via autonomous decisions based on observation of surroundings (autonomous robots, self-driving vehicles, autonomous drones)	☐	☐

E1 a) to g) = "No" then go to E6 (if optional included) else go to X1

E2.	**Does your enterprise use Artificial Intelligence software or systems for any of the following purposes?**	**Yes**	**No**
	a) **Use of AI for marketing or sales** some of the examples may be: • customer profiling, price optimisation, personalised marketing offers, market analysis based on machine learning • chatbots based on natural language processing for customer support • autonomous robots for orders processing	☐	☐

	b) **Use of AI for production or service processes** some of the examples may be: • predictive maintenance or process optimization based on machine learning • tools to classify products or find defects in products based on computer vision • autonomous drones for production surveillance, security or inspection tasks • assembly works performed by autonomous robots	☐	☐
	c) Use of **AI for organisation of business administration processes or management** some of the examples may be: • business virtual assistants based on machine learning and/or natural language processing, e.g. for document drafting • data analysis or strategic decision making based on machine learning, e.g. risk assessment based on machine learning • planning or business forecasting based on machine learning • human resources management based on machine learning or natural language processing, e.g. candidates pre-selection screening, employee profiling or performance analysis	☐	☐
	d) Use of **AI for logistics** some of the examples may be: • autonomous robots for pick-and-pack solutions in warehouses for parcel shipping, tracing, distribution or sorting • route optimization based on machine learning	☐	☐
	e) Use of **AI for ICT security** some of the examples may be: • face recognition based on computer vision for authentication of ICT users • detection and prevention of cyber-attacks based on machine learning	☐	☐
	f) Use of **AI for accounting, controlling or finance management** some of the examples may be: • machine learning to analyse data that helps to make financial decisions • invoice processing based on machine learning • machine learning or natural language processing used for bookkeeping tasks	☐	☐
	g) Use of **AI for research and development (R&D) or innovation activity** (excluding research on AI) some of the examples may be: • analysis of data for conducting research, solving research problems, developing a new or significantly improved product/service based on machine learning	☐	☐

E3.	**How did you enterprise acquire the Artificial Intelligence software or systems that it uses?** ***– Optional***	**Yes**	**No**
	a) They were developed by own employees (including those employed in parent or affiliate enterprise)	☐	☐
	b) Commercial software or systems were modified by own employees (including those employed in parent or affiliate enterprise)	☐	☐
	c) Open-source software or systems were modified by own employees (including those employed in parent or affiliate enterprise)	☐	☐
	d) Commercial software or systems ready to use were purchased (including examples where it was already incorporated in a purchased item or system)	☐	☐
	e) External providers were contracted to develop or modify them	☐	☐

E4.	**Does your enterprise process data on individuals using AI technologies?** Some of examples of data on individuals (e.g. employees, job applicants or customers) are: sex, age, racial or ethnic origin, disability, religion or belief, sexual orientation, facial images, record of purchases, occupation or address. ***– Optional***	**Yes** ☐	**No** ☐ -> go to **E6** (if optional included) else go to **X1**

E5.	**Does your enterprise have any measures to check the results generated by AI technologies for possible biases towards individuals based on sex, age, racial or ethnic origin, disability, religion or belief, sexual orientation?** Some of examples of measures to check for possible bias: • analysing the results of various machine learning models • examining the dataset that was used to train the machine learning model • data augmentation (which involves techniques to artificially generate additional data points from existing data, i.e. synthetic data) ***– Optional***	**Yes** ☐	**No** ☐

Questions E6 is presented only to respondents who answered 'No' to E1a)-g) i.e. enterprises that did not use any of the Artificial Intelligence technologies listed in question E1.

E6.	**Has your enterprise ever considered using any of the Artificial Intelligence technologies listed in question E1? *– Optional*** *(Filter question)*	**Yes** ☐	**No** ☐ -> go to **X1**
E7.	**What are the reasons for your enterprise not to use any of the Artificial Intelligence technologies listed in question E1? *– Optional***	**Yes**	**No**
	a) The costs seem too high	☐	☐
	b) There is a lack of relevant expertise in the enterprise	☐	☐

	c) Incompatibility with existing equipment, software or systems	☐	☐
	d) Difficulties with availability or quality of the necessary data	☐	☐
	e) Concerns regarding violation of data protection and privacy	☐	☐
	f) Lack of clarity about the legal consequences (e.g. liability in case of damage caused by the use of Artificial Intelligence)	☐	☐
	g) Ethical considerations	☐	☐
	h) Artificial Intelligence technologies are not useful for the enterprise	☐	☐

	Module X: Background information **(X1-X3) available in some countries from SBS, the business register or administrative data and thus not to be included; latest available information should be provided**	
X1.	**Main economic activity of the enterprise, during 2023**	…
X2.	**Average number of employees and self-employed persons (persons employed), during 2023**	…
X3.	**Total turnover (in monetary terms, excluding VAT), for 2023**	…

EU Survey on ICT Usage and e-Commerce in Enterprises

Glossary

App(s)	A mobile app, short for mobile application or just app, is application software designed for a specific purpose (e.g. entertainment, shopping), downloaded and used on computers depending on their operating system. (e.g. portable devices such as tablets, Smartphones) Further information: http://en.wikipedia.org/wiki/Mobile_app; http://www.techopedia.com/definition/2953/mobile-application-mobile-app
Authentication methods	Authentication is a way to ascertain that a user is who they claim to be. This is usually performed by presenting one or more challenges to the user. There are three broad categories of challenges: 1) Something the user knows. The user is asked for a secret, known only to her. Typical examples are passwords and PINs, but can also take the form of security questions. 2) Something the user has. The user is in possession of a unique token, like a key. In the case of computer tokens, this can take the form of an NFC tag, or a device. 3) Something the user is. Aka biometrics. The user is asked to present a part of her body that forms unique and repeatable patterns, like fingerprints, voice, or face recognition. Source:https://www.enisa.europa.eu/topics/csirts-in-europe/glossary/authentication-methods
Biometric authentication	Biometric authentication is a security process that relies on the unique biological characteristics of an individual to verify that he is who is says he is. Biometric authentication systems compare a biometric data capture to stored, confirmed authentic data in a database. If both samples of the biometric data match, authentication is confirmed. Typically, biometric authentication is used to manage access to physical and digital resources such as buildings, rooms and computing devices. Types of biometric authentication technologies: Retina scans produce an image of the blood vessel pattern in the light-sensitive surface lining the individual's inner eye. Iris recognition is used to identify individuals based on unique patterns within the ring-shaped region surrounding the pupil of the eye. Fingerscanning, the digital version of the ink-and-paper fingerprinting process, works with details in the pattern of raised areas and branches in a human finger image. Finger vein ID is based on the unique vascular pattern in an individual's finger.

Facial recognition systems work with numeric codes called faceprints, which identify 80 nodal points on a human face.

Voice identification systems rely on characteristics created by the shape of the speaker's mouth and throat, rather than more variable conditions.

Source: https://searchsecurity.techtarget.com/definition/biometric-authentication

Business process

A business process or business method is a collection of related, structured activities or tasks that produce a specific service or product (serve a particular goal) for a particular customer or customers. Business processes can be of three types: *Management processes* (e.g. corporate governance, strategic management), *Operational processes* (e.g. purchasing, manufacturing, marketing and sale) and *Supporting processes* (e.g. accounting, recruitment, technical support).

Source: http://en.wikipedia.org/wiki/Business_process

Chatbots or Virtual agent

A chatbot or virtual agent is a computer generated, animated, artificial intelligence virtual character that serves as an online customer service representative.

CRM

Customer Relationship Management (CRM) is a management methodology which places the customer at the centre of the business activity, based in an intensive use of information technologies to collect, integrate, process and analyse information related to the customers.

One can distinguish between:

1. Operational CRM – Integration of the front office business processes that are in contact with the customer.

2. Analytical CRM – Analysis, through data mining, of the information available in the enterprise on its customers. This aims to gather in depth knowledge of the customer and how to answer to its needs.

Denial of service attack

A denial-of-service attack (DoS attack) or distributed denial-of-service attack (DDoS attack) is an attempt to make a computer resource unavailable to its intended users. Although the means to carry out, motives for, and targets of a DoS attack may vary, it generally consists of the concerted efforts of a person or persons to prevent an internet site or service from functioning efficiently or at all, temporarily or indefinitely. Perpetrators of DoS attacks typically target sites or services hosted on high-profile web servers such as banks, credit card payment gateways, and even root name servers.

One common method of attack involves saturating the target (victim) machine with external communications requests, such that it cannot respond to legitimate traffic, or responds so slowly as to be rendered effectively unavailable. In general terms, DoS attacks are implemented by either forcing the targeted computer(s) to reset, or consuming its resources so that it can no longer provide its intended service or obstructing the communication

media between the intended users and the victim so that they can no longer communicate adequately.

DSL

Digital Subscriber Line (DSL) is a family of technologies that provides digital data transmission over the wires of a local telephone network. DSL is widely understood to mean Asymmetric Digital Subscriber Line (ADSL), the most commonly installed technical varieties of DSL. DSL service is delivered simultaneously with regular telephone on the same telephone line as it uses a higher frequency band that is separated by filtering.

Source: http://en.wikipedia.org/wiki/DSL

EDI, EDI-type

Electronic Data Interchange (EDI) refers to the structured transmission of data or documents between organizations or enterprises by electronic means. It also refers specifically to a family of standards (EDI-type) and EDI-type messages suitable for automated processing.

Source: http://en.wikipedia.org/wiki/Electronic_Data_Interchange

EDI e-commerce

Orders initiated with EDI-type messages. EDI (electronic data interchange) is an e-business tool for exchanging different kinds of business messages. EDI is here used as a generic term for sending or receiving business information in an agreed format suitable for automated processing (e.g. EDIFACT, XML) and without the individual message being manually typed. “EDI e-commerce” is limited to EDI messages placing an order.

Source: OECD, DSTI/ICCP/IIS(2009)5/FINAL

Electronic commerce

(e-Commerce)

An e-commerce transaction is the sale or purchase of goods or services, conducted over computer networks by methods specifically designed for the purpose of receiving or placing of orders. The goods or services are ordered by those methods, but the payment and the ultimate delivery of the goods or services do not have to be conducted online. An e-commerce transaction can be between enterprises, households, individuals, governments, and other public or private organisations. e-Commerce comprises orders made in Web pages or apps, extranet or EDI and excludes orders made by telephone calls, facsimile, or manually typed e-mail. The type is defined by the method of making the order.

Source: OECD, DSTI/ICCP/IIS(2009)5/FINAL

E-mail

Electronic transmission of messages, including text and attachments, from one computer to another located within or outside of the organisation. This includes electronic mail by internet or other computer networks.

ERP

Enterprise Resource Planning (ERP) consists of one or of a set of software applications that integrate information and processes across the several business functions of the enterprise. Typically ERP integrates planning, procurement, sales, marketing, customer relationship, finance and human resources.

ERP software can be customised or package software. These latter are single-vendor, enterprise wide, software packages, but they are built in a modular way allowing enterprises to customise the system to their specific activity implementing only some of those modules.

ERP systems typically have the following characteristics:

1. are designed for client server environment (traditional or web-based);
2. integrate the majority of a business's processes;
3. process a large majority of an organization's transactions;
4. use enterprise-wide database that stores each piece of data only once;
5. allow access to the data in real time.

Extranet

A closed network that uses internet protocols to securely share enterprise's information with suppliers, vendors, customers or other businesses partners. It can take the form of a secure extension of an Intranet that allows external users to access some parts of the enterprise's Intranet. It can also be a private part of the enterprise's website, where business partners can navigate after being authenticated in a login page.

GPS

The Global Positioning System (GPS) is a satellite-based radionavigation system. Is one of the global network of satellites that enable satellite navigation through GPS signals. GPS receiver anywhere on or near the Earth where there is an unobstructed line of sight to four or more GPS satellites can determine location, time, and velocity using this information.

Source: https://en.wikipedia.org/wiki/Global_Positioning_System, https://en.wikipedia.org/wiki/GPS_signals

Internet

The internet is a global system of interconnected computer networks that use the standard internet Protocol Suite (TCP/IP) to serve billions of users worldwide. It is a network of networks that consists of millions of private, public, academic, business, and government networks of local to global scope that are linked by a broad array of electronic and optical networking technologies. The internet carries a vast array of information resources and services, most notably the inter-linked hypertext documents of the World Wide Web (WWW) and the infrastructure to support electronic mail.

Source: http://en.wikipedia.org/wiki/internet

Relates to internet Protocol based networks: www, Extranet over the internet, EDI over the internet, internet-enabled mobile phones.

Intrusion

An intrusion is an attempt to bypass security controls on a information system. Means of intrusion can be eavesdropping, viruses, worms, trojan horses, logic or time bombs, brute force attacks, etc.

Intrusion detection is a process with the purpose of detecting intrusions or attempts of intrusions into a computer or network to compromise the confidentiality, integrity or availability by observation of system, application and

	user activity as well as network traffic. Intrusion detection systems take preventive actions against intrusions without direct human intervention.
Malicious software	Malicious software, also known as "malware" is any piece of software that performs undesirable operations such as data theft or some other type of computer compromise. Source:https://www.enisa.europa.eu/topics/csirts-in-europe/glossary/malware
Marketplace(s) **(e-Commerce marketplaces)**	The term "e-commerce marketplaces" refers to websites or apps used by several enterprises for trading products e.g. Booking, eBay, Amazon, Amazon Business, Alibaba, Rakuten). e-Commerce marketplaces are different from e-commerce platforms. The latter provide scalable, self-made online solutions for business that would like to set up their own e-commerce website.
Machine learning (incl. deep learning)	Machine learning (e.g. deep learning) involves 'training' a computer model to better perform an automated task, e.g. pattern recognition.
Natural language generation (NLG)	Natural language generation is the ability for a computer program to convert data into natural language representation.
Natural language processing (NLP)	Natural language processing is the ability for a computer program to understand human language as it is spoken.
Online payment	An online payment is an integrated ordering-payment transaction
Pharming	The term "pharming" connotes an attack to redirect the traffic of a website to another, bogus website in order to acquire sensitive information.
Phishing	Phishing is a criminally fraudulent attempt to acquire sensitive information such as usernames, passwords and credit card details by masquerading as a trustworthy entity in an electronic communication.
Ransomware	Ransomware is a type of malware (like Viruses, Trojans, etc.) that infect the computer systems of users and manipulates the infected system in a way, that the victim can not (partially or fully) use it and the data stored on it. The victim usually shortly after receives a blackmail note by pop-up, pressing the victim to pay a ransom (hence the name) to regain full access to system and files. Source:https://www.enisa.europa.eu/topics/csirts-in-europe/glossary/ransomware
Robotic process automation (Artificial Intelligence based)	Artificial Intelligence based robotic process automation refers to software that automates business processes (e.g. workflows automation) based on Artificial Intelligence technologies.

Sales via website (web sales)

Web sales are sales made via an online store (web shop), via web forms on a website or extranet, or apps. Web sales are distinguished from EDI sales. In particular, the type of e-commerce transaction is defined by the method of making the order. This approach should mitigate the interpretation problems where both types, EDI and Web, are used in the process. An example is a situation where an order is made by the customer through a web application but the information is transmitted to the seller as an EDI-type message. Here the type of selling application is however web; EDI is only a business application to transmit information about the sale. Web sales can be done by mobile phones using an internet browser.

Source: OECD, DSTI/ICCP/IIS(2009)5/FINAL

Speech recognition

Speech recognition is the ability of a machine or program to identify words and phrases in spoken language and convert them to a machine-readable format.

VPN

A virtual private network (VPN) extends a private network across a public network, and enables users to send and receive data across shared or public networks as if their computing devices were directly connected to the private network. Applications running on a computing device, e.g., a laptop, desktop, smartphone, across a VPN may therefore benefit from the functionality, security, and management of the private network. Encryption is a common, though not an inherent, part of a VPN connection.

Source: https://en.wikipedia.org/wiki/Virtual_private_network

Web e-commerce

Web (e-commerce) sales are sales made via an online store (web shop), via web forms on a website or extranet, or apps regardless of how the web is accessed (computer, laptop, mobile phone etc.)

Source: OECD, DSTI/ICCP/IIS(2009)5/FINAL

Website

Location on the World Wide Web identified by a Web address. Collection of Web files on a particular subject that includes a beginning file called a home page. Information is encoded with specific languages (Hypertext mark-up language (HTML), XML, Java) readable with a Web browser, like Netscape's Navigator or Microsoft's internet Explorer.

Wi-Fi

Wi-Fi (or Wi-fi, WiFi, Wifi, wifi), short for ‘Wireless Fidelity’, is a set of ethernet standards for wireless local area networks (WLAN) currently based on the IEEE 802.11 specifications. New standards beyond the 802.11 specifications, such as 802.16 have been developed. Wi-Fi was intended to be used for wireless devices and LANs, but is now often used for internet access (one of the main international standards for wireless broadband internet access and networking, with widespread use in business, homes and public spaces). It is based on radio signals with a frequency of 2.4 GHz and theoretically capable of speeds of over 54 Mbit/s. It enables a person with a wireless-enabled computer or personal digital assistant to connect to the internet when close to an access point called a hotspot.

xDSL	Digital Subscriber Line. DSL technologies are designed to increase bandwidth available over standard copper telephone wires. Includes IDSL, HDSL, SDSL, ADSL, RADSL, VDSL, DSL-Lite.
XML	The Extensible Markup Language is a markup language for documents containing structured information. Structured information contains both content (words, pictures, etc.) and some indication of what role that content plays (for example, content in a section heading has a different meaning from content in a footnote, which means something different than content in a figure caption or content in a database table, etc.). Almost all documents have some structure. A markup language is a mechanism to identify structures in a document. The XML specification defines a standard way to add markup to documents. Source: http://www.xml.com/

Annex 2 – Validation rules

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	ACTIVITY	byNaceEU	C10T12 <= C10T18
Sdmxe	*	ACTIVITY	byNaceEU	C10T12 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	C10T12 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	C13T15 <= C10T18
Sdmxe	*	ACTIVITY	byNaceEU	C13T15 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	C13T15 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	C16T18 <= C10T18
Sdmxe	*	ACTIVITY	byNaceEU	C16T18 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	C16T18 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	C19 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	C19 <= C19T23
Sdmxe	*	ACTIVITY	byNaceEU	C19 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	C19T23 = C19 + C20 + C21 + C22_23
Sdmxe	*	ACTIVITY	byNaceEU	C20 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	C20 <= C19T23
Sdmxe	*	ACTIVITY	byNaceEU	C20 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	C21 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	C21 <= C19T23
Sdmxe	*	ACTIVITY	byNaceEU	C21 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	C22_23 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	C22_23 <= C19T23
Sdmxe	*	ACTIVITY	byNaceEU	C22_23 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	C26 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	C26 <= C26T33
Sdmxe	*	ACTIVITY	byNaceEU	C26 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	C27 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	C27 <= C27_28

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	ACTIVITY	byNaceEU	C27 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	C27_28 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	C27_28 <= C26T33
Sdmxe	*	ACTIVITY	byNaceEU	C27_28 = C27 + C28
Sdmxe	*	ACTIVITY	byNaceEU	C27_28 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	C28 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	C28 <= C27_28
Sdmxe	*	ACTIVITY	byNaceEU	C28 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	C29_30 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	C29_30 <= C26T33
Sdmxe	*	ACTIVITY	byNaceEU	C29_30 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	C31T33 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	C31T33 <= C26T33
Sdmxe	*	ACTIVITY	byNaceEU	C31T33 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	D35 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	D35 <= D35TE39
Sdmxe	*	ACTIVITY	byNaceEU	D35 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	D35TE39 = D35 + E36T39
Sdmxe	*	ACTIVITY	byNaceEU	E36T39 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	E36T39 <= D35TE39
Sdmxe	*	ACTIVITY	byNaceEU	E36T39 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	G45 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	G45 <= G45T47
Sdmxe	*	ACTIVITY	byNaceEU	G45 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	G46 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	G46 <= G45T47
Sdmxe	*	ACTIVITY	byNaceEU	G46 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	G47 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	G47 <= G45T47

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	ACTIVITY	byNaceEU	G47 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	J58T60 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	J58T60 <= J58T63
Sdmxe	*	ACTIVITY	byNaceEU	J58T60 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	J61 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	J61 <= J58T63
Sdmxe	*	ACTIVITY	byNaceEU	J61 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	J62_63 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	J62_63 <= J58T63
Sdmxe	*	ACTIVITY	byNaceEU	J62_63 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	M69T71 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	M69T71 <= M69T75
Sdmxe	*	ACTIVITY	byNaceEU	M69T71 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	M69T75 = M69T71 + M72 + M73T75
Sdmxe	*	ACTIVITY	byNaceEU	M72 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	M72 <= M69T75
Sdmxe	*	ACTIVITY	byNaceEU	M72 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	M73T75 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	M73T75 <= M69T75
Sdmxe	*	ACTIVITY	byNaceEU	M73T75 >= 0
Sdmxe	*	ACTIVITY	byNaceEU	N77T82X79 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceEU	N77T82X79 <= N77T82
Sdmxe	*	ACTIVITY	byNaceEU	N77T82X79 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	C10T18 <= C10T33
Sdmxe	*	ACTIVITY	byNaceNat	C10T18 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	C10T18 = C10T12 + C13T15 + C16T18
Sdmxe	*	ACTIVITY	byNaceNat	C10T18 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	C10T33 <= C10TF43

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	ACTIVITY	byNaceNat	C10T33 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	C10T33 = C10T18 + C19T23 + C24_25 + C26T33
Sdmxe	*	ACTIVITY	byNaceNat	C10T33 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	C10TE39 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	C10TE39 = C10T33 + D35TE39
Sdmxe	*	ACTIVITY	byNaceNat	C10TE39 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	C10TF43 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	C10TF43 = C10T33 + D35TE39 + F41T43
Sdmxe	*	ACTIVITY	byNaceNat	C10TF43 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	C10TS951XK >= 0
Sdmxe	*	ACTIVITY	byNaceNat	C19T23 <= C10T33
Sdmxe	*	ACTIVITY	byNaceNat	C19T23 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	C19T23 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	C24_25 <= C10T33
Sdmxe	*	ACTIVITY	byNaceNat	C24_25 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	C24_25 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	C26T33 <= C10T33
Sdmxe	*	ACTIVITY	byNaceNat	C26T33 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	C26T33 = C26 + C27_28 + C29_30 + C31T33
Sdmxe	*	ACTIVITY	byNaceNat	C26T33 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	D35TE39 <= C10TF43
Sdmxe	*	ACTIVITY	byNaceNat	D35TE39 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	D35TE39 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	F41T43 <= C10TF43
Sdmxe	*	ACTIVITY	byNaceNat	F41T43 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	F41T43 >= 0

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	ACTIVITY	byNaceNat	G45T47 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	G45T47 <= G45TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	G45T47 = G45 + G46 + G47
Sdmxe	*	ACTIVITY	byNaceNat	G45T47 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	H49T53 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	H49T53 <= G45TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	H49T53 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	I55 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	I55 <= G45TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	I55 <= I55_56
Sdmxe	*	ACTIVITY	byNaceNat	I55 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	I55_56 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	I55_56 <= G45TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	I55_56 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	ICT_T <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	ICT_T <= C26T33 + G45T47 + J58T63 + S951
Sdmxe	*	ACTIVITY	byNaceNat	ICT_T >= 0
Sdmxe	*	ACTIVITY	byNaceNat	J58T63 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	J58T63 <= G45TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	J58T63 = J58T60 + J61 + J62_63
Sdmxe	*	ACTIVITY	byNaceNat	J58T63 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	L68 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	L68 <= G45TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	L68 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	L68TM75 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	L68TM75 <= G45TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	L68TM75 = L68 + M69T75
Sdmxe	*	ACTIVITY	byNaceNat	L68TM75 >= 0

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	ACTIVITY	byNaceNat	M69T75 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	M69T75 <= G45TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	M69T75 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	N77T82 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	N77T82 <= G45TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	N77T82 = N77T82X79 + N79
Sdmxe	*	ACTIVITY	byNaceNat	N77T82 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	N79 <= N77T82
Sdmxe	*	ACTIVITY	byNaceNat	N79 >= 0
Sdmxe	*	ACTIVITY	byNaceNat	S951 <= C10TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	S951 <= G45TS951XK
Sdmxe	*	ACTIVITY	byNaceNat	S951 >= 0
Sdmxe	*	ACTIVITY	Total	C10TS951XK = C10T18 + C19T23 + C24_25 + C26T33 + D35TE39 + F41T43 + G45T47 + H49T53 + I55_56 + J58T63 + L68 + M69T75 + N77T82 + S951
Sdmxe	*	ACTIVITY	Total	C10TS951XK = C10TF43 + G45TS951XK
Sdmxe	*	ACTIVITY	Total	G45TS951XK = G45T47 + H49T53 + I55_56 + J58T63 + L68 + M69T75 + N77T82 + S951
Sdmxe	*	ACTIVITY	Total	G45TS951XK >= 0
Sdmxe	*	INDICATOR	back	empl <= 25000000
Sdmxe	*	INDICATOR	back	ent <= 5000000
Sdmxe	*	INDICATOR	back	ent_sample <= ent
Sdmxe	*	INDICATOR	back	tovt <= 10000000
Sdmxe	*	INDICATOR	a1	e_iuse <= ent
Sdmxe	*	INDICATOR	a1	e_iuse_ge10a <= e_iuse
Sdmxe	*	INDICATOR	a1	e_iuse_gt10 <= e_iuse

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	a1	e_iuse_gt50 <= e_iuse_gt10
Sdmxe	*	INDICATOR	a2	e_fixbb <= e_iuse
Sdmxe	*	INDICATOR	a2	e_fixbbx <= e_iuse
Sdmxe	*	INDICATOR	a2	e_fixbbz <= e_iuse
Sdmxe	*	INDICATOR	a2	e_iuse = e_fixbb + e_fixbbx + e_fixbbz
Sdmxe	*	INDICATOR	a2	e_iuse >= e_fixbb + e_fixbbx + e_fixbbz
Sdmxe	*	INDICATOR	a3	e_fixbb = e_ispdf_lt30 + e_ispdf1_30_100 + e_ispdf_100_500 + e_ispdf_500_1G + e_ispdf_ge1G
Sdmxe	*	INDICATOR	a3	e_fixbb >= e_ispdf_lt30 + e_ispdf1_30_100 + e_ispdf_100_500 + e_ispdf_500_1G + e_ispdf_ge1G
Sdmxe	*	INDICATOR	a3	e_ispdf_100_500 <= e_fixbb
Sdmxe	*	INDICATOR	a3	e_ispdf_500_1G <= e_fixbb
Sdmxe	*	INDICATOR	a3	e_ispdf_ge1G <= e_fixbb
Sdmxe	*	INDICATOR	a3	e_ispdf_lt30 <= e_fixbb
Sdmxe	*	INDICATOR	a3	e_ispdf1_30_100 <= e_fixbb
Sdmxe	*	INDICATOR	a3	e_ispdf1_ge100 <= e_fixbb
Sdmxe	*	INDICATOR	a3	e_ispdf1_ge100 = e_ispdf_100_500 + e_ispdf_500_1G + e_ispdf_ge1G
Sdmxe	*	INDICATOR	a3	e_ispdf1_ge30 <= e_fixbb
Sdmxe	*	INDICATOR	a3	e_ispdf1_ge30 = e_ispdf1_30_100 + e_ispdf_100_500 + e_ispdf_500_1G + e_ispdf_ge1G
Sdmxe	*	INDICATOR	a4	e_fixbb = e_ispdf1ok + e_ispdf1okx

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	a4	e_fixbb >= e_ispdf1ok + e_ispdf1okx
Sdmxe	*	INDICATOR	a4	e_ispdf1ok <= e_fixbb
Sdmxe	*	INDICATOR	a4	e_ispdf1okx <= e_fixbb
Sdmxe	*	INDICATOR	a4a3	e_ispdf1ok_ge30 <= e_ispdf1_ge30
Sdmxe	*	INDICATOR	a4a3	e_ispdf1ok_ge30 <= e_ispdf1ok
Sdmxe	*	INDICATOR	a4a3	e_ispdf1okx_30_100 <= e_ispdf1_30_100
Sdmxe	*	INDICATOR	a4a3	e_ispdf1okx_30_100 <= e_ispdf1okx
Sdmxe	*	INDICATOR	a4a3	e_ispdf1okx_ge100 <= e_ispdf1_ge100
Sdmxe	*	INDICATOR	a4a3	e_ispdf1okx_ge100 <= e_ispdf1okx
Sdmxe	*	INDICATOR	a4a3	e_ispdf1okx_ge30 <= e_ispdf1_ge30
Sdmxe	*	INDICATOR	a4a3	e_ispdf1okx_ge30 <= e_ispdf1okx
Sdmxe	*	INDICATOR	a4a3	e_ispdfokx_100_500 <= e_ispdf_100_500
Sdmxe	*	INDICATOR	a4a3	e_ispdfokx_100_500 <= e_ispdf1okx
Sdmxe	*	INDICATOR	a4a3	e_ispdfokx_500_1G <= e_ispdf_500_1G
Sdmxe	*	INDICATOR	a4a3	e_ispdfokx_500_1G <= e_ispdf1okx
Sdmxe	*	INDICATOR	a4a3	e_ispdfokx_ge1G <= e_ispdf_ge1G
Sdmxe	*	INDICATOR	a4a3	e_ispdfokx_ge1G <= e_ispdf1okx
Sdmxe	*	INDICATOR	a4a3	e_ispdfokx_lt30 <= e_ispdf_lt30
Sdmxe	*	INDICATOR	a4a3	e_ispdfokx_lt30 <= e_ispdf1okx
Sdmxe	*	INDICATOR	a5	e_empmd2_gt0 <= e_iuse

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	a5	e_empmd2_gt0 <= e_pmd
Sdmxe	*	INDICATOR	a5	e_empmd2_gt0 = e_pmd
Sdmxe	*	INDICATOR	a5	e_empmd2_gt10 <= e_empmd2_gt0
Sdmxe	*	INDICATOR	a5	e_empmd2_gt10 <= e_iuse
Sdmxe	*	INDICATOR	a5	e_empmd2_gt20 <= e_empmd2_gt10
Sdmxe	*	INDICATOR	a5	e_empmd2_gt20 <= e_iuse
Sdmxe	*	INDICATOR	a5	e_empmd2_gt50 <= e_empmd2_gt20
Sdmxe	*	INDICATOR	a5	e_empmd2_gt50 <= e_iuse
Sdmxe	*	INDICATOR	a5	e_iuse = e_pmd + e_pmdx
Sdmxe	*	INDICATOR	a5	e_iuse >= e_pmd + e_pmdx
Sdmxe	*	INDICATOR	a5	e_pmd <= e_iuse
Sdmxe	*	INDICATOR	a5	e_pmdx <= e_iuse
Sdmxe	*	INDICATOR	a5a2	e_broad3 <= e_fixbb + e_pmd
Sdmxe	*	INDICATOR	a5a2	e_broad3 <= e_iuse
Sdmxe	*	INDICATOR	a5a2	e_broad3x <= e_iuse
Sdmxe	*	INDICATOR	a5a2	e_fixbb <= e_broad3
Sdmxe	*	INDICATOR	a5a2	e_iuse = e_broad3 + e_broad3x
Sdmxe	*	INDICATOR	a5a2	e_iuse >= e_broad3 + e_broad3x
Sdmxe	*	INDICATOR	a5a2	e_pmd <= e_broad3
Sdmxe	*	INDICATOR	a6	e_iuse = e_ra_d + e_ra_dx
Sdmxe	*	INDICATOR	a6	e_iuse = e_ra_m + e_ra_mx
Sdmxe	*	INDICATOR	a6	e_iuse = e_ra_s + e_ra_sx
Sdmxe	*	INDICATOR	a6	e_iuse >= e_ra_d + e_ra_dx
Sdmxe	*	INDICATOR	a6	e_iuse >= e_ra_m + e_ra_mx
Sdmxe	*	INDICATOR	a6	e_iuse >= e_ra_s + e_ra_sx
Sdmxe	*	INDICATOR	a6	e_ra <= e_iuse

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	a6	e_ra <= e_ra_m + e_ra_d + e_ra_s
Sdmxe	*	INDICATOR	a6	e_ra_all <= e_iuse
Sdmxe	*	INDICATOR	a6	e_ra_all <= e_ra_d
Sdmxe	*	INDICATOR	a6	e_ra_all <= e_ra_m
Sdmxe	*	INDICATOR	a6	e_ra_all <= e_ra_s
Sdmxe	*	INDICATOR	a6	e_ra_d <= e_iuse
Sdmxe	*	INDICATOR	a6	e_ra_d <= e_ra
Sdmxe	*	INDICATOR	a6	e_ra_d <= e_ra_ds
Sdmxe	*	INDICATOR	a6	e_ra_ds <= e_iuse
Sdmxe	*	INDICATOR	a6	e_ra_ds <= e_ra_d + e_ra_s
Sdmxe	*	INDICATOR	a6	e_ra_dx <= e_iuse
Sdmxe	*	INDICATOR	a6	e_ra_m <= e_iuse
Sdmxe	*	INDICATOR	a6	e_ra_m <= e_ra
Sdmxe	*	INDICATOR	a6	e_ra_mx <= e_iuse
Sdmxe	*	INDICATOR	a6	e_ra_s <= e_iuse
Sdmxe	*	INDICATOR	a6	e_ra_s <= e_ra
Sdmxe	*	INDICATOR	a6	e_ra_s <= e_ra_ds
Sdmxe	*	INDICATOR	a6	e_ra_sx <= e_iuse
Sdmxe	*	INDICATOR	a7	e_iuse = e_rm + e_rmx
Sdmxe	*	INDICATOR	a7	e_iuse >= e_rm + e_rmx
Sdmxe	*	INDICATOR	a7	e_rm <= e_iuse
Sdmxe	*	INDICATOR	a7	e_rmx <= e_iuse
Sdmxe	*	INDICATOR	a8	e_ads <= e_iuse
Sdmxe	*	INDICATOR	a8	e_adsx <= e_iuse
Sdmxe	*	INDICATOR	a8	e_iuse = e_ads + e_adsx
Sdmxe	*	INDICATOR	a8	e_iuse >= e_ads + e_adsx
Sdmxe	*	INDICATOR	a9	e_ads = e_ads_kw + e_ads_kwx
Sdmxe	*	INDICATOR	a9	e_ads = e_ads_loc + e_ads_locx

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	a9	e_ads = e_ads_oth + e_ads_othx
Sdmxe	*	INDICATOR	a9	e_ads = e_ads_trk + e_ads_trkx
Sdmxe	*	INDICATOR	a9	e_ads >= e_ads_kw + e_ads_kwx
Sdmxe	*	INDICATOR	a9	e_ads >= e_ads_loc + e_ads_locx
Sdmxe	*	INDICATOR	a9	e_ads >= e_ads_oth + e_ads_othx
Sdmxe	*	INDICATOR	a9	e_ads >= e_ads_trk + e_ads_trkx
Sdmxe	*	INDICATOR	a9	e_ads_kw <= e_ads
Sdmxe	*	INDICATOR	a9	e_ads_kw <= e_ads3
Sdmxe	*	INDICATOR	a9	e_ads_kwx <= e_ads
Sdmxe	*	INDICATOR	a9	e_ads_loc <= e_ads
Sdmxe	*	INDICATOR	a9	e_ads_loc <= e_ads3
Sdmxe	*	INDICATOR	a9	e_ads_locx <= e_ads
Sdmxe	*	INDICATOR	a9	e_ads_oth <= e_ads
Sdmxe	*	INDICATOR	a9	e_ads_othx <= e_ads
Sdmxe	*	INDICATOR	a9	e_ads_trk <= e_ads
Sdmxe	*	INDICATOR	a9	e_ads_trk <= e_ads3
Sdmxe	*	INDICATOR	a9	e_ads_trkx <= e_ads
Sdmxe	*	INDICATOR	a9	e_ads3 <= e_ads
Sdmxe	*	INDICATOR	a9	e_ads3 <= e_ads_kw + e_ads_trk + e_ads_loc
Sdmxe	*	INDICATOR	b1	e_aws_cboth <= e_aws_cmp
Sdmxe	*	INDICATOR	b1	e_aws_cboth <= e_aws_cown
Sdmxe	*	INDICATOR	b1	e_aws_cboth <= e_iuse
Sdmxe	*	INDICATOR	b1	e_aws_cmp <= e_awsell
Sdmxe	*	INDICATOR	b1	e_aws_cmp <= e_iuse

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	b1	e_aws_cmpq <= e_aws_cmp
Sdmxe	*	INDICATOR	b1	e_aws_cmpq <= e_aws_cownx
Sdmxe	*	INDICATOR	b1	e_aws_cmpq <= e_iuse
Sdmxe	*	INDICATOR	b1	e_aws_cmpx <= e_iuse
Sdmxe	*	INDICATOR	b1	e_aws_cown <= e_awsell
Sdmxe	*	INDICATOR	b1	e_aws_cown <= e_iuse
Sdmxe	*	INDICATOR	b1	e_aws_cownx <= e_iuse
Sdmxe	*	INDICATOR	b1	e_awsell <= e_aws_cmp + e_aws_cown
Sdmxe	*	INDICATOR	b1	e_awsell <= e_iuse
Sdmxe	*	INDICATOR	b1	e_awsell = e_aws_cmp + e_aws_cown - e_aws_cboth
Sdmxe	*	INDICATOR	b1	e_awsellx <= e_iuse
Sdmxe	*	INDICATOR	b1	e_iuse = e_aws_cmp + e_aws_cmpx
Sdmxe	*	INDICATOR	b1	e_iuse = e_aws_cown + e_aws_cownx
Sdmxe	*	INDICATOR	b1	e_iuse = e_awsell + e_awsellx
Sdmxe	*	INDICATOR	b1	e_iuse >= e_aws_cmp + e_aws_cmpx
Sdmxe	*	INDICATOR	b1	e_iuse >= e_aws_cown + e_aws_cownx
Sdmxe	*	INDICATOR	b1	e_iuse >= e_awsell + e_awsellx
Sdmxe	*	INDICATOR	b2	e_awsell = e_wsel0 + e_wselz
Sdmxe	*	INDICATOR	b2	e_awsell >= e_wsel0 + e_wselz
Sdmxe	*	INDICATOR	b2	e_awsval <= tovt
Sdmxe	*	INDICATOR	b2	e_awsval = e_awsvalb + e_awsvals
Sdmxe	*	INDICATOR	b2	e_awsvalb <= tovt

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	b2	e_awsvals <= tovt
Sdmxe	*	INDICATOR	b2	e_wsel0 <= e_awsell
Sdmxe	*	INDICATOR	b2	e_wsel1 <= e_wsel0
Sdmxe	*	INDICATOR	b2	e_wsel10 <= e_wsel5
Sdmxe	*	INDICATOR	b2	e_wsel2 <= e_wsel1
Sdmxe	*	INDICATOR	b2	e_wsel25 <= e_wsel10
Sdmxe	*	INDICATOR	b2	e_wsel5 <= e_wsel2
Sdmxe	*	INDICATOR	b2	e_wsel50 <= e_wsel25
Sdmxe	*	INDICATOR	b2	e_wselz <= e_awsell
Sdmxe	*	INDICATOR	b2b1	e_awsval_cmpq <= e_awsval
Sdmxe	*	INDICATOR	b3	e_awsval = e_awsval_cown + e_awsval_cmp
Sdmxe	*	INDICATOR	b3	e_awsval >= e_awsval_cown + e_awsval_cmp
Sdmxe	*	INDICATOR	b3	e_awsval_cmp <= e_awsval
Sdmxe	*	INDICATOR	b3	e_awsval_cown <= e_awsval
Sdmxe	*	INDICATOR	b3b2b1	e_aws_cmp_ge20 <= e_aws_cmp
Sdmxe	*	INDICATOR	b3b2b1	e_aws_cmp_ge50 <= e_aws_cmp
Sdmxe	*	INDICATOR	b3b2b1	e_aws_cmp_ge50 <= e_aws_cmp_ge20
Sdmxe	*	INDICATOR	b3b2b1	e_awsval_cmpq <= e_awsval_cmp
Sdmxe	*	INDICATOR	b3b2b1	t_aws_cmp_ge20 <= tovt
Sdmxe	*	INDICATOR	b3b2b1	t_aws_cmp_ge50 <= t_aws_cmp_ge20
Sdmxe	*	INDICATOR	b3b2b1	t_aws_cmp_ge50 <= tovt
Sdmxe	*	INDICATOR	b4	e_aws_b2bg <= e_awsell
Sdmxe	*	INDICATOR	b4	e_aws_b2c <= e_awsell

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	b4	e_awsell <= e_aws_b2c + e_aws_b2bg
Sdmxe	*	INDICATOR	b4	e_awsell >= (e_aws_b2c + e_aws_b2bg) / 2
Sdmxe	*	INDICATOR	b4	e_awsval = e_awsval_b2c + e_awsval_b2bg
Sdmxe	*	INDICATOR	b4	e_awsval >= e_awsval_b2c + e_awsval_b2bg
Sdmxe	*	INDICATOR	b4	e_awsval_b2bg <= e_awsval
Sdmxe	*	INDICATOR	b4	e_awsval_b2bg <= e_eturn
Sdmxe	*	INDICATOR	b4	e_awsval_b2bg <= tovt
Sdmxe	*	INDICATOR	b4	e_awsval_b2c <= e_awsval
Sdmxe	*	INDICATOR	b4	e_awsval_b2c <= e_eturn
Sdmxe	*	INDICATOR	b4	e_awsval_b2c <= tovt
Sdmxe	*	INDICATOR	b4b1	e_aws_b2c_cmp <= e_aws_b2c
Sdmxe	*	INDICATOR	b4b1	e_aws_b2c_cmp <= e_aws_cmp
Sdmxe	*	INDICATOR	b4b2	e_aws_b2c_gt1ws <= e_aws_b2c
Sdmxe	*	INDICATOR	b4b2	e_aws_b2c_gt1ws <= e_awsell
Sdmxe	*	INDICATOR	b4b2	e_aws_gt1_b2c_gt10ws <= e_aws_b2c
Sdmxe	*	INDICATOR	b4b2	e_aws_gt1_b2c_gt10ws <= e_awsval_b2c_ge10ws
Sdmxe	*	INDICATOR	b4b2	e_awsval_b2c_ge10ws <= e_aws_b2c
Sdmxe	*	INDICATOR	b4b2	e_awsval_b2c_ge10ws <= e_awsell
Sdmxe	*	INDICATOR	b4b2	e_awsval_b2c_ge10ws <= e_awsval_b2c_ge5ws
Sdmxe	*	INDICATOR	b4b2	e_awsval_b2c_ge5ws <= e_aws_b2c
Sdmxe	*	INDICATOR	b4b2	e_awsval_b2c_ge5ws <= e_aws_b2c_gt1ws

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	b4b2	e_awsval_b2c_ge5ws <= e_awsell
Sdmxe	*	INDICATOR	b4b2	e_awsval_gt1_b2c_gt10ws <= e_awsval_b2c
Sdmxe	*	INDICATOR	b4b2	e_awsval_gt1_b2c_gt10ws <= t_aws_gt1_b2c_gt10ws
Sdmxe	*	INDICATOR	b4b2	t_aws_gt1_b2c_gt10ws <= tovt
Sdmxe	*	INDICATOR	b4b2b1	e_awsval_b2c_ge10ws_cmp <= e_aws_cmp
Sdmxe	*	INDICATOR	b4b2b1	e_awsval_b2c_ge10ws_cmp <= e_awsval_b2c_ge10ws
Sdmxe	*	INDICATOR	b5	e_awseu <= e_awsell
Sdmxe	*	INDICATOR	b5	e_awseu <= e_awsfor
Sdmxe	*	INDICATOR	b5	e_awsfor <= e_awseu + e_awsww
Sdmxe	*	INDICATOR	b5	e_awshm <= e_awsell
Sdmxe	*	INDICATOR	b5	e_awsww <= e_awsell
Sdmxe	*	INDICATOR	b5	e_awsww <= e_awsfor
Sdmxe	*	INDICATOR	b5b2b1	e_awsval_eu_cmpq <= e_awsval_eu_cmp
Sdmxe	*	INDICATOR	b5b2b1	e_awsval_hm_cmpq <= e_awsval_hm_cmp
Sdmxe	*	INDICATOR	b5b2b1	e_awsval_ww_cmpq <= e_awsval_ww_cmp
Sdmxe	*	INDICATOR	b5b3	e_awsval_eu_cmp <= e_awsval_cmp
Sdmxe	*	INDICATOR	b5b3	e_awsval_eu_cown <= e_awsval_cown
Sdmxe	*	INDICATOR	b5b3	e_awsval_hm_cmp <= e_awsval_cmp
Sdmxe	*	INDICATOR	b5b3	e_awsval_hm_cown <= e_awsval_cown
Sdmxe	*	INDICATOR	b5b3	e_awsval_ww_cmp <= e_awsval_cmp
Sdmxe	*	INDICATOR	b5b3	e_awsval_ww_cown <= e_awsval_cown

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	b6	e_awsval = e_awsval_hm + e_awsval_eu + e_awsval_ww
Sdmxe	*	INDICATOR	b6	e_awsval >= e_awsval_hm + e_awsval_eu + e_awsval_ww
Sdmxe	*	INDICATOR	b6	e_awsval_eu <= e_awsval
Sdmxe	*	INDICATOR	b6	e_awsval_hm <= e_awsval
Sdmxe	*	INDICATOR	b6	e_awsval_ww <= e_awsval
Sdmxe	*	INDICATOR	b6b5b2	e_awsfor_ge20 <= e_awsfor
Sdmxe	*	INDICATOR	b6b5b2	e_awsfor_ge50 <= e_awsfor
Sdmxe	*	INDICATOR	b6b5b2	e_awsfor_ge50 <= e_awsfor_ge20
Sdmxe	*	INDICATOR	b6b5b2b1	e_awsfor_ge20_cmp <= e_aws_cmp
Sdmxe	*	INDICATOR	b6b5b2b1	e_awsfor_ge20_cmp <= e_awsfor_ge20
Sdmxe	*	INDICATOR	b6b5b2b1	e_awsfor_ge50_cmp <= e_aws_cmp
Sdmxe	*	INDICATOR	b6b5b2b1	e_awsfor_ge50_cmp <= e_awsfor_ge50
Sdmxe	*	INDICATOR	b7	e_axsell <= e_iuse
Sdmxe	*	INDICATOR	b7	e_axsell = e_xsel0 + e_xselz
Sdmxe	*	INDICATOR	b7	e_axsell >= e_xsel0 + e_xselz
Sdmxe	*	INDICATOR	b7	e_axsellx <= e_iuse
Sdmxe	*	INDICATOR	b7	e_axsellz <= e_iuse
Sdmxe	*	INDICATOR	b7	e_iuse = e_axsell + e_axsellx + e_axsellz
Sdmxe	*	INDICATOR	b7	e_iuse >= e_axsell + e_axsellx + e_axsellz
Sdmxe	*	INDICATOR	b7b1	e_aesell <= e_awsell + e_axsell
Sdmxe	*	INDICATOR	b7b1	e_aesell <= e_iuse
Sdmxe	*	INDICATOR	b7b1	e_awsell <= e_aesell

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	b7b1	e_axsell <= e_aesell
Sdmxe	*	INDICATOR	b7b2	e_wsel1q <= e_axsellx
Sdmxe	*	INDICATOR	b7b2	e_wsel1q <= e_wsel1
Sdmxe	*	INDICATOR	b8	e_axsval <= tovt
Sdmxe	*	INDICATOR	b8	e_axsval = e_axsvalb + e_axsvals
Sdmxe	*	INDICATOR	b8	e_axsvalb <= e_axsval
Sdmxe	*	INDICATOR	b8	e_axsvalb <= tovt
Sdmxe	*	INDICATOR	b8	e_axsvals <= e_axsval
Sdmxe	*	INDICATOR	b8	e_axsvals <= tovt
Sdmxe	*	INDICATOR	b8	e_xsel0 <= e_axsell
Sdmxe	*	INDICATOR	b8	e_xsel1 <= e_xsel0
Sdmxe	*	INDICATOR	b8	e_xsel10 <= e_xsel5
Sdmxe	*	INDICATOR	b8	e_xsel2 <= e_xsel1
Sdmxe	*	INDICATOR	b8	e_xsel25 <= e_xsel10
Sdmxe	*	INDICATOR	b8	e_xsel5 <= e_xsel2
Sdmxe	*	INDICATOR	b8	e_xsel50 <= e_xsel25
Sdmxe	*	INDICATOR	b8	e_xselz <= e_axsell
Sdmxe	*	INDICATOR	b8b2	e_awsval <= e_eturn
Sdmxe	*	INDICATOR	b8b2	e_awsvalb <= e_eturn
Sdmxe	*	INDICATOR	b8b2	e_awsvals <= e_eturn
Sdmxe	*	INDICATOR	b8b2	e_axsval <= e_eturn
Sdmxe	*	INDICATOR	b8b2	e_axsvalb <= e_eturn
Sdmxe	*	INDICATOR	b8b2	e_axsvals <= e_eturn
Sdmxe	*	INDICATOR	b8b2	e_esell <= e_aesell
Sdmxe	*	INDICATOR	b8b2	e_eturn <= tovt
Sdmxe	*	INDICATOR	b8b2	e_eturn = e_awsval + e_axsval
Sdmxe	*	INDICATOR	b8b2	e_eturn = e_awsvals + e_awsvalb + e_axsvals + e_axsvalb

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	b8b4b2	e_awsval_b2c_ge10ec <= e_aws_b2c
Sdmxe	*	INDICATOR	c1	e_itsp2 <= ent
Sdmxe	*	INDICATOR	c1	e_itsp2x <= ent
Sdmxe	*	INDICATOR	c1	ent = e_itsp2 + e_itsp2x
Sdmxe	*	INDICATOR	c1	ent >= e_itsp2 + e_itsp2x
Sdmxe	*	INDICATOR	c2	e_itspt2 <= e_itt2
Sdmxe	*	INDICATOR	c2	e_itspt2 <= ent
Sdmxe	*	INDICATOR	c2	e_itspt2x <= ent
Sdmxe	*	INDICATOR	c2	e_itt2 <= e_itspt2 + e_itust2
Sdmxe	*	INDICATOR	c2	e_itt2 <= ent
Sdmxe	*	INDICATOR	c2	e_itust2 <= e_itt2
Sdmxe	*	INDICATOR	c2	e_itust2 <= ent
Sdmxe	*	INDICATOR	c2	e_itust2x <= ent
Sdmxe	*	INDICATOR	c2	ent = e_itspt2 + e_itspt2x
Sdmxe	*	INDICATOR	c2	ent = e_itust2 + e_itust2x
Sdmxe	*	INDICATOR	c2	ent >= e_itspt2 + e_itspt2x
Sdmxe	*	INDICATOR	c2	ent >= e_itust2 + e_itust2x
Sdmxe	*	INDICATOR	c2c1	e_itsp2_itt2 <= e_itsp2
Sdmxe	*	INDICATOR	c2c1	e_itsp2_itt2 <= e_itt2
Sdmxe	*	INDICATOR	c2c1	e_itsp2_itt2 <= ent
Sdmxe	*	INDICATOR	c2c1	e_itsp2x_itt2 <= e_itsp2x
Sdmxe	*	INDICATOR	c2c1	e_itsp2x_itt2 <= e_itust2
Sdmxe	*	INDICATOR	c2c1	e_itsp2x_itt2 <= ent
Sdmxe	*	INDICATOR	c2c1	e_itspt2 <= e_itsp2
Sdmxe	*	INDICATOR	c3	e_itsprcr2 <= ent
Sdmxe	*	INDICATOR	c3	e_itsprcr2x <= ent
Sdmxe	*	INDICATOR	c3	ent = e_itsprcr2 + e_itsprcr2x
Sdmxe	*	INDICATOR	c3	ent >= e_itsprcr2 + e_itsprcr2x

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	c3c1	e_itsp2 <= e_itsp2_or_rcr2
Sdmxe	*	INDICATOR	c3c1	e_itsp2_or_rcr2 <= e_itsp2 + e_itsprcr2
Sdmxe	*	INDICATOR	c3c1	e_itsprcr2 <= e_itsp2_or_rcr2
Sdmxe	*	INDICATOR	c4	e_itsprcr2 = e_itspvac2 + e_itspvac2x
Sdmxe	*	INDICATOR	c4	e_itsprcr2 >= e_itspvac2 + e_itspvac2x
Sdmxe	*	INDICATOR	c4	e_itspvac2 <= e_itsprcr2
Sdmxe	*	INDICATOR	c4	e_itspvac2x <= e_itsprcr2
Sdmxe	*	INDICATOR	c4c1	e_itsp2_vac2 <= e_itsp2
Sdmxe	*	INDICATOR	c4c1	e_itsp2_vac2 <= e_itsprcr2
Sdmxe	*	INDICATOR	c4c1	e_itsp2_vac2 <= e_itspvac2
Sdmxe	*	INDICATOR	c5	e_itspdla <= e_itspvac2
Sdmxe	*	INDICATOR	c5	e_itspdlax <= e_itspvac2
Sdmxe	*	INDICATOR	c5	e_itspdlet <= e_itspvac2
Sdmxe	*	INDICATOR	c5	e_itspdletx <= e_itspvac2
Sdmxe	*	INDICATOR	c5	e_itspdlwe <= e_itspvac2
Sdmxe	*	INDICATOR	c5	e_itspdlwex <= e_itspvac2
Sdmxe	*	INDICATOR	c5	e_itspdsal <= e_itspvac2
Sdmxe	*	INDICATOR	c5	e_itspdsalx <= e_itspvac2
Sdmxe	*	INDICATOR	c5	e_itspvac2 = e_itspdla + e_itspdlax
Sdmxe	*	INDICATOR	c5	e_itspvac2 = e_itspdlet + e_itspdletx
Sdmxe	*	INDICATOR	c5	e_itspvac2 = e_itspdlwe + e_itspdlwex
Sdmxe	*	INDICATOR	c5	e_itspvac2 = e_itspdsal + e_itspdsalx
Sdmxe	*	INDICATOR	c5	e_itspvac2 >= e_itspdla + e_itspdlax
Sdmxe	*	INDICATOR	c5	e_itspvac2 >= e_itspdlet + e_itspdletx

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	c5	e_itspvac2 >= e_itspdlwe + e_itspdlwex
Sdmxe	*	INDICATOR	c5	e_itspvac2 >= e_itspdsal + e_itspdsalx
Sdmxe	*	INDICATOR	c6	e_it_ext <= ent
Sdmxe	*	INDICATOR	c6	e_it_extq <= ent
Sdmxe	*	INDICATOR	c6	e_it_extx <= ent
Sdmxe	*	INDICATOR	c6	e_it_own <= ent
Sdmxe	*	INDICATOR	c6	e_it_ownq <= ent
Sdmxe	*	INDICATOR	c6	e_it_ownx <= ent
Sdmxe	*	INDICATOR	c6	ent = e_it_ext + e_it_extx
Sdmxe	*	INDICATOR	c6	ent = e_it_own + e_it_ownx
Sdmxe	*	INDICATOR	c6	ent >= e_it_ext + e_it_extx
Sdmxe	*	INDICATOR	c6	ent >= e_it_extq + e_it_own
Sdmxe	*	INDICATOR	c6	ent >= e_it_own + e_it_ownx
Sdmxe	*	INDICATOR	c6	ent >= e_it_ownq + e_it_ext
Sdmxe	*	INDICATOR	c6c1	e_it_ext_itsp2 <= e_it_ext
Sdmxe	*	INDICATOR	c6c1	e_it_ext_itsp2 <= e_itsp2
Sdmxe	*	INDICATOR	c6c1	e_it_ext_itsp2 <= ent
Sdmxe	*	INDICATOR	c6c1	e_it_own_itsp2 <= e_it_own
Sdmxe	*	INDICATOR	c6c1	e_it_own_itsp2 <= e_itsp2
Sdmxe	*	INDICATOR	c6c1	e_it_own_itsp2 <= ent
Sdmxe	*	INDICATOR	c6c2	e_it_ext_itt2 <= e_it_ext
Sdmxe	*	INDICATOR	c6c2	e_it_ext_itt2 <= e_itt2
Sdmxe	*	INDICATOR	c6c2	e_it_ext_itt2 <= ent
Sdmxe	*	INDICATOR	d1	e_iuse = e_secmdenc + e_secmdencx
Sdmxe	*	INDICATOR	d1	e_iuse = e_secmduo + e_secmduox
Sdmxe	*	INDICATOR	d1	e_iuse = e_secmlog + e_secmlogx

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	d1	e_iuse = e_secmnac + e_secmnacx
Sdmxe	*	INDICATOR	d1	e_iuse = e_secmosbu + e_secmosbux
Sdmxe	*	INDICATOR	d1	e_iuse = e_secmrass + e_secmrassx
Sdmxe	*	INDICATOR	d1	e_iuse = e_secmsms1 + e_secmsms1x
Sdmxe	*	INDICATOR	d1	e_iuse = e_secmspsw + e_secmspswx
Sdmxe	*	INDICATOR	d1	e_iuse = e_secmtst + e_secmtstx
Sdmxe	*	INDICATOR	d1	e_iuse = e_secmuibm + e_secmuibmx
Sdmxe	*	INDICATOR	d1	e_iuse = e_secmvpn + e_secmvpnx
Sdmxe	*	INDICATOR	d1	e_iuse >= e_secmdenc + e_secmdencx
Sdmxe	*	INDICATOR	d1	e_iuse >= e_secmduo + e_secmduox
Sdmxe	*	INDICATOR	d1	e_iuse >= e_secmlog + e_secmlogx
Sdmxe	*	INDICATOR	d1	e_iuse >= e_secmnac + e_secmnacx
Sdmxe	*	INDICATOR	d1	e_iuse >= e_secmosbu + e_secmosbux
Sdmxe	*	INDICATOR	d1	e_iuse >= e_secmrass + e_secmrassx
Sdmxe	*	INDICATOR	d1	e_iuse >= e_secmsms1 + e_secmsms1x
Sdmxe	*	INDICATOR	d1	e_iuse >= e_secmspsw + e_secmspswx
Sdmxe	*	INDICATOR	d1	e_iuse >= e_secmtst + e_secmtstx
Sdmxe	*	INDICATOR	d1	e_iuse >= e_secmuibm + e_secmuibmx
Sdmxe	*	INDICATOR	d1	e_iuse >= e_secmvpn + e_secmvpnx

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	d1	e_secmall <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmall <= e_secmge7
Sdmxe	*	INDICATOR	d1	e_secmdenc <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmdenc <= e_secmge1
Sdmxe	*	INDICATOR	d1	e_secmdencx <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmduo <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmduo <= e_secmge1
Sdmxe	*	INDICATOR	d1	e_secmduox <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmge1 <= e_secmspsw + e_secmuibm + e_secmduo + e_secmdenc + e_secmosbu + e_secmnac + e_secmvpn + e_secmsms1 + e_secmlog + e_secmrass + e_secmtst
Sdmxe	*	INDICATOR	d1	e_secmge3 <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmge3 <= e_secmge1
Sdmxe	*	INDICATOR	d1	e_secmge5 <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmge5 <= e_secmge3
Sdmxe	*	INDICATOR	d1	e_secmge7 <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmge7 <= e_secmge5
Sdmxe	*	INDICATOR	d1	e_secmlog <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmlog <= e_secmge1
Sdmxe	*	INDICATOR	d1	e_secmlogx <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmnac <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmnac <= e_secmge1
Sdmxe	*	INDICATOR	d1	e_secmnacx <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmosbu <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmosbu <= e_secmge1
Sdmxe	*	INDICATOR	d1	e_secmosbux <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmrass <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmrass <= e_secmge1

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	d1	e_secmrassx <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmsms1 <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmsms1 <= e_secmge1
Sdmxe	*	INDICATOR	d1	e_secmsms1x <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmspsw <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmspsw <= e_secmge1
Sdmxe	*	INDICATOR	d1	e_secmspswx <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmtst <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmtst <= e_secmge1
Sdmxe	*	INDICATOR	d1	e_secmtstx <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmuibm <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmuibm <= e_secmge1
Sdmxe	*	INDICATOR	d1	e_secmuibmx <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmvpn <= e_iuse
Sdmxe	*	INDICATOR	d1	e_secmvpn <= e_secmge1
Sdmxe	*	INDICATOR	d1	e_secmvpnx <= e_iuse
Sdmxe	*	INDICATOR	d2	e_iuse = e_secawany + e_secawnone
Sdmxe	*	INDICATOR	d2	e_iuse = e_secawcont + e_secawcontx
Sdmxe	*	INDICATOR	d2	e_iuse = e_secawctp + e_secawctpx
Sdmxe	*	INDICATOR	d2	e_iuse = e_secawvtgi + e_secawvtgix
Sdmxe	*	INDICATOR	d2	e_iuse >= e_secawany + e_secawnone
Sdmxe	*	INDICATOR	d2	e_iuse >= e_secawcont + e_secawcontx
Sdmxe	*	INDICATOR	d2	e_iuse >= e_secawcont + e_secawnone
Sdmxe	*	INDICATOR	d2	e_iuse >= e_secawctp + e_secawctpx
Sdmxe	*	INDICATOR	d2	e_iuse >= e_secawctp + e_secawnone

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	d2	e_iuse >= e_secawvtgi + e_secawnone
Sdmxe	*	INDICATOR	d2	e_iuse >= e_secawvtgi + e_secawvtgix
Sdmxe	*	INDICATOR	d2	e_secawany <= e_iuse
Sdmxe	*	INDICATOR	d2	e_secawany <= e_secawvtgi + e_secawctp + e_secawcont
Sdmxe	*	INDICATOR	d2	e_secawcont <= e_iuse
Sdmxe	*	INDICATOR	d2	e_secawcont <= e_secawany
Sdmxe	*	INDICATOR	d2	e_secawcontx <= e_iuse
Sdmxe	*	INDICATOR	d2	e_secawctp <= e_iuse
Sdmxe	*	INDICATOR	d2	e_secawctp <= e_secawany
Sdmxe	*	INDICATOR	d2	e_secawctpx <= e_iuse
Sdmxe	*	INDICATOR	d2	e_secawnone <= e_iuse
Sdmxe	*	INDICATOR	d2	e_secawvtgi <= e_iuse
Sdmxe	*	INDICATOR	d2	e_secawvtgi <= e_secawany
Sdmxe	*	INDICATOR	d2	e_secawvtgix <= e_iuse
Sdmxe	*	INDICATOR	d2c2	e_secawany_itspt2 <= e_itspt2
Sdmxe	*	INDICATOR	d2c2	e_secawany_itspt2 <= e_iuse
Sdmxe	*	INDICATOR	d2c2	e_secawany_itspt2 <= e_secawany
Sdmxe	*	INDICATOR	d2c2	e_secawany_itust2 <= e_itust2
Sdmxe	*	INDICATOR	d2c2	e_secawany_itust2 <= e_iuse
Sdmxe	*	INDICATOR	d2c2	e_secawany_itust2 <= e_secawany
Sdmxe	*	INDICATOR	d2c2	e_secawcont_itspt2 <= e_itspt2
Sdmxe	*	INDICATOR	d2c2	e_secawcont_itspt2 <= e_iuse

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	d2c2	e_secawcont_itspt2 <= e_secawcont
Sdmxe	*	INDICATOR	d2c2	e_secawcont_itust2 <= e_itust2
Sdmxe	*	INDICATOR	d2c2	e_secawcont_itust2 <= e_iuse
Sdmxe	*	INDICATOR	d2c2	e_secawcont_itust2 <= e_secawcont
Sdmxe	*	INDICATOR	d2c2	e_secawctp_itspt2 <= e_itspt2
Sdmxe	*	INDICATOR	d2c2	e_secawctp_itspt2 <= e_iuse
Sdmxe	*	INDICATOR	d2c2	e_secawctp_itspt2 <= e_secawctp
Sdmxe	*	INDICATOR	d2c2	e_secawctp_itust2 <= e_itust2
Sdmxe	*	INDICATOR	d2c2	e_secawctp_itust2 <= e_iuse
Sdmxe	*	INDICATOR	d2c2	e_secawctp_itust2 <= e_secawctp
Sdmxe	*	INDICATOR	d2c2	e_secawvtgi_itspt2 <= e_itspt2
Sdmxe	*	INDICATOR	d2c2	e_secawvtgi_itspt2 <= e_iuse
Sdmxe	*	INDICATOR	d2c2	e_secawvtgi_itspt2 <= e_secawvtgi
Sdmxe	*	INDICATOR	d2c2	e_secawvtgi_itust2 <= e_itust2
Sdmxe	*	INDICATOR	d2c2	e_secawvtgi_itust2 <= e_iuse
Sdmxe	*	INDICATOR	d2c2	e_secawvtgi_itust2 <= e_secawvtgi
Sdmxe	*	INDICATOR	d3	e_iuse = e_secpol2 + e_secpol2x
Sdmxe	*	INDICATOR	d3	e_iuse >= e_secpol2 + e_secpol2x
Sdmxe	*	INDICATOR	d3	e_secpol2 <= e_iuse
Sdmxe	*	INDICATOR	d3	e_secpol2x <= e_iuse

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	d3c1	e_secpol2_itsp2 <= e_itsp2
Sdmxe	*	INDICATOR	d3c1	e_secpol2_itsp2 <= e_iuse
Sdmxe	*	INDICATOR	d3c1	e_secpol2_itsp2 <= e_secpol2
Sdmxe	*	INDICATOR	d3d2	e_secawany_pol2 <= e_iuse
Sdmxe	*	INDICATOR	d3d2	e_secawany_pol2 <= e_secawany
Sdmxe	*	INDICATOR	d3d2	e_secawany_pol2 <= e_secpol2
Sdmxe	*	INDICATOR	d4	e_secpol2 = e_secprev_cy + e_secprev_1_2 + e_secprev_mt2
Sdmxe	*	INDICATOR	d4	e_secpol2 >= e_secprev_cy + e_secprev_1_2 + e_secprev_mt2
Sdmxe	*	INDICATOR	d4	e_secprev_1_2 <= e_secpol2
Sdmxe	*	INDICATOR	d4	e_secprev_1_2 <= e_secprev_le2
Sdmxe	*	INDICATOR	d4	e_secprev_cy <= e_secpol2
Sdmxe	*	INDICATOR	d4	e_secprev_cy <= e_secprev_le2
Sdmxe	*	INDICATOR	d4	e_secprev_le2 <= e_secpol2
Sdmxe	*	INDICATOR	d4	e_secprev_le2 = e_secprev_cy + e_secprev_1_2
Sdmxe	*	INDICATOR	d4	e_secprev_mt2 <= e_secpol2
Sdmxe	*	INDICATOR	d5	e_iuse = e_sec2icnfa + e_sec2icnfax
Sdmxe	*	INDICATOR	d5	e_iuse = e_sec2icnff + e_sec2icnffx
Sdmxe	*	INDICATOR	d5	e_iuse = e_sec2idcda + e_sec2idcdax
Sdmxe	*	INDICATOR	d5	e_iuse = e_sec2idcdf + e_sec2idcdfx

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	d5	e_iuse = e_sec2iusva + e_sec2iusvax
Sdmxe	*	INDICATOR	d5	e_iuse = e_sec2iusvf + e_sec2iusvfx
Sdmxe	*	INDICATOR	d5	e_iuse >= e_sec2icnfa + e_sec2icnfax
Sdmxe	*	INDICATOR	d5	e_iuse >= e_sec2icnff + e_sec2icnffx
Sdmxe	*	INDICATOR	d5	e_iuse >= e_sec2idcda + e_sec2idcdax
Sdmxe	*	INDICATOR	d5	e_iuse >= e_sec2idcdf + e_sec2idcdfx
Sdmxe	*	INDICATOR	d5	e_iuse >= e_sec2iusva + e_sec2iusvax
Sdmxe	*	INDICATOR	d5	e_iuse >= e_sec2iusvf + e_sec2iusvfx
Sdmxe	*	INDICATOR	d5	e_sec2iany <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2iany <= e_sec2icnfa + e_sec2icnff + e_sec2idcda + e_sec2idcdf + e_sec2iusva + e_sec2iusvf
Sdmxe	*	INDICATOR	d5	e_sec2icnf <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2icnf <= e_sec2icnff + e_sec2icnfa
Sdmxe	*	INDICATOR	d5	e_sec2icnfa <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2icnfa <= e_sec2iany
Sdmxe	*	INDICATOR	d5	e_sec2icnfa <= e_sec2icnf
Sdmxe	*	INDICATOR	d5	e_sec2icnfax <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2icnff <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2icnff <= e_sec2iany
Sdmxe	*	INDICATOR	d5	e_sec2icnff <= e_sec2icnf
Sdmxe	*	INDICATOR	d5	e_sec2icnffx <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2idcd <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2idcd <= e_sec2idcdf + e_sec2idcda

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	d5	e_sec2idcda <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2idcda <= e_sec2iany
Sdmxe	*	INDICATOR	d5	e_sec2idcda <= e_sec2idcd
Sdmxe	*	INDICATOR	d5	e_sec2idcdax <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2idcdf <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2idcdf <= e_sec2iany
Sdmxe	*	INDICATOR	d5	e_sec2idcdf <= e_sec2idcd
Sdmxe	*	INDICATOR	d5	e_sec2idcdfx <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2ige3 <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2ige3 <= e_sec2iany
Sdmxe	*	INDICATOR	d5	e_sec2iusv <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2iusv <= e_sec2iusvf + e_sec2iusva
Sdmxe	*	INDICATOR	d5	e_sec2iusva <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2iusva <= e_sec2iany
Sdmxe	*	INDICATOR	d5	e_sec2iusva <= e_sec2iusv
Sdmxe	*	INDICATOR	d5	e_sec2iusvax <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2iusvf <= e_iuse
Sdmxe	*	INDICATOR	d5	e_sec2iusvf <= e_sec2iany
Sdmxe	*	INDICATOR	d5	e_sec2iusvf <= e_sec2iusv
Sdmxe	*	INDICATOR	d5	e_sec2iusvfx <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tany <= e_ai_ttm + e_ai_tsr + e_ai_tnlg + e_ai_tir + e_ai_tml + e_ai_tpa + e_ai_tar
Sdmxe	*	INDICATOR	e1	e_ai_tany <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tar <= e_ai_tany
Sdmxe	*	INDICATOR	e1	e_ai_tar <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tarx <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tge2 <= e_ai_tany
Sdmxe	*	INDICATOR	e1	e_ai_tge2 <= e_iuse

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	e1	e_ai_tge3 <= e_ai_tge2
Sdmxe	*	INDICATOR	e1	e_ai_tge3 <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tir <= e_ai_tany
Sdmxe	*	INDICATOR	e1	e_ai_tir <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tirx <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tml <= e_ai_tany
Sdmxe	*	INDICATOR	e1	e_ai_tml <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tmlx <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tnlg <= e_ai_tany
Sdmxe	*	INDICATOR	e1	e_ai_tnlg <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tnlgx <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tpa <= e_ai_tany
Sdmxe	*	INDICATOR	e1	e_ai_tpa <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tpax <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tsr <= e_ai_tany
Sdmxe	*	INDICATOR	e1	e_ai_tsr <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tsrx <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_ttm <= e_ai_tany
Sdmxe	*	INDICATOR	e1	e_ai_ttm <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_ttmx <= e_iuse
Sdmxe	*	INDICATOR	e1	e_ai_tx <= e_iuse
Sdmxe	*	INDICATOR	e1	e_iuse = e_ai_tany + e_ai_tx
Sdmxe	*	INDICATOR	e1	e_iuse = e_ai_tar + e_ai_tarx
Sdmxe	*	INDICATOR	e1	e_iuse = e_ai_tir + e_ai_tirx
Sdmxe	*	INDICATOR	e1	e_iuse = e_ai_tml + e_ai_tmlx
Sdmxe	*	INDICATOR	e1	e_iuse = e_ai_tnlg + e_ai_tnlgx
Sdmxe	*	INDICATOR	e1	e_iuse = e_ai_tpa + e_ai_tpax

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	e1	e_iuse = e_ai_tsr + e_ai_tsrx
Sdmxe	*	INDICATOR	e1	e_iuse = e_ai_ttm + e_ai_ttmx
Sdmxe	*	INDICATOR	e1	e_iuse >= e_ai_tany + e_ai_tx
Sdmxe	*	INDICATOR	e1	e_iuse >= e_ai_tar + e_ai_tarx
Sdmxe	*	INDICATOR	e1	e_iuse >= e_ai_tir + e_ai_tirx
Sdmxe	*	INDICATOR	e1	e_iuse >= e_ai_tml + e_ai_tmlx
Sdmxe	*	INDICATOR	e1	e_iuse >= e_ai_tnlg + e_ai_tnlgx
Sdmxe	*	INDICATOR	e1	e_iuse >= e_ai_tpa + e_ai_tpax
Sdmxe	*	INDICATOR	e1	e_iuse >= e_ai_tsr + e_ai_tsrx
Sdmxe	*	INDICATOR	e1	e_iuse >= e_ai_ttm + e_ai_ttmx
Sdmxe	*	INDICATOR	e2	e_ai_p1any <= e_ai_pms + e_ai_ppp + e_ai_pbam + e_ai_plog + e_ai_pits + e_ai_pfin + e_ai_prdi
Sdmxe	*	INDICATOR	e2	e_ai_p1any <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_p1ge2 <= e_ai_p1any
Sdmxe	*	INDICATOR	e2	e_ai_p1ge2 <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_p1ge3 <= e_ai_p1ge2
Sdmxe	*	INDICATOR	e2	e_ai_p1ge3 <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_pbam <= e_ai_p1any
Sdmxe	*	INDICATOR	e2	e_ai_pbam <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_pbamx <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_pfin <= e_ai_p1any
Sdmxe	*	INDICATOR	e2	e_ai_pfin <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_pfinx <= e_ai_tany

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	e2	e_ai_pits <= e_ai_p1any
Sdmxe	*	INDICATOR	e2	e_ai_pits <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_pitsx <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_plog <= e_ai_p1any
Sdmxe	*	INDICATOR	e2	e_ai_plog <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_plogx <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_pms <= e_ai_p1any
Sdmxe	*	INDICATOR	e2	e_ai_pms <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_pmsx <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_ppp <= e_ai_p1any
Sdmxe	*	INDICATOR	e2	e_ai_ppp <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_pppx <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_prdi <= e_ai_p1any
Sdmxe	*	INDICATOR	e2	e_ai_prdi <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_prdix <= e_ai_tany
Sdmxe	*	INDICATOR	e2	e_ai_tany = e_ai_pbam + e_ai_pbamx
Sdmxe	*	INDICATOR	e2	e_ai_tany = e_ai_pfin + e_ai_pfinx
Sdmxe	*	INDICATOR	e2	e_ai_tany = e_ai_pits + e_ai_pitsx
Sdmxe	*	INDICATOR	e2	e_ai_tany = e_ai_plog + e_ai_plogx
Sdmxe	*	INDICATOR	e2	e_ai_tany = e_ai_pms + e_ai_pmsx
Sdmxe	*	INDICATOR	e2	e_ai_tany = e_ai_ppp + e_ai_pppx
Sdmxe	*	INDICATOR	e2	e_ai_tany = e_ai_prdi + e_ai_prdix
Sdmxe	*	INDICATOR	e2	e_ai_tany >= e_ai_pbam + e_ai_pbamx
Sdmxe	*	INDICATOR	e2	e_ai_tany >= e_ai_pfin + e_ai_pfinx

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	e2	e_ai_tany >= e_ai_pits + e_ai_pitsx
Sdmxe	*	INDICATOR	e2	e_ai_tany >= e_ai_plog + e_ai_plogx
Sdmxe	*	INDICATOR	e2	e_ai_tany >= e_ai_pms + e_ai_pmsx
Sdmxe	*	INDICATOR	e2	e_ai_tany >= e_ai_ppp + e_ai_pppx
Sdmxe	*	INDICATOR	e2	e_ai_tany >= e_ai_prdi + e_ai_prdix
Sdmxe	*	INDICATOR	e3	e_ai_adown <= e_ai_tany
Sdmxe	*	INDICATOR	e3	e_ai_adownx <= e_ai_tany
Sdmxe	*	INDICATOR	e3	e_ai_aext <= e_ai_tany
Sdmxe	*	INDICATOR	e3	e_ai_aextx <= e_ai_tany
Sdmxe	*	INDICATOR	e3	e_ai_amown <= e_ai_tany
Sdmxe	*	INDICATOR	e3	e_ai_amownx <= e_ai_tany
Sdmxe	*	INDICATOR	e3	e_ai_aos <= e_ai_tany
Sdmxe	*	INDICATOR	e3	e_ai_aosx <= e_ai_tany
Sdmxe	*	INDICATOR	e3	e_ai_ardy <= e_ai_tany
Sdmxe	*	INDICATOR	e3	e_ai_ardyx <= e_ai_tany
Sdmxe	*	INDICATOR	e3	e_ai_tany = e_ai_adown + e_ai_adownx
Sdmxe	*	INDICATOR	e3	e_ai_tany = e_ai_aext + e_ai_aextx
Sdmxe	*	INDICATOR	e3	e_ai_tany = e_ai_amown + e_ai_amownx
Sdmxe	*	INDICATOR	e3	e_ai_tany = e_ai_aos + e_ai_aosx
Sdmxe	*	INDICATOR	e3	e_ai_tany = e_ai_ardy + e_ai_ardyx
Sdmxe	*	INDICATOR	e3	e_ai_tany >= e_ai_adown + e_ai_adownx
Sdmxe	*	INDICATOR	e3	e_ai_tany >= e_ai_aext + e_ai_aextx

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	e3	e_ai_tany >= e_ai_amown + e_ai_amownx
Sdmxe	*	INDICATOR	e3	e_ai_tany >= e_ai_aos + e_ai_aosx
Sdmxe	*	INDICATOR	e3	e_ai_tany >= e_ai_ardy + e_ai_ardyx
Sdmxe	*	INDICATOR	e4	e_ai_pdi <= e_ai_tany
Sdmxe	*	INDICATOR	e4	e_ai_pdix <= e_ai_tany
Sdmxe	*	INDICATOR	e4	e_ai_tany = e_ai_pdi+e_ai_pdix
Sdmxe	*	INDICATOR	e4	e_ai_tany >= e_ai_pdi+e_ai_pdix
Sdmxe	*	INDICATOR	e5	e_ai_bias <= e_ai_pdi
Sdmxe	*	INDICATOR	e5	e_ai_biasx <= e_ai_pdi
Sdmxe	*	INDICATOR	e5	e_ai_pdi = e_ai_bias+e_ai_biasx
Sdmxe	*	INDICATOR	e5	e_ai_pdi >= e_ai_bias+e_ai_biasx
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_adown <= e_ai_adown
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_adown <= e_ai_bias
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_adown <= e_ai_pdi
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_aext <= e_ai_aext
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_aext <= e_ai_bias
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_aext <= e_ai_pdi
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_amown <= e_ai_amown
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_amown <= e_ai_bias
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_amown <= e_ai_pdi
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_aos <= e_ai_aos
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_aos <= e_ai_bias

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_aos <= e_ai_pdi
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_ardy <= e_ai_ardy
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_ardy <= e_ai_bias
Sdmxe	*	INDICATOR	e5e3	e_ai_bias_ardy <= e_ai_pdi
Sdmxe	*	INDICATOR	e6	e_ai_ec <= e_ai_tx
Sdmxe	*	INDICATOR	e6	e_ai_ecx <= e_ai_tx
Sdmxe	*	INDICATOR	e6	e_ai_tx = e_ai_ec + e_ai_ecx
Sdmxe	*	INDICATOR	e6	e_ai_tx >= e_ai_ec + e_ai_ecx
Sdmxe	*	INDICATOR	e7	e_ai_bcdp <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_bcdpx <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_bcst <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_bcstx <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_bddt <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_bddtx <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_bec <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_becx <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_binc <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_bincx <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_ble <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_bleg <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_blegx <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_blex <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_bnu <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_bnux <= e_ai_ec
Sdmxe	*	INDICATOR	e7	e_ai_ec = e_ai_bcdp + e_ai_bcdpx
Sdmxe	*	INDICATOR	e7	e_ai_ec = e_ai_bcst + e_ai_bcstx

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	e7	e_ai_ec = e_ai_bddt + e_ai_bddtx
Sdmxe	*	INDICATOR	e7	e_ai_ec = e_ai_bec + e_ai_becx
Sdmxe	*	INDICATOR	e7	e_ai_ec = e_ai_binc + e_ai_bincx
Sdmxe	*	INDICATOR	e7	e_ai_ec = e_ai_ble + e_ai_blex
Sdmxe	*	INDICATOR	e7	e_ai_ec = e_ai_bleg + e_ai_blegx
Sdmxe	*	INDICATOR	e7	e_ai_ec = e_ai_bnu + e_ai_bnux
Sdmxe	*	INDICATOR	e7	e_ai_ec >= e_ai_bcdp + e_ai_bcdpx
Sdmxe	*	INDICATOR	e7	e_ai_ec >= e_ai_bcst + e_ai_bcstx
Sdmxe	*	INDICATOR	e7	e_ai_ec >= e_ai_bddt + e_ai_bddtx
Sdmxe	*	INDICATOR	e7	e_ai_ec >= e_ai_bec + e_ai_becx
Sdmxe	*	INDICATOR	e7	e_ai_ec >= e_ai_binc + e_ai_bincx
Sdmxe	*	INDICATOR	e7	e_ai_ec >= e_ai_ble + e_ai_blex
Sdmxe	*	INDICATOR	e7	e_ai_ec >= e_ai_bleg + e_ai_blegx
Sdmxe	*	INDICATOR	e7	e_ai_ec >= e_ai_bnu + e_ai_bnux
Sdmxe	*	INDICATOR	p_a1	empl = p_iuse + p_iusex
Sdmxe	*	INDICATOR	p_a1	empl >= p_iuse + p_iusex
Sdmxe	*	INDICATOR	p_a1	p_iuse <= empl
Sdmxe	*	INDICATOR	p_a1	p_iusex <= empl
Sdmxe	*	INDICATOR	p_b1	p_awsell <= empl
Sdmxe	*	INDICATOR	p_b7	p_axsell <= empl
Sdmxe	*	INDICATOR	p_b7b1	p_aesell <= empl

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	p_b7b1	p_aesell <= p_awsell + p_axsell
Sdmxe	*	INDICATOR	p_b7b1	p_awsell <= p_aesell
Sdmxe	*	INDICATOR	p_b7b1	p_axsell <= p_aesell
Sdmxe	*	INDICATOR	xc3	e_daown <= e_iuse
Sdmxe	*	INDICATOR	xc3	e_daown = e_dascrm + e_dascrmx
Sdmxe	*	INDICATOR	xc3	e_daown = e_daserp + e_daserpx
Sdmxe	*	INDICATOR	xc3	e_daown = e_dasgov + e_dasgovx
Sdmxe	*	INDICATOR	xc3	e_daown = e_dasloc + e_daslocx
Sdmxe	*	INDICATOR	xc3	e_daown = e_dassat + e_dassatx
Sdmxe	*	INDICATOR	xc3	e_daown = e_dassds + e_dassdsx
Sdmxe	*	INDICATOR	xc3	e_daown = e_dassm + e_dassmx
Sdmxe	*	INDICATOR	xc3	e_daown = e_dasweb + e_daswebx
Sdmxe	*	INDICATOR	xc3	e_daown >= e_dascrm + e_dascrmx
Sdmxe	*	INDICATOR	xc3	e_daown >= e_daserp + e_daserpx
Sdmxe	*	INDICATOR	xc3	e_daown >= e_dasgov + e_dasgovx
Sdmxe	*	INDICATOR	xc3	e_daown >= e_dasloc + e_daslocx
Sdmxe	*	INDICATOR	xc3	e_daown >= e_dassat + e_dassatx
Sdmxe	*	INDICATOR	xc3	e_daown >= e_dassds + e_dassdsx
Sdmxe	*	INDICATOR	xc3	e_daown >= e_dassm + e_dassmx
Sdmxe	*	INDICATOR	xc3	e_daown >= e_dasweb + e_daswebx

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	xc3	e_daownx <= e_iuse
Sdmxe	*	INDICATOR	xc3	e_iuse = e_daown + e_daownx
Sdmxe	*	INDICATOR	xc3	e_iuse >= e_daown + e_daownx
Sdmxe	*	INDICATOR	xc4	e_dasany <= e_daserp + e_dascrm + e_dassm + e_dasweb + e_dasloc + e_dassds + e_dasgov + e_dassat
Sdmxe	*	INDICATOR	xc4	e_dasany <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dasany2 <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dasany2 <= e_dasany
Sdmxe	*	INDICATOR	xc4	e_dasany2 <= e_dasweb + e_dasloc + e_dassds + e_dasgov + e_dassat
Sdmxe	*	INDICATOR	xc4	e_dasany2 <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dascrm <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dascrm <= e_dasany
Sdmxe	*	INDICATOR	xc4	e_dascrm <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dascrmx <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dascrmx <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_daserp <= e_daown
Sdmxe	*	INDICATOR	xc4	e_daserp <= e_dasany
Sdmxe	*	INDICATOR	xc4	e_daserp <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_daserpx <= e_daown
Sdmxe	*	INDICATOR	xc4	e_daserpx <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dasge3 <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dasge3 <= e_dasany
Sdmxe	*	INDICATOR	xc4	e_dasge3 <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dasgov <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dasgov <= e_dasany
Sdmxe	*	INDICATOR	xc4	e_dasgov <= e_dasany2

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	xc4	e_dasgov <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dasgovx <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dasgovx <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dasloc <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dasloc <= e_dasany
Sdmxe	*	INDICATOR	xc4	e_dasloc <= e_dasany2
Sdmxe	*	INDICATOR	xc4	e_dasloc <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_daslocx <= e_daown
Sdmxe	*	INDICATOR	xc4	e_daslocx <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dassat <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dassat <= e_dasany
Sdmxe	*	INDICATOR	xc4	e_dassat <= e_dasany2
Sdmxe	*	INDICATOR	xc4	e_dassat <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dassatx <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dassatx <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dassds <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dassds <= e_dasany
Sdmxe	*	INDICATOR	xc4	e_dassds <= e_dasany2
Sdmxe	*	INDICATOR	xc4	e_dassds <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dassdsx <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dassdsx <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dassm <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dassm <= e_dasany
Sdmxe	*	INDICATOR	xc4	e_dassm <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dassmx <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dassmx <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_dasweb <= e_daown
Sdmxe	*	INDICATOR	xc4	e_dasweb <= e_dasany
Sdmxe	*	INDICATOR	xc4	e_dasweb <= e_dasany2

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	xc4	e_dasweb <= e_iuse
Sdmxe	*	INDICATOR	xc4	e_daswebx <= e_daown
Sdmxe	*	INDICATOR	xc4	e_daswebx <= e_iuse
Sdmxe	*	INDICATOR	xc4a8	e_dasany_ads <= e_ads
Sdmxe	*	INDICATOR	xc4a8	e_dasany_ads <= e_dasany
Sdmxe	*	INDICATOR	xc4a8	e_dasany_ads <= e_iuse
Sdmxe	*	INDICATOR	xc5	e_daext <= e_iuse
Sdmxe	*	INDICATOR	xc5	e_daextx <= e_iuse
Sdmxe	*	INDICATOR	xc5	e_iuse = e_daext + e_daextx
Sdmxe	*	INDICATOR	xc5	e_iuse >= e_daext + e_daextx
Sdmxe	*	INDICATOR	xc5xc3	e_da <= e_daown + e_daext
Sdmxe	*	INDICATOR	xc5xc3	e_da >= e_daext
Sdmxe	*	INDICATOR	xc5xc3	e_da >= e_daown
Sdmxe	*	INDICATOR	xc5xc3e1	e_ai_da <= e_ai_tany
Sdmxe	*	INDICATOR	xc5xc3e1	e_ai_da <= e_da
Sdmxe	*	INDICATOR	xd1	e_cc <= e_iuse
Sdmxe	*	INDICATOR	xd1	e_ccx <= e_iuse
Sdmxe	*	INDICATOR	xd1	e_iuse = e_cc + e_ccx
Sdmxe	*	INDICATOR	xd1	e_iuse >= e_cc + e_ccx
Sdmxe	*	INDICATOR	xd1e1	e_ai_cc <= e_ai_tany
Sdmxe	*	INDICATOR	xd1e1	e_ai_cc <= e_cc
Sdmxe	*	INDICATOR	xd1xc5xc3	e_cc_da <= e_cc
Sdmxe	*	INDICATOR	xd1xc5xc3	e_cc_da <= e_da
Sdmxe	*	INDICATOR	xd2	e_cc = e_cc_pcpu + e_cc_pcpux
Sdmxe	*	INDICATOR	xd2	e_cc = e_cc_pcrm + e_cc_pcrmx
Sdmxe	*	INDICATOR	xd2	e_cc = e_cc_pdb + e_cc_pdbx

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	xd2	e_cc = e_cc_pdev + e_cc_pdevx
Sdmxe	*	INDICATOR	xd2	e_cc = e_cc_pem + e_cc_pemx
Sdmxe	*	INDICATOR	xd2	e_cc = e_cc_perp + e_cc_perpx
Sdmxe	*	INDICATOR	xd2	e_cc = e_cc_pfacc + e_cc_pfaccx
Sdmxe	*	INDICATOR	xd2	e_cc = e_cc_pfil + e_cc_pfilx
Sdmxe	*	INDICATOR	xd2	e_cc = e_cc_psec + e_cc_psecx
Sdmxe	*	INDICATOR	xd2	e_cc = e_cc_psoft + e_cc_psoftx
Sdmxe	*	INDICATOR	xd2	e_cc = e_cc1_pany + e_cc1_pnone
Sdmxe	*	INDICATOR	xd2	e_cc >= e_cc_pcpu + e_cc_pcpux
Sdmxe	*	INDICATOR	xd2	e_cc >= e_cc_pcrm + e_cc_pcrmx
Sdmxe	*	INDICATOR	xd2	e_cc >= e_cc_pdb + e_cc_pdbx
Sdmxe	*	INDICATOR	xd2	e_cc >= e_cc_pdev + e_cc_pdevx
Sdmxe	*	INDICATOR	xd2	e_cc >= e_cc_pem + e_cc_pemx
Sdmxe	*	INDICATOR	xd2	e_cc >= e_cc_perp + e_cc_perpx
Sdmxe	*	INDICATOR	xd2	e_cc >= e_cc_pfacc + e_cc_pfaccx
Sdmxe	*	INDICATOR	xd2	e_cc >= e_cc_pfil + e_cc_pfilx
Sdmxe	*	INDICATOR	xd2	e_cc >= e_cc_psec + e_cc_psecx
Sdmxe	*	INDICATOR	xd2	e_cc >= e_cc_psoft + e_cc_psoftx
Sdmxe	*	INDICATOR	xd2	e_cc >= e_cc1_pany + e_cc1_pnone
Sdmxe	*	INDICATOR	xd2	e_cc_pcpu <= e_cc

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	xd2	e_cc_pcpu <= e_cc_phw
Sdmxe	*	INDICATOR	xd2	e_cc_pcpu <= e_cc1_pany
Sdmxe	*	INDICATOR	xd2	e_cc_pcpux <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_pcrm <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_pcrm <= e_cc1_pany
Sdmxe	*	INDICATOR	xd2	e_cc_pcrm <= e_cc1_si
Sdmxe	*	INDICATOR	xd2	e_cc_pcrmx <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_pdb <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_pdb <= e_cc_pdbfil
Sdmxe	*	INDICATOR	xd2	e_cc_pdb <= e_cc_phw
Sdmxe	*	INDICATOR	xd2	e_cc_pdb <= e_cc1_pany
Sdmxe	*	INDICATOR	xd2	e_cc_pdb <= e_cc1_s
Sdmxe	*	INDICATOR	xd2	e_cc_pdb <= e_cc1_si
Sdmxe	*	INDICATOR	xd2	e_cc_pdbfil <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_pdbfil <= e_cc_pdb + e_cc_pfil
Sdmxe	*	INDICATOR	xd2	e_cc_pdbx <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_pdev <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_pdev <= e_cc1_pany
Sdmxe	*	INDICATOR	xd2	e_cc_pdev <= e_cc1_s
Sdmxe	*	INDICATOR	xd2	e_cc_pdev <= e_cc1_si
Sdmxe	*	INDICATOR	xd2	e_cc_pdevx <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_pem <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_pem <= e_cc1_pany
Sdmxe	*	INDICATOR	xd2	e_cc_pemx <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_perp <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_perp <= e_cc1_pany
Sdmxe	*	INDICATOR	xd2	e_cc_perp <= e_cc1_si
Sdmxe	*	INDICATOR	xd2	e_cc_perpx <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_pfacc <= e_cc

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	xd2	e_cc_pfacc <= e_cc1_pany
Sdmxe	*	INDICATOR	xd2	e_cc_pfacc <= e_cc1_si
Sdmxe	*	INDICATOR	xd2	e_cc_pfaccx <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_pfil <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_pfil <= e_cc_pdbfil
Sdmxe	*	INDICATOR	xd2	e_cc_pfil <= e_cc_phw
Sdmxe	*	INDICATOR	xd2	e_cc_pfil <= e_cc1_pany
Sdmxe	*	INDICATOR	xd2	e_cc_pfilx <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_phw <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_phw <= e_cc_pdb + e_cc_pfil + e_cc_pcpu
Sdmxe	*	INDICATOR	xd2	e_cc_psec <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_psec <= e_cc1_pany
Sdmxe	*	INDICATOR	xd2	e_cc_psec <= e_cc1_s
Sdmxe	*	INDICATOR	xd2	e_cc_psec <= e_cc1_si
Sdmxe	*	INDICATOR	xd2	e_cc_psecx <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_psoft <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc_psoft <= e_cc1_pany
Sdmxe	*	INDICATOR	xd2	e_cc_psoftx <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc1_b <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc1_b <= e_cc_pcrmx
Sdmxe	*	INDICATOR	xd2	e_cc1_b <= e_cc_pdbx
Sdmxe	*	INDICATOR	xd2	e_cc1_b <= e_cc_pdevx
Sdmxe	*	INDICATOR	xd2	e_cc1_b <= e_cc_pem + e_cc_psoft + e_cc_pfil + e_cc_pcpu
Sdmxe	*	INDICATOR	xd2	e_cc1_b <= e_cc_perpx
Sdmxe	*	INDICATOR	xd2	e_cc1_b <= e_cc_pfaccx
Sdmxe	*	INDICATOR	xd2	e_cc1_b <= e_cc_psecx
Sdmxe	*	INDICATOR	xd2	e_cc1_bi <= e_cc

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	xd2	e_cc1_bi <= e_cc_pdbx
Sdmxe	*	INDICATOR	xd2	e_cc1_bi <= e_cc_pdevx
Sdmxe	*	INDICATOR	xd2	e_cc1_bi <= e_cc_pem + e_cc_psoft + e_cc_pfil + e_cc_pcpu + e_cc_pfacc + e_cc_perp + e_cc_pcrm
Sdmxe	*	INDICATOR	xd2	e_cc1_bi <= e_cc_psecx
Sdmxe	*	INDICATOR	xd2	e_cc1_bi = e_cc1_b + e_cc1_i
Sdmxe	*	INDICATOR	xd2	e_cc1_i <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc1_i <= e_cc_pdbx
Sdmxe	*	INDICATOR	xd2	e_cc1_i <= e_cc_pdevx
Sdmxe	*	INDICATOR	xd2	e_cc1_i <= e_cc_pfacc + e_cc_perp + e_cc_pcrm
Sdmxe	*	INDICATOR	xd2	e_cc1_i <= e_cc_psecx
Sdmxe	*	INDICATOR	xd2	e_cc1_pany <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc1_pany <= e_cc_pem + e_cc_psoft + e_cc_pfacc + e_cc_perp + e_cc_pcrm + e_cc_psec + e_cc_pdb + e_cc_pfil + e_cc_pcpu + e_cc_pdev
Sdmxe	*	INDICATOR	xd2	e_cc1_pnone <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc1_pnone <= e_cc_pcpux
Sdmxe	*	INDICATOR	xd2	e_cc1_pnone <= e_cc_pcrmx
Sdmxe	*	INDICATOR	xd2	e_cc1_pnone <= e_cc_pdbx
Sdmxe	*	INDICATOR	xd2	e_cc1_pnone <= e_cc_pdevx
Sdmxe	*	INDICATOR	xd2	e_cc1_pnone <= e_cc_pemx
Sdmxe	*	INDICATOR	xd2	e_cc1_pnone <= e_cc_perpx
Sdmxe	*	INDICATOR	xd2	e_cc1_pnone <= e_cc_pfaccx

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	xd2	e_cc1_pnone <= e_cc_pfilx
Sdmxe	*	INDICATOR	xd2	e_cc1_pnone <= e_cc_psecx
Sdmxe	*	INDICATOR	xd2	e_cc1_pnone <= e_cc_psoftx
Sdmxe	*	INDICATOR	xd2	e_cc1_ps = e_cc_pdev
Sdmxe	*	INDICATOR	xd2	e_cc1_s <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc1_s <= e_cc_psec + e_cc_pdb + e_cc_pdev
Sdmxe	*	INDICATOR	xd2	e_cc1_si <= e_cc
Sdmxe	*	INDICATOR	xd2	e_cc1_si <= e_cc_pfacc + e_cc_perp + e_cc_pcrm + e_cc_psec + e_cc_pdb + e_cc_pdev
Sdmxe	*	INDICATOR	xd2	e_cc1_si = e_cc1_s + e_cc1_i
Sdmxe	*	INDICATOR	xd2xc4e1	e_ai_cc1si_dasany <= e_ai_tany + e_cc1_si + e_dasany
Sdmxe	*	INDICATOR	xd2xc4e1	e_ai_cc1si_dasany2 <= e_ai_cc1si_dasany
Sdmxe	*	INDICATOR	xd2xc4e1	e_ai_cc1si_dasany2 <= e_ai_tany + e_cc1_si + e_dasany2
Sdmxe	*	INDICATOR	xd2xc4e1	e_ai_cc1si_dasge3 <= e_ai_cc1si_dasany
Sdmxe	*	INDICATOR	xd2xc4e1	e_ai_cc1si_dasge3 <= e_ai_tany + e_cc1_si + e_dasge3
Sdmxe	*	INDICATOR	xd2xc4e1	e_ai_tany <= e_ai_cc1si_dasany
Sdmxe	*	INDICATOR	xd2xc4e1	e_ai_tany <= e_ai_cc1si_dasany2
Sdmxe	*	INDICATOR	xd2xc4e1	e_ai_tany <= e_ai_cc1si_dasge3
Sdmxe	*	INDICATOR	xd2xc4e1	e_cc1_si <= e_ai_cc1si_dasany

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	xd2xc4e1	e_cc1_si <= e_ai_cc1si_dasany2
Sdmxe	*	INDICATOR	xd2xc4e1	e_cc1_si <= e_ai_cc1si_dasge3
Sdmxe	*	INDICATOR	xd2xc4e1	e_dasany <= e_ai_cc1si_dasany
Sdmxe	*	INDICATOR	xd2xc4e1	e_dasany2 <= e_ai_cc1si_dasany2
Sdmxe	*	INDICATOR	xd2xc4e1	e_dasge3 <= e_ai_cc1si_dasge3
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1si_da <= e_ai_tany
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1si_da <= e_cc1_si
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1si_da <= e_da
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1si_da_any <= e_ai_tany + e_cc1_si + e_da
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1si_da_any = e_ai_cc1si_da + e_aix_cc1si_da + e_ai_cc1six_da + e_ai_cc1si_dax + e_aix_cc1six_da + e_aix_cc1si_dax + e_ai_cc1six_dax
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1si_dax <= e_ai_tany
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1si_dax <= e_cc1_si
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1si_dax <= e_iuse - e_da
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1six_da <= e_ai_tany
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1six_da <= e_da
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1six_da <= e_iuse - e_cc1_si
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1six_dax <= e_ai_tany
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1six_dax <= e_iuse - e_cc1_si

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_cc1six_dax <= e_iuse - e_da
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_tany <= e_ai_cc1si_da_any
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_ai_tany = e_ai_cc1si_da + e_ai_cc1six_da + e_ai_cc1si_dax + e_ai_cc1six_dax
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_aix_cc1si_da <= e_ai_tx
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_aix_cc1si_da <= e_cc1_si
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_aix_cc1si_da <= e_da
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_aix_cc1si_dax <= e_ai_tx
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_aix_cc1si_dax <= e_cc1_si
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_aix_cc1si_dax <= e_iuse - e_da
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_aix_cc1six_da <= e_ai_tx
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_aix_cc1six_da <= e_da
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_aix_cc1six_da <= e_iuse - e_cc1_si
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_cc1_si <= e_ai_cc1si_da_any
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_cc1_si = e_ai_cc1si_da + e_aix_cc1si_da + e_ai_cc1si_dax + e_aix_cc1si_dax
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_da <= e_ai_cc1si_da_any
Sdmxe	*	INDICATOR	xd2xc5xc3e1	e_da = e_ai_cc1si_da + e_aix_cc1si_da + e_ai_cc1six_da + e_aix_cc1six_da
Sdmxe	*	INDICATOR	zdi	e_di4_gelo <= ent
Sdmxe	*	INDICATOR	zdi	e_di4_gelo = e_di4_lo + e_di4_hi + e_di4_vhi
Sdmxe	*	INDICATOR	zdi	e_di4_gelo >= e_di4_lo + e_di4_hi + e_di4_vhi
Sdmxe	*	INDICATOR	zdi	e_di4_hi <= e_di4_gelo

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	zdi	e_di4_hi <= ent
Sdmxe	*	INDICATOR	zdi	e_di4_lo <= e_di4_gelo
Sdmxe	*	INDICATOR	zdi	e_di4_lo <= ent
Sdmxe	*	INDICATOR	zdi	e_di4_vhi <= e_di4_gelo
Sdmxe	*	INDICATOR	zdi	e_di4_vhi <= ent
Sdmxe	*	INDICATOR	zdi	e_di4_vlo <= ent
Sdmxe	*	INDICATOR	zdi	ent = e_di4_vlo + e_di4_lo + e_di4_hi + e_di4_vhi
Sdmxe	*	INDICATOR	zdi	ent >= e_di4_vlo + e_di4_lo + e_di4_hi + e_di4_vhi
Sdmxe	*	INDICATOR	zdid1	e_di4_hi_secmge1 <= e_di4_hi
Sdmxe	*	INDICATOR	zdid1	e_di4_hi_secmge1 <= e_secmge1
Sdmxe	*	INDICATOR	zdid1	e_di4_hi_secmge3 <= e_di4_hi
Sdmxe	*	INDICATOR	zdid1	e_di4_hi_secmge3 <= e_secmge3
Sdmxe	*	INDICATOR	zdid1	e_di4_hi_secmge5 <= e_di4_hi
Sdmxe	*	INDICATOR	zdid1	e_di4_hi_secmge5 <= e_secmge5
Sdmxe	*	INDICATOR	zdid1	e_di4_hi_secmge7 <= e_di4_hi
Sdmxe	*	INDICATOR	zdid1	e_di4_hi_secmge7 <= e_secmge7
Sdmxe	*	INDICATOR	zdid1	e_di4_lo_secmge1 <= e_di4_lo
Sdmxe	*	INDICATOR	zdid1	e_di4_lo_secmge1 <= e_secmge1
Sdmxe	*	INDICATOR	zdid1	e_di4_lo_secmge3 <= e_di4_lo
Sdmxe	*	INDICATOR	zdid1	e_di4_lo_secmge3 <= e_secmge3
Sdmxe	*	INDICATOR	zdid1	e_di4_lo_secmge5 <= e_di4_lo

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	zdid1	e_di4_lo_secmge5 <= e_secmge5
Sdmxe	*	INDICATOR	zdid1	e_di4_lo_secmge7 <= e_di4_lo
Sdmxe	*	INDICATOR	zdid1	e_di4_lo_secmge7 <= e_secmge7
Sdmxe	*	INDICATOR	zdid1	e_di4_vhi_secmge1 <= e_di4_vhi
Sdmxe	*	INDICATOR	zdid1	e_di4_vhi_secmge1 <= e_secmge1
Sdmxe	*	INDICATOR	zdid1	e_di4_vhi_secmge3 <= e_di4_vhi
Sdmxe	*	INDICATOR	zdid1	e_di4_vhi_secmge3 <= e_secmge3
Sdmxe	*	INDICATOR	zdid1	e_di4_vhi_secmge5 <= e_di4_vhi
Sdmxe	*	INDICATOR	zdid1	e_di4_vhi_secmge5 <= e_secmge5
Sdmxe	*	INDICATOR	zdid1	e_di4_vhi_secmge7 <= e_di4_vhi
Sdmxe	*	INDICATOR	zdid1	e_di4_vhi_secmge7 <= e_secmge7
Sdmxe	*	INDICATOR	zdid1	e_di4_vlo_secmge1 <= e_di4_vlo
Sdmxe	*	INDICATOR	zdid1	e_di4_vlo_secmge1 <= e_secmge1
Sdmxe	*	INDICATOR	zdid1	e_di4_vlo_secmge3 <= e_di4_vlo
Sdmxe	*	INDICATOR	zdid1	e_di4_vlo_secmge3 <= e_secmge3
Sdmxe	*	INDICATOR	zdid1	e_di4_vlo_secmge5 <= e_di4_vlo
Sdmxe	*	INDICATOR	zdid1	e_di4_vlo_secmge5 <= e_secmge5
Sdmxe	*	INDICATOR	zdid1	e_di4_vlo_secmge7 <= e_di4_vlo
Sdmxe	*	INDICATOR	zdid1	e_di4_vlo_secmge7 <= e_secmge7

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	zdid2	e_di4_hi_secawany <= e_di4_hi
Sdmxe	*	INDICATOR	zdid2	e_di4_hi_secawany <= e_secawany
Sdmxe	*	INDICATOR	zdid2	e_di4_lo_secawany <= e_di4_lo
Sdmxe	*	INDICATOR	zdid2	e_di4_lo_secawany <= e_secawany
Sdmxe	*	INDICATOR	zdid2	e_di4_vhi_secawany <= e_di4_vhi
Sdmxe	*	INDICATOR	zdid2	e_di4_vhi_secawany <= e_secawany
Sdmxe	*	INDICATOR	zdid2	e_di4_vlo_secawany <= e_di4_vlo
Sdmxe	*	INDICATOR	zdid2	e_di4_vlo_secawany <= e_secawany
Sdmxe	*	INDICATOR	zdid3	e_di4_hi_secpol2 <= e_di4_hi
Sdmxe	*	INDICATOR	zdid3	e_di4_hi_secpol2 <= e_secpol2
Sdmxe	*	INDICATOR	zdid3	e_di4_lo_secpol2 <= e_di4_lo
Sdmxe	*	INDICATOR	zdid3	e_di4_lo_secpol2 <= e_secpol2
Sdmxe	*	INDICATOR	zdid3	e_di4_vhi_secpol2 <= e_di4_vhi
Sdmxe	*	INDICATOR	zdid3	e_di4_vhi_secpol2 <= e_secpol2
Sdmxe	*	INDICATOR	zdid3	e_di4_vlo_secpol2 <= e_di4_vlo
Sdmxe	*	INDICATOR	zdid3	e_di4_vlo_secpol2 <= e_secpol2
Sdmxe	*	INDICATOR	zdid4	e_di4_hi_secprev_cy <= e_di4_hi
Sdmxe	*	INDICATOR	zdid4	e_di4_hi_secprev_cy <= e_secprev_cy
Sdmxe	*	INDICATOR	zdid4	e_di4_lo_secprev_cy <= e_di4_lo

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	INDICATOR	zdid4	e_di4_lo_secprev_cy <= e_secprev_cy
Sdmxe	*	INDICATOR	zdid4	e_di4_vhi_secprev_cy <= e_di4_vhi
Sdmxe	*	INDICATOR	zdid4	e_di4_vhi_secprev_cy <= e_secprev_cy
Sdmxe	*	INDICATOR	zdid4	e_di4_vlo_secprev_cy <= e_di4_vlo
Sdmxe	*	INDICATOR	zdid4	e_di4_vlo_secprev_cy <= e_secprev_cy
Sdmxe	*	INDICATOR	zdie1	e_ai_tany = e_di4_vlo_ai_tany + e_di4_lo_ai_tany + e_di4_hi_ai_tany + e_di4_vhi_ai_tany
Sdmxe	*	INDICATOR	zdie1	e_ai_tany >= e_di4_vlo_ai_tany + e_di4_lo_ai_tany + e_di4_hi_ai_tany + e_di4_vhi_ai_tany
Sdmxe	*	INDICATOR	zdie1	e_di4_hi_ai_tany <= e_ai_tany
Sdmxe	*	INDICATOR	zdie1	e_di4_hi_ai_tany <= e_di4_hi
Sdmxe	*	INDICATOR	zdie1	e_di4_lo_ai_tany <= e_ai_tany
Sdmxe	*	INDICATOR	zdie1	e_di4_lo_ai_tany <= e_di4_lo
Sdmxe	*	INDICATOR	zdie1	e_di4_vhi_ai_tany <= e_ai_tany
Sdmxe	*	INDICATOR	zdie1	e_di4_vhi_ai_tany <= e_di4_vhi
Sdmxe	*	INDICATOR	zdie1	e_di4_vlo_ai_tany <= e_ai_tany
Sdmxe	*	INDICATOR	zdie1	e_di4_vlo_ai_tany <= e_di4_vlo
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E0T1 <= E0T9
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E0T1 >= 0
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E0T9 = E0T1 + E2T9

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E0T9 >= 0
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E10T249 <= EGE10
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E10T249 = E10T49 + E50T249
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E10T249 >= 0
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E10T49 <= E10T249
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E10T49 <= EGE10
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E10T49 >= 0
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E2T9 <= E0T9
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E2T9 >= 0
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E50T249 <= E10T249
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E50T249 <= EGE10
Sdmxe	*	NUMBER_EMPLOYEES	bySize	E50T249 >= 0
Sdmxe	*	NUMBER_EMPLOYEES	bySize	EGE10 = E10T49 + E50T249 + EGE250
Sdmxe	*	NUMBER_EMPLOYEES	bySize	EGE10 >= 0
Sdmxe	*	NUMBER_EMPLOYEES	bySize	EGE250 <= EGE10
Sdmxe	*	NUMBER_EMPLOYEES	bySize	EGE250 >= 0
Sdmxe	AT	REGIONAL	AT	_Z = AT1 + AT2 + AT3
Sdmxe	AT	REGIONAL	AT	AT1 <= _Z
Sdmxe	AT	REGIONAL	AT	AT1 = AT11 + AT12 + AT13
Sdmxe	AT	REGIONAL	AT	AT11 <= AT1
Sdmxe	AT	REGIONAL	AT	AT12 <= AT1
Sdmxe	AT	REGIONAL	AT	AT13 <= AT1
Sdmxe	AT	REGIONAL	AT	AT2 <= _Z
Sdmxe	AT	REGIONAL	AT	AT2 = AT21 + AT22
Sdmxe	AT	REGIONAL	AT	AT21 <= AT2
Sdmxe	AT	REGIONAL	AT	AT22 <= AT2
Sdmxe	AT	REGIONAL	AT	AT3 <= _Z
Sdmxe	AT	REGIONAL	AT	AT3 = AT31 + AT32 + AT33 + AT34

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	AT	REGIONAL	AT	AT31 <= AT3
Sdmxe	AT	REGIONAL	AT	AT32 <= AT3
Sdmxe	AT	REGIONAL	AT	AT33 <= AT3
Sdmxe	AT	REGIONAL	AT	AT34 <= AT3
Sdmxe	BE	REGIONAL	BE	_Z = BE1 + BE2 + BE3
Sdmxe	BE	REGIONAL	BE	BE1 <= _Z
Sdmxe	BE	REGIONAL	BE	BE1 = BE10
Sdmxe	BE	REGIONAL	BE	BE10 <= BE1
Sdmxe	BE	REGIONAL	BE	BE2 <= _Z
Sdmxe	BE	REGIONAL	BE	BE2 = BE21 + BE22 + BE23 + BE24 + BE25
Sdmxe	BE	REGIONAL	BE	BE21 <= BE2
Sdmxe	BE	REGIONAL	BE	BE22 <= BE2
Sdmxe	BE	REGIONAL	BE	BE23 <= BE2
Sdmxe	BE	REGIONAL	BE	BE24 <= BE2
Sdmxe	BE	REGIONAL	BE	BE25 <= BE2
Sdmxe	BE	REGIONAL	BE	BE3 <= _Z
Sdmxe	BE	REGIONAL	BE	BE3 = BE31 + BE32 + BE33 + BE34 + BE35
Sdmxe	BE	REGIONAL	BE	BE31 <= BE3
Sdmxe	BE	REGIONAL	BE	BE32 <= BE3
Sdmxe	BE	REGIONAL	BE	BE33 <= BE3
Sdmxe	BE	REGIONAL	BE	BE34 <= BE3
Sdmxe	BE	REGIONAL	BE	BE35 <= BE3
Sdmxe	BG	REGIONAL	BG	_Z = BG3 + BG4
Sdmxe	BG	REGIONAL	BG	BG3 <= _Z
Sdmxe	BG	REGIONAL	BG	BG3 = BG31 + BG32 + BG33 + BG34
Sdmxe	BG	REGIONAL	BG	BG31 <= BG3
Sdmxe	BG	REGIONAL	BG	BG32 <= BG3
Sdmxe	BG	REGIONAL	BG	BG33 <= BG3

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	BG	REGIONAL	BG	BG34 <= BG3
Sdmxe	BG	REGIONAL	BG	BG4 <= _Z
Sdmxe	BG	REGIONAL	BG	BG4 = BG41 + BG42
Sdmxe	BG	REGIONAL	BG	BG41 <= BG4
Sdmxe	BG	REGIONAL	BG	BG42 <= BG4
Sdmxe	CY	REGIONAL	CY	CY0 = _Z
Sdmxe	CY	REGIONAL	CY	CY00 = CY0
Sdmxe	CZ	REGIONAL	CZ	CZ0 = _Z
Sdmxe	CZ	REGIONAL	CZ	CZ0 = CZ01 + CZ02 + CZ03 + CZ04 + CZ05 + CZ06 + CZ07 + CZ08
Sdmxe	CZ	REGIONAL	CZ	CZ01 <= CZ0
Sdmxe	CZ	REGIONAL	CZ	CZ02 <= CZ0
Sdmxe	CZ	REGIONAL	CZ	CZ03 <= CZ0
Sdmxe	CZ	REGIONAL	CZ	CZ04 <= CZ0
Sdmxe	CZ	REGIONAL	CZ	CZ05 <= CZ0
Sdmxe	CZ	REGIONAL	CZ	CZ06 <= CZ0
Sdmxe	CZ	REGIONAL	CZ	CZ07 <= CZ0
Sdmxe	CZ	REGIONAL	CZ	CZ08 <= CZ0
Sdmxe	DE	REGIONAL	DE	_Z = DE1 + DE2 + DE3 + DE4 + DE5 + DE6 + DE7 + DE8 + DE9 + DEA + DEB + DEC + DED + DEE + DEF + DEG
Sdmxe	DE	REGIONAL	DE	DE1 <= _Z
Sdmxe	DE	REGIONAL	DE	DE1 = DE11 + DE12 + DE13 + DE14
Sdmxe	DE	REGIONAL	DE	DE11 <= DE1
Sdmxe	DE	REGIONAL	DE	DE12 <= DE1
Sdmxe	DE	REGIONAL	DE	DE13 <= DE1
Sdmxe	DE	REGIONAL	DE	DE14 <= DE1
Sdmxe	DE	REGIONAL	DE	DE2 <= _Z

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	DE	REGIONAL	DE	DE2 = DE21 + DE22 + DE23 + DE24 + DE25 + DE26 + DE27
Sdmxe	DE	REGIONAL	DE	DE21 <= DE2
Sdmxe	DE	REGIONAL	DE	DE22 <= DE2
Sdmxe	DE	REGIONAL	DE	DE23 <= DE2
Sdmxe	DE	REGIONAL	DE	DE24 <= DE2
Sdmxe	DE	REGIONAL	DE	DE25 <= DE2
Sdmxe	DE	REGIONAL	DE	DE26 <= DE2
Sdmxe	DE	REGIONAL	DE	DE27 <= DE2
Sdmxe	DE	REGIONAL	DE	DE3 <= _Z
Sdmxe	DE	REGIONAL	DE	DE3 = DE30
Sdmxe	DE	REGIONAL	DE	DE30 <= DE3
Sdmxe	DE	REGIONAL	DE	DE4 <= _Z
Sdmxe	DE	REGIONAL	DE	DE4 = DE40
Sdmxe	DE	REGIONAL	DE	DE40 <= DE4
Sdmxe	DE	REGIONAL	DE	DE5 <= _Z
Sdmxe	DE	REGIONAL	DE	DE5 = DE50
Sdmxe	DE	REGIONAL	DE	DE50 <= DE5
Sdmxe	DE	REGIONAL	DE	DE6 <= _Z
Sdmxe	DE	REGIONAL	DE	DE6 = DE60
Sdmxe	DE	REGIONAL	DE	DE60 <= DE6
Sdmxe	DE	REGIONAL	DE	DE7 <= _Z
Sdmxe	DE	REGIONAL	DE	DE7 = DE71 + DE72 + DE73
Sdmxe	DE	REGIONAL	DE	DE71 <= DE7
Sdmxe	DE	REGIONAL	DE	DE72 <= DE7
Sdmxe	DE	REGIONAL	DE	DE73 <= DE7
Sdmxe	DE	REGIONAL	DE	DE8 <= _Z
Sdmxe	DE	REGIONAL	DE	DE8 = DE80

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	DE	REGIONAL	DE	DE80 <= DE8
Sdmxe	DE	REGIONAL	DE	DE9 <= _Z
Sdmxe	DE	REGIONAL	DE	DE9 = DE91 + DE92 + DE93 + DE94
Sdmxe	DE	REGIONAL	DE	DE91 <= DE9
Sdmxe	DE	REGIONAL	DE	DE92 <= DE9
Sdmxe	DE	REGIONAL	DE	DE93 <= DE9
Sdmxe	DE	REGIONAL	DE	DE94 <= DE9
Sdmxe	DE	REGIONAL	DE	DEA <= _Z
Sdmxe	DE	REGIONAL	DE	DEA = DEA1 + DEA2 + DEA3 + DEA4 + DEA5
Sdmxe	DE	REGIONAL	DE	DEA1 <= DEA
Sdmxe	DE	REGIONAL	DE	DEA2 <= DEA
Sdmxe	DE	REGIONAL	DE	DEA3 <= DEA
Sdmxe	DE	REGIONAL	DE	DEA4 <= DEA
Sdmxe	DE	REGIONAL	DE	DEA5 <= DEA
Sdmxe	DE	REGIONAL	DE	DEB <= _Z
Sdmxe	DE	REGIONAL	DE	DEB = DEB1 + DEB2 + DEB3
Sdmxe	DE	REGIONAL	DE	DEB1 <= DEB
Sdmxe	DE	REGIONAL	DE	DEB2 <= DEB
Sdmxe	DE	REGIONAL	DE	DEB3 <= DEB
Sdmxe	DE	REGIONAL	DE	DEC <= _Z
Sdmxe	DE	REGIONAL	DE	DEC = DEC0
Sdmxe	DE	REGIONAL	DE	DEC0 <= DEC
Sdmxe	DE	REGIONAL	DE	DED <= _Z
Sdmxe	DE	REGIONAL	DE	DED = DED2 + DED4 + DED5
Sdmxe	DE	REGIONAL	DE	DED2 <= DED
Sdmxe	DE	REGIONAL	DE	DED4 <= DED
Sdmxe	DE	REGIONAL	DE	DED5 <= DED

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	DE	REGIONAL	DE	DEE <= _Z
Sdmxe	DE	REGIONAL	DE	DEE = DEE0
Sdmxe	DE	REGIONAL	DE	DEE0 <= DEE
Sdmxe	DE	REGIONAL	DE	DEF <= _Z
Sdmxe	DE	REGIONAL	DE	DEF = DEF0
Sdmxe	DE	REGIONAL	DE	DEF0 <= DEF
Sdmxe	DE	REGIONAL	DE	DEG <= _Z
Sdmxe	DE	REGIONAL	DE	DEG = DEG0
Sdmxe	DE	REGIONAL	DE	DEG0 <= DEG
Sdmxe	DK	REGIONAL	DK	DK0 = _Z
Sdmxe	DK	REGIONAL	DK	DK0 = DK01 + DK02 + DK03 + DK04 + DK05
Sdmxe	DK	REGIONAL	DK	DK01 <= DK0
Sdmxe	DK	REGIONAL	DK	DK02 <= DK0
Sdmxe	DK	REGIONAL	DK	DK03 <= DK0
Sdmxe	DK	REGIONAL	DK	DK04 <= DK0
Sdmxe	DK	REGIONAL	DK	DK05 <= DK0
Sdmxe	EE	REGIONAL	EE	EE0 = _Z
Sdmxe	EE	REGIONAL	EE	EE00 = EE0
Sdmxe	EL	REGIONAL	EL	_Z = EL3 + EL4 + EL5 + EL6
Sdmxe	EL	REGIONAL	EL	EL3 <= _Z
Sdmxe	EL	REGIONAL	EL	EL3 = EL30
Sdmxe	EL	REGIONAL	EL	EL30 <= EL3
Sdmxe	EL	REGIONAL	EL	EL4 <= _Z
Sdmxe	EL	REGIONAL	EL	EL4 = EL41 + EL42 + EL43
Sdmxe	EL	REGIONAL	EL	EL41 <= EL4
Sdmxe	EL	REGIONAL	EL	EL42 <= EL4
Sdmxe	EL	REGIONAL	EL	EL43 <= EL4
Sdmxe	EL	REGIONAL	EL	EL5 <= _Z

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	EL	REGIONAL	EL	EL5 = EL51 + EL52 + EL53 + EL54
Sdmxe	EL	REGIONAL	EL	EL51 <= EL5
Sdmxe	EL	REGIONAL	EL	EL52 <= EL5
Sdmxe	EL	REGIONAL	EL	EL53 <= EL5
Sdmxe	EL	REGIONAL	EL	EL54 <= EL5
Sdmxe	EL	REGIONAL	EL	EL6 <= _Z
Sdmxe	EL	REGIONAL	EL	EL6 = EL61 + EL62 + EL63 + EL64 + EL65
Sdmxe	EL	REGIONAL	EL	EL61 <= EL6
Sdmxe	EL	REGIONAL	EL	EL62 <= EL6
Sdmxe	EL	REGIONAL	EL	EL63 <= EL6
Sdmxe	EL	REGIONAL	EL	EL64 <= EL6
Sdmxe	EL	REGIONAL	EL	EL65 <= EL6
Sdmxe	ES	REGIONAL	ES	_Z = ES1 + ES2 + ES3 + ES4 + ES5 + ES6 + ES7
Sdmxe	ES	REGIONAL	ES	ES1 <= _Z
Sdmxe	ES	REGIONAL	ES	ES1 = ES11 + ES12 + ES13
Sdmxe	ES	REGIONAL	ES	ES11 <= ES1
Sdmxe	ES	REGIONAL	ES	ES12 <= ES1
Sdmxe	ES	REGIONAL	ES	ES13 <= ES1
Sdmxe	ES	REGIONAL	ES	ES2 <= _Z
Sdmxe	ES	REGIONAL	ES	ES2 = ES21 + ES22 + ES23 + ES24
Sdmxe	ES	REGIONAL	ES	ES21 <= ES2
Sdmxe	ES	REGIONAL	ES	ES22 <= ES2
Sdmxe	ES	REGIONAL	ES	ES23 <= ES2
Sdmxe	ES	REGIONAL	ES	ES24 <= ES2
Sdmxe	ES	REGIONAL	ES	ES3 <= _Z
Sdmxe	ES	REGIONAL	ES	ES3 = ES30
Sdmxe	ES	REGIONAL	ES	ES30 <= ES3

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	ES	REGIONAL	ES	ES4 <= _Z
Sdmxe	ES	REGIONAL	ES	ES4 = ES41 + ES42 + ES43
Sdmxe	ES	REGIONAL	ES	ES41 <= ES4
Sdmxe	ES	REGIONAL	ES	ES42 <= ES4
Sdmxe	ES	REGIONAL	ES	ES43 <= ES4
Sdmxe	ES	REGIONAL	ES	ES5 <= _Z
Sdmxe	ES	REGIONAL	ES	ES5 = ES51 + ES52 + ES53
Sdmxe	ES	REGIONAL	ES	ES51 <= ES5
Sdmxe	ES	REGIONAL	ES	ES52 <= ES5
Sdmxe	ES	REGIONAL	ES	ES53 <= ES5
Sdmxe	ES	REGIONAL	ES	ES6 <= _Z
Sdmxe	ES	REGIONAL	ES	ES6 = ES61 + ES62 + ES63 + ES64
Sdmxe	ES	REGIONAL	ES	ES61 <= ES6
Sdmxe	ES	REGIONAL	ES	ES62 <= ES6
Sdmxe	ES	REGIONAL	ES	ES63 <= ES6
Sdmxe	ES	REGIONAL	ES	ES64 <= ES6
Sdmxe	ES	REGIONAL	ES	ES7 <= _Z
Sdmxe	ES	REGIONAL	ES	ES7 = ES70
Sdmxe	ES	REGIONAL	ES	ES70 <= ES7
Sdmxe	FI	REGIONAL	FI	_Z = FI1 + FI2
Sdmxe	FI	REGIONAL	FI	FI1 <= _Z
Sdmxe	FI	REGIONAL	FI	FI1 = FI19 + FI1B + FI1C + FI1D
Sdmxe	FI	REGIONAL	FI	FI19 <= FI1
Sdmxe	FI	REGIONAL	FI	FI1B <= FI1
Sdmxe	FI	REGIONAL	FI	FI1C <= FI1
Sdmxe	FI	REGIONAL	FI	FI1D <= FI1
Sdmxe	FI	REGIONAL	FI	FI2 <= _Z
Sdmxe	FI	REGIONAL	FI	FI2 = FI20

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	FI	REGIONAL	FI	FI20 <= FI2
Sdmxe	FR	REGIONAL	FR	_Z = FR1 + FRB + FRC + FRD + FRE + FRF + FRG + FRH + FRI + FRJ + FRK + FRL + FRM + FRY
Sdmxe	FR	REGIONAL	FR	FR1 <= _Z
Sdmxe	FR	REGIONAL	FR	FR1 = FR10
Sdmxe	FR	REGIONAL	FR	FR10 <= FR1
Sdmxe	FR	REGIONAL	FR	FRB <= _Z
Sdmxe	FR	REGIONAL	FR	FRB = FRB0
Sdmxe	FR	REGIONAL	FR	FRB0 <= FRB
Sdmxe	FR	REGIONAL	FR	FRC <= _Z
Sdmxe	FR	REGIONAL	FR	FRC = FRC1 + FRC2
Sdmxe	FR	REGIONAL	FR	FRC1 <= FRC
Sdmxe	FR	REGIONAL	FR	FRC2 <= FRC
Sdmxe	FR	REGIONAL	FR	FRD <= _Z
Sdmxe	FR	REGIONAL	FR	FRD = FRD1 + FRD2
Sdmxe	FR	REGIONAL	FR	FRD1 <= FRD
Sdmxe	FR	REGIONAL	FR	FRD2 <= FRD
Sdmxe	FR	REGIONAL	FR	FRE <= _Z
Sdmxe	FR	REGIONAL	FR	FRE = FRE1 + FRE2
Sdmxe	FR	REGIONAL	FR	FRE1 <= FRE
Sdmxe	FR	REGIONAL	FR	FRE2 <= FRE
Sdmxe	FR	REGIONAL	FR	FRF <= _Z
Sdmxe	FR	REGIONAL	FR	FRF = FRF1 + FRF2 + FRF3
Sdmxe	FR	REGIONAL	FR	FRF1 <= FRF
Sdmxe	FR	REGIONAL	FR	FRF2 <= FRF
Sdmxe	FR	REGIONAL	FR	FRF3 <= FRF
Sdmxe	FR	REGIONAL	FR	FRG <= _Z
Sdmxe	FR	REGIONAL	FR	FRG = FRG0

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	FR	REGIONAL	FR	FRG0 <= FRG
Sdmxe	FR	REGIONAL	FR	FRH <= _Z
Sdmxe	FR	REGIONAL	FR	FRH = FRH0
Sdmxe	FR	REGIONAL	FR	FRH0 <= FRH
Sdmxe	FR	REGIONAL	FR	FRI <= _Z
Sdmxe	FR	REGIONAL	FR	FRI = FRI1 + FRI2 + FRI3
Sdmxe	FR	REGIONAL	FR	FRI1 <= FRI
Sdmxe	FR	REGIONAL	FR	FRI2 <= FRI
Sdmxe	FR	REGIONAL	FR	FRI3 <= FRI
Sdmxe	FR	REGIONAL	FR	FRJ <= _Z
Sdmxe	FR	REGIONAL	FR	FRJ = FRJ1 + FRJ2
Sdmxe	FR	REGIONAL	FR	FRJ1 <= FRJ
Sdmxe	FR	REGIONAL	FR	FRJ2 <= FRJ
Sdmxe	FR	REGIONAL	FR	FRK <= _Z
Sdmxe	FR	REGIONAL	FR	FRK = FRK1 + FRK2
Sdmxe	FR	REGIONAL	FR	FRK1 <= FRK
Sdmxe	FR	REGIONAL	FR	FRK2 <= FRK
Sdmxe	FR	REGIONAL	FR	FRL <= _Z
Sdmxe	FR	REGIONAL	FR	FRL = FRL0
Sdmxe	FR	REGIONAL	FR	FRL0 <= FRL
Sdmxe	FR	REGIONAL	FR	FRM <= _Z
Sdmxe	FR	REGIONAL	FR	FRM = FRM0
Sdmxe	FR	REGIONAL	FR	FRM0 <= FRM
Sdmxe	FR	REGIONAL	FR	FRY <= _Z
Sdmxe	FR	REGIONAL	FR	FRY = FRY1 + FRY2 + FRY3 + FRY4 + FRY5
Sdmxe	FR	REGIONAL	FR	FRY1 <= FRY
Sdmxe	FR	REGIONAL	FR	FRY2 <= FRY
Sdmxe	FR	REGIONAL	FR	FRY3 <= FRY
Sdmxe	FR	REGIONAL	FR	FRY4 <= FRY

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	FR	REGIONAL	FR	FRY5 <= FRY
Sdmxe	HR	REGIONAL	HR	HR0 = _Z
Sdmxe	HR	REGIONAL	HR	HR0 = HR02 + HR03 + HR05 + HR06
Sdmxe	HR	REGIONAL	HR	HR02 <= HR0
Sdmxe	HR	REGIONAL	HR	HR03 <= HR0
Sdmxe	HR	REGIONAL	HR	HR05 <= HR0
Sdmxe	HR	REGIONAL	HR	HR06 <= HR0
Sdmxe	HU	REGIONAL	HU	_Z = HU1 + HU2 + HU3
Sdmxe	HU	REGIONAL	HU	HU1 <= _Z
Sdmxe	HU	REGIONAL	HU	HU1 = HU11 + HU12
Sdmxe	HU	REGIONAL	HU	HU11 <= HU1
Sdmxe	HU	REGIONAL	HU	HU12 <= HU1
Sdmxe	HU	REGIONAL	HU	HU2 <= _Z
Sdmxe	HU	REGIONAL	HU	HU2 = HU21 + HU22 + HU23
Sdmxe	HU	REGIONAL	HU	HU21 <= HU2
Sdmxe	HU	REGIONAL	HU	HU22 <= HU2
Sdmxe	HU	REGIONAL	HU	HU23 <= HU2
Sdmxe	HU	REGIONAL	HU	HU3 <= _Z
Sdmxe	HU	REGIONAL	HU	HU3 = HU31 + HU32 + HU33
Sdmxe	HU	REGIONAL	HU	HU31 <= HU3
Sdmxe	HU	REGIONAL	HU	HU32 <= HU3
Sdmxe	HU	REGIONAL	HU	HU33 <= HU3
Sdmxe	IE	REGIONAL	IE	IE0 = _Z
Sdmxe	IE	REGIONAL	IE	IE0 = IE04 + IE05 + IE06
Sdmxe	IE	REGIONAL	IE	IE04 <= IE0
Sdmxe	IE	REGIONAL	IE	IE05 <= IE0
Sdmxe	IE	REGIONAL	IE	IE06 <= IE0
Sdmxe	IS	REGIONAL	IS	IS0 = _Z

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	IS	REGIONAL	IS	IS00 = IS0
Sdmxe	IT	REGIONAL	IT	_Z = ITC + ITF + ITG + ITH + ITI
Sdmxe	IT	REGIONAL	IT	ITC <= _Z
Sdmxe	IT	REGIONAL	IT	ITC = ITC1 + ITC2 + ITC3 + ITC4
Sdmxe	IT	REGIONAL	IT	ITC1 <= ITC
Sdmxe	IT	REGIONAL	IT	ITC2 <= ITC
Sdmxe	IT	REGIONAL	IT	ITC3 <= ITC
Sdmxe	IT	REGIONAL	IT	ITC4 <= ITC
Sdmxe	IT	REGIONAL	IT	ITF <= _Z
Sdmxe	IT	REGIONAL	IT	ITF = ITF1 + ITF2 + ITF3 + ITF4 + ITF5 + ITF6
Sdmxe	IT	REGIONAL	IT	ITF1 <= ITF
Sdmxe	IT	REGIONAL	IT	ITF2 <= ITF
Sdmxe	IT	REGIONAL	IT	ITF3 <= ITF
Sdmxe	IT	REGIONAL	IT	ITF4 <= ITF
Sdmxe	IT	REGIONAL	IT	ITF5 <= ITF
Sdmxe	IT	REGIONAL	IT	ITF6 <= ITF
Sdmxe	IT	REGIONAL	IT	ITG <= _Z
Sdmxe	IT	REGIONAL	IT	ITG = ITG1 + ITG2
Sdmxe	IT	REGIONAL	IT	ITG1 <= ITG
Sdmxe	IT	REGIONAL	IT	ITG2 <= ITG
Sdmxe	IT	REGIONAL	IT	ITH <= _Z
Sdmxe	IT	REGIONAL	IT	ITH = ITH1 + ITH2 + ITH3 + ITH4 + ITH5
Sdmxe	IT	REGIONAL	IT	ITH1 <= ITH
Sdmxe	IT	REGIONAL	IT	ITH2 <= ITH
Sdmxe	IT	REGIONAL	IT	ITH3 <= ITH
Sdmxe	IT	REGIONAL	IT	ITH4 <= ITH
Sdmxe	IT	REGIONAL	IT	ITH5 <= ITH

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	IT	REGIONAL	IT	ITI <= _Z
Sdmxe	IT	REGIONAL	IT	ITI = ITI1 + ITI2 + ITI3 + ITI4
Sdmxe	IT	REGIONAL	IT	ITI1 <= ITI
Sdmxe	IT	REGIONAL	IT	ITI2 <= ITI
Sdmxe	IT	REGIONAL	IT	ITI3 <= ITI
Sdmxe	IT	REGIONAL	IT	ITI4 <= ITI
Sdmxe	LT	REGIONAL	LT	LT0 = _Z
Sdmxe	LT	REGIONAL	LT	LT0 = LT01 + LT02
Sdmxe	LT	REGIONAL	LT	LT01 <= LT0
Sdmxe	LT	REGIONAL	LT	LT02 <= LT0
Sdmxe	LU	REGIONAL	LU	LU0 = _Z
Sdmxe	LU	REGIONAL	LU	LU00 = LU0
Sdmxe	LV	REGIONAL	LV	LV0 = _Z
Sdmxe	LV	REGIONAL	LV	LV00 = LV0
Sdmxe	MT	REGIONAL	MT	MT0 = _Z
Sdmxe	MT	REGIONAL	MT	MT00 = MT0
Sdmxe	NL	REGIONAL	NL	_Z = NL1 + NL2 + NL3 + NL4
Sdmxe	NL	REGIONAL	NL	NL1 <= _Z
Sdmxe	NL	REGIONAL	NL	NL1 = NL11 + NL12 + NL13
Sdmxe	NL	REGIONAL	NL	NL11 <= NL1
Sdmxe	NL	REGIONAL	NL	NL12 <= NL1
Sdmxe	NL	REGIONAL	NL	NL13 <= NL1
Sdmxe	NL	REGIONAL	NL	NL2 <= _Z
Sdmxe	NL	REGIONAL	NL	NL2 = NL21 + NL22 + NL23
Sdmxe	NL	REGIONAL	NL	NL21 <= NL2
Sdmxe	NL	REGIONAL	NL	NL22 <= NL2
Sdmxe	NL	REGIONAL	NL	NL23 <= NL2
Sdmxe	NL	REGIONAL	NL	NL3 <= _Z

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	NL	REGIONAL	NL	NL3 = NL31 + NL32 + NL33 + NL34
Sdmxe	NL	REGIONAL	NL	NL31 <= NL3
Sdmxe	NL	REGIONAL	NL	NL32 <= NL3
Sdmxe	NL	REGIONAL	NL	NL33 <= NL3
Sdmxe	NL	REGIONAL	NL	NL34 <= NL3
Sdmxe	NL	REGIONAL	NL	NL4 <= _Z
Sdmxe	NL	REGIONAL	NL	NL4 = NL41 + NL42
Sdmxe	NL	REGIONAL	NL	NL41 <= NL4
Sdmxe	NL	REGIONAL	NL	NL42 <= NL4
Sdmxe	NO	REGIONAL	NO	NO0 = _Z
Sdmxe	NO	REGIONAL	NO	NO0 = NO02 + NO06 + NO07 + NO08 + NO09 + NO0A + NO0B
Sdmxe	NO	REGIONAL	NO	NO02 <= NO0
Sdmxe	NO	REGIONAL	NO	NO06 <= NO0
Sdmxe	NO	REGIONAL	NO	NO07 <= NO0
Sdmxe	NO	REGIONAL	NO	NO08 <= NO0
Sdmxe	NO	REGIONAL	NO	NO09 <= NO0
Sdmxe	NO	REGIONAL	NO	NO0A <= NO0
Sdmxe	NO	REGIONAL	NO	NO0B <= NO0
Sdmxe	PL	REGIONAL	PL	_Z = PL2 + PL4 + PL5 + PL6 + PL7 + PL8 + PL9
Sdmxe	PL	REGIONAL	PL	PL2 <= _Z
Sdmxe	PL	REGIONAL	PL	PL2 = PL21 + PL22
Sdmxe	PL	REGIONAL	PL	PL21 <= PL2
Sdmxe	PL	REGIONAL	PL	PL22 <= PL2
Sdmxe	PL	REGIONAL	PL	PL4 <= _Z
Sdmxe	PL	REGIONAL	PL	PL4 = PL41 + PL42 + PL43
Sdmxe	PL	REGIONAL	PL	PL41 <= PL4
Sdmxe	PL	REGIONAL	PL	PL42 <= PL4

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	PL	REGIONAL	PL	PL43 <= PL4
Sdmxe	PL	REGIONAL	PL	PL5 <= _Z
Sdmxe	PL	REGIONAL	PL	PL5 = PL51 + PL52
Sdmxe	PL	REGIONAL	PL	PL51 <= PL5
Sdmxe	PL	REGIONAL	PL	PL52 <= PL5
Sdmxe	PL	REGIONAL	PL	PL6 <= _Z
Sdmxe	PL	REGIONAL	PL	PL6 = PL61 + PL62 + PL63
Sdmxe	PL	REGIONAL	PL	PL61 <= PL6
Sdmxe	PL	REGIONAL	PL	PL62 <= PL6
Sdmxe	PL	REGIONAL	PL	PL63 <= PL6
Sdmxe	PL	REGIONAL	PL	PL7 <= _Z
Sdmxe	PL	REGIONAL	PL	PL7 = PL71 + PL72
Sdmxe	PL	REGIONAL	PL	PL71 <= PL7
Sdmxe	PL	REGIONAL	PL	PL72 <= PL7
Sdmxe	PL	REGIONAL	PL	PL8 <= _Z
Sdmxe	PL	REGIONAL	PL	PL8 = PL81 + PL82 + PL84
Sdmxe	PL	REGIONAL	PL	PL81 <= PL8
Sdmxe	PL	REGIONAL	PL	PL82 <= PL8
Sdmxe	PL	REGIONAL	PL	PL84 <= PL8
Sdmxe	PL	REGIONAL	PL	PL9 <= _Z
Sdmxe	PL	REGIONAL	PL	PL9 = PL91 + PL92
Sdmxe	PL	REGIONAL	PL	PL91 <= PL9
Sdmxe	PL	REGIONAL	PL	PL92 <= PL9
Sdmxe	PT	REGIONAL	PT	_Z = PT1 + PT2 + PT3
Sdmxe	PT	REGIONAL	PT	PT1 <= _Z
Sdmxe	PT	REGIONAL	PT	PT1 = PT11 + PT15 + PT16 + PT17 + PT18
Sdmxe	PT	REGIONAL	PT	PT11 <= PT1
Sdmxe	PT	REGIONAL	PT	PT15 <= PT1
Sdmxe	PT	REGIONAL	PT	PT16 <= PT1

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	PT	REGIONAL	PT	PT17 <= PT1
Sdmxe	PT	REGIONAL	PT	PT18 <= PT1
Sdmxe	PT	REGIONAL	PT	PT2 <= _Z
Sdmxe	PT	REGIONAL	PT	PT2 = PT20
Sdmxe	PT	REGIONAL	PT	PT20 <= PT2
Sdmxe	PT	REGIONAL	PT	PT3 <= _Z
Sdmxe	PT	REGIONAL	PT	PT3 = PT30
Sdmxe	PT	REGIONAL	PT	PT30 <= PT3
Sdmxe	RO	REGIONAL	RO	_Z = RO1 + RO2 + RO3 + RO4
Sdmxe	RO	REGIONAL	RO	RO1 <= _Z
Sdmxe	RO	REGIONAL	RO	RO1 = RO11 + RO12
Sdmxe	RO	REGIONAL	RO	RO11 <= RO1
Sdmxe	RO	REGIONAL	RO	RO12 <= RO1
Sdmxe	RO	REGIONAL	RO	RO2 <= _Z
Sdmxe	RO	REGIONAL	RO	RO2 = RO21 + RO22
Sdmxe	RO	REGIONAL	RO	RO21 <= RO2
Sdmxe	RO	REGIONAL	RO	RO22 <= RO2
Sdmxe	RO	REGIONAL	RO	RO3 <= _Z
Sdmxe	RO	REGIONAL	RO	RO3 = RO31 + RO32
Sdmxe	RO	REGIONAL	RO	RO31 <= RO3
Sdmxe	RO	REGIONAL	RO	RO32 <= RO3
Sdmxe	RO	REGIONAL	RO	RO4 <= _Z
Sdmxe	RO	REGIONAL	RO	RO4 = RO41 + RO42
Sdmxe	RO	REGIONAL	RO	RO41 <= RO4
Sdmxe	RO	REGIONAL	RO	RO42 <= RO4
Sdmxe	SE	REGIONAL	SE	_Z = SE1 + SE2 + SE3
Sdmxe	SE	REGIONAL	SE	SE1 <= _Z
Sdmxe	SE	REGIONAL	SE	SE1 = SE11 + SE12
Sdmxe	SE	REGIONAL	SE	SE11 <= SE1

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	SE	REGIONAL	SE	SE12 <= SE1
Sdmxe	SE	REGIONAL	SE	SE2 <= _Z
Sdmxe	SE	REGIONAL	SE	SE2 = SE21 + SE22 + SE23
Sdmxe	SE	REGIONAL	SE	SE21 <= SE2
Sdmxe	SE	REGIONAL	SE	SE22 <= SE2
Sdmxe	SE	REGIONAL	SE	SE23 <= SE2
Sdmxe	SE	REGIONAL	SE	SE3 <= _Z
Sdmxe	SE	REGIONAL	SE	SE3 = SE31 + SE32 + SE33
Sdmxe	SE	REGIONAL	SE	SE31 <= SE3
Sdmxe	SE	REGIONAL	SE	SE32 <= SE3
Sdmxe	SE	REGIONAL	SE	SE33 <= SE3
Sdmxe	SI	REGIONAL	SI	SI0 = _Z
Sdmxe	SI	REGIONAL	SI	SI0 = SI03 + SI04
Sdmxe	SI	REGIONAL	SI	SI03 <= SI0
Sdmxe	SI	REGIONAL	SI	SI04 <= SI0
Sdmxe	SK	REGIONAL	SK	SK0 = _Z
Sdmxe	SK	REGIONAL	SK	SK0 = SK01 + SK02 + SK03 + SK04
Sdmxe	SK	REGIONAL	SK	SK01 <= SK0
Sdmxe	SK	REGIONAL	SK	SK02 <= SK0
Sdmxe	SK	REGIONAL	SK	SK03 <= SK0
Sdmxe	SK	REGIONAL	SK	SK04 <= SK0
Sdmxe	BA	UNIT_MEASURE	Cur	EUR = XDC / 1.95583
Sdmxe	BG	UNIT_MEASURE	Cur	EUR = XDC / 1.9558
Sdmxe	CZ	UNIT_MEASURE	Cur	EUR = XDC / 24.004
Sdmxe	DK	UNIT_MEASURE	Cur	EUR = XDC / 7.4509
Sdmxe	HR	UNIT_MEASURE	Cur	EUR = XDC / 7.5345
Sdmxe	HU	UNIT_MEASURE	Cur	EUR = XDC / 381.85
Sdmxe	IS	UNIT_MEASURE	Cur	EUR = XDC / 149.13
Sdmxe	MK	UNIT_MEASURE	Cur	EUR = XDC / 61.5587

Survey	Country	KeyName	KeyGroup	Message
Sdmxe	NO	UNIT_MEASURE	Cur	EUR = XDC / 11.4248
Sdmxe	PL	UNIT_MEASURE	Cur	EUR = XDC / 4.5420
Sdmxe	RO	UNIT_MEASURE	Cur	EUR = XDC / 4.9467
Sdmxe	RS	UNIT_MEASURE	Cur	EUR = XDC / 117.2513
Sdmxe	SE	UNIT_MEASURE	Cur	EUR = XDC / 11.4788
Sdmxe	TR	UNIT_MEASURE	Cur	EUR = XDC / 25.7597
Sdmxe	UK	UNIT_MEASURE	Cur	EUR = XDC / 0.86979

Annex 3 – Metadata reporting template

- *Full view* -

INFOSOC_ETNSI_A_2024

National Reference Metadata in SIMS structure for INFOSOC Enterprises

Compiling agency:

Eurostat metadata

Reference metadata

1. Contact
2. Metadata update
3. Statistical presentation
4. Unit of measure
5. Reference Period
6. Institutional Mandate
7. Confidentiality
8. Release policy
9. Frequency of dissemination
10. Accessibility and clarity
11. Quality management
12. Relevance
13. Accuracy
14. Timeliness and punctuality
15. Coherence and comparability
16. Cost and Burden
17. Data revision
18. Statistical processing
19. Comment
Related Metadata
Annexes (including footnotes)

For any question on data and metadata, please contact: Eurostat user support

1. Contact

1.1. Contact organisation

1.2. Contact organisation unit	
1.3. Contact name	
1.4. Contact person function	
1.5. Contact mail address	
1.6. Contact email address	
1.7. Contact phone number	
1.8. Contact fax number	

2. Metadata update

2.1. Metadata last certified	
2.2. Metadata last posted	
2.3. Metadata last update	

3. Statistical presentation

3.1. Data description

Data on the Information and Communication Technologies (ICT) usage and e-commerce in enterprises are survey data. They are collected by the National Statistical Institutes or Ministries and are in principle based on Eurostat's **annual model questionnaires** on **ICT usage and e-commerce in enterprises**.

The legal basis for ICT enterprise statistics for survey year 2024 is Commission Implementing Regulation (EU) **2023/1507** of **20 July 2023** laying down the technical specifications of data requirements for the topic 'ICT usage and e-commerce' for the reference year 2024.Large part of the data collected is used to measure the progress in the implementation of one of the main political priorities of the European Commission for 2019 to 2024 – A Europe fit for the digital age.

Furthermore, ICT data facilitate the monitoring of the EU's digital targets for 2030, set by the Digital Decade Policy Programme. Four of the key performance indicators (KPIs) of the current programme stem from the statistics for which the implementing and

delegated acts are enclosed for adoption: Artificial Intelligence, cloud, big data (data analytics) and the digital intensity index for businesses (DII) - a composite indicator reflecting the digital transformation of business. **The aim of the European survey on ICT usage and e-commerce in enterprises is to collect and disseminate harmonised and comparable information at European level.**

Name of data collection

Please complete for the Metadata report.

3.2. Classification system

NACE Rev.2 2008 - Statistical classification of economic activities

3.3. Coverage - sector

All economic activities in the scope of Annex of the Commission Regulation are intended to be included in the general survey, covering enterprises with 10 or more employees and self-employed persons. These activities are:

- Section C – “Manufacturing”
- Section D, E – “Electricity, gas, steam and air conditioning supply”, “Water supply, sewerage, waste management and remediation activities”
- Section F – “Construction”
- Section G – “Wholesale and retail trade; repair of motor vehicles and motorcycles”
- Section H – “Transportation and storage”
- Section I – “Accommodation and food service activities”
- Section J – “Information and communication”
- Section L – “Real estate activities”
- Section M – “Professional, scientific and technical activities”
- Section N – "Administrative and support service activities"
- Group 95.1 – “Repair of computers and communication equipment”.

For micro-enterprises see the sub-concepts below.

3.3.1. Coverage-sector economic activity for micro-enterprises - All NACE Rev. 2 categories are covered

3.3.2. Coverage sector economic activity for micro-enterprises - If not all activities were covered, which ones were covered?

3.4. Statistical concepts and definitions

The model questionnaire on ICT usage and e-commerce in enterprises provides a large variety of variables covering among others the following topics:

- Access to and use of the Internet
- E-commerce and e-business
- ICT specialists and skills
- ICT security
- Artificial Intelligence.

The annual model questionnaires and the European businesses statistics compliers' manual for ICT usage and e-commerce in enterprises comprise definitions and explanations regarding the topics of the survey.

3.5. Statistical unit

Please complete for the Metadata report.

3.6. Statistical population

Target Population

As required by Annex of the Commission Implementing Regulation, enterprises with 10 or more employees and self-employed persons are covered by the survey.

For micro-enterprises see the sub-concepts below.

3.6.1. Coverage of micro-enterprises

3.6.2. Breakdown between size classes [0 to 1] and [2 to 9]

3.6.3. If for micro-enterprises different size delimitation was used, please indicate it.

Please complete for the Metadata report.

3.7. Reference area

Detailed information on the provision of data on NUTS 2 regional level is available in "Annex I. Completeness".

Please complete for the Metadata report.

3.8. Coverage - Time

Years 2023 and 2024.

3.9. Base period

Not applicable

4. Unit of measure

Percentages of enterprises, Percentages of turnover, Percentages of employees and self-employed persons, Million euro (for selected indicators in some countries).

5. Reference period

Please complete for the Metadata report.

6. Institutional Mandate

6.1. Institutional Mandate - legal acts and other agreements

Complementary national legislation constituting the legal basis for the survey on the use of ICT in enterprises:

Please complete for the Metadata report.

6.2. Institutional Mandate - data sharing

Please complete for the Metadata report.

7. Confidentiality

7.1. Confidentiality - policy

Regulation (EC) No 223/2009 on European statistics (recital 24 and Article 20(4)) of 11 March 2009 (OJ L 87, p. 164), stipulates the need to establish common principles and guidelines ensuring the confidentiality of data used for the production of European statistics and the access to those confidential data with due account for technical developments and the requirements of users in a democratic society.

At national level:

Please complete for the Quality report.

7.2. Confidentiality - data treatment

Data are transmitted via eDAMIS (encrypted) and delivered to a secure environment where they are treated. Flags are added for confidentiality in case results must not be disclosed.

At national level:

Please complete for the Quality report.

8. Release policy

8.1. Release calendar

Please complete for the Quality report.

8.2. Release calendar access

Please complete for the Quality report.

8.3. Release policy - user access

Please complete for the Quality report.

9. Frequency of dissemination

Annual

10. Accessibility and clarity

Top

10.1. Dissemination format - News release

Please complete for the Quality report.

10.2. Dissemination format - Publications

Please complete for the Quality report.

10.3. Dissemination format - online database

See detailed section 10.3.1.

10.3.1. Data tables - consultations

Results for selected variables collected in the framework of this survey are available for all participating countries on Digital economy and society of Eurostat website.

At national level:

Please complete for the Quality report.

10.4. Dissemination format - microdata access

Please complete for the Quality report.

10.5. Dissemination format - other

Not requested

10.5.1. Metadata - consultations

Not requested

10.6. Documentation on methodology

The European businesses statistics compilers' manual for ICT usage and e-commerce in enterprises provides guidelines and clarifications for the implementation of the surveys.

At national level:

Please complete for the Quality report.

10.6.1. Metadata completeness - rate

Not requested

10.7. Quality management - documentation

Please complete for the Quality report.

11. Quality management

Top

11.1. Quality assurance

The European businesses statistics compliers' manual for ICT usage and e-commerce in enterprises provides guidelines and standards for the implementation of the surveys. It is updated every year according to the changed contents of the model questionnaires.

At national level:

Please complete for the Metadata report.

11.2. Quality management - assessment

At European level, the recommended use of the annual Eurostat model questionnaire aims at improving comparability of the results among the countries that conduct the survey on ICT usage and e-commerce in enterprises. Moreover, the European businesses statistics compilers' manual for ICT usage and e-commerce in enterprises provides guidelines and clarifications for the implementation of the surveys.

At national level:

Please complete for the Metadata report.

12. Relevance

Top

12.1. Relevance - User Needs

Please complete for the Metadata report.

12.2. Relevance - User Satisfaction

Please complete for the Metadata report.

12.3. Completeness

Detailed information is available in " Annex I. Completeness " - related to questionnaire, coverage, additional questions, regional data.

12.3.1. Data completeness - rate

Not requested

13. Accuracy

13.1. Accuracy - overall

Comments on reliability and representativeness of results and completeness of dataset

These comments reflect overall standard errors reported for the indicators and breakdowns in section 13.2.1 (Sampling error - indicators) and the rest of the breakdowns for national and European aggregates, as well as other accuracy measurements. The estimated standard error should not exceed 2pp for the overall proportions and should not exceed 5pp for the proportions related to the different subgroups of the population (for those NACE aggregates for the calculation and dissemination of national aggregates). If problems were found, these could have implications for future surveys (e.g. need to improve sampling design, to increase sample sizes, to increase the response rates).

More detailed information is available in " Annex II. Accuracy " - related to European aggregates, comments on reliability and use of flag.

13.2. Sampling error

For calculation of the standard error see 13.2.1.1.

13.2.1. Sampling error - indicators

Standard error (for selected indicators and breakdowns)

Precision measures related to variability due to sampling, unit non-response (the size of the subset of respondents is smaller than the size of the original sample) and other (imputation for item non-response, calibration etc.) are not (yet) required from the MS for all indicators. Eurostat will make basic assumptions to compute these measures for all indicators produced (e.g. stratified random sampling assuming as strata the crossing of the variables "Number of employees and self-employed persons" and "Economic Activity" as it was defined in the 3 tables of section 18.1).

More detailed information is available in" Annex III. Sample and standard error tables 2024 " – worksheets starting with "Standard error".

13.2.1.1. Sampling error indicator calculation

a) Name and brief description of the applied estimation approach

b) Basic formula

c) Main reference in the literature

d) How has the stratification been taken into account?

e) Which strata have been considered?

13.3. Non-sampling error
See detailed sections below.
13.3.1. Coverage error
See concept 18.1.1. A) Description of frame population.
13.3.1.1. Over-coverage - rate
Please complete for the Quality report.
13.3.1.2. Common units - proportion
Not requested
13.3.2. Measurement error
Please complete for the Quality report.
13.3. Non-sampling error
See detailed sections below.
13.3.1. Coverage error
See concept 18.1.1. A) Description of frame population.
13.3.1.1. Over-coverage - rate
Please complete for the Quality report.
13.3.1.2. Common units - proportion
Not requested
13.3.2. Measurement error
Please complete for the Quality report.
13.3.3. Non response error

See detailed sections below.

13.3.3.1. Unit non-response - rate

See detailed sub-concepts below.

13.3.3.1.1. Unit response

The following table contains the number of units (i.e. enterprises), by type of response to the survey and by the percentage of these values in relation to the gross sample size.

Type of response	Enterprises			
	0-9 (or 2-9) employees and self-employed persons		10 or more employees and self-employed persons	
	Number	%	Number	%
Gross sample size (as in section 3.1 C)		**100%**		**100%**
1. Response (questionnaires returned by the enterprise)				
1.1 Used for tabulation and grossing up (Net sample or Final Sample; as in section 3.1 D)				
1.2 Not used for tabulation				
1.2.1 Out of scope (deaths, misclassified originally in the target population, etc.)				
1.2.2 Other reasons (e.g. unusable questionnaire)				
2. Non-response (e.g. non returned mail, returned mail by post office)				

Comments on unit response, if unit response is below 60%

13.3.3.1.2. Methods used for minimizing unit non-response
Please complete for the Quality report.

13.3.3.1.3. Methods used for unit non-response treatment	
1. No treatment for unit non-response	
2. Treatment by re-weighting	
2.1 Re-weighting by the sampling design strata considering that non-response is ignorable inside each stratum (the naïve model)	
2.2 Re-weighting by identified response homogeneity groups (created using sample-level information)	
2.3 Re-weighting through calibration/post-stratification (performed using population information) by the groups used for calibration/post-stratification	
3. Treatment by imputation (done distinctly for each variable/item)	
4. Method(s) and the model(s) corresponding to the above or other method(s) used for the treatment of unit non-response. (e.g. Re-weighting using Horvitz-Thompson estimator, ratio estimator or regression estimator, auxiliary variables)	

13.3.3.1.4. Assessment of unit non-response bias
Please complete for the Quality report.

13.3.3.2. Item non-response - rate
Please complete for the Quality report.

13.3.3.2.1. Methods used for item non-response treatment	
1. No treatment for item non-response	
2. Deductive imputation An exact value can be derived as a known function of other characteristics.	
3. Deterministic imputation (e.g. mean/median, mean/median by class, ratio-based, regression-	

based, single donor nearest-neighbour) Deterministic imputation leads to estimators with no random component, that is, if the imputation were to be reconducted, the outcome would be the same.	
4. Random imputation (e.g. hot-deck, cold-deck) Random imputation leads to estimators with a random component, that is, if the imputation were re-conducted, it would have led to a different result.	
5. Re-weighting	
6. Multiple imputation In multiple imputation each missing value is replaced (instead of a single value) with a set of plausible values that represent the uncertainty of the right value to impute. Multiple imputation methods offer the possibility of deriving variance estimators by taking imputation into account. The incorporation of imputation into the variance can be easily derived based on variability of estimates among the multiply imputed data sets.	
7. Method(s) and the model(s) corresponding to the above or other method(s) used for the treatment of item non-response.	

13.3.3.2.2. Questions or items with item response rates below 90% and other comments

Other comments relating to the item non-response

Additional issues concerning "item non-response" calculation (e.g. method used in nation

Questions and items with low response rates (cut-off value is 90%) and item non-respons

13.3.4. Processing error

Please complete for the Quality report.

13.3.5. Model assumption error

Not requested

14. Timeliness and punctuality

Top

14.1. Timeliness

See detailed section below.

14.1.1. Time lag - first result

Not applicable

14.1.2. Time lag - final result

Data are to be delivered to Eurostat in the fourth quarter of the reference year (due date for the finalised dataset is 5th October). European results are released before the end of Data are to be delivered to Eurostat in the fourth quarter of the reference year (due date for the finalised dataset is 5th October). European results are released before the end of the survey year or in the beginning of the year following the survey year (T=reference year, T+0 for indicators referring to the current year, T+12 months for other indicators referring to the previous year e.g. e-commerce).

At national level:

Please complete for the Quality report.

14.2. Punctuality

See detailed section below.

14.2.1. Punctuality - delivery and publication

Please complete for the Quality report.

Data were delivered to Eurostat on ……; ……. days before (after) the deadline.

15. Coherence and comparability

Top

15.1. Comparability - geographical

The model questionnaire is generally used by the countries that conduct the survey on ICT usage and e-commerce in enterprises. Due to (small) differences in translation, in the used survey vehicle, in non-response treatment or different routing through the questionnaire, some results for some countries may be of reduced comparability. In these cases, notes are added in the data**.**

Detailed information on differences in the wording of the questions in the national questionnaires is available in " Annex I. Completeness " - worksheets related to questionnaire, coverage, additional questions.

Comparability between regions:

Please complete for the Quality report.

Detailed information on the provision of data on NUTS 2 regional level is available in "Annex I. Completeness" – worksheets related to regional data.

15.1.1. Asymmetry for mirror flow statistics - coefficient

Not applicable

15.2. Comparability - over time

See section below.

15.2.1. Length of comparable time series

The length of comparable time series depends on the module and the variable considered within each survey module. Additional information is available in annexes attached to the European metadata.

Please complete for the Quality report.

15.3. Coherence - cross domain

Not applicable

15.3.1. Coherence - sub annual and annual statistics

Not applicable

15.3.2. Coherence - National Accounts

Not applicable

15.4. Coherence - internal

Not applicable

16. Cost and Burden

Top

Cost:

Please complete for the Quality report.

Burden:

Please complete for the Quality report

17. Data revision

17.1. Data revision - policy

Please complete for the Quality report.

17.2. Data revision - practice

Please complete for the Quality report.

17.2.1. Data revision - average size

Not requested

18. Statistical processing

Top

18.1. Source data

A) Frame population description and distribution

For more information see concept 18.1.1.

B) Sampling design - Sampling method

Description of the sampling method used (e.g. stratified random sample, quota sampling, cluster sampling; one-stage or two-stage sampling) and information which variables were used to stratify, the categories of those variables, in particular for the NACE Rev. 2 categories related to the "possible calculation of European aggregates", and the final number of strata:

Please complete for the Metadata report.

***C)* Gross sample distribution**

More detailed information is available in " Annex III. Sample and standard error tables 2024 " (Worksheet: GROSS SAMPLE)

D) Net sample distribution

More detailed information is available in " Annex III. Sample and standard error tables 2024 " (Worksheet: NET SAMPLE)

18.1.1. Population frame

A) Description of frame population

a) When was the sample for the ICT usage and e-commerce in enterprise survey drawn?	
b) Last update of the Business register that was used for drawing the sample of enterprises for the survey:	
c) Indication if the frame population is the same as, or is in some way coordinated with, the one used for the Structural Business Statistics (different snapshots):	
d) Description if different frames are used during different stages of the statistical process (e.g. frame used for sampling vs. frame used for grossing up):	
e) Indication the shortcomings in terms of timeliness (e.g. time lag between last update e of the sampling frame and the moment of the actual sampling), geographical coverage, coverage of different subpopulations, data available etc., and any measures taken to correct it, for this survey.	

B) Frame population distribution

More detailed information is available in " Annex III. Sample and standard error tables 2024 " (Worksheet: FRAME POPULATION)

18.2. Frequency of data collection

Annual

18.3. Data collection

See detailed sections below.

18.3.1. Survey period

Survey / Collection	Date of sending out questionnaires	Date of reception of the last questionnaire treated
General survey		
Micro-enterprises		

18.3.2. Survey vehicle – general survey
General survey - Stand-alone survey
18.3.3. Survey vehicle – micro-enterprises
The collection of micro-enterprises was integrated with the general survey
18.3.4. Survey type
Please complete for the Metadata report.
18.3.5. Survey participation
Mandatory
18.4. Data validation
Please complete for the Quality report.
18.5. Data compilation
Grossing-up procedures
Please complete for the Quality report.
18.5.1. Imputation - rate
Please complete for the Quality report
18.6. Adjustment
Not applicable
18.6.1. Seasonal adjustment
Not applicable

19. Comment

Top

Problems encountered and lessons to be learnt:

Please complete for the Quality report.

19.1. Documents
Questionnaire in national language
Questionnaire in English (if available)

National reports on methodology (if available)
Analysis of key results, backed up by tables and graphs in English (if available)
Other Annexes

Related metadata

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Annexes

Top

Annex I. Completeness 2024

Annex II. Accuracy 2024

Annex III. Sample and standard error tables 2024

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PDF: KS-01-25-012-EN-N

European business statistics compiler's manualf or ICT usage and e-commerce in enterprises

The purpose of this publication is to provide the compilers of European statistics on information and communication technology (ICT) usage and e-commerce in enterprises with clarifications on how to apply the EU legal provisions. With the help of explanations and legal references, the Manual is meant to serve as a practical reference document for National Statistical Authorities.

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